

Report



SPDC-BFPC UPSTREAM GAS SUPPLY, GATHERING, EVACUATION & PROCESSING (UGS-GEP) PROJECT

Process Simulation Study – Hysys

Gas Gathering & Evacuation Pipelines

Prepared for:

Brass Fertilizer & Petrochemical Company
Limited

Doc. No. EA/SPDC-BFPC UGS-GEP/PRC-RPT-001-A01

Endorsed by:

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28th February 2017

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SUMMARY PAGE

Client: BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED.		
Project: SPDC-BFPC UGS-GEP	Title: Process Simulation Study – Hysys	
Facilities: Gas Gathering & Evacuation Pipelines	Doc No.: EA/SPDC-BFPC UGS-GEP/PRC-RPT-001-A01	

The administration of this document is in respect of the Hysys Simulation of the Gas Gathering & Evacuation Pipelines aspect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project.

The document defines the basic design concepts, issues, and considerations in the subject matter of process simulation, and the attendant specifications of line sizes that will form the basis of the project's Front End and Engineering Design (FEED).

It is also an input & accompaniment to the conceptual study and basic design reports & presentations to DPR as procedural and statutory requirements in the process of securing regulators' approval to progress to the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and if further information becomes available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEY WORDS: BASIS FOR DESIGN, PIPELINE, PIPELINE MANIFOLD PLATFORM, GAS PROCESSING PLANT, ROW

REFERENCES

Document No.	Description
TO-D-AA-00001	Basis for Design (Trewan)
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Gas Gathering & Surface Facilities Concept Screening Study
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-002	Process Simulation Study – PIPESIM
EA/SPDC-BFPC UGS-GEP/PRC-RPT-003	Process Simulation Study – OLGA
EA/SPDC-BFPC UGS-GEP/PRC-RPT-004	Process Simulation Study - Gas Processing Plant

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organisational

AKAB	Akabel
BBOU	Bubouwe
BFPC	Brass Fertilizer Petrochemical Company Limited
EA	Epic Atlantic Limited
ELEP	Elepa
NUNR	Nun River
OPUG	Opugbene
SEIB	Seibou
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

B/L	Bulkline
CF-MH	Central Field MF Header
CTP	Custody Transfer Point
E-MF	Evacuation Manifold
EOS	Equation of State
F-MF	Field Manifold
GEP	Gathering, Evacuation and Processing
GSPA	Gas Supply and Purchase Agreement
NGL	Natural Gas Liquids
NAG	Non Associated Gas

Technical Terms (Cont'd)

OML	Oil Mining License
OSD	On-Stream Date
PR	Peng-Robinson
S-MF	Satellite Manifold
T/L	Trunkline
UGS	Upstream Gas Supply

Units of Measurements

km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day
°C	Degree Celsius
bar	Bar

1. Administration

This Process Simulation Study is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project. It stems from the Concept Select Study on the Gas Gathering Concept, and presents the selected design concept as basis for the Basic Design of the project.

It is also an input and accompaniment to the Conceptual Study & Basic Design Memoranda to be presented to DPR as statutory requirements in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

1.1. Validity & Change Management

The report has relied and used data and reference documents provided by client. Some of the data are preliminary, based on limited information, but to a large extent, however, are adequate for this phase of work: - conceptual study and basic design.

For example, analogue PVT data was used in the absence of actual PVT, and preliminary pipeline distances were estimated based on desktop route selection study. These design data, and the consequent simulation results will invariably need to be updated during FEED, by which time actual PVT data would have been obtained, and the FP survey done to establish actual distances and elevations of the pipeline routes.

Similarly, the project timelines as set out in the execution schedule was predicated on FID date of June 2018, and that will correspondingly change with actual FID date.

Also, the security situation in the region with its wide spate of vandalism may dictate that certain design provisions be changed during the FEED. The situation will continue to be monitored, and if still perceived to be a cog in the wheel of innovation, some of the ideal designs will be replaced with what will be practically suitable for the situation.

For example, in as much as it is feasible to lay cable from the IPP to the remote manifold platforms to transmit electricity to motorise the platform valves in the Manifold Platforms Power Supply Options, or install solar systems on the platforms, there will be no need to implement such options if the facilities (cables & solar systems) will eventually be vandalised in no time. Instead, it will be rather more appealing to limit the process option to manual valve operations, without the features of valve motorisation that will require electricity.

Epic Atlantic therefore reserves the right to update accordingly, the opinion set out in this report, as further information and / or refinement become available.

1.2. Background & Introduction

The SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project is constituted by Brass Fertilizer & Petrochemical Company Limited (BFPC), in partnership with The Shell Petroleum Development Company of Nigeria (SPDC).

The project scope includes Gas Gathering Pipelines, and a Gas Process Processing Plant that will provide processed gas as feedstock to a Petrochemical Plant and a Power Plant. The Gas Processing Plant will process natural gas gathered from five SPDC fields – Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene - in OMLs 33, 32, & 36.

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. In this respect, BFPC have concluded the Gas Supply & Processing Agreement (GSPA) with SPDC for 270 MMscfd of the natural gas feedstock. The Plants (Gas Plant, Petrochemical Plant, and Power Plant) will all be located in the same complex at Akabel, Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean.

The Petrochemical Plant will comprise a Methanol Plant of 1.75 MTPA capacity, and a Urea Plant of 2.6 MTPA total capacity, built in two phases of 1.3 MTPA each. It will also have storage capacity for 150,000 Metric Tonnes of Methanol and 100,000 Metric Tonnes of Urea. The facilities complex will also comprise a captive export jetty of 4 berths for up to 35,000 DWT vessels.

The current on-stream date for the project is December 2020, and to achieve that, FID is planned for June, 2018. In pursuit of these business objectives, BFPC have virtually concluded Project Management & EPC contracts with EIL, Saipem, and TCC, who are internationally renowned contractors in the implementation of the subject projects type.

The Gas Processing Plant, and the associated Gas Gathering System / Pipelines which will transport wellhead gas from the SPDC fields to the Gas Processing Plant, constitute the key upstream enabler facilities for the realisation of the objective supply of processed gas supply as feedstock to the Petrochemical Plant for the product manufacturing of methanol & urea, and to the Power Plant for power generation.

1.3. Scope of Document

The scope of the document covers the Process Simulation Study with Hysys and its evolution of results for the various pipeline segments that will be involved.

✚ Phase 1 Gas Delivery: BBOU – ELEP – AKAB Pipeline Network

✚ Phase 2 Gas Delivery Simulation: NUNR – ELEP – AKAB Pipeline Network

- ✚ Phase 3 Gas Delivery Simulation: SEIB – NUNR – ELEP – AKAB Pipeline Network
- ✚ Phase 4 Gas Delivery Simulation: OPUG – BBOU – ELEP / SEIB – NUNR –ELEP – AKAB Pipeline Network
- ✚ Flow Assurance for each production phase
- ✚ Pressure Profiles

2. Introduction

This Conceptual Study captures Upstream Gas Supply (UGS) Scope by SPDC and Gathering, Evacuation & Processing (GEP) by Brass fertilizer & Petrochemical Company.

The Gas Supply, is expected from SPDC fields i.e. Bubouwe, Elepa, Nun River, Seibou & Opugbene which are within OML 33, 32, & 36 and are phased (phase 1-4) with respective maturing on-stream dates (OSD). Phase 1 – Bubouwe & Elepa with 2020 OSD, Phase 2 - Nun River with 2033 OSD, Phase 3 – Seibou with 2039 OSD, Phase 4 – Opugbene with an OSD beyond 2045.

Gathering and Evacuation; shall be achieved at the custody transfer point (CTP) through an interconnection between SPDC field manifold (F-MF) and BFPC field manifold (F-MF), which is then routed to a BFPC satellite manifold and further to an evacuation manifold for onward delivery to the gas plant.

Processing operations projected at the Akabel Gas plant includes; fluid separation, NGL fractionation, and condensate stabilization. Also inclusive, is a petrochemical and power plants.

2.1. Objective

The objective of this simulation; is to study the gas gathering concept(s), line sizing, flow assurance, and gas deliverability to Akabel Gas plant at required pressure.

2.2. Simulation Scope

-  Generate mass and material balances for flow sources in-situ the pipelines and at the delivery outlets.
-  Establish appropriate line sizes suitable for the anticipated peak flow from the fields.
-  Flow assurance (parameter checks; which includes pressure, temperature, velocity and hydrate formation).
-  Develop a suitable operating flow conditions (pressure and temperature).

3. Process Description

The purpose of BFPC pipeline gathering networks is to deliver gas to AKAB Gas Plant for further processing.

Non Associated Gas (NAG) will be produced from BBOU, ELEP, NUNR, SEIB and OPUG fields in phases and delivered to the gas plant.

The overall process description is as shown below.

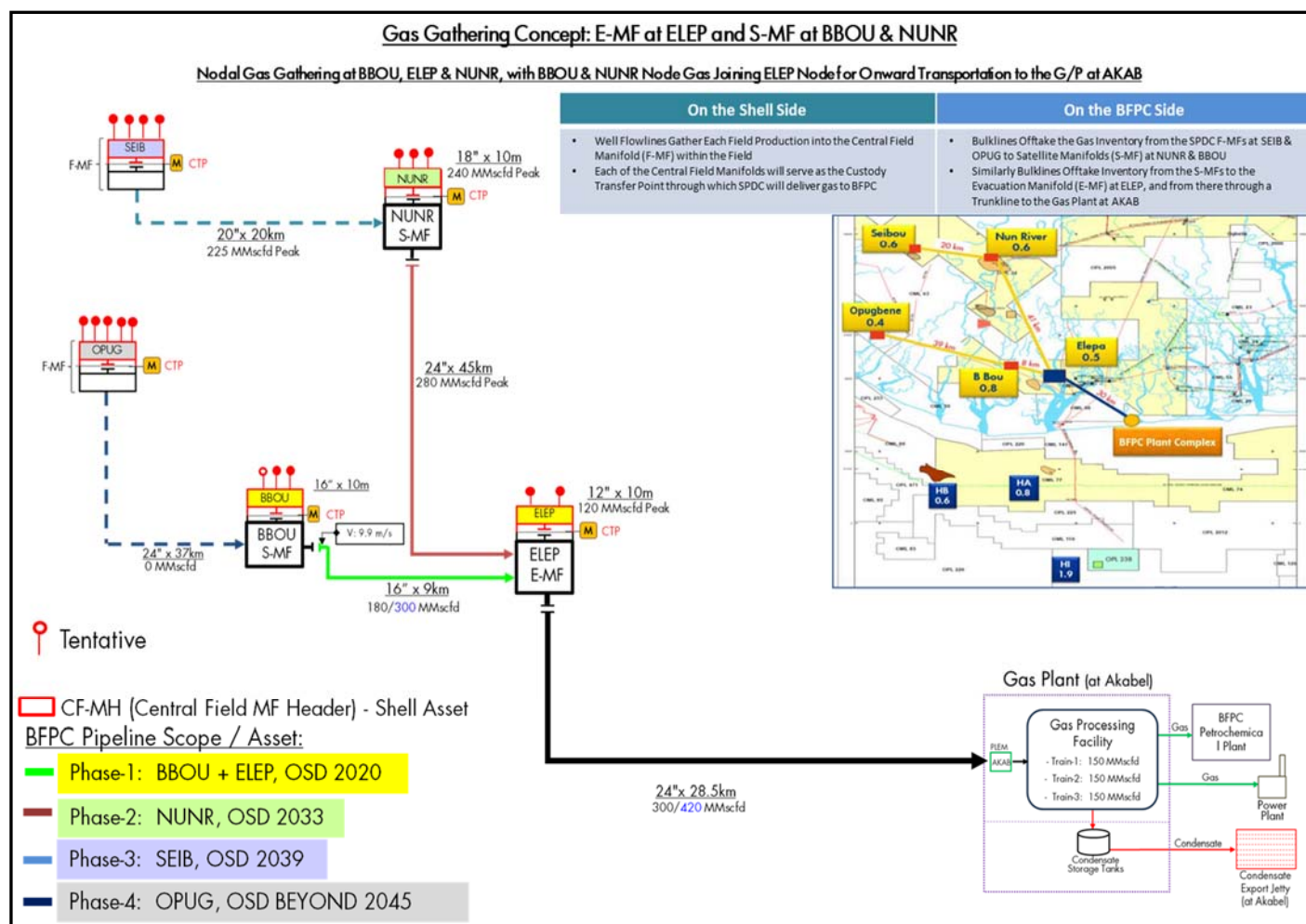


Figure 3.1: Overall Phase Flow Schematics

The line sizing was based on the peak production for all scenarios except for the sensitivity case that tested the robustness of the line to handle 420 MMscfd.

Phase-1 Gas Supply Plan; is expected from BBOU and ELEP fields. Three scenarios were explored:

- ✚ A scenario whereby BBOU field supply wholly 300 MMscfd of gas needed at AKAB.
- ✚ And second scenario; whereby BBOU and ELEP fields make joint contribution of 180 MMscfd & 120 MMscfd of gas respectively for AKAB delivery.
- ✚ In addition, is a third scenario which was undertaken to examine ELEP E-MF to AKAB T/L capacity to support up to 420 MMscfd of gas flow.

At BBOU field, BFPC F-MF off-takes gas from SPDC F-MF with a short spool at the CTP and deliver it through a 16" x 9 km bulk-line (B/L) to ELEP evacuation manifold (E-MF).

Also at the ELEP field, ELEP E-MF off-takes gas from SPDC F-MF with a short 12" x 10 m spool at the CTP. The combined BBOU and ELEP field produced gas at ELEP E-MF is then routed to AKAB Gas Plant through a 24" x 28.5 km trunk-line with minimum delivery pressure of 60 - 65 barg as required for the Gas Plant operations.

Phase-2 Gas Supply Plan; at this stage, NUNR field has matured and it is projected to produce 240 MMscfd of gas at the peak production.

The matured NUNR field contributes 240 MMscfd of gas and the balance of 60 MMscfd of gas is expected from ELEP field wells to complement up to the required 300 MMscfd of gas needed at AKAB.

At NUNR field, a BFPC satellite manifold (S-MF) off-takes gas from SPDC F-MF with a short 18" x 10 m spool at the CTP and deliver's it through a 24" x 45 km trunk-line to ELEP E-MF.

ELEP field; flows into ELEP E-MF as detailed above in phase-1. Both combined streams of NUNR and ELEP then, flows to Akabel Gas plant through the existing 24" x 28.5 km trunk-line at 60 - 65 barg.

Phase-3 Gas Supply Plan; SEIB field wells in-flow signalled the phase-3 gas supply, with a volume of 225 MMscfd as its peak production.

At SEIB field, a BFPC F-MF off-takes gas from SPDC F-MF with a short spool at the CTP and delivers it through a 20" x 20 km bulk-line (B/L) to NUNR S-MF.

NUNR field contribute 55 MMscfd of gas to commingle with in-coming SEIB Field flow at NUNR S-MF. A total 280 MMscfd combined SEIB and NUNR gas, flows through 24" x 45 km trunk-line to ELEP E-MF. ELEP field then contributes 20 MMscfd of gas into ELEP E-MF to make-up 300 MMscfd to be delivered to AKAB Gas Plant.

Phase-4 Gas Supply Plan: at maturity, OPUG field is expected to flow 200 MMscfd of gas at the peak production.

The existing, BBOU F-MF to ELEP E-MF bulk-line is employed to service OPUG field flow.

At OPUG field, BFPC F-MF off-takes gas from SPDC F-MF with a short spool at the CTP and deliver's it through a 24" x 37 km bulk-line (B/L) which is to be adjoined/linked to the existing 16" bulk-line which runs from BBOU F-MF to ELEP E-MF.

This leg (i.e. BBOU F-MF to ELEP E-MF B/L) which may have been redundant before the OSD of OPUG is deemed to be re-used to service OPUG field flow to ELEP E-MF.

SEIB and NUNR fields have a make-up contribution of 80 MMscfd and 20 MMscfd respectively. A total sum-up of 300 MMscfd of gas is anticipated to be delivered to AKAB.

2033 Gas Production/Supply: (for Water Production Sensitivity)

This is a water sensitivity study, in-which 2033 (on-set of phase 2) with its associated water production is explored for studying fluid behaviour under water production circumstance. BBOU, ELEP and NUNR fields contributes 207 MMscfd, 70 MMscfd, 22 MMscfd of gas and 9,727 stb/day, 12,600 stb/day, 0 stb/day of produced water respectively for the sum total flow in this scenario.

4. Simulation Basis and Assumption

Base Assumption

The pipeline temperature ambient medium was taken as 25 °C. Also, it is assumed for this conceptual study, that there are no elevation changes across the pipeline route.

Software

The hydraulics process simulation study was undertaken using Aspen Hysys v7.3.

Water saturation extension was explored to determine the saturated water in the gas.

Simulation Property Package

The thermodynamic equation of state (EOS) used for the process simulation is; Peng-Robinson (PR) fluid property package.

Production Forecast

The phase flow distribution used for this simulation study; is as highlighted in the production forecast below.

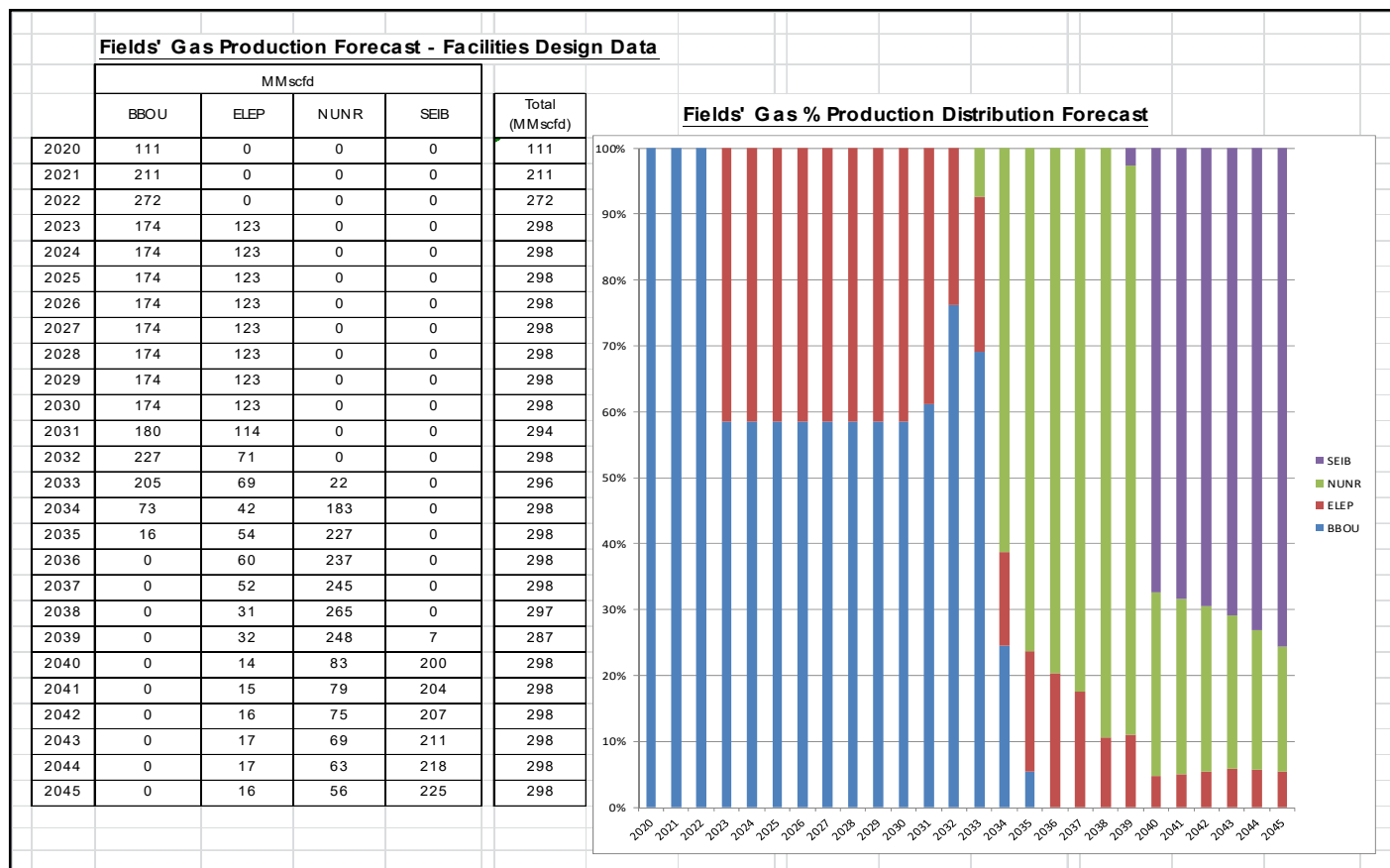


Figure 4.1: Production Forecast

Input Data and Reservoir Conditions

Table 4.1 shows BBOU, ELEP, NUNR, SEIB, and OPUG field reservoir conditions and fluid composition in mole fraction basis as provided by SPDC.

Similarly, table 4.2 shows water saturated compositions of BBOU, ELEP, NUNR, SEIB, and OPUG field compositions.

For this simulation purpose, Bonny PVT was used for BBOU stream. For ELEP stream, ELEP Upper U4 sand was used. NUNR H4 was used for NUNR stream. The average of SEIB G1.1M & F9.3M was used for SEIB stream. Similarly, the average of OPUG F2000 & F2001 was used for OPUG stream.

Table 4.1: BBOU, ELEP, NUNR, SEIB, and OPUG Reservoir Compositions & Conditions

Field's Gas Compositions																			
BBO U				ELEP				NUNR				SEIB				O PUG			
Component	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid				
N2	0.07	0.00	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.07	0.12	0.00	0.10				
CO2	0.59	0.00	0.59	2.30	0.00	2.22	0.00	0.00	0.77	1.63	0.00	1.55	2.53	0.00	2.23				
H2S				0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00				
C1	93.26	0.00	92.92	84.63	0.00	81.66	0.00	0.00	72.83	83.24	0.00	78.62	77.45	0.00	68.00				
C2	3.17	0.01	3.16	6.35	0.03	6.13	0.00	0.00	7.68	6.27	0.28	5.93	9.06	0.13	7.97				
C3	1.01	0.03	1.01	2.67	0.27	2.59	0.00	0.00	4.85	3.49	1.88	3.40	5.40	0.53	4.81				
iC4	0.61	0.07	0.61	0.76	0.37	0.75	0.00	0.00	1.46	1.20	1.55	1.22	2.13	0.76	1.96				
nC4	0.44	0.07	0.44	0.82	0.54	0.81	0.00	0.00	1.97	1.31	2.36	1.38	1.00	1.59	1.07				
iC5	0.25	0.16	0.25	0.43	0.87	0.45	0.00	0.00	0.95	0.73	2.34	0.83	1.07	1.79	1.16				
nC5	0.19	0.16	0.19	0.32	0.68	0.33	0.00	0.00	0.76	0.50	1.96	0.58	0.53	1.97	0.69				
C6	0.22	1.88	0.23	0.59	2.26	0.65	0.00	0.00	1.03	0.68	4.76	0.92	0.53	5.23	1.07				
C7	0.17	6.21	0.19	0.69	6.75	0.90	0.00	0.00	1.41	0.62	9.71	1.13	0.17	8.16	1.07				
C8	0.02	6.61	0.04	0.44	12.76	0.87	0.00	0.00	1.61	0.28	15.65	1.14	0.05	16.75	1.92				
C9	0.00	7.13	0.03	0.00	10.87	0.38	0.00	0.00	0.90	0.00	11.71	0.65	0.00	8.81	0.99				
C10	0.00	5.70	0.02	0.00	8.62	0.30	0.00	0.00	0.65	0.00	9.14	0.51	0.00	5.87	0.67				
C11	0.00	7.83	0.03	0.00	6.64	0.23	0.00	0.00	0.47	0.00	5.68	0.30	0.00	6.44	0.79				
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	0.00	2.66	0.00	33.00	1.82	0.00	42.00	5.55				
Total	100.00	100.00	100.00	100.00	100.00	100.00	0.00	0.00	100.00	100.00	100.00	100.01	100.00	100.01	100.00				
Molar ratio				0.9650	0.0350	1.0000				0.0720	0.9280	1.0000							

BBO U				ELEP				NUNR				SEIB				ORUG AVG.			
	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid	Flashed Gas	Flashed Liq	Monophasic Reservoir Fluid				
Reservoir Pressure, psia			3,246			6,910			7,259			6,533			5,176				
Reservoir Temp, °C			78			123			102			117			117				
Mole %																			
C7+	0.19	97.62	0.53	1.13	94.98	4.41	0.00	0.00	7.70	0.90	84.89	5.54	0.21	88.01	10.98				
C12+	0.00	64.14	0.22	0.00	49.34	1.73	0.00	0.00	2.66	0.00	33.00	1.82	0.00	42.00	5.55				
Mola mass	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol	g/ mol				
C7+	100.89	167.69	143.17	105.67	186.72	166.70	104.82	176.38	169.26	104.48	174.90	164.20	98.81	159.87	158.76				
C12+	-	191.60	191.60																
fluid	17.97	165.61	18.47	20.84	181.08	26.44	21.82	160.91	31.84	21.38	158.85	29.00	22.29	149.53	38.11				
Density	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³	kg/ m³				
C7+						832			830			837							
C12+									813										
fluid		811			828						817			805					
Rel Density (air = 1)	0.62			0.719		0.913	0.753		1.099	0.738			0.770						
CGR			stb/ M M scfd			stb/ M M scfd			stb/ M M scfd			stb/ M M scfd			stb/ M M scfd				
fluid			4 - 36			62 - 158			62 - 158			62 - 158							

Table 4.2: BBOU, ELEM, NUNR, SEIB, and OPUG Water-Saturated Reservoir Compositions

Saturated Gas Compositions															
Component	BBO U			ELEM			NUNR			SEIB			OPUG		
	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid	Flashed Gas	Flashed Liq	Saturated Fluid
N ₂			0.07			0.00			0.00			0.07			0.10
CO ₂			0.59			2.17			0.77			1.53			2.21
H ₂ S			0.00			0.00			0.00			0.00			0.00
C ₁			92.61			79.87			72.37			77.84			67.24
C ₂			3.15			6.00			7.63			5.87			7.88
C ₃			1.01			2.53			4.82			3.37			4.76
iC ₄			0.61			0.73			1.45			1.21			1.94
nC ₄			0.44			0.79			1.96			1.37			1.06
iC ₅			0.25			0.44			0.94			0.82			1.15
nC ₅			0.19			0.32			0.76			0.57			0.68
C ₆			0.23			0.64			1.02			0.91			1.06
C ₇₊			0.53			4.31			7.65			5.48			10.86
H ₂ O			0.34			2.19			0.63			0.95			1.08
Total			100.00			100.00			100.00			100.00			100.00

Product Specification

The delivery pressure at AKAB Gas Plant is projected to be between 60 - 65 barg.

5. Phase-1 Gas Production Simulation Results

For phase-1 gas production, three scenarios were explored:

- ✚ Scenario-1 is BBOU-only field flow to AKAB at 300 MMscfd.
- ✚ Scenario-2 is a 180 MMscfd BBOU field flow, with ELEM field flow at 120 MMscfd and AKAB total gas delivery of 300 MMscfd.
- ✚ Scenario-3, examines the capacity of ELEM E-MF to AKAB T/L to support the delivery of a total of 420 MMscfd of gas gathered from BBOU field flow at 300 MMscfd, and ELEM field flow at 120 MMscfd to make an AKAB gas delivery of 420 MMscfd.

5.1. Phase-1/Scenario-1 Gas Production

This scenario shows the deliverability of an anticipated peak production of 300 MMscfd of gas from Bubouwe field; if it were to be the only source of gas supply to AKAB at the onset of production.

i. BBOU to ELEM E-MF B/L Flow (at 300 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 300 MMscfd of gas flow with minimal pressure loss across the 9 km BBOU-ELEM E-MF bulk-line is 92.0 bar. And the corresponding temperature is 50.3 °C.

Delivery Pressure; at ELEM E-MF is 73.4 bar, which is sufficient to support ELEM E-MF to AKAB trunk-line delivery at required pressure spec. And the corresponding temperature is 43.9 °C.

The Line Size; which also supported 300 MMscfd of gas flow across the 9 km BBOU-ELEM E-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 16". And the pressure drop is 2.1 bar/km.

See below; fig. 5.2, table 5.1, table 5.3, & fig. 5.3 for more details

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.5 m/s and 3.9 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 9.9 m/s and 12.4 m/s at both inlet and outlet of the B/L. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the B/L is 16.8 m/s and 18.8 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See below; fig. 5.2, table 5.1, table 5.3, & fig. 5.5 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 92.0 bar, the formation temperature is 17.7 °C. While the prevailing flow conditions at the inlet is 92.0 bar pressure and 50.3 °C temperature.

Similarly for outlet stream, at stream pressure of 73.4 bar, the formation temperature is 16.3 °C. While the prevailing flow conditions at the outlet is, pressure 73.37 bar and temperature 43.9 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.7 for more details

ii. ELEP E-MF to AKAB B/L Flow (at 300 MMscfd)

Pressure and Temperature Regime

Source Pressure at the ELEP E-MF outlet; the incoming BBOU field flow exited ELEP E-MF at a pressure of 73.4 bar and with a corresponding temperature of 43.9 °C.

Delivery Pressure; at Akabel is 65.1 bar and with a corresponding temperature of 39.7 °C.

The Line Size; which supported 300 MMscfd of gas flow across the 28.5 km ELEP E-MF AKAB trunk-line with minimal pressure loss and with a velocity below erosional velocity mark is 24". The pressure drop is 0.3 bar/km.

See below; fig. 5.2, table 5.2, table 5.4, & fig. 5.4 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.9 m/s; while at the outlet is 4.1 m/s.

Prevailing Fluid Flow Velocity; at inlet is 5.5 m/s and outlet is 6.1 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 18.6 m/s and outlet is 19.8 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.2, table 5.2, table 5.4, & fig. 5.6 for more details

See below; fig. 5.8 for more details

5.1.1. Hysys PFD

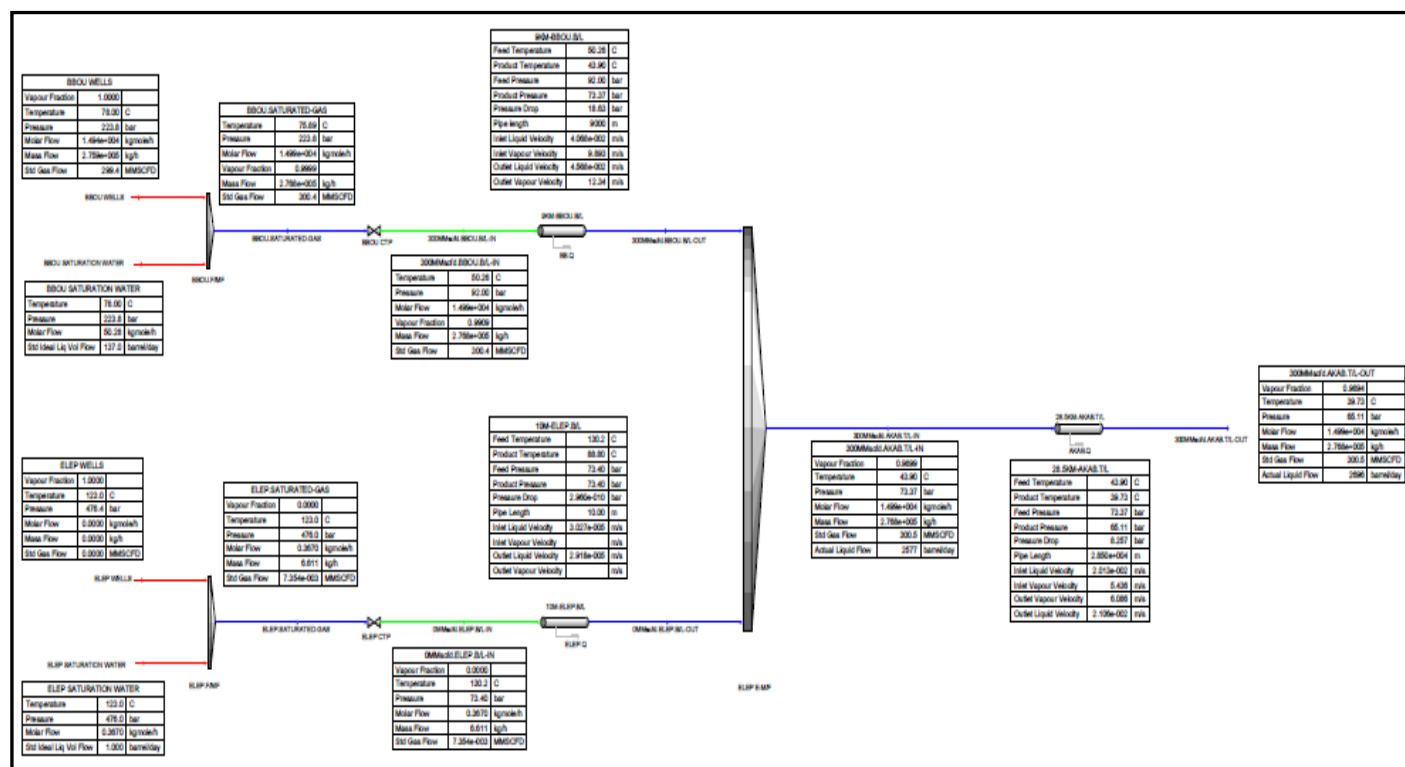


Figure 5.1: Phase-1/Scenario-1 Gas Production; Hysys PFD

5.1.2. Flow Schematics

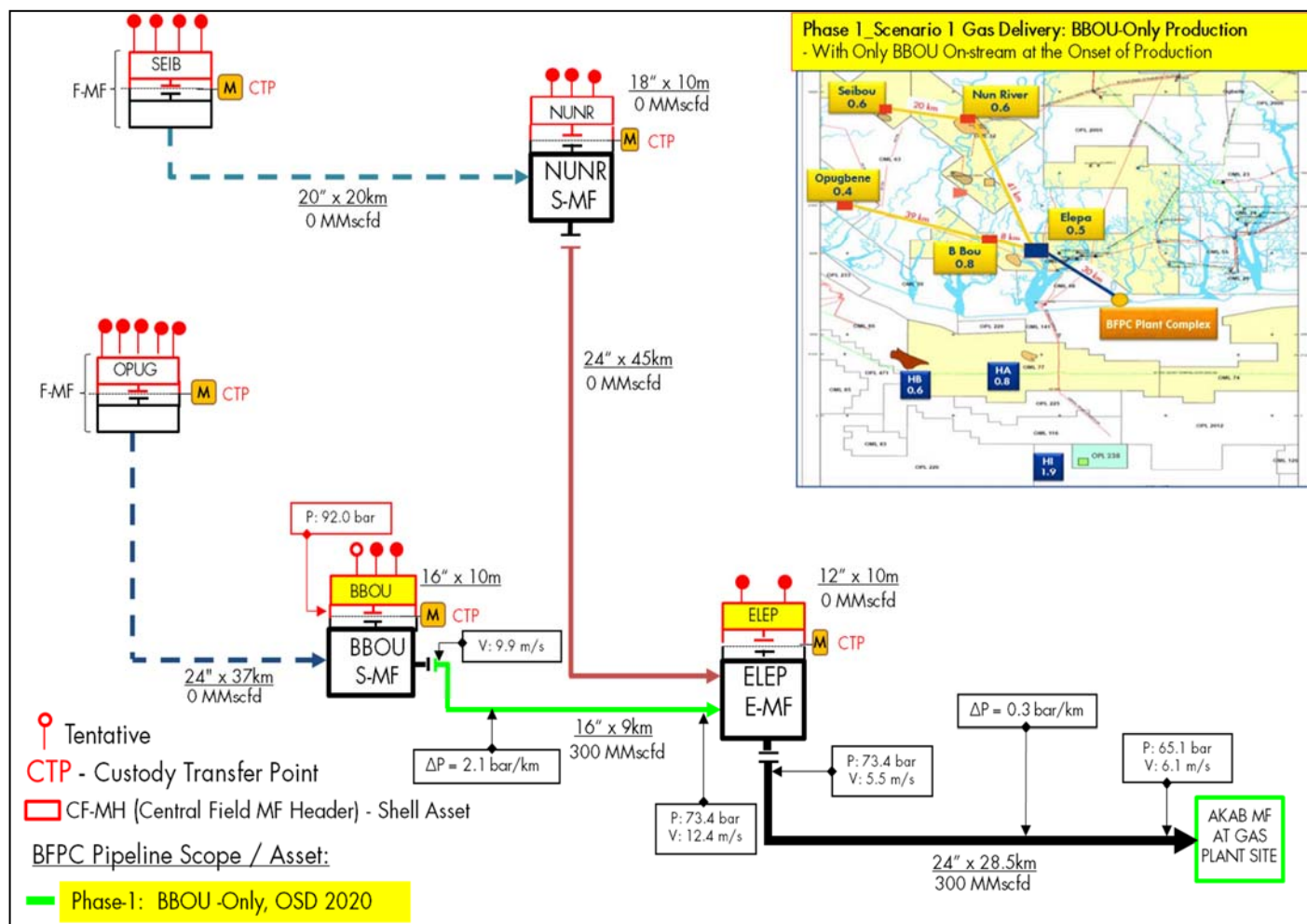


Figure 5.2: Phase-1/Scenario-1 Gas Production; Flow Schematics

5.1.3. Calculation Notes

i. BBOU to ELEP E-MF B/L

Table 5.1: Phase-1/Scenario-1; BBOU to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from BB to ELEP E-MF																				
Line Dimension 16" SCH 80 (ID)		in	mm	m																
		14.33	364.00	0.36																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	92.00	50.26	276,837	3,712	61.87	1.03	74.58	265,982	3,697	61.62	1.03	71.95	0.01	0.85	10,449	14.79	0.25	0.004	706.53	0.36
Outlet	73.37	43.90	276,837	4,627	77.12	1.29	59.83	264,429	4,610	76.83	1.28	57.36	0.01	0.86	11,932	16.59	0.28	0.005	719.31	0.40
Calc. Parameters																				
	X-sectional Area (m ²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V ² overall	V _G ²	V _L ²	Overall Phase Momentu m (Pa)	Gas Phase Momentu m (Pa)	Liq. Phase Momentu m (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d ² (π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V ² overall	ρ _G * V _G ²	ρ _L * V _L ²	X = m _G /m _L √(ρ _G /ρ _L)	Ve = c/(Vρ _G)	Ve = c/(Vρ _L)							
Inlet	0.10	9.90	9.86	0.04	98.10	97.31	0.002	7,317	7,002	1.10	0.0125	16.79	3.47							
Outlet	0.10	12.35	12.30	0.04	152.43	151.31	0.002	9,119	8,679	1.41	0.0127	18.75	3.88							

ii. ELEP E-MF to AKAB T/L

Table 5.2: Phase-1/Scenario-1; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
trunk-line from ELEP E-MF to AKABEL																				
Line Dimension	in	mm	m																	
24" SCH 80 (ID)	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	73.37	43.90	276,844	4,627	77.12	1.29	59.83	264,429	4,610	76.83	1.28	57.36	0.01	0.86	11,932	16.59	0.28	0.005	719.31	0.40
Outlet	65.11	39.73	276,844	5,179	86.32	1.44	53.45	263,749	5,161	86.02	1.43	51.10	0.01	0.87	12,562	17.33	0.29	0.005	724.93	0.42
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X = m _G /m _L √(ρ _G /ρ _L)	Ve = c/(Vp _m)	Ve = c/(Vp _m)							
Inlet	0.24	5.45	5.43	0.02	29.67	29.45	0.000	1,775	1,689	0.27	0.0127	18.75	3.88							
Outlet	0.24	6.10	6.08	0.02	37.17	36.92	0.000	1,987	1,886	0.30	0.0126	19.83	4.10							

5.1.4. Flow Parameters

i. BBOU to ELEP E-MF B/L

Table 5.3: Phase-1/Scenario-1; BBOU to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	92.0000	50.2566	0.9909	<empty>	Segregated	0.0512	0.2276	<empty>	0.00	<empty>	0.000181	<empty>	26,504	17,541,773	0.0407	9.8935
1,800	0	88.5870	49.1427	0.9907	12.0454	Segregated	0.0506	0.2363	3.4752	0.00	0.00	0.000203	0.002985	26,827	17,721,589	0.0418	10.2606
3,600	0	85.0382	47.9614	0.9905	11.4853	Segregated	0.0499	0.2461	3.6193	0.00	0.00	0.000229	0.003369	27,085	17,912,463	0.0428	10.6753
5,400	0	81.3349	46.7027	0.9903	10.8897	Segregated	0.0490	0.2573	3.7841	0.00	0.00	0.000262	0.003847	27,275	18,115,946	0.0438	11.1489
7,200	0	77.4539	45.3538	0.9901	10.2532	Segregated	0.0480	0.2702	3.9748	0.00	0.00	0.000303	0.004452	27,391	18,333,965	0.0448	11.6962
9,000	0	73.3655	43.8986	0.9899	9.5685	Segregated	0.0468	0.2855	4.1990	0.00	0.00	0.000356	0.005239	27,427	18,568,992	0.0457	12.3384

ii. ELEP E-MF to AKAB T/L

Table 5.4: Phase-1/Scenario-1; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	73.3655	43.9002	0.9899	<empty>	Segregated	0.0559	0.0328	<empty>	0.00	<empty>	0.000008	<empty>	18,190	12,325,340	0.0201	5.4360
5,700	0	71.8140	43.0951	0.9898	11.0573	Segregated	0.0555	0.0338	1.5763	0.00	0.00	0.000009	0.000397	18,209	12,388,853	0.0203	5.5453
11,400	0	70.2105	42.2748	0.9897	10.5712	Segregated	0.0549	0.0350	1.6307	0.00	0.00	0.000009	0.000430	18,219	12,455,101	0.0205	5.6641
17,100	0	68.5486	41.4366	0.9896	10.0753	Segregated	0.0544	0.0361	1.6798	0.00	0.00	0.000010	0.000464	18,217	12,524,401	0.0207	5.7939
22,800	0	66.8489	40.5873	0.9895	9.5706	Segregated	0.0538	0.0369	1.7192	0.00	0.00	0.000011	0.000498	18,206	12,595,990	0.0209	5.9340
28,500	0	65.1081	39.7252	0.9894	9.0588	Segregated	0.0531	0.0378	1.7620	0.00	0.00	0.000012	0.000538	18,183	12,670,041	0.0211	6.0860

5.1.5. Pressure Profile Plots

i. BBOU to ELEP E-MF B/L

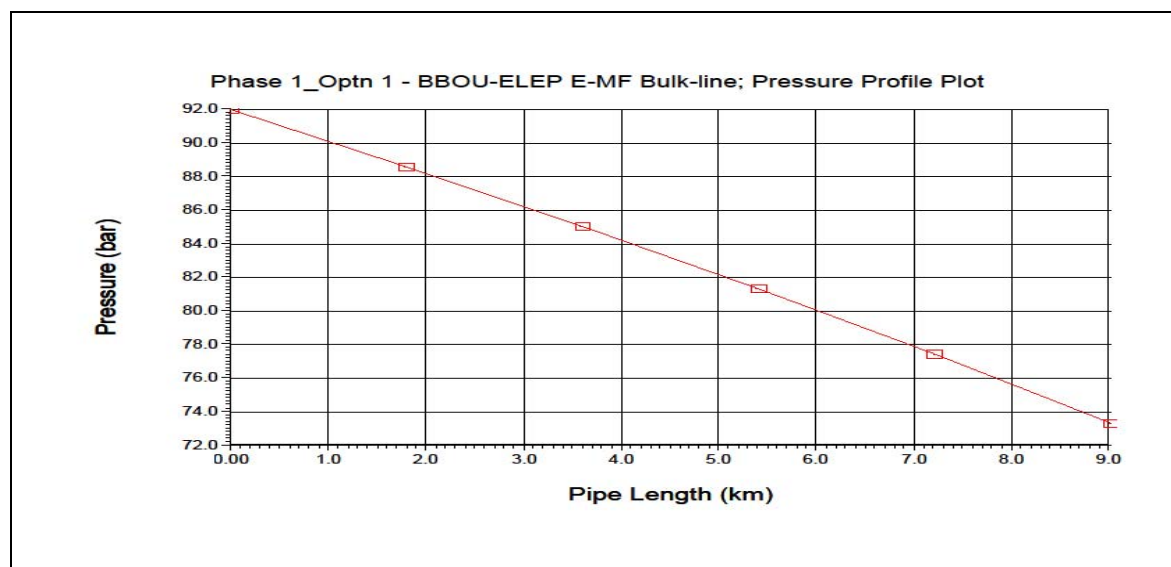


Figure 5.3: Phase-1/Scenario-1; BBOU to ELEP E-MF B/L Pressure Profile Plot

ii. ELEP E-MF to AKAB T/L

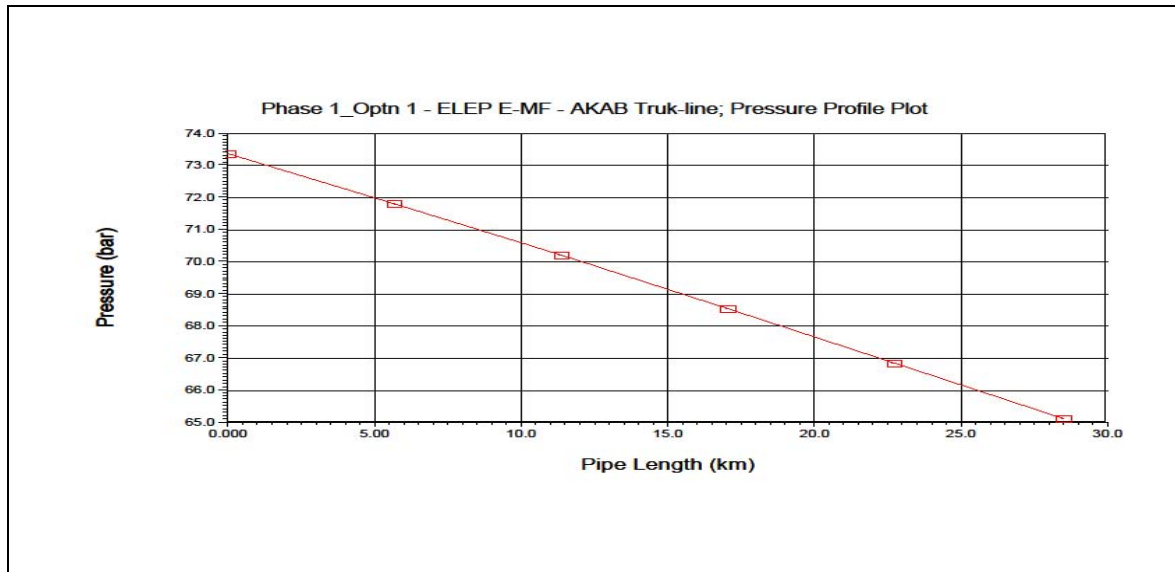


Figure 5.4: Phase-1/Scenario-1; ELEP E-MF to AKAB T/L Pressure Profile Plot

5.1.6. Velocity Profile Plots

i. BBOU to ELEP E-MF B/L

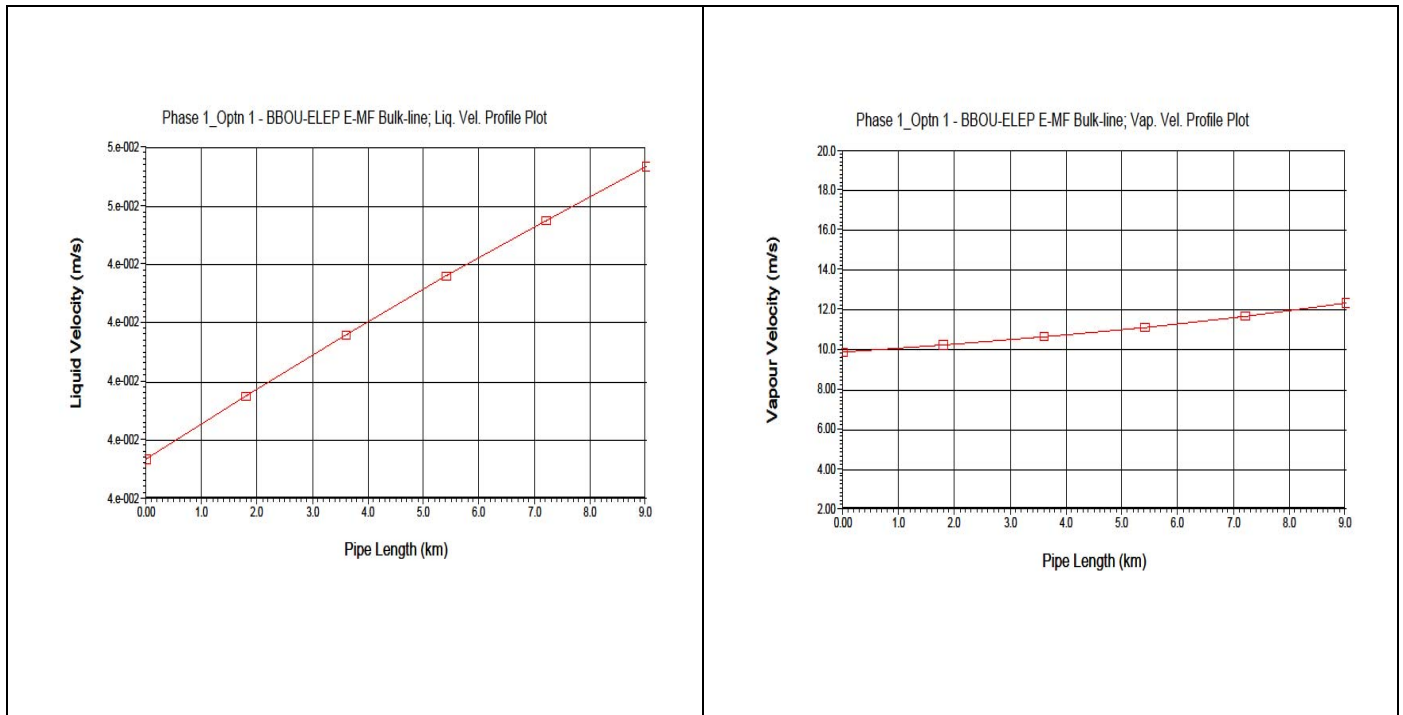


Figure 5.5: Phase-1/Scenario-1; BBOU to ELEP E-MF B/L Velocity Profile Plot

ii. ELEP E-MF to AKAB T/L

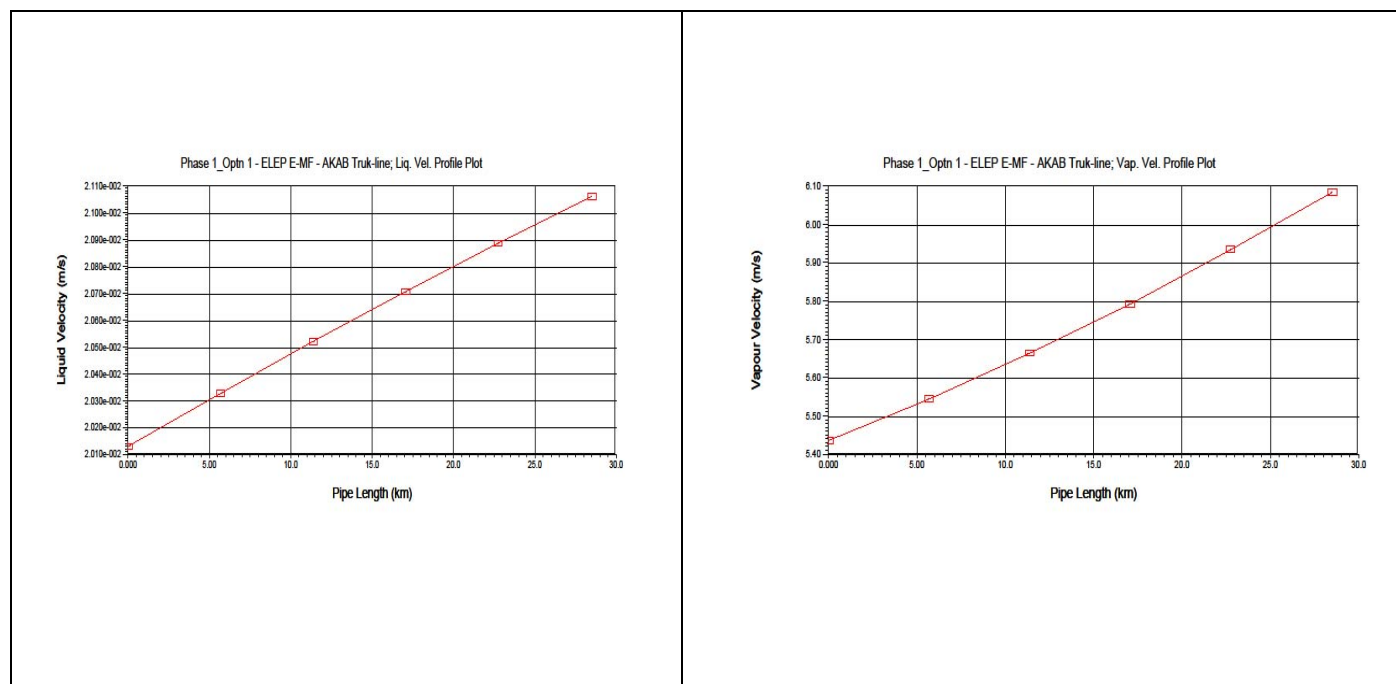


Figure 5.6: Phase-1/Scenario-1; ELEP E-MF to AKAB T/L Velocity Profile Plot

5.1.7. Phase Envelopes/Hydrate Curves

i. BBOU to ELEP E-MF B/L

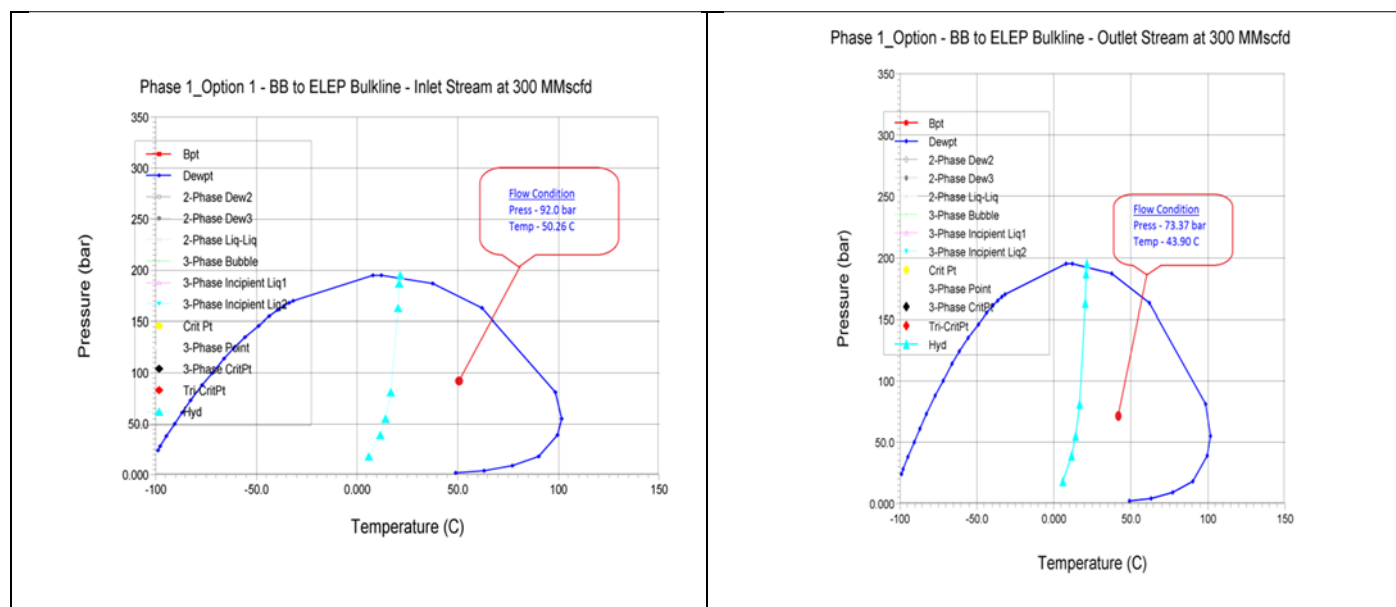


Figure 5.7: Phase-1/Scenario-1; BBOU to ELEP E-MF B/L Phase Envelope/Hydrate Curve

ii. ELEP E-MF to AKAB T/L

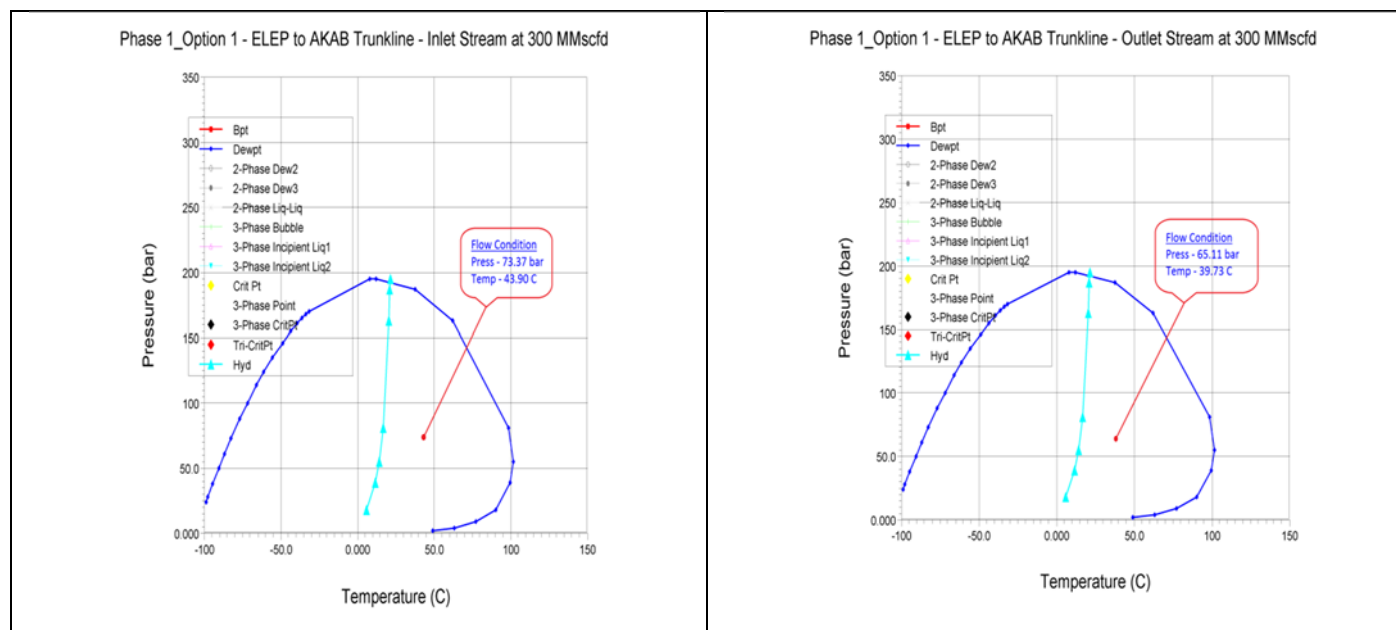


Figure 5.8: Phase-1/Scenario-1, ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curve

5.2. Phase-1/Scenario-2 Gas Production

This scenario examines the deliverability of Bubouwe and Elepa field peak flow in a situation when Elepa field comes on-stream nine (9) year later than Bubouwe field. The flow contributions are, Bubouwe 180 MMscfd and Elepa 120 MMscfd of gas.

For this scenario, the existing 16" BBOU-ELEP E-MF bulk-line rated for a peak flow of 300 MMscfd in scenario-1 simulation is still employed for 180 MMscfd flow. This is as a result of expectation that, BBOU field flow could peak up to 300 MMscfd.

The variation (on BBOU - ELEP bulk-line) is significantly on flow conditions (i.e. flow rates, pressure and temperature). While line sizes is unchanged.

i. BBOU to ELEP E-MF B/L (at 180 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; for the 180 MMscfd flow across the 16" x 9 km BBOU - ELEP bulk-line, a source pressure of 83.0 bar gives a suitable head flow. The corresponding temperature is 47.5 °C.

Delivery Pressure; at ELEP E-MF is 76.1 bar and with a corresponding temperature of 44.6 °C.

The Line Size; for this phase-1/scenario-2 study, 16" rated BBOU - ELEP line was employed in reviewing the flow conditions. For this scenario, pressure drop is 0.8 bar/km

See below; fig. 5.10, table 5.5, table 5.8, & fig. 5.11 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.7 m/s; while the outlet is 3.8 m/s

Prevailing Fluid Velocity; at the inlet is 6.6 m/s; while the outlet is 7.1 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at the inlet is 17.7 m/s; while the outlet is 18.4 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.10, table 5.5, table 5.8, & fig. 5.14 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 83.0 bar, hydrate formation temperature is 17.1 °C. While the prevailing flow conditions at the inlet is 83.0 bar pressure and 47.5 °C temperature.

Similarly for outlet stream, at stream pressure of 76.1 bar, the formation temperature is 16.6 °C. While the prevailing flow conditions at the outlet is; pressure at 76.1 bar and temperature at 44.6 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.17 for more detail

ii. ELEP to ELEP E-MF (at 120 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; for the 120 MMscfd flow across the 12" x 10 m ELEP F-MF – ELEP E-MF bulk-line, a source pressure of 76.0 bar gives a suitable head flow. The corresponding temperature is 90.0 °C.

Delivery Pressure; at ELEP E-MF is 76.0 bar and with a corresponding temperature of 90.0 °C.

The Line Size; which supported 120 MMscfd of gas flow across the 10 m ELEP F-MF – ELEP E-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity mark is 12". The pressure drop is 0.0 bar.

See below; fig. 5.10, table 5.6, table 5.9, & fig. 5.12 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at both inlet and outlet is 3.4 m/s.

Prevailing Fluid Velocity; at both inlet and outlet is 8.8 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at both inlet and outlet is 16.5 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.10, table 5.6, table 5.9, & fig. 5.15 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 76.0 bar, the formation temperature is 17.6 °C. While the prevailing flow conditions at the inlet is 76.0 bar pressure and 90.0 °C temperature.

Similarly for outlet stream, at stream pressure of 76.0 bar, the formation temperature is also 17.6 °C. While the prevailing flow conditions at the outlet is; pressure at 76.0 bar and temperature at 90.0 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.18 for more details

iii. ELEP E-MF to AKAB (at 300 MMscfd)

Pressure and Temperature Regime

Source Pressure at the ELEP E-MF Outlet; the mixed BBOU field flow and ELEP field flow exited ELEP E-MF at a pressure of 76.0 bar and with a corresponding temperature of 66.8 °C.

Delivery Pressure; at AKAB is 65.5 bar and with a corresponding temperature of 61.4 °C.

The Line Size; which supported 300 MMscfd of gas flow across the 28.5 Km ELEP E-MF AKAB trunk-line with minimal pressure loss and with a velocity below erosional velocity mark is 24". For this case, pressure drop is 0.4 bar/km

See below; fig. 5.10, table 5.7, table 5.10, & fig. 5.13 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.7 m/s; while outlet is 3.9 m/s.

Prevailing Fluid Velocity; at the inlet is 5.7 m/s; while the outlet is 6.6 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at the inlet is 17.7 m/s and outlet is 18.9 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.10, table 5.7, table 5.10, & fig. 5.16 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 76.0 bar, the formation temperature is 17.1 °C. While the prevailing flow conditions at the inlet is 76.0 bar pressure and 66.8 °C temperature.

Similarly for outlet stream, at stream pressure of 65.5 bar, the formation temperature is 16.2 °C. While the prevailing flow conditions at the outlet is; pressure at 65.5 bar and temperature at 61.4 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.19 for more details

5.2.1. Hysys PFD

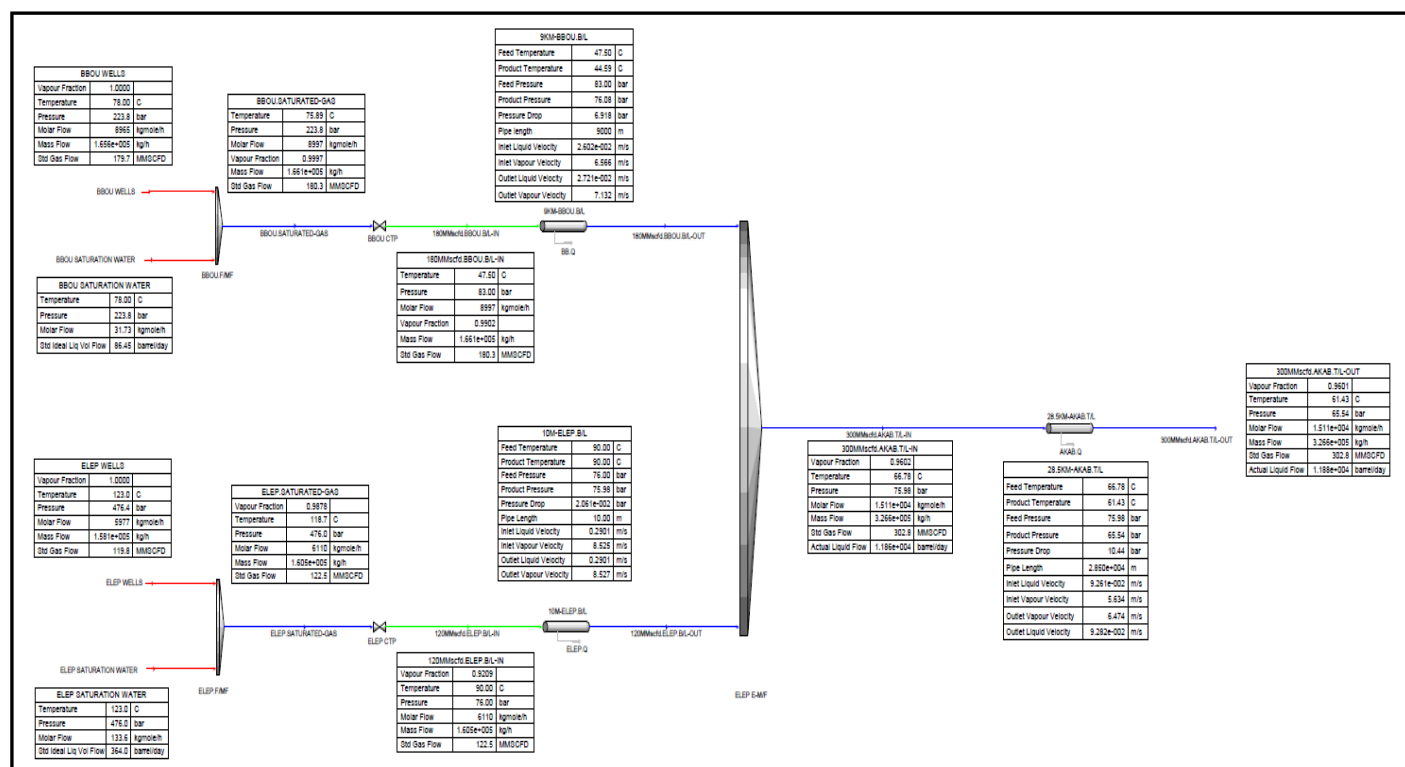


Figure 5.9: Phase-1/Scenario-2 Gas Production; Hysys PFD

5.2.2. Flow Schematics

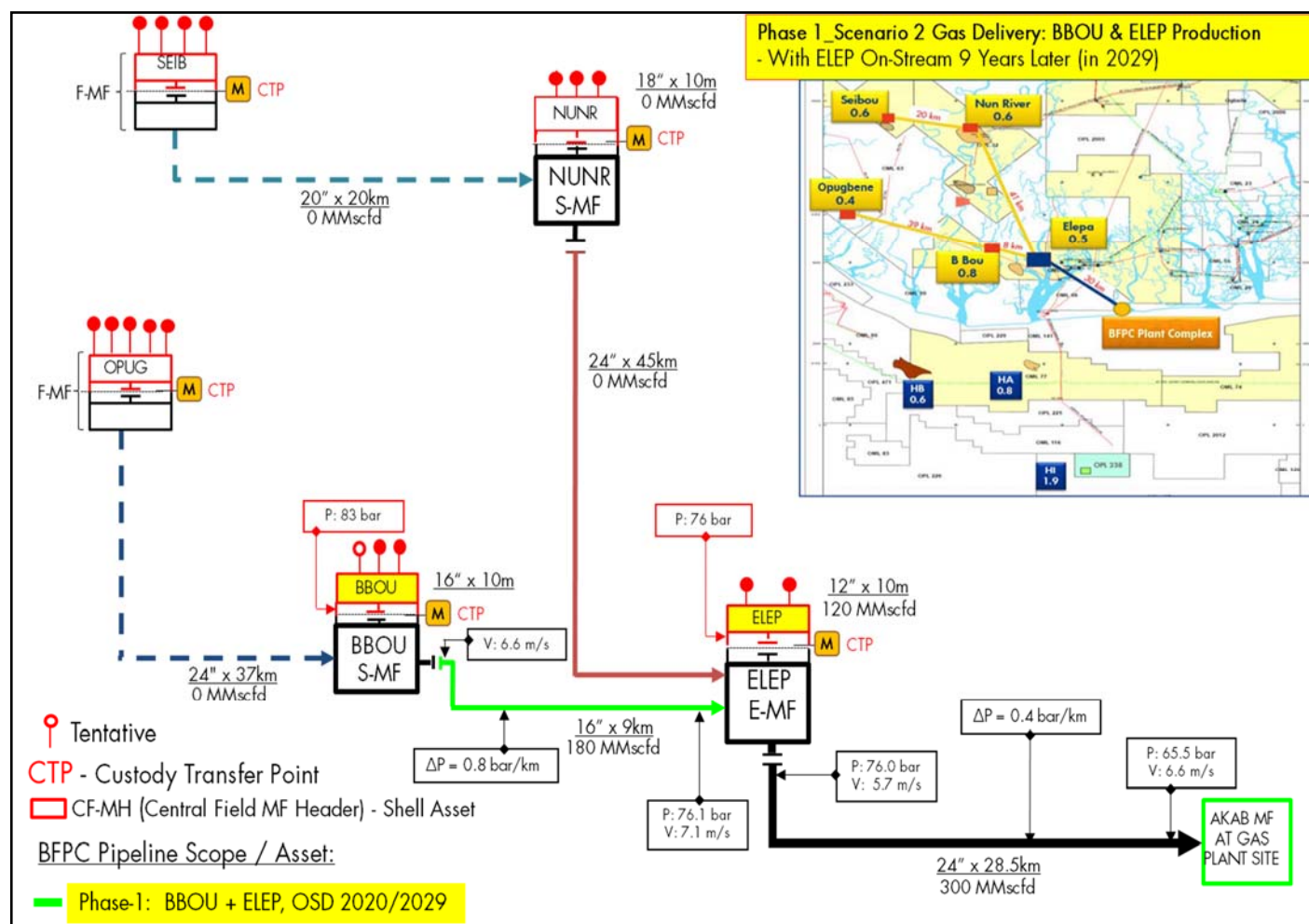


Figure 5.10: Phase-1/Scenario-2 Gas Production; Flow Schematics

5.2.3. Calculation Notes

i. BBOU to ELEP E-MF B/L

Table 5.5: Phase-1 /Scenario-2; BBOU to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from BB to ELEP E-MF																				
Line Dimension		in	mm	m																
16" SCH 80 (ID)		14.33	364.00	0.36																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	83.00	47.50	166,131	2,463	41.05	0.68	67.45	159,120	2,453	40.88	0.68	64.86	0.01	0.86	6,721	9.43	0.16	0.003	712.70	0.38
Outlet	76.08	44.59	166,131	2,675	44.58	0.74	62.10	158,749	2,665	44.42	0.74	59.57	0.01	0.86	7,070	9.86	0.16	0.003	717.43	0.39
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² overall	ρ _G * V _G ²	ρ _L * V _L ²	X _{tt} = m ₁ /m ₂ √(ρ ₁ /ρ ₂)	Ve = c/(Vp _{no})	Ve = c/(Vp _{no})							
Inlet	0.10	6.57	6.55	0.03	43.19	42.84	0.001	2,913	2,779	0.45	0.0127	17.66	3.65							
Outlet	0.10	7.14	7.11	0.03	50.95	50.57	0.001	3,164	3,012	0.50	0.0128	18.40	3.81							

ii. ELEP F-MF to ELEP E-MF B/L

Table 5.6: Phase-1 /Scenario-2; ELEP F-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from ELEP to ELEP E-MF																				
Line Dimension	in	mm	m																	
12" SCH 80 (ID)	11.38	289.00	0.29																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	76.00	90.00	160,459	2,080	34.67	0.58	77.14	110,845	2,012	33.53	0.56	55.10	0.02	0.90	48,406	67.20	1.12	0.019	720.29	0.38
Outlet	75.98	90.00	160,459	2,081	34.68	0.58	77.11	110,845	2,012	33.53	0.56	55.08	0.02	0.90	48,406	67.20	1.12	0.019	720.31	0.38
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$\rho_G \cdot V^2_{overall}$	$\rho_G \cdot V_G^2$	$\rho_L \cdot V_L^2$	$X_{tt} = \frac{m_1}{m_2} \cdot \sqrt{(\rho_1/\rho_2)}$	$Ve = c/(Vp_{-})$	$Ve = c/(Vp_{-})$							
Inlet	0.07	8.80	8.52	0.28	77.52	72.53	0.081	5,979	3,996	58.28	0.1208	16.51	3.42							
Outlet	0.07	8.81	8.52	0.28	77.59	72.53	0.081	5,983	3,995	58.28	0.1208	16.51	3.42							

iii. ELEP E-MF to AKAB T/L

Table 5.7: Phase-1/Scenario-2; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
trunk-line from ELEP E-MF to AKABEL																				
Line Dimension		in	mm	m																
24" SCH 80 (ID)		21.57	548.00	0.55																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	75.98	66.78	326,590	4,857	80.95	1.35	67.24	268,507	4,778	79.63	1.33	56.19	0.01	0.89	56,306	76.72	1.28	0.021	733.89	0.45
Outlet	65.54	61.43	326,590	5,569	92.82	1.55	58.64	267,708	5,491	91.52	1.53	48.76	0.01	0.89	56,968	76.77	1.28	0.021	742.06	0.49
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X _{tt} = m _G /m _L ⁰.⁵(p _G /p _L)	Ve = c/(Vp _{tt})	Ve = c/(Vp _{min})							
Inlet	0.24	5.72	5.62	0.09	32.69	31.64	0.008	2,199	1,778	5.99	0.0580	17.68	3.66							
Outlet	0.24	6.56	6.46	0.09	42.98	41.79	0.008	2,521	2,037	6.06	0.0545	18.93	3.92							

5.2.4. Flow Parameters

i. BBOU to ELEP E-MF B/L

Table 5.8: Phase-1/Scenario-2; BBOU to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (°C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	83.0000	47.5018	0.9902	<empty>	Segregated	0.0540	0.0903	<empty>	0.00	<empty>	0.000032	<empty>	16,194	10,812,324	0.0260	6.5661
1,800	0	81.6614	46.9388	0.9901	10.8391	Segregated	0.0537	0.0917	1.3486	0.00	0.00	0.000033	0.000490	16,244	10,857,554	0.0263	6.6675
3,600	0	80.3015	46.3670	0.9901	10.5622	Segregated	0.0534	0.0932	1.3704	0.00	0.00	0.000035	0.000514	16,287	10,903,914	0.0265	6.7742
5,400	0	78.9191	45.7858	0.9900	10.2810	Segregated	0.0530	0.0947	1.3934	0.00	0.00	0.000037	0.000541	16,324	10,951,462	0.0268	6.8869
7,200	0	77.5131	45.1946	0.9899	9.9950	Segregated	0.0526	0.0964	1.4178	0.00	0.00	0.000039	0.000570	16,354	11,000,259	0.0270	7.0059
9,000	0	76.0820	44.5927	0.9898	9.7039	Segregated	0.0522	0.0981	1.4436	0.00	0.00	0.000041	0.000602	16,378	11,050,376	0.0272	7.1320

ii. ELEP F-MF to ELEP E-MF B/L

Table 5.9: Phase-1/Scenario-2; ELEP F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	76.0	90.0	0.9209	<empty>	Intermittent	0.1284	0.2520	<empty>	0.00	<empty>	0.000192	<empty>	149,218	8,903,444	0.29	8.52
2	0	76.0	90.0	0.9209	28.79	Intermittent	0.1284	0.2520	0.0041	0.00	0.00	0.000192	0.000003	149,214	8,903,543	0.29	8.53
4	0	76.0	90.0	0.9209	28.79	Intermittent	0.1284	0.2520	0.0041	0.00	0.00	0.000192	0.000003	149,210	8,903,642	0.29	8.53
6	0	76.0	90.0	0.9209	28.79	Intermittent	0.1284	0.2521	0.0041	0.00	0.00	0.000192	0.000003	149,206	8,903,740	0.29	8.53
8	0	76.0	90.0	0.9209	28.79	Intermittent	0.1284	0.2521	0.0041	0.00	0.00	0.000192	0.000003	149,203	8,903,840	0.29	8.53
10	0	76.0	90.0	0.9209	28.79	Intermittent	0.1284	0.2521	0.0041	0.00	0.00	0.000193	0.000003	149,199	8,903,939	0.29	8.53

iii. ELEP E-MF to AKAB T/L

Table 5.10: Phase-1/Scenario-2; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	75.9794	66.7763	0.9602	<empty>	Segregated	0.1135	0.0420	<empty>	0.00	<empty>	0.000012	<empty>	76,229	11,918,879	0.0926	5.6342
5,700	0	73.9999	65.7388	0.9602	24.6653	Segregated	0.1118	0.0430	2.0028	0.00	0.00	0.000013	0.000602	75,273	11,989,295	0.0927	5.7734
11,400	0	71.9712	64.6879	0.9601	24.0410	Segregated	0.1099	0.0441	2.0542	0.00	0.00	0.000014	0.000651	74,287	12,062,439	0.0928	5.9249
17,100	0	69.8887	63.6216	0.9601	23.4080	Segregated	0.1080	0.0453	2.1104	0.00	0.00	0.000015	0.000708	73,271	12,138,543	0.0928	6.0906
22,800	0	67.7470	62.5375	0.9601	22.7650	Segregated	0.1060	0.0466	2.1724	0.00	0.00	0.000017	0.000775	72,224	12,217,873	0.0928	6.2727
28,500	0	65.5401	61.4329	0.9601	22.1106	Segregated	0.1038	0.0481	2.2411	0.00	0.00	0.000018	0.000853	71,142	12,300,738	0.0928	6.4741

5.2.5. Pressure Profile Plots

i. BBOU to ELEP E-MF B/L

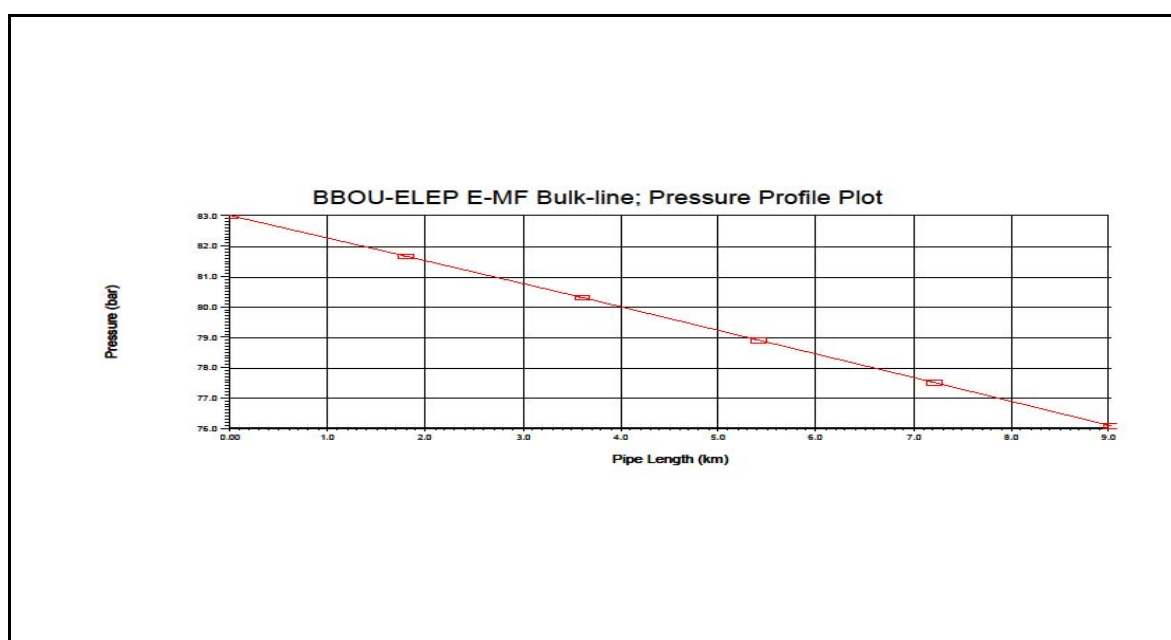


Figure 5.11: Phase-1/scenario-2; BBOU to ELEP E-MF Pressure Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

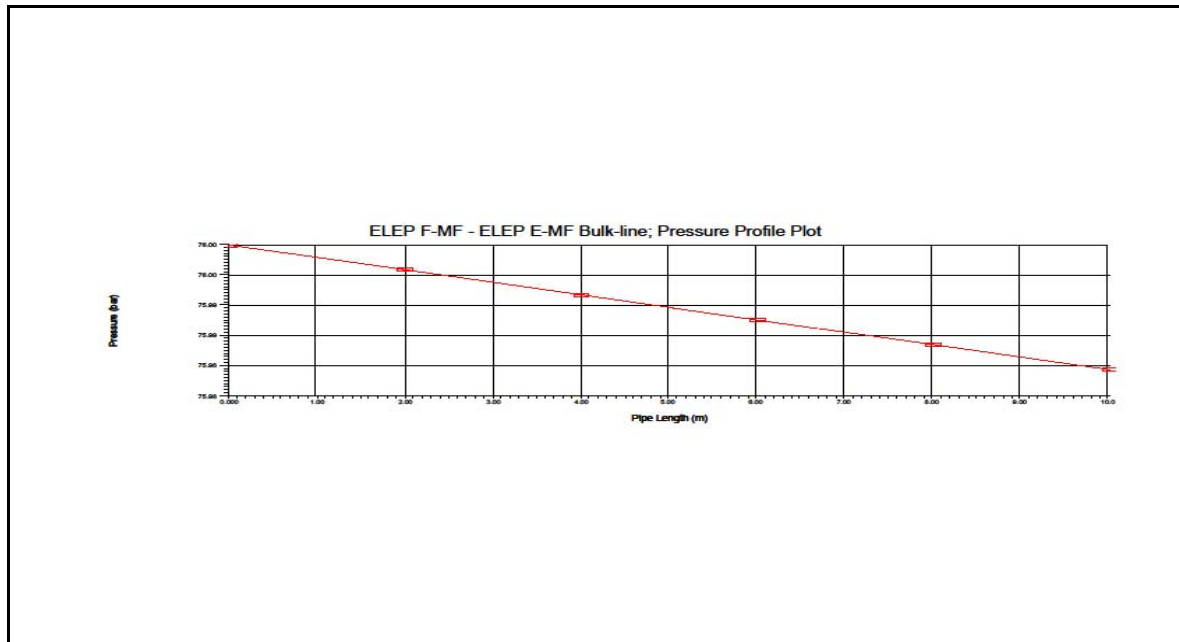


Figure 5.12: Phase-1 /Scenario-2; ELEP F-MF to ELEP E-MF B/L Pressure Profile Plot

iii. ELEP E-MF to AKAB T/L Pressure Profile Plot

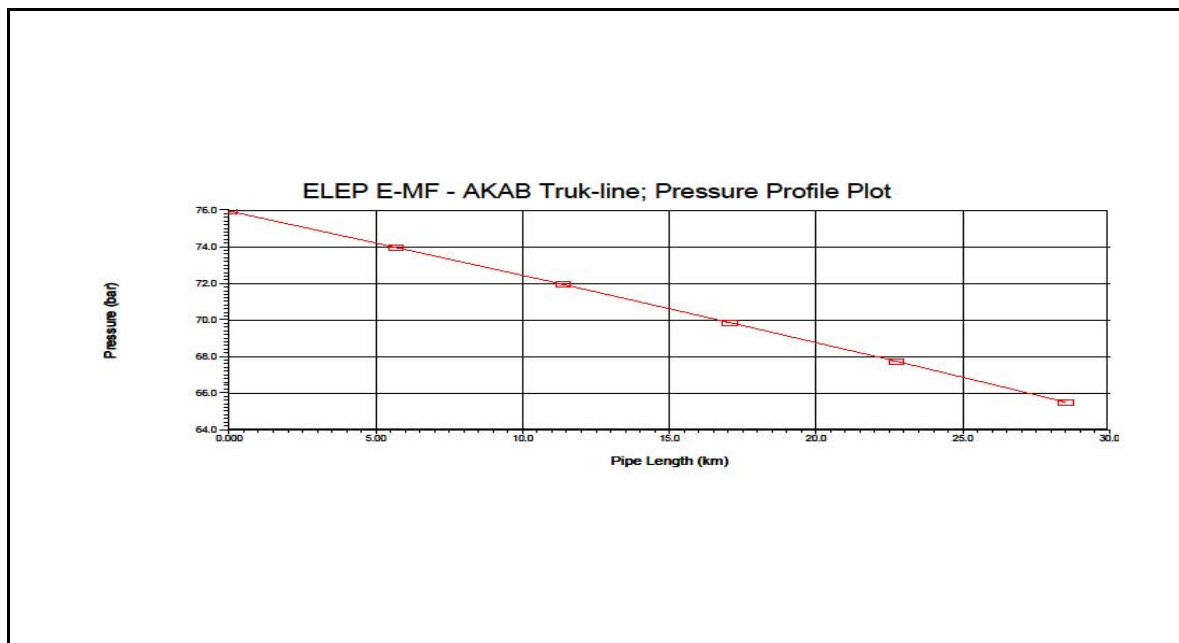


Figure 5.13: Phase-1 /Scenario-2; ELEP E-MF to AKAB T/L Pressure Profile Plot

5.2.6. Velocity Profile Plots

i. BBOU to ELEP E-MF B/L

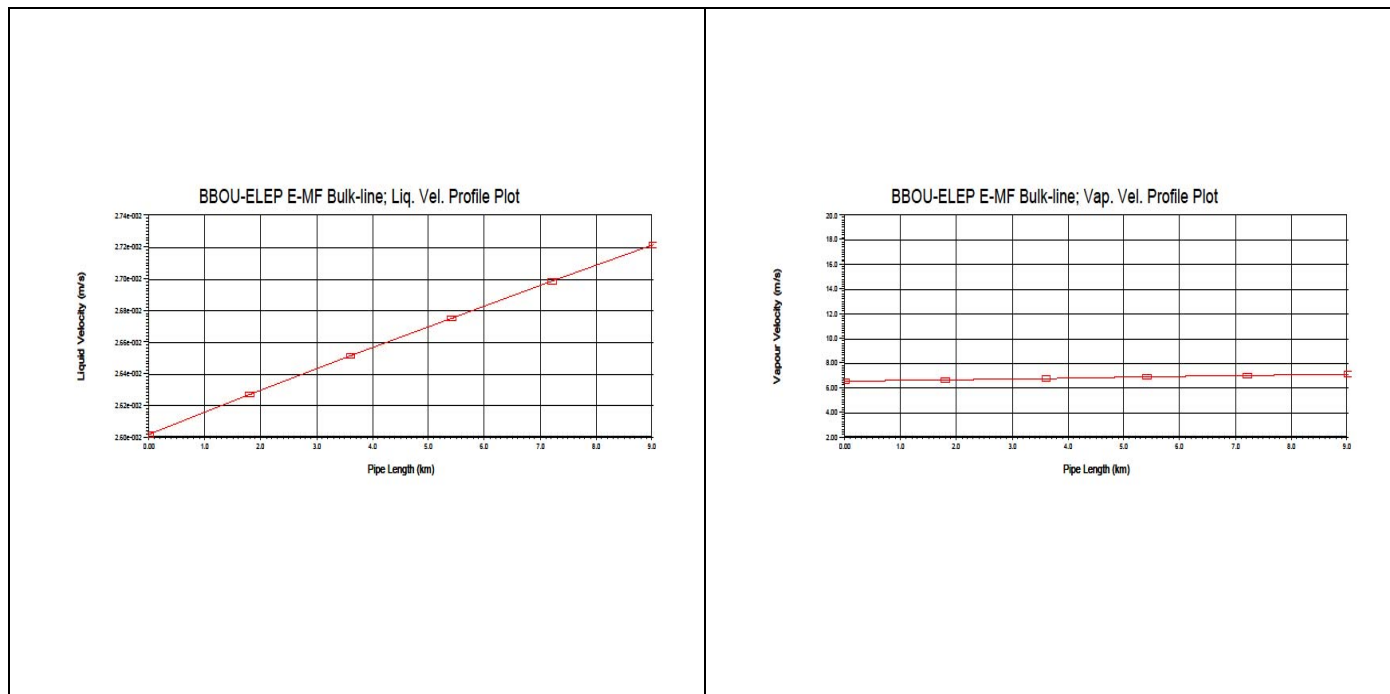


Figure 5.14: Phase-1/Scenario-2; BBOU to ELEP E-MF B/L Velocity Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

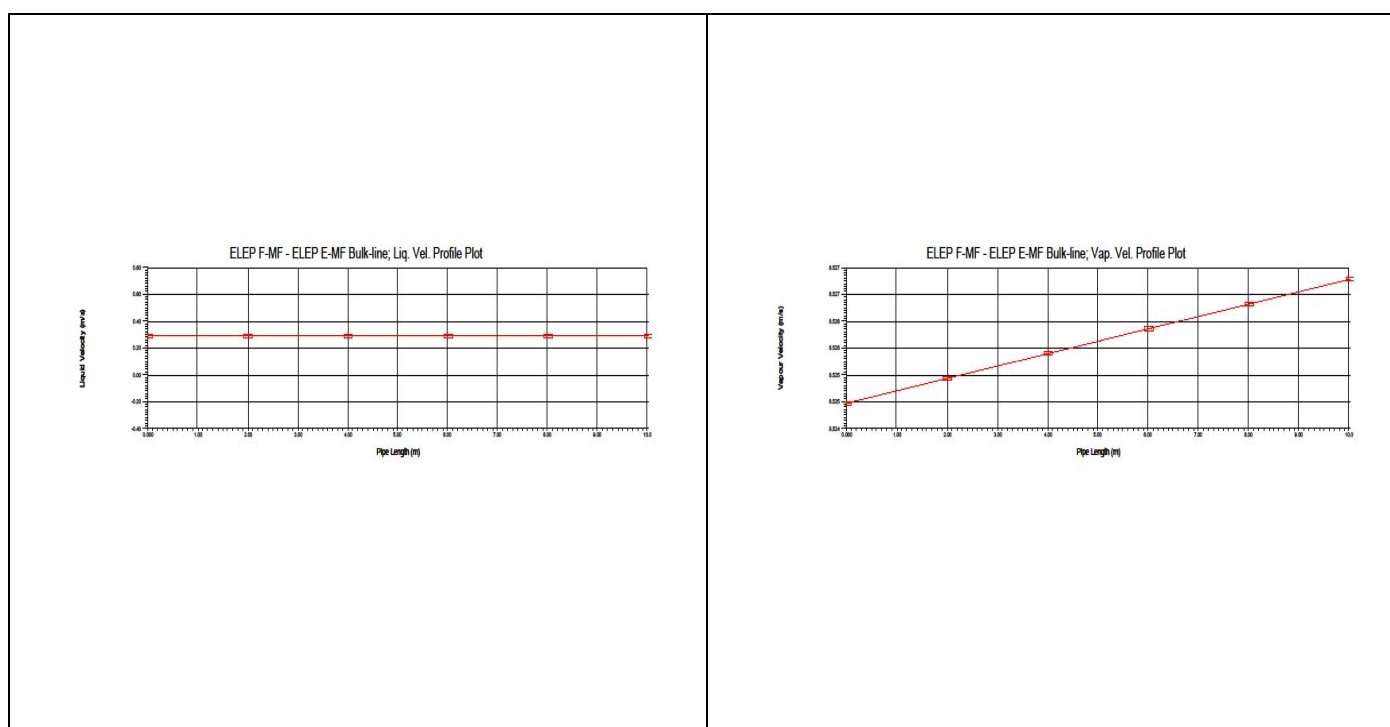


Figure 5.15: Phase-1/Scenario-2; ELEP F-MF to ELEP E-MF B/L Velocity Profile Plot

iii. ELEP E-MF to AKAB T/L

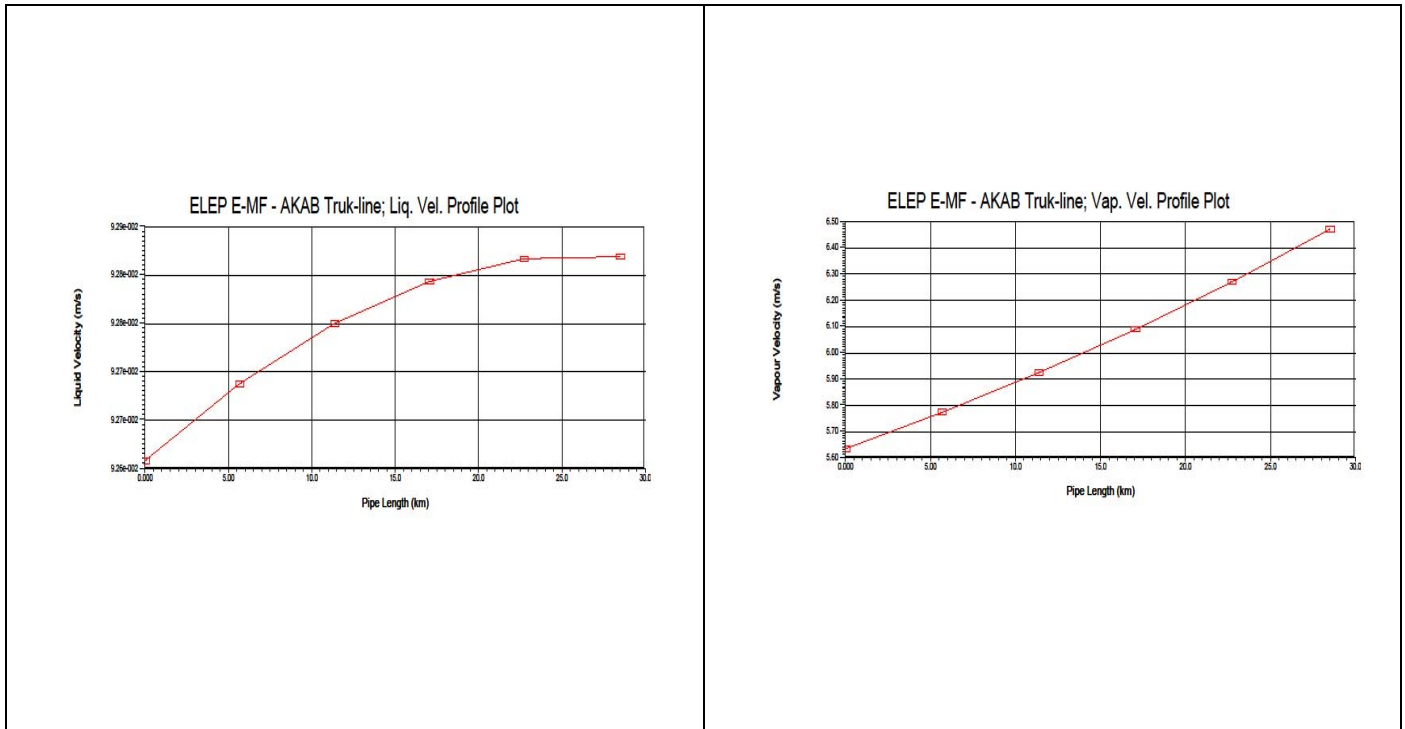


Figure 5.16: Phase-1/Scenario-2; ELEP E-MF to AKAB T/L Velocity Profile Plot

5.2.7. Phase Envelopes/Hydrate Curves

i. BBOU to ELEP E-MF B/L

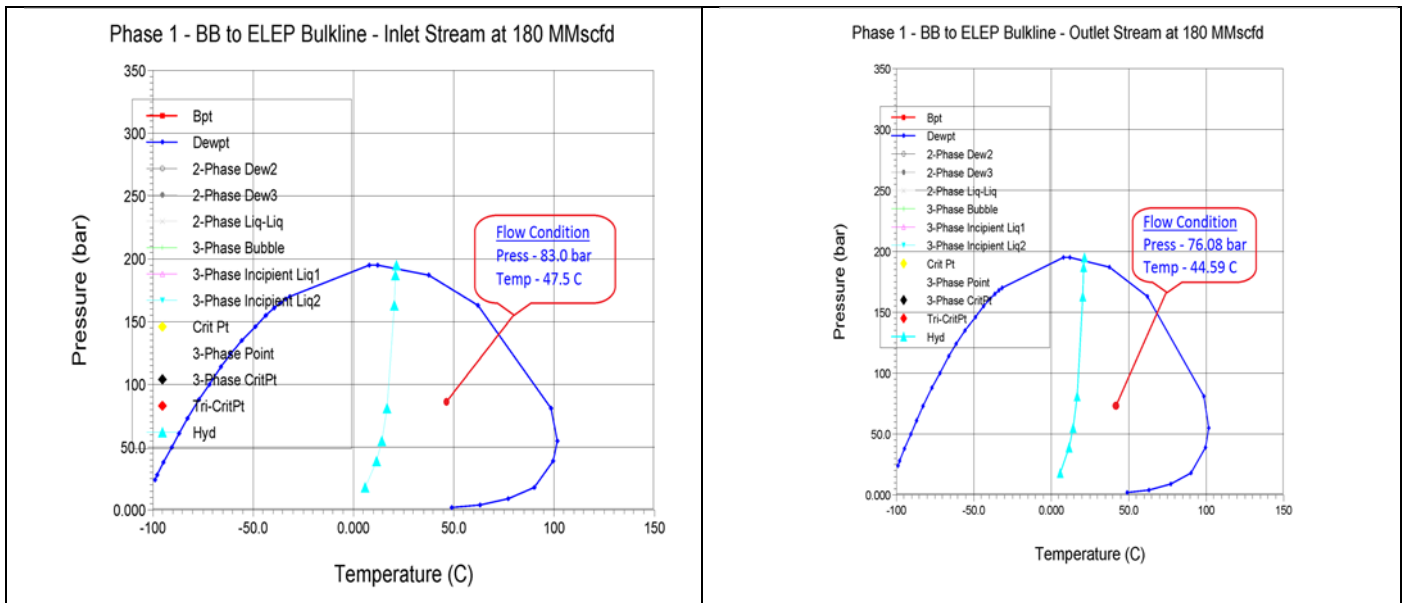


Figure 5.17: Phase-1/Scenario-2; BBOU to ELEP E-MF B/L Phase Envelope/Hydrate Curve

ii. ELEP F-MF to ELEP E-MF B/L

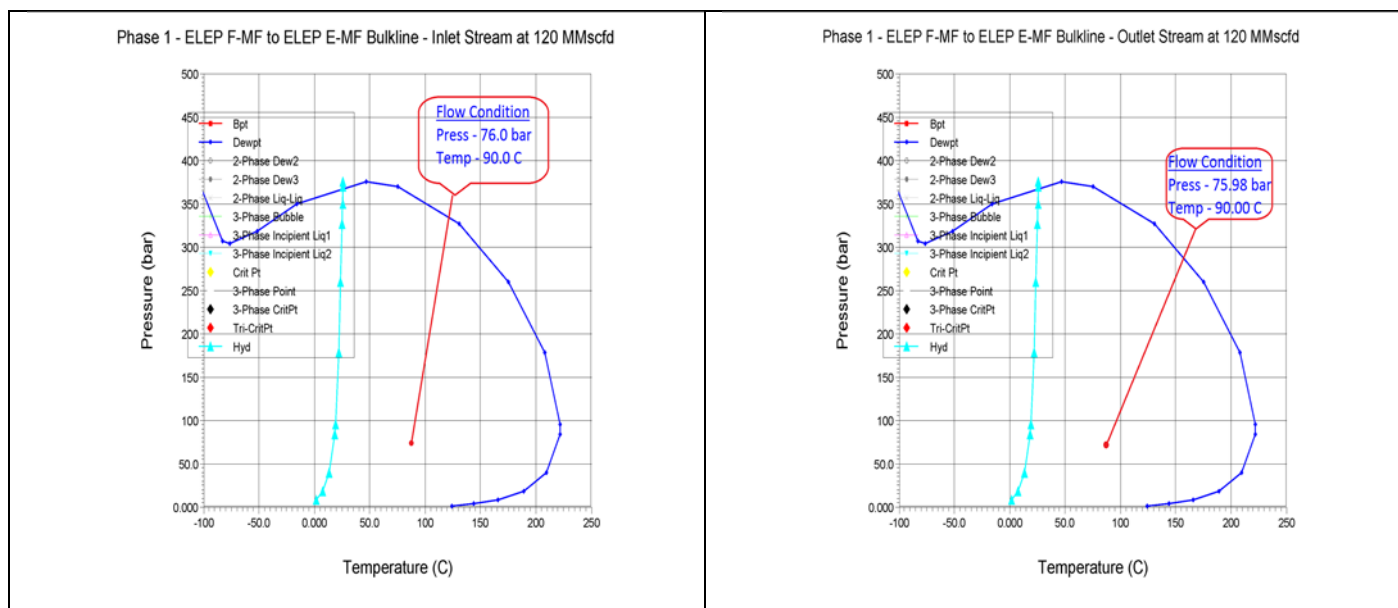


Figure 5.18: Phase-1/Scenario-2; ELEP F-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curve

iii. ELEP E-MF to AKAB T/L

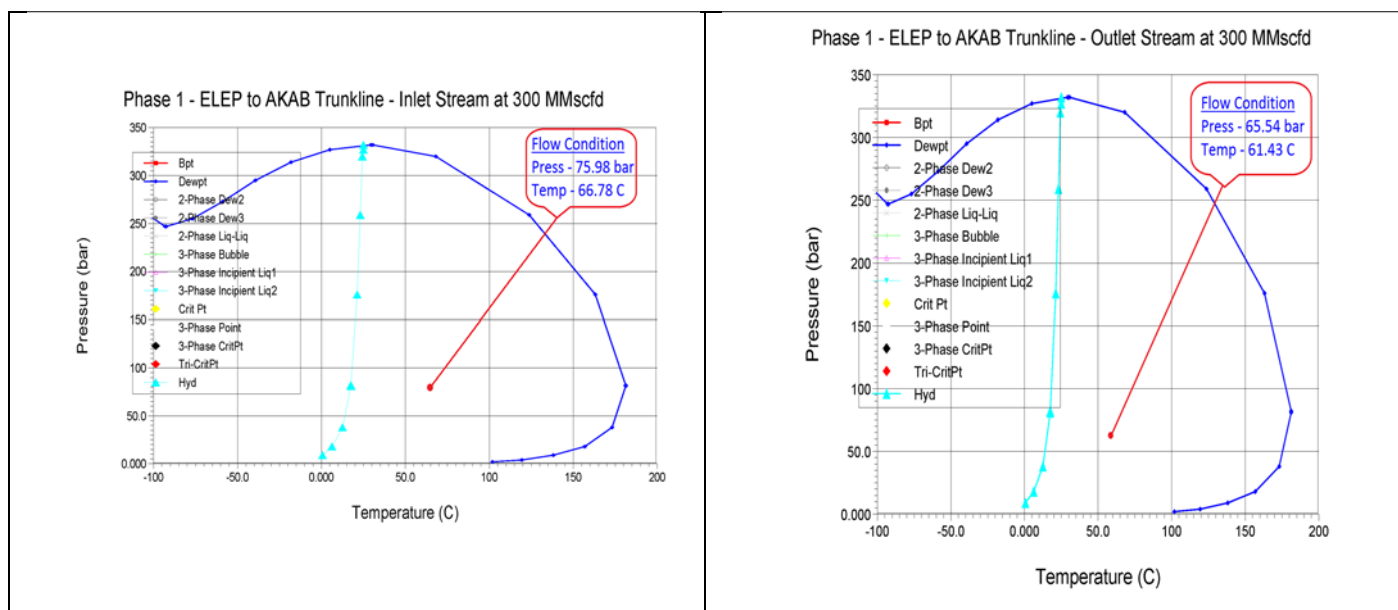


Figure 5.19: Phase-1/Scenario-2; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curve

5.3. Phase-1/Scenario-3 Gas Production (ELEP-AKAB T/L Capacity Sensitivity)

This scenario was undertaken, to check the capacity of ELEP E-MF – AKAB trunk-line if it could support gas flow up to 420 MMscfd cap.

The existing rated 16" BBOU-ELEP E-MF bulk-line, 12" ELEP F-MF to ELEP E-MF bulk-line and 24" ELEP E-MF – AKAB trunk-line was employed to deduce the flow parameters (i.e. flow rates, pressure and temperature) for this scenario.

i. BBOU to ELEP E-MF (at 300 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 300 MMscfd of gas flow with minimal pressure loss across the 9 km BBOU - ELEP E-MF bulk-line is 100.0 bar. And the corresponding temperature is 52.6 °C.

Delivery Pressure; at ELEP E-MF is 83.2 bar, which is sufficient to support ELEP E-MF to AKAB trunk-line delivery at required pressure spec. And the corresponding temperature is 47.1 °C.

The Line Size; for this phase-1/scenario-3 study, 16" rated BBOU F-MF – ELEP E-MF bulk-line was employed in reviewing the flow conditions. Pressure drop, for this case is 1.9 bar/km

See below; fig. 5.21, table 5.11, table 5.14, & fig. 5.22 for more details

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.3 m/s and 3.6 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 9.1 m/s and 10.9 m/s at both inlet and outlet of the B/L. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the bulk-line is 16.1 m/s and 17.6 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See below; fig. 5.21, table 5.11, table 5.14, & fig. 5.25 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 100.0 bar, the formation temperature is 18.2 °C. While the prevailing flow conditions at the inlet is 100.0 bar pressure and 52.6 °C temperature.

Similarly for outlet stream, at stream pressure of 83.2 bar, the formation temperature is 17.1 °C. While the prevailing flow conditions at the outlet is; pressure 83.12 bar and temperature 47.1 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.28 for more details

ii. ELEP to ELEP E-MF (at 120 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; is 85.0 bar which satisfy a good head flow for 120 MMscfd of gas across 10m ELEP F-MF - ELEP E-MF bulk-line. The corresponding temperature is 92.0 °C.

Delivery Pressure; at ELEP E-MF is 85.0 bar and with a corresponding temperature of 92.0 °C.

The Line Size; for this phase-1/scenario-3 study, 12" rated ELEP F-MF – ELEP E-MF bulk-line was employed in reviewing the flow conditions. Pressure drop, for this case is 0.0 bar.

See below; fig. 5.21, table 5.12, table 5.15, & fig. 5.23 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet and outlet is 3.2 m/s.

Prevailing Fluid Flow Velocity; at both inlet and outlet is 7.9 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at both inlet and outlet is 15.6 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.21, table 5.12, table 5.15, & fig. 5.26 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 85.0 bar, the formation temperature is 18.3 °C. While the prevailing flow conditions at the inlet is 85.0 bar pressure and 92.0 °C temperature.

Similarly for outlet stream, at stream pressure of 85.0 bar, the formation temperature is also 18.3 °C. While the prevailing flow conditions at the outlet is; pressure 85.0 bar and temperature 92.0 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.29 for more details

iii. ELEP E-MF to AKAB (at 420 MMscfd)

Pressure and Temperature Regime

Source Pressure at the ELEP E-MF outlet; the mixed BBOU field flow and ELEP field flow exited ELEP E-MF at a pressure of 83.2 bar and with a corresponding temperature of 63.3 °C.

Delivery Pressure; at AKAB is 64.9 bar and with a corresponding temperature of 56.3 °C.

The Line Size; the existing rated 24" ELEP E-MF – AKAB trunk-line shows capacity to support 420 MMscfd flow, with minimal pressure loss and with a velocity below erosional velocity mark. For this case, the pressure drop is 0.6 bar/km.

See below; fig. 5.21, table 5.13, table 5.16, & fig. 5.24 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.5 m/s; while at the outlet is 4.0 m/s.

Prevailing Fluid Flow Velocity; at inlet is 7.2 m/s and outlet is 9.1 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 17.1 m/s and outlet is 19.3 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 5.21, table 5.13, table 5.16, & fig. 5.27 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 83.2 bar, the formation temperature is 17.5 °C. While the prevailing flow conditions at the inlet is 83.2 bar pressure and 63.3 °C temperature.

Similarly for outlet stream, at stream pressure of 64.9 bar, the formation temperature is 16.0 °C. While the prevailing flow conditions at the outlet is; pressure 64.9 bar and temperature 56.3 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 5.30 for more details

5.3.1. Hysys PFD

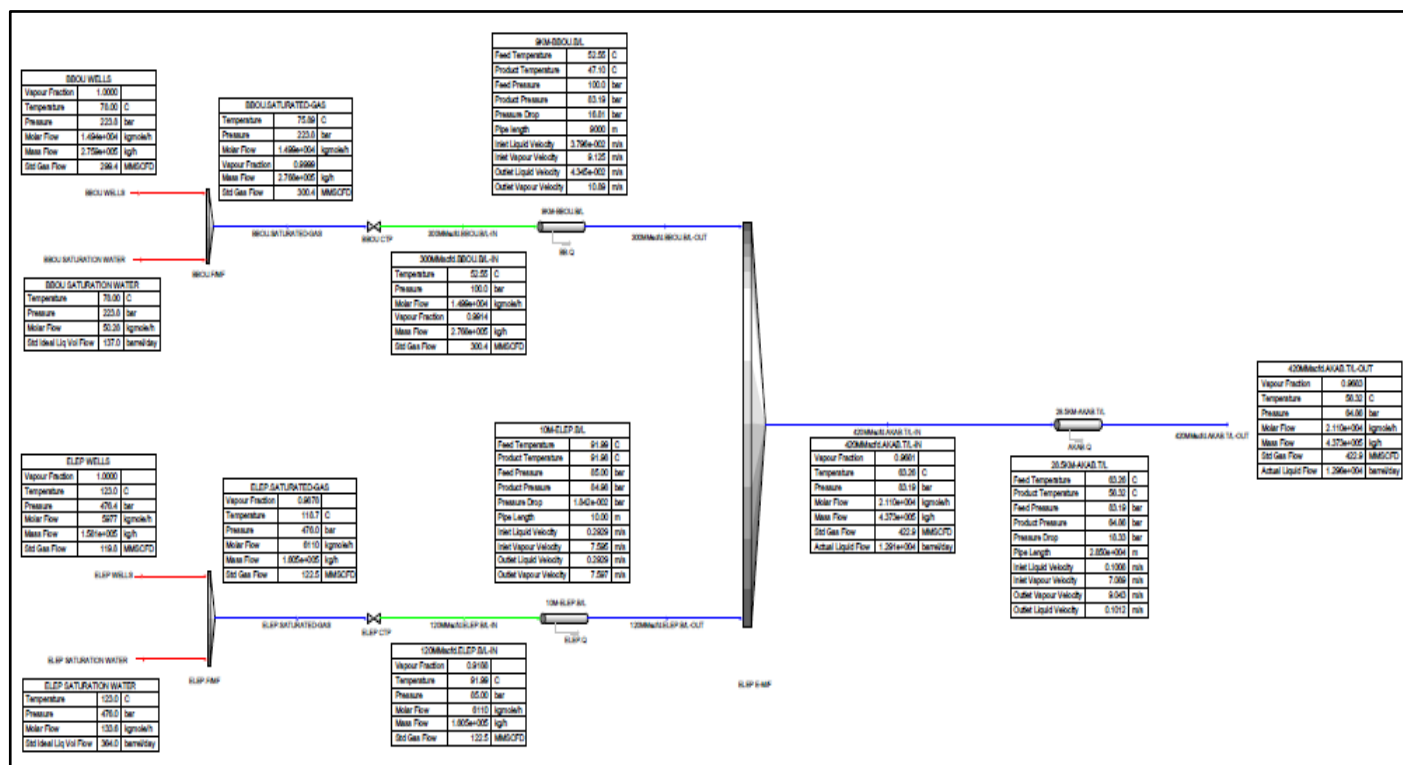


Figure 5.20: Phase-1/Scenario-3 Gas Production; Hysys PFD

5.3.2. Flow Schematics

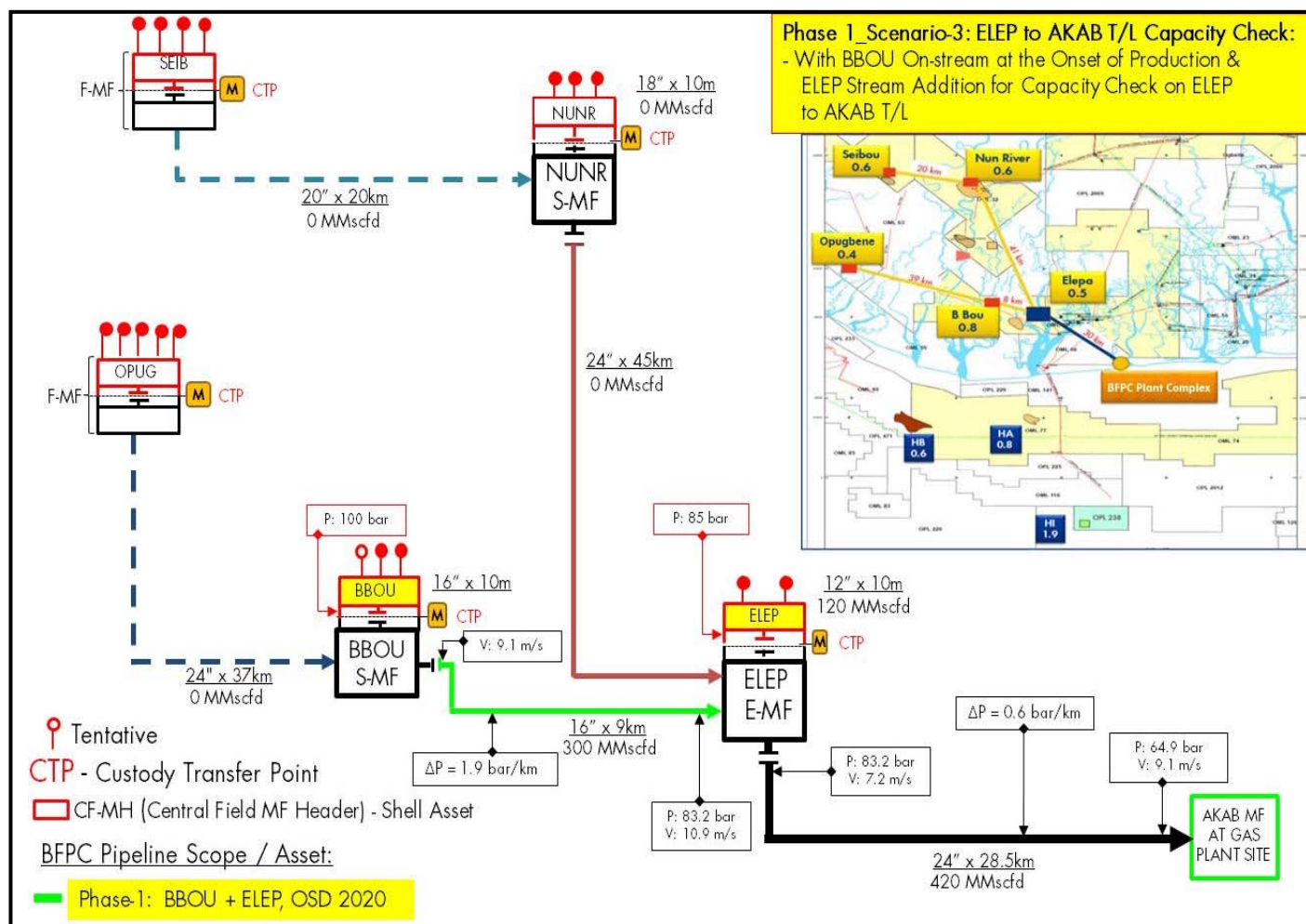


Figure 5.21: Phase-1/Scenario-3 Gas Production; Flow Schematics

5.3.3. Calculation Notes

i. BBOU to ELEP E-MF B/L

Table 5.11: Phase-1/Scenario-3; BBOU to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from BB to ELEP E-MF																				
Line Dimension		in	mm	m																
16" SCH 80 (ID)		14.33	364.00	0.36																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	100.00	52.55	276,837	3,424	57.07	0.95	80.85	266,784	3,410	56.83	0.95	78.24	0.02	0.85	9,673	13.80	0.23	0.004	701.04	0.35
Outlet	83.19	47.10	276,837	4,086	68.10	1.14	67.75	265,143	4,070	67.83	1.13	65.15	0.01	0.86	11,249	15.79	0.26	0.004	712.56	0.38
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _o ²	V _i ²	Overall Phase Momentu m (Pa)	Gas Phase Momentu m (Pa)	Liq. Phase Momentu m (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d² (π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X = m/m _g * (ρ _g /ρ _L)	Ve = c/(Vp-m)	Ve = c/(Vp-m)							
Inlet	0.10	9.14	9.10	0.04	83.47	82.79	0.001	6,749	6,478	0.95	0.0121	16.13	3.34							
Outlet	0.10	10.90	10.86	0.04	118.87	117.94	0.002	8,053	7,683	1.26	0.0128	17.62	3.64							

ii. ELEP F-MF to ELEP E-MF B/L

Table 5.12: Phase-1/Scenario-3; ELEP F-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from ELEP to ELEP E-MF																				
Line Dimension		in	mm	m																
12" SCH 80 (ID)		11.38	289.00	0.29																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	85.00	91.99	160,459	1,861	31.02	0.52	86.20	110,824	1,792	29.87	0.50	61.83	0.02	0.89	48,410	67.85	1.13	0.019	713.49	0.36
Outlet	84.98	91.98	160,459	1,862	31.03	0.52	86.18	110,824	1,793	29.88	0.50	61.82	0.02	0.89	48,410	67.85	1.13	0.019	713.51	0.36
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _o ²	V _i ²	Overall Phase Momentu m (Pa)	Gas Phase Momentu m (Pa)	Liq. Phase Momentu m (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d² (π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² overall	ρ _G * V _G ²	ρ _L * V _L ²	X _c m/m _g *(ρ _g /ρ _L)	Ve = c/(Vp-m)	Ve = c/(Vp-m)							
Inlet	0.07	7.88	7.59	0.29	62.05	57.54	0.082	5,349	3,558	58.85	0.1286	15.62	3.23							
Outlet	0.07	7.88	7.59	0.29	62.12	57.60	0.082	5,354	3,561	58.85	0.1286	15.62	3.23							

iii. ELEP E-MF to AKAB T/L

Table 5.13: Phase-1/Scenario-3; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
trunk-line from ELEP E-MF to AKABEL																				
Line Dimension		in	mm	m																
24" SCH 80 (ID)		21.57	548.00	0.55																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	83.19	63.26	437,297	6,098	101.63	1.69	71.71	374,365	6,012	100.20	1.67	62.27	0.01	0.88	60,972	83.50	1.39	0.023	730.21	0.45
Outlet	64.86	56.32	437,297	7,755	129.25	2.15	56.39	372,961	7,669	127.82	2.13	48.63	0.01	0.89	62,218	83.70	1.40	0.023	743.34	0.50
Calc. Parameters																				
	X-sectional Area (m ²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V ² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d ² (π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V _G ²	ρ _G * V _G ²	ρ _L * V _L ²	X = m _L /m _G *(ρ _L /ρ _G)	Ve = c/(Vρ _L)	Ve = c/(Vρ _G)							
Inlet	0.24	7.18	7.08	0.10	51.54	50.09	0.010	3,696	3,119	7.06	0.0476	17.12	3.54							
Outlet	0.24	9.13	9.03	0.10	83.35	81.51	0.010	4,700	3,964	7.22	0.0427	19.31	4.00							

5.3.4. Flow Parameters

i. BBOU to ELEP E-MF B/L

Table 5.14: Phase-1/Scenario-3; BBOU to ELEP E-MF Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (°C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	100.0000	52.5543	0.9914	<empty>	Segregated	0.0522	0.2095	<empty>	0.00	<empty>	0.000142	<empty>	25,486	17,135,874	0.0380	9.1255
1,800	0	96.8685	51.5761	0.9912	13.1999	Segregated	0.0519	0.2161	3.1787	0.00	0.00	0.000156	0.002288	25,960	17,293,328	0.0391	9.4055
3,600	0	93.6346	50.5485	0.9910	12.7105	Segregated	0.0515	0.2234	3.2862	0.00	0.00	0.000172	0.002528	26,373	17,459,097	0.0403	9.7157
5,400	0	90.2869	49.4652	0.9907	12.1954	Segregated	0.0510	0.2316	3.4062	0.00	0.00	0.000191	0.002815	26,725	17,634,126	0.0414	10.0616
7,200	0	86.8116	48.3187	0.9905	11.6512	Segregated	0.0504	0.2407	3.5413	0.00	0.00	0.000215	0.003161	27,012	17,819,568	0.0424	10.4507
9,000	0	83.1919	47.1001	0.9903	11.0740	Segregated	0.0496	0.2512	3.6949	0.00	0.00	0.000244	0.003588	27,233	18,016,807	0.0435	10.8924

ii. ELEP F-MF to ELEP E-MF B/L

Table 5.15: Phase-1/Scenario-3; ELEP F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (°C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	85.0000	91.9894	0.9188	<empty>	Intermittent	0.1375	0.2252	<empty>	0.00	<empty>	0.000137	<empty>	156,580	8,693,219	0.2929	7.5950
2	0	84.9963	91.9883	0.9188	29.6669	Intermittent	0.1375	0.2253	0.0037	0.00	0.00	0.000137	0.000002	156,576	8,693,303	0.2929	7.5953
4	0	84.9926	91.9871	0.9188	29.6663	Intermittent	0.1375	0.2253	0.0037	0.00	0.00	0.000137	0.000002	156,573	8,693,387	0.2929	7.5956
6	0	84.9889	91.9859	0.9188	29.6658	Intermittent	0.1375	0.2253	0.0037	0.00	0.00	0.000137	0.000002	156,570	8,693,472	0.2929	7.5960
8	0	84.9853	91.9847	0.9188	29.6653	Intermittent	0.1375	0.2253	0.0037	0.00	0.00	0.000137	0.000002	156,566	8,693,556	0.2929	7.5963
10	0	84.9816	91.9836	0.9188	29.6648	Intermittent	0.1375	0.2253	0.0037	0.00	0.00	0.000137	0.000002	156,563	8,693,640	0.2929	7.5966

iii. ELEP E-MF to AKAB T/L

Table 5.16: Phase-1/Scenario-3; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	83.1919	63.2618	0.9681	<empty>	Segregated	0.1018	0.0702	<empty>	0.00	<empty>	0.000031	<empty>	83,845	16,436,185	0.1008	7.0895
5,700	0	79.8546	62.0072	0.9681	22.4996	Segregated	0.0993	0.0731	3.4028	0.00	0.00	0.000035	0.001621	82,473	16,595,352	0.1010	7.3745
11,400	0	76.3770	60.6966	0.9681	21.7324	Segregated	0.0966	0.0763	3.5526	0.00	0.00	0.000040	0.001848	81,007	16,764,416	0.1011	7.6999
17,100	0	72.7379	59.3203	0.9681	20.9288	Segregated	0.0937	0.0800	3.7261	0.00	0.00	0.000046	0.002134	79,431	16,944,843	0.1012	8.0760
22,800	0	68.9106	57.8657	0.9682	20.0820	Segregated	0.0906	0.0844	3.9299	0.00	0.00	0.000054	0.002506	77,726	17,138,506	0.1013	8.5169
28,500	0	64.8599	56.3159	0.9683	19.1832	Segregated	0.0871	0.0896	4.1739	0.00	0.00	0.000064	0.003003	75,878	17,347,864	0.1012	9.0433

5.3.5. Pressure Profile Plots

i. BBOU to ELEP E-MF B/L

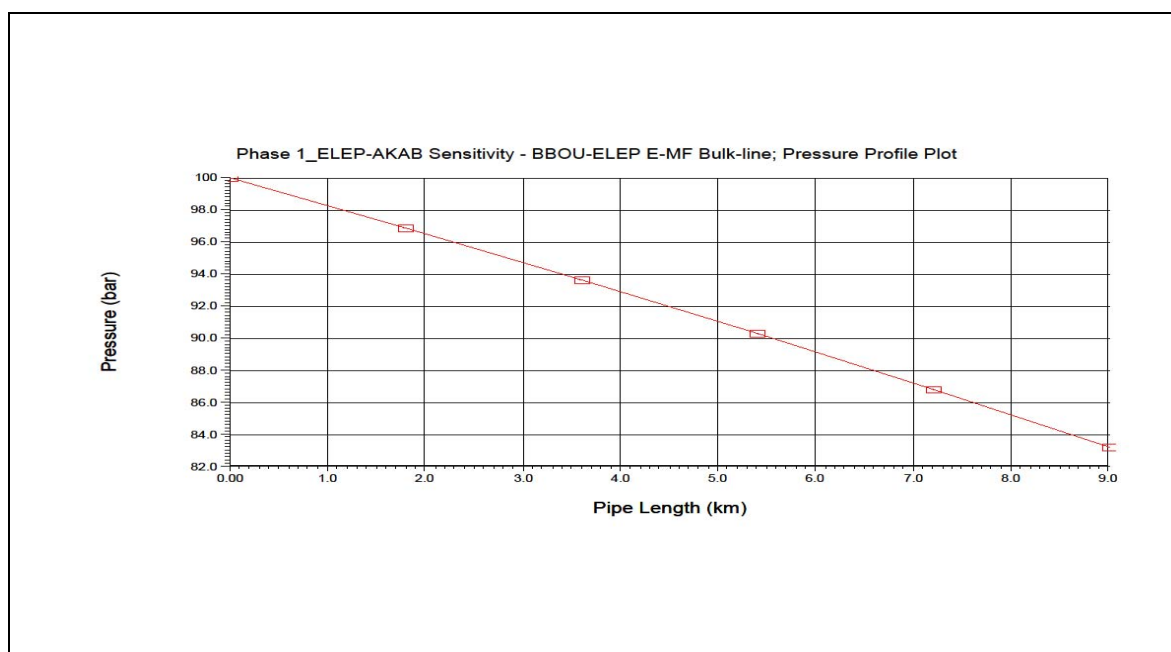


Figure 5.22: Phase-1/Scenario-3; BBOU to ELEP E-MF B/L Pressure Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

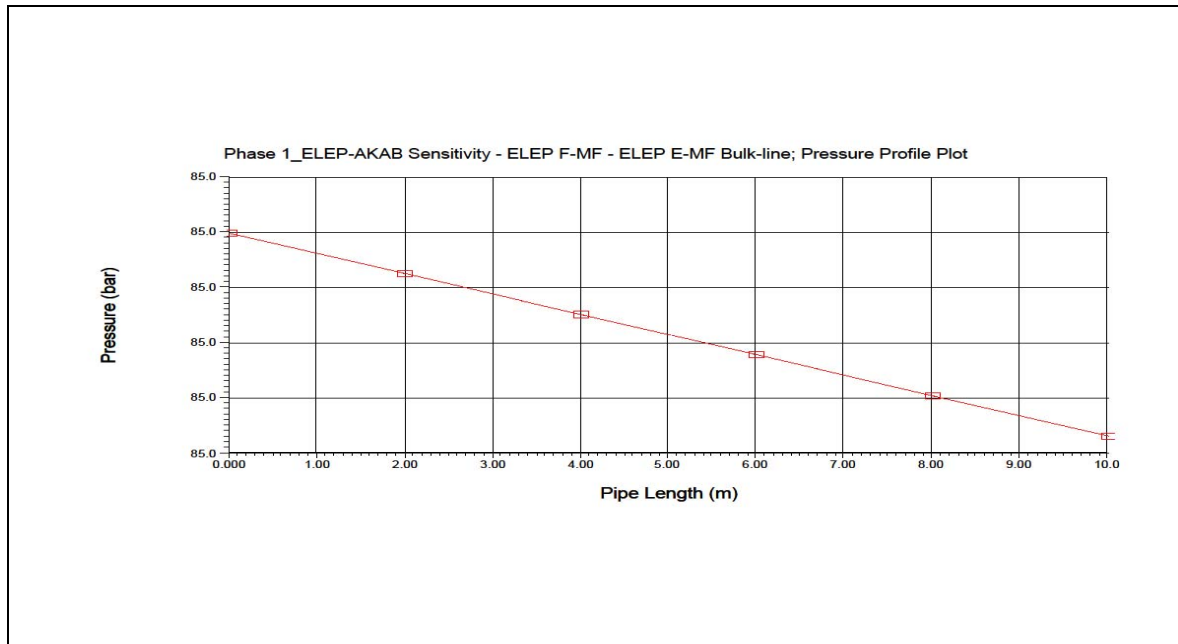


Figure 5.23: Phase-1/Scenario-3; ELEP F-MF to ELEP E-MF B/L Pressure Profile Plot

iii. ELEP E-MF to AKAB T/L

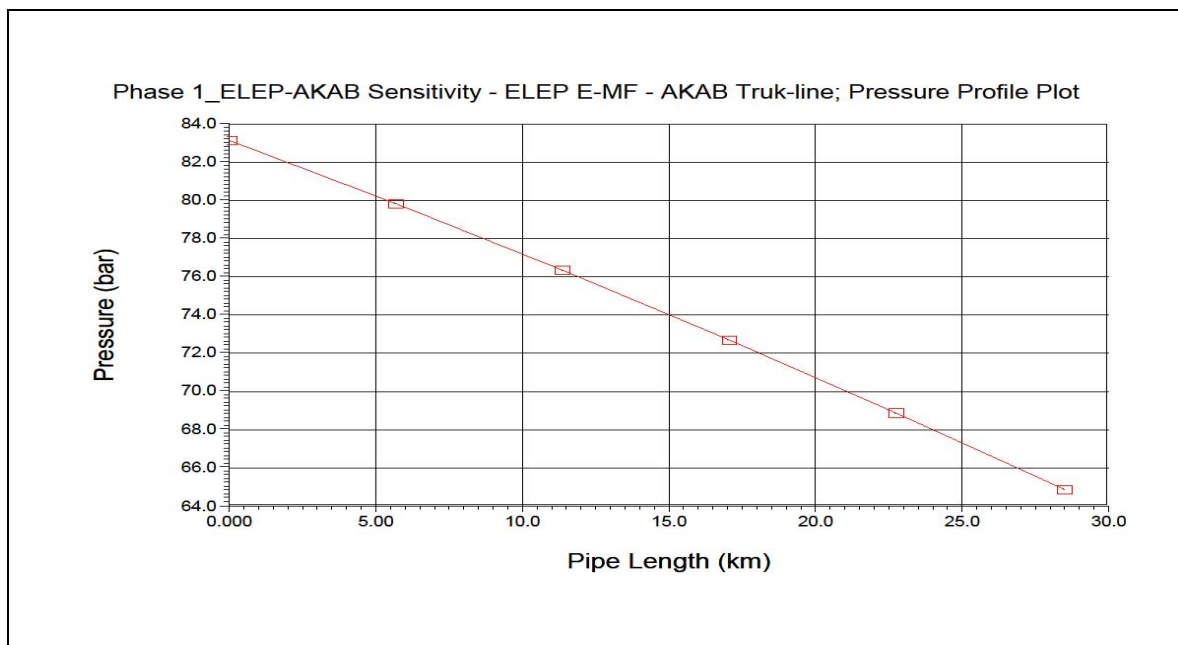


Figure 5.24: Phase-1/Scenario-3; ELEP E-MF to AKAB T/L Pressure Profile Plot

5.3.6. Velocity Profile Plots

i. BBOU to ELEP E-MF B/L

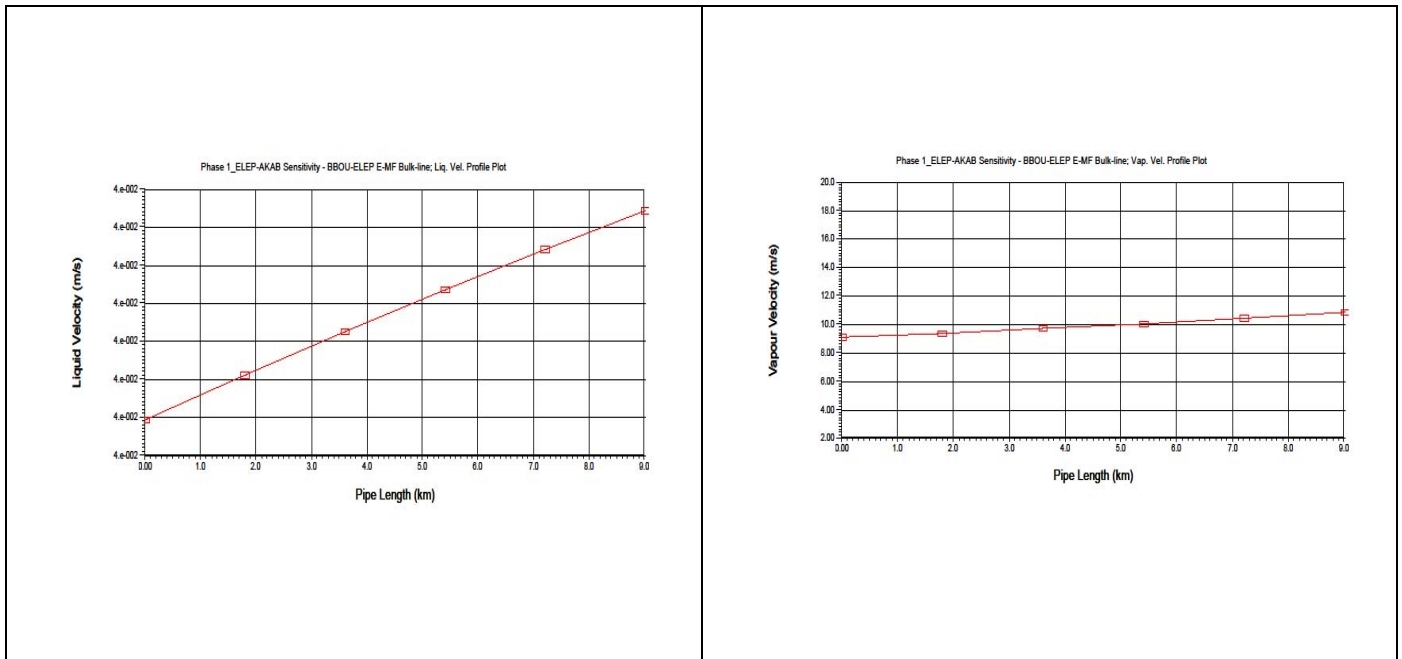


Figure 5.25: Phase-1/Scenario-3; BBOU to ELEP E-MF B/L Velocity Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

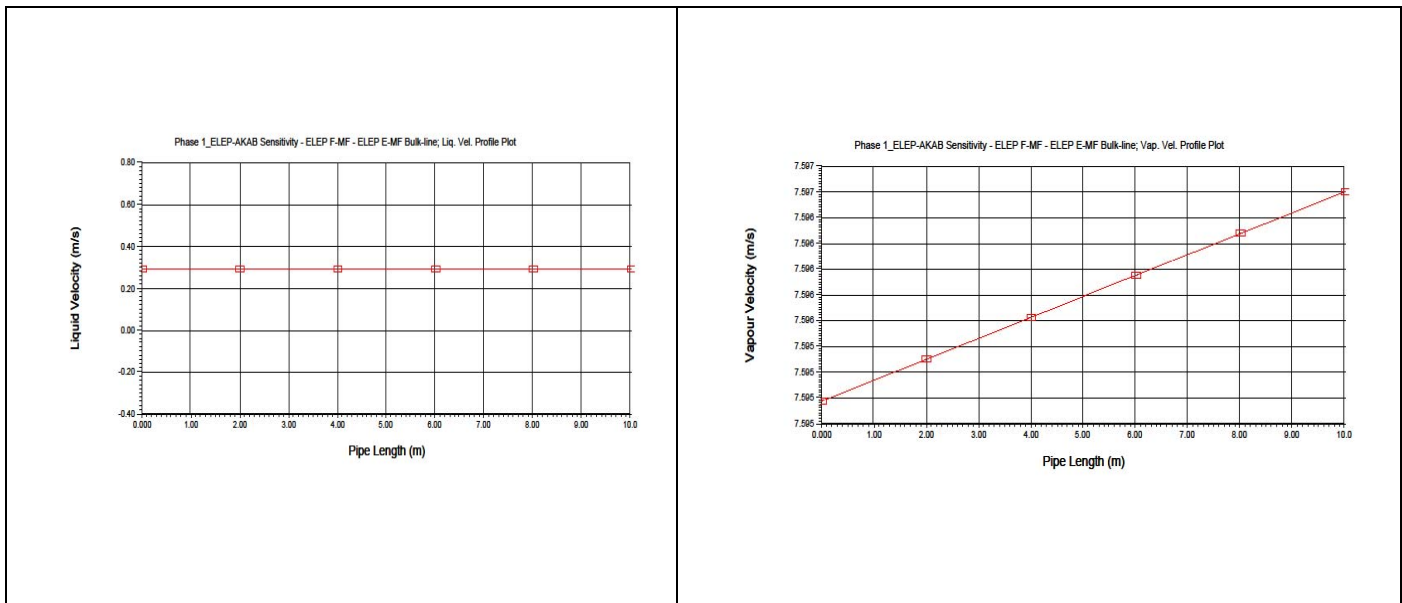


Figure 5.26: Phase-1/Scenario-3; ELEP F-MF to ELEP E-MF B/L Velocity Profile Plot

iii. ELEP E-MF to AKAB T/L

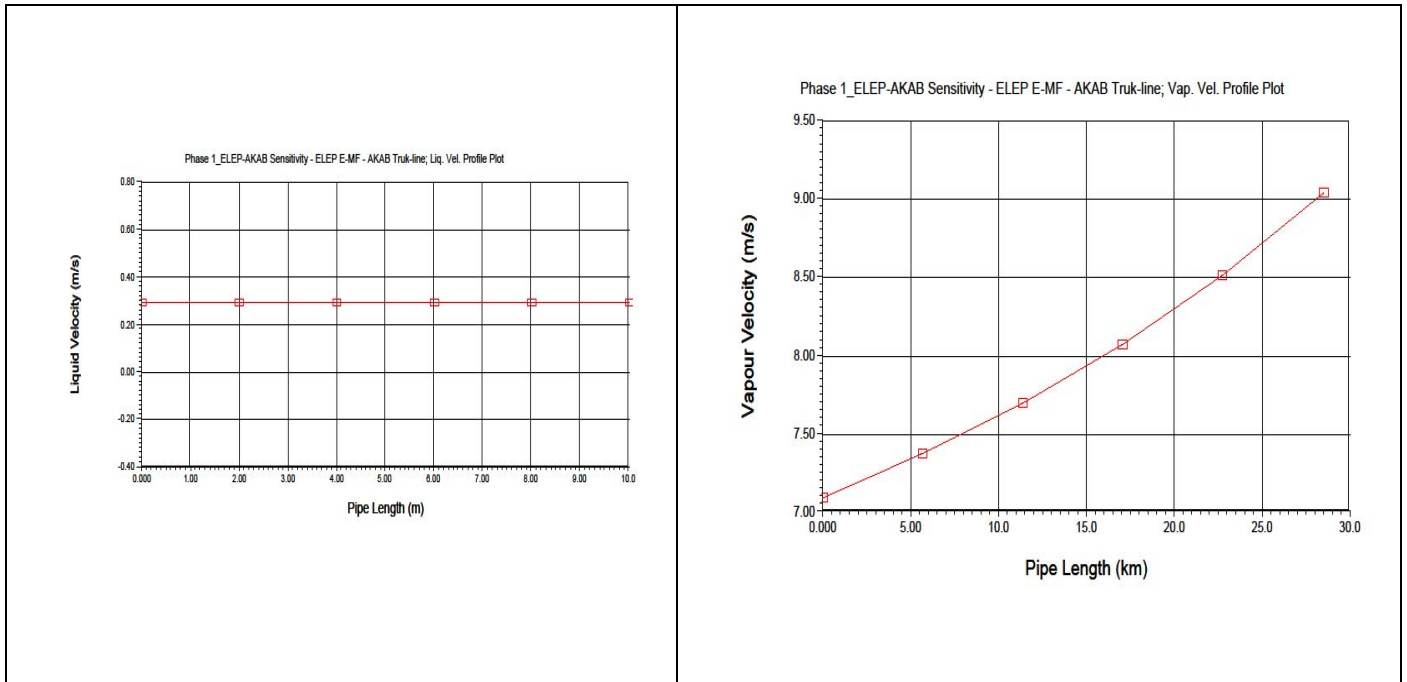


Figure 5.27: Phase-1/Scenario-3; ELEP E-MF to AKAB T/L Velocity Profile Plot

5.3.7. Phase Envelopes/Hydrate Curves

i. BBOU to ELEP E-MF B/L

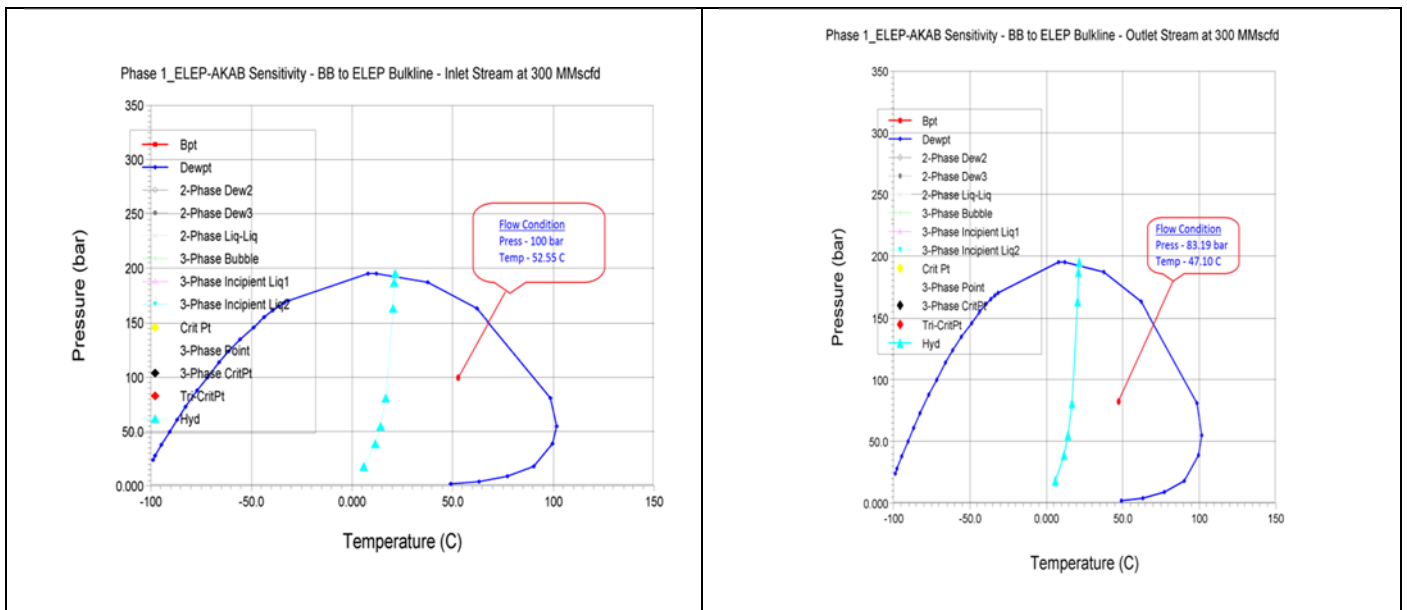


Figure 5.28: Phase-1/Scenario-3; BBOU to ELEP E-MF B/L Phase Envelope/Hydrate Curve

ii. ELEP F-MF to ELEP E-MF B/L

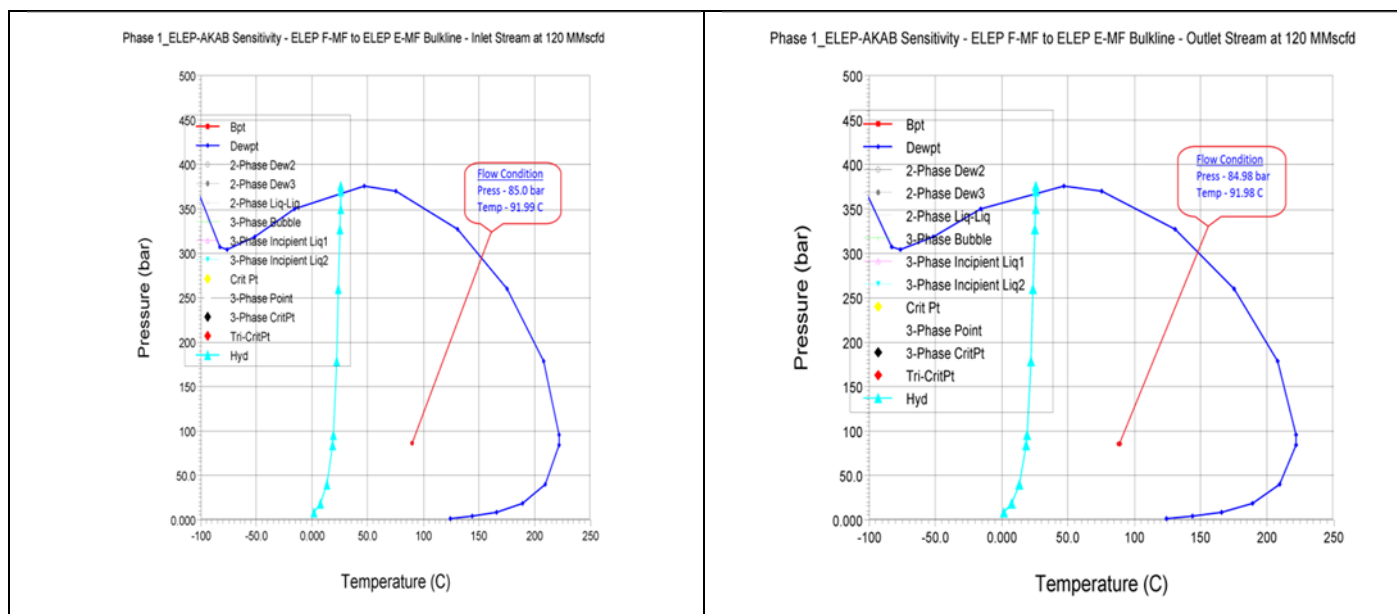


Figure 5.29: Phase-1/Scenario-3; ELEP F-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curve

iii. ELEP E-MF to AKAB T/L

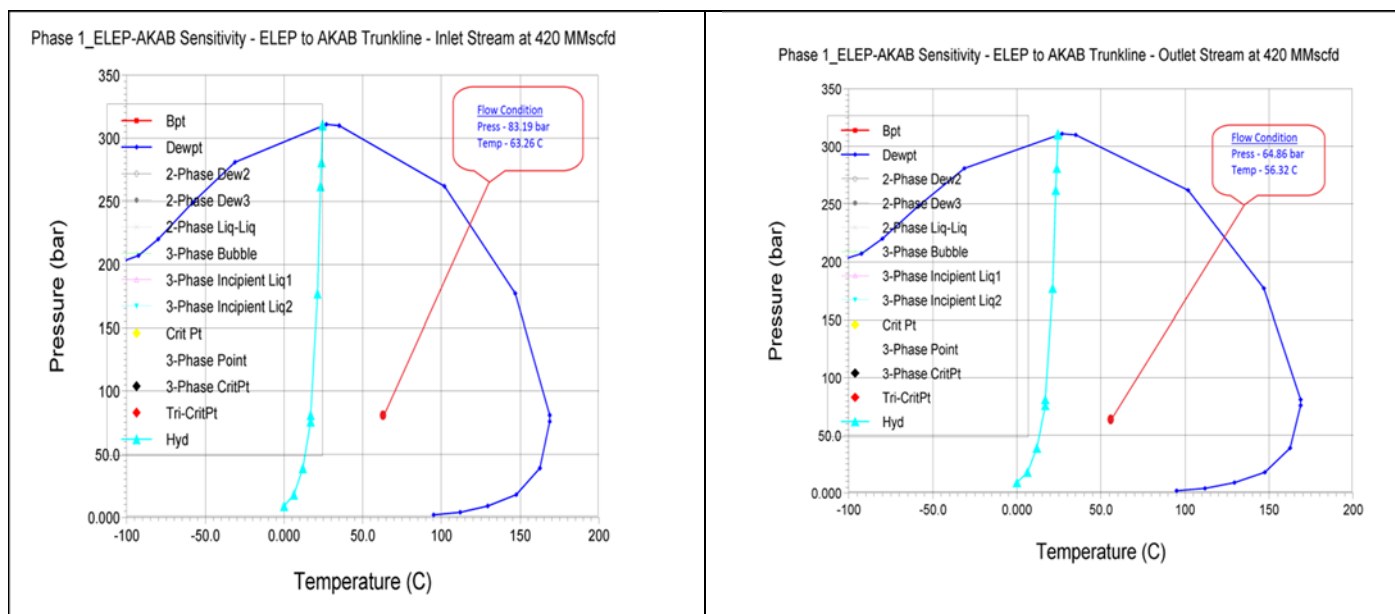


Figure 5.30: Phase-1/Scenario-3; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curve

6. Phase-2 Gas Production Simulation Results

At phase-2 gas delivery, NUNR field flow has come on-stream. The peak flow anticipated is 240 MMscfd of gas which this simulation is based on.

The sum-total 300 MMscfd of gas delivery gathering is anticipated from ELEP field flow at 60 MMscfd, and NUNR field flow at 240 MMscfd.

It is expected at the on-stream of NUNR field flow, that BBOU field flow would have been depleted.

The account of NUNR field flow delivery includes; NUNR F-MF to NUNR S-MF bulk-line flow and NUNR S-MF to ELEP E-MF trunk-line flow.

Note; the ELEP F-MF to ELEP E-MF bulk-line is an existing 12" rated pipeline. The study only reviews flow-conditions under the 60 MMscfd flow.

i. NUNR to NUNR S-MF B/L

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 240 MMscfd of gas flow with minimal pressure loss across the 10 m NUNR F-MF – NUNR S-MF bulk-line is 93.5 bar. The corresponding temperature is 81.8 °C.

Delivery Pressure; at NUNR S-MF is 93.5 bar, which is sufficient to support NUNR S-MF to ELEP E-MF trunk-line flow. The corresponding temperature is 81.8 °C.

The Line Size; which also supported 240 MMscfd of gas flow across the 10 m NUNR F-MF – NUNR S-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity boundary is 18". And the pressure drop is 0.0 bar.

See below; fig. 6.2, table 6.1, table 6.5, & fig. 6.3 for more details

Flow Velocity Profile

Minimum Fluid Flow Velocity; at the inlet and outlet is 2.6 m/s.

Prevailing Fluid Flow Velocity; at both inlet and outlet is 6.3 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at both inlet and outlet is 12.5 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 6.2, table 6.1, table 6.5, & fig. 6.7 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 93.5 bar, the formation temperature is 20.0 °C. While the prevailing flow conditions at the inlet is 93.5 bar pressure and 81.8 °C temperature.

Similarly for outlet stream, at stream pressure of 93.5 bar, the formation temperature is also 19.9 °C. While the prevailing flow conditions at the outlet are; pressure 93.5 bar and temperature 81.8 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 6.11 for more details

ii. NUNR S-MF to ELEP E-MF B/L***Pressure and Temperature Regime***

Source Pressure at NUNR S-MF; is 93.5 bar which satisfy a good head flow for 240 MMscfd of gas across 45 km NUNR S-MF to ELEP E-MF bulk-line. The corresponding temperature is 81.8 °C.

Delivery Pressure; at ELEP E-MF is 80.8 bar, which is sufficient to support ELEP E-MF to AKAB bulk-line flow. The corresponding temperature is 74.9 °C.

The Line Size; which supported 240 MMscfd of gas flow across the 45 km NUNR S-MF to ELEP E-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity boundary is 24". And the pressure drop is 0.28 bar/km.

See below; fig. 6.2, table 6.2, table 6.6, & fig. 6.4 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.6 m/s; while at the outlet is 2.7 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 3.5 m/s; while at the outlet is 4.0 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 12.5 m/s, and outlet is 13.3 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 6.2, table 6.2, table 6.6, & fig. 6.8 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 93.5 bar, the formation temperature is 19.9 °C. While the prevailing flow conditions at the inlet is 93.5 bar pressure and 81.8 °C temperature.

Similarly for outlet stream, at stream pressure of 80.8 bar, the formation temperature is also 19.1 °C. While the prevailing flow conditions at the outlet are; pressure 80.8 bar and temperature 74.9 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 6.12 for more details

iii. ELEP F-MF to ELEP E-MF B/L

Pressure and Temperature Regime

Source Pressure at the CTP; is 80 bar which satisfy a good head flow for 60 MMscfd of gas across 10m ELEP F-MF - ELEP E-MF bulk-line. The corresponding temperature is 91.7 °C.

Delivery Pressure; at ELEP E-MF is 80.0 bar and with a corresponding temperature of 91.7 °C.

The Line Size; existing 12" x 10m ELEP F-MF to ELEP E-MF bulk-line was employed in reviewing the flow condition for a 60 MMscfd flow. The pressure drop is 0.0 bar.

See below; fig. 6.2, table 6.3, table 6.7, & fig. 6.5 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at both inlet and outlet is 3.3 m/s.

Prevailing Fluid Velocity; at both inlet and outlet is 4.2 m/s.

Erosional Velocity; at both inlet and outlet is 16.0 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 6.2, table 6.3, table 6.7, & fig. 6.9 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.0 bar, the formation temperature is 17.9 °C. While the prevailing flow conditions at the inlet is 80.0 bar pressure and 91.7 °C temperature.

Similarly for outlet stream, at stream pressure of 80.0 bar, the formation temperature is 17.9 °C. While the prevailing flow conditions at the outlet are; pressure at 80.0 bar and temperature at 91.7 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 6.13 for more details

*iv. ELEP E-MF to AKAB T/L****Pressure and Temperature Regime***

Source Pressure at the ELEP E-MF Outlet; the mixed NUNR field flow and ELEP field flow exited ELEP E-MF at a pressure of 80.0 bar and with a corresponding temperature of 77.6 °C.

Delivery Pressure; at Akabel is 65.2 bar and with a corresponding temperature of 72.1 °C.

The Line Size; existing 24" x 28.5 km ELEP E-MF to AKAB trunk-line was employed in reviewing the flow condition for the 300MMscfd flow. The pressure drop is 0.5 bar/km.

See below; fig. 6.2, table 6.4, table 6.8, & fig. 6.6 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.9 m/s; while at the outlet is 3.2 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 5.2 m/s; while at the outlet is 6.3 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 13.8 m/s, and outlet is 15.2 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 6.2, table 6.4, table 6.8, & fig. 6.10 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.0 bar, the formation temperature is 18.9 °C. While the prevailing flow conditions at the inlet is 80.0 bar pressure and 77.6 °C temperature.

Similarly for outlet stream, at stream pressure of 65.1 bar, the formation temperature is 17.7 °C. While the prevailing flow conditions at the outlet are; pressure at 65.2 bar and temperature at 72.1 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 6.14 for more details

6.1. Hysys PFD

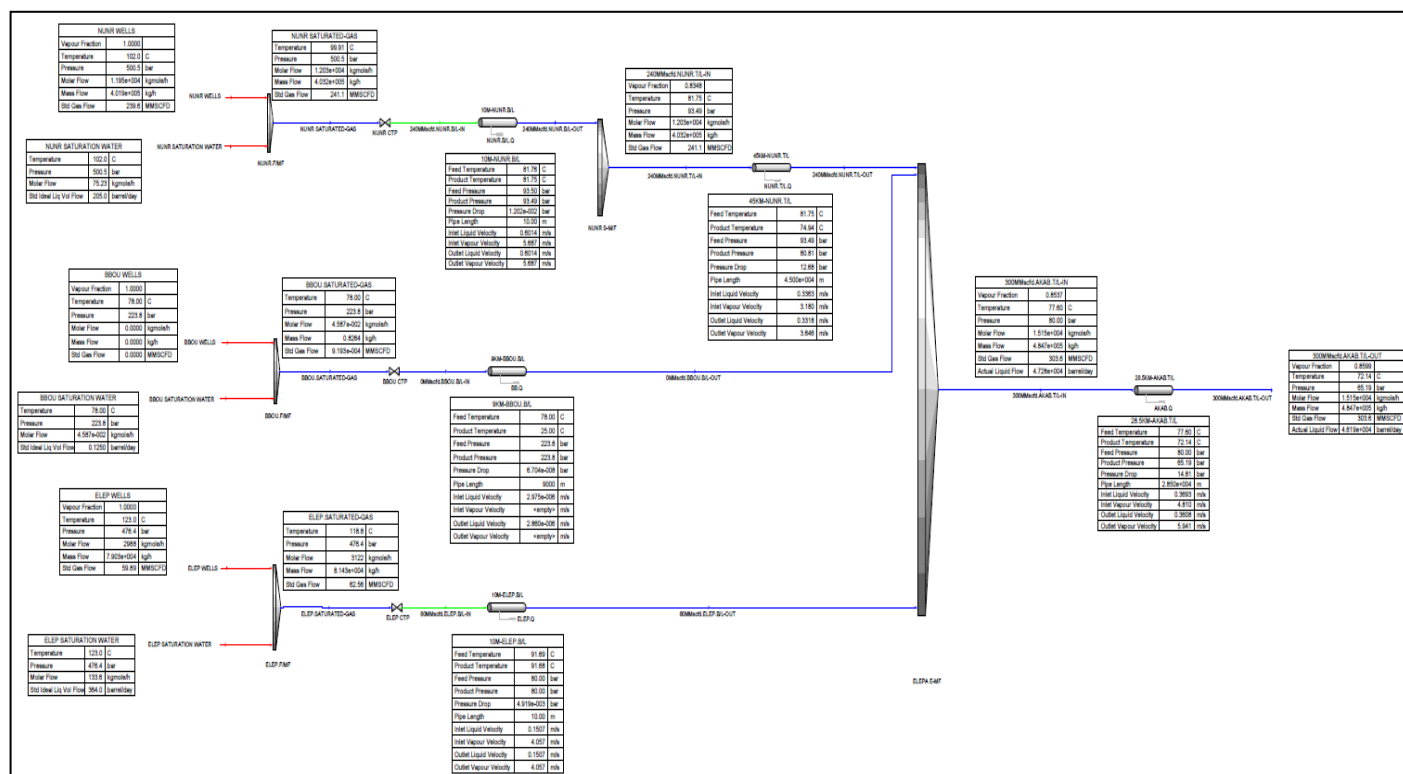


Figure 6.1: Phase-2 Gas Production; Hysys PFD

6.2. Flow Schematics

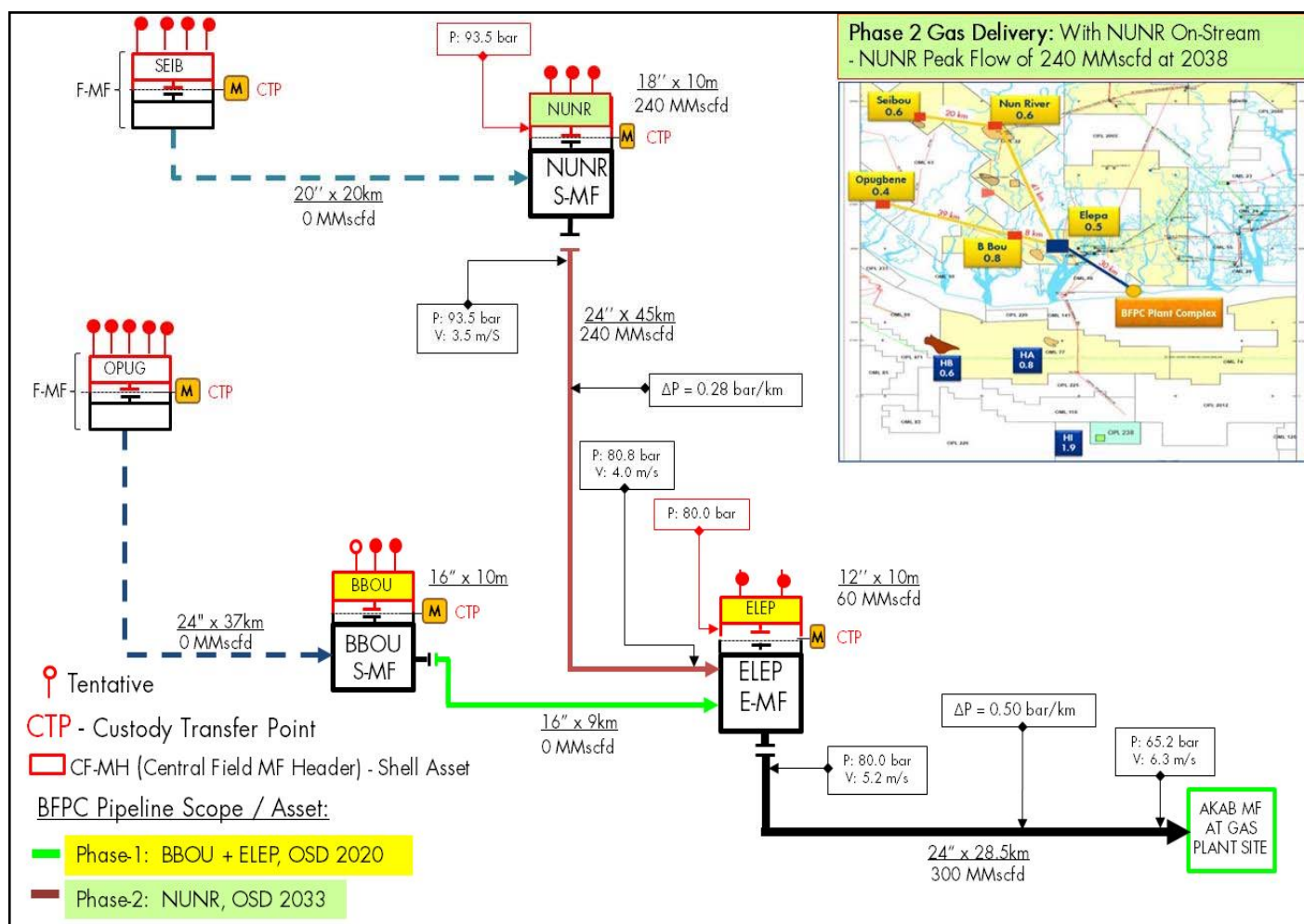


Figure 6.2: Phase-2 Gas Production; Flow Schematics

6.3. Calculation Notes

i. NUNR F-MF to NUNR S-MF B/L

Table 6.1: Phase-2; NUNR F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from NUNR to NUNR S-MF																				
Line Dimension 18" SCH 80 (ID)	in	mm	m																	
	16.14	410.00	0.41																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	93.50	81.76	403,228	2,982	49.70	0.83	135.22	208,671	2,697	44.95	0.75	77.38	0.02	0.85	194,556	285.20	4.75	0.079	682.10	0.32
Outlet	93.49	81.75	403,228	2,982	49.70	0.83	135.20	208,672	2,697	44.95	0.75	77.37	0.02	0.85	194,556	285.20	4.75	0.079	682.11	0.32
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _o * V² overall	ρ _G * V _G ²	ρ _L * V _L ²	X _c (m/m _G)*(ρ _G /ρ _L)	Ve = c/(Vp _m)	Ve = c/(Vp _n)							
Inlet	0.13	6.27	5.67	0.60	39.33	32.17	0.360	5,318	2,489	245.40	0.3140	12.47	2.58							
Outlet	0.13	6.27	5.67	0.60	39.33	32.17	0.360	5,318	2,489	245.41	0.3140	12.47	2.58							

ii. NUNR S-MF to ELEP E-MF B/L

Table 6.2: Phase-2; NUNR S-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Trunk-line from NUNR S-MF to ELEP E-MF																				
Line Dimension 24" SCH 80 (ID)	in	mm	m																	
	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	93.49	81.75	403,228	2,982	49.70	0.83	135.20	208,672	2,697	44.95	0.75	77.37	0.02	0.85	194,556	285.20	4.75	0.079	682.11	0.32
Outlet	80.81	74.94	403,228	3,374	56.23	0.94	119.53	207,776	3,092	51.53	0.86	67.19	0.02	0.86	195,313	281.20	4.69	0.078	694.48	0.35
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² overall	ρ _G * V _G ²	ρ _L * V _L ²	X = m _G /m _L (ρ _L /ρ _G)	Ve = c/(Vp _{min})	Ve = c/(Vp _{min})							
Inlet	0.24	3.51	3.18	0.34	12.32	10.08	0.113	1,666	780	76.90	0.3140	12.47	2.58							
Outlet	0.24	3.97	3.64	0.33	15.78	13.25	0.110	1,886	890	76.11	0.2924	13.26	2.74							

iii. ELEP F-MF to ELEP E-MF B/L

Table 6.3: Phase-2; ELEP F-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from ELEP to ELEP E-MF																				
Line Dimension 12" SCH 80 (ID)		in	mm	m																
		11.38	289.00	0.29																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	80.00	91.69	81,433	993	16.55	0.28	82.01	55,479	957	15.96	0.27	57.95	0.02	0.90	24,159	33.70	0.56	0.009	716.86	0.37
Outlet	80.00	91.68	81,433	993	16.55	0.28	82.00	55,479	958	15.96	0.27	57.94	0.02	0.90	24,159	33.70	0.56	0.009	716.87	0.37
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q)/A	V = (Q)/A	V = (Q)/A				ρ _G * V² _{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X = m _G /m _L (ρ _G /ρ _L)	Ve = c/(Vρ _G)	Ve = c/(Vρ _L)							
Inlet	0.07	4.20	4.05	0.14	17.67	16.42	0.020	1,449	952	14.59	0.1238	16.01	3.31							
Outlet	0.07	4.20	4.05	0.14	17.67	16.43	0.020	1,449	952	14.59	0.1238	16.01	3.31							

iv. ELEP E-MF to AKAB T/L

Table 6.4: Phase-2; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
Trunk-line from ELEP E-MF to AKABEL																				
Line Dimension 24" SCH 80 (ID)	in	mm	m																	
	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	80.00	77.60	484,661	4,392	73.20	1.22	110.35	264,909	4,079	67.98	1.13	64.95	0.02	0.87	217,713	311.10	5.19	0.086	699.82	0.36
Outlet	65.19	72.14	484,661	5,344	89.07	1.48	90.69	265,788	5,038	83.97	1.40	52.75	0.01	0.88	216,721	303.70	5.06	0.084	713.51	0.40
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$\rho_G \cdot V_{Overall}^2$	$\rho_G \cdot V_G^2$	$\rho_L \cdot V_L^2$	$X = m_G/m_L \sqrt{\rho_G/\rho_L}$	$Ve = c/(Vp_{min})$	$Ve = c/(Vp_{min})$							
Inlet	0.24	5.17	4.80	0.37	26.73	23.06	0.134	2,950	1,498	93.87	0.2504	13.80	2.86							
Outlet	0.24	6.29	5.93	0.36	39.58	35.18	0.128	3,589	1,856	91.21	0.2217	15.23	3.15							

6.4. Flow Parameters

i. NUNR F-MF to NUNR S-MF B/L

Table 6.5: Phase-2; NUNR F-MF to NUNR S-MF Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	93.5	81.8	0.8348	<empty>	Intermittent	0.2313	0.1469	<empty>	0.00	<empty>	0.000076	<empty>	521,786	11,312,216	0.60	5.69
2	0	93.5	81.8	0.8348	29.23	Intermittent	0.2313	0.1469	0.0024	0.00	0.00	0.000076	0.000001	521,779	11,312,313	0.60	5.69
4	0	93.5	81.8	0.8348	29.23	Intermittent	0.2313	0.1470	0.0024	0.00	0.00	0.000076	0.000001	521,771	11,312,411	0.60	5.69
6	0	93.5	81.8	0.8348	29.23	Intermittent	0.2313	0.1470	0.0024	0.00	0.00	0.000076	0.000001	521,763	11,312,507	0.60	5.69
8	0	93.5	81.8	0.8348	29.23	Intermittent	0.2313	0.1470	0.0024	0.00	0.00	0.000076	0.000001	521,756	11,312,606	0.60	5.69
10	0	93.5	81.8	0.8348	29.23	Intermittent	0.2313	0.1470	0.0024	0.00	0.00	0.000076	0.000001	521,748	11,312,702	0.60	5.69

ii. NUNR S-MF to ELEP E-MF B/L

Table 6.6: Phase-2; NUNR S-MF to ELEP E-MF Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	93.5	81.8	0.8348	<empty>	Transition	0.2517	0.0325	<empty>	0.00	<empty>	0.000005	<empty>	390,162	8,459,605	0.34	3.18
9,000	0	91.1	80.3	0.8354	33.50	Transition	0.2479	0.0332	2.4403	0.00	0.00	0.000006	0.000415	383,591	8,527,077	0.34	3.26
18,000	0	88.6	79.0	0.8360	32.67	Transition	0.2439	0.0340	2.4975	0.00	0.00	0.000006	0.000448	376,959	8,597,269	0.33	3.34
27,000	0	86.1	77.6	0.8365	31.86	Transition	0.2397	0.0348	2.5605	0.00	0.00	0.000007	0.000486	370,244	8,670,198	0.33	3.43
36,000	0	83.5	76.3	0.8372	31.06	Transition	0.2352	0.0358	2.6301	0.00	0.00	0.000007	0.000529	363,419	8,747,060	0.33	3.53
45,000	0	80.8	74.9	0.8380	30.26	Transition	0.2304	0.0368	2.7072	0.00	0.00	0.000008	0.000581	356,460	8,828,243	0.33	3.65

iii. ELEP F-MF to ELEP E-MF B/L

Table 6.7: Phase-2; ELEP F-MF to ELEP E-MF Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	80.0	91.7	0.9008	<empty>	Transition	0.1577	0.0602	<empty>	0.00	<empty>	0.000011	<empty>	70,899	4,404,160	0.15	4.06
2	0	80.0	91.7	0.9008	29.53	Transition	0.1577	0.0602	0.0010	0.00	0.00	0.000011	0.000000	70,898	4,404,171	0.15	4.06
4	0	80.0	91.7	0.9008	29.53	Transition	0.1577	0.0602	0.0010	0.00	0.00	0.000011	0.000000	70,898	4,404,183	0.15	4.06
6	0	80.0	91.7	0.9008	29.53	Transition	0.1577	0.0602	0.0010	0.00	0.00	0.000011	0.000000	70,897	4,404,194	0.15	4.06
8	0	80.0	91.7	0.9008	29.53	Transition	0.1577	0.0602	0.0010	0.00	0.00	0.000011	0.000000	70,896	4,404,206	0.15	4.06
10	0	80.0	91.7	0.9008	29.53	Transition	0.1577	0.0602	0.0010	0.00	0.00	0.000011	0.000000	70,896	4,404,218	0.15	4.06

iv. ELEP E-MF to AKAB T/L

Table 6.8: Phase-2; ELEP E-MF to AKAB Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	80.0	77.6	0.8537	<empty>	Intermittent	0.2000	0.0579	<empty>	0.00	<empty>	0.000020	<empty>	384,389	11,254,037	0.37	4.81
5,700	0	77.3	76.6	0.8548	31.14	Intermittent	0.1959	0.0598	2.7877	0.00	0.00	0.000022	0.001028	377,153	11,361,210	0.37	4.98
11,400	0	74.4	75.5	0.8559	30.52	Intermittent	0.1915	0.0620	2.8891	0.00	0.00	0.000025	0.001150	369,675	11,474,484	0.37	5.18
17,100	0	71.5	74.4	0.8571	29.89	Intermittent	0.1869	0.0645	3.0037	0.00	0.00	0.000028	0.001297	361,901	11,594,672	0.36	5.40
22,800	0	68.4	73.3	0.8584	29.22	Intermittent	0.1820	0.0673	3.1344	0.00	0.00	0.000032	0.001481	353,796	11,722,788	0.36	5.65
28,500	0	65.2	72.1	0.8599	28.53	Intermittent	0.1766	0.0705	3.2854	0.00	0.00	0.000037	0.001712	345,309	11,860,120	0.36	5.94

6.5. Pressure Profile Plots

i. NUNR F-MF to NUNR S-MF B/L

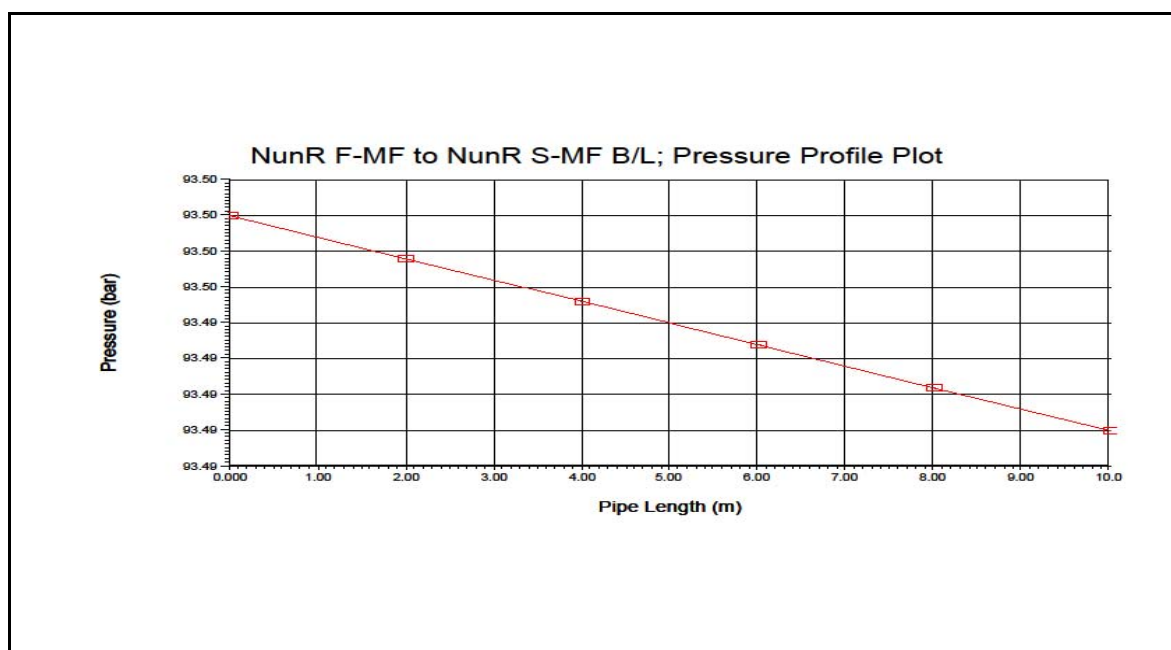


Figure 6.3: Phase-2; NUNR F-MF to NUNR S-MF Pressure Profile Plot

ii. NUNR S-MF to ELEP E-MF B/L

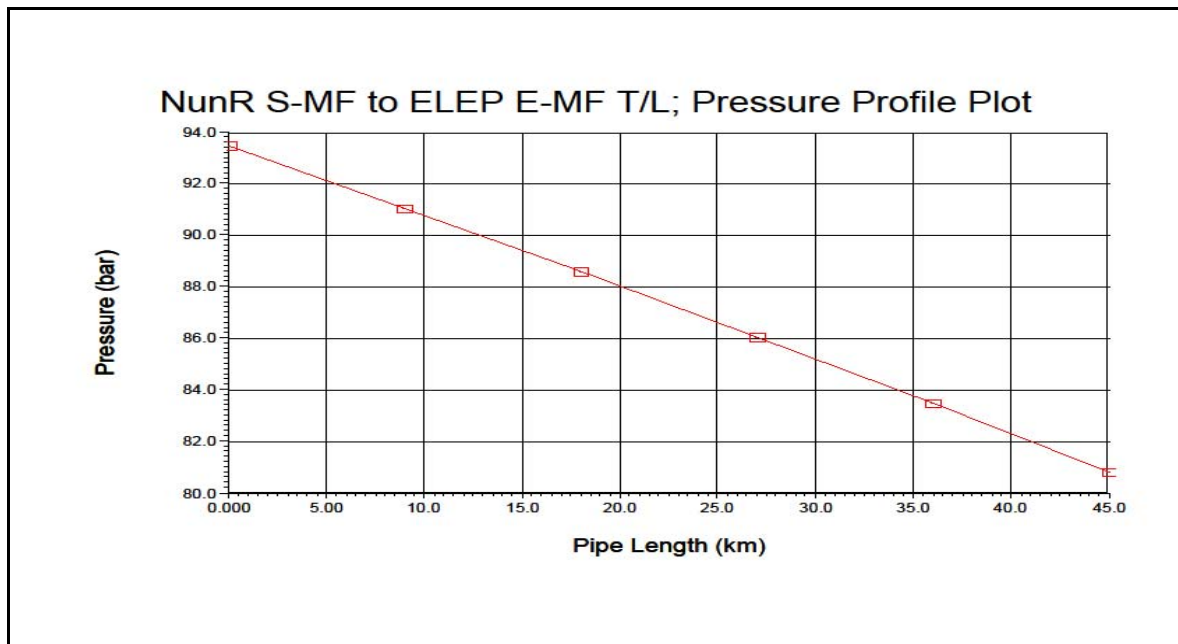


Figure 6.4: Phase-2; NUNR S-MF to ELEP E-MF Pressure Profile Plot

iii. ELEP F-MF to ELEP E-MF B/L

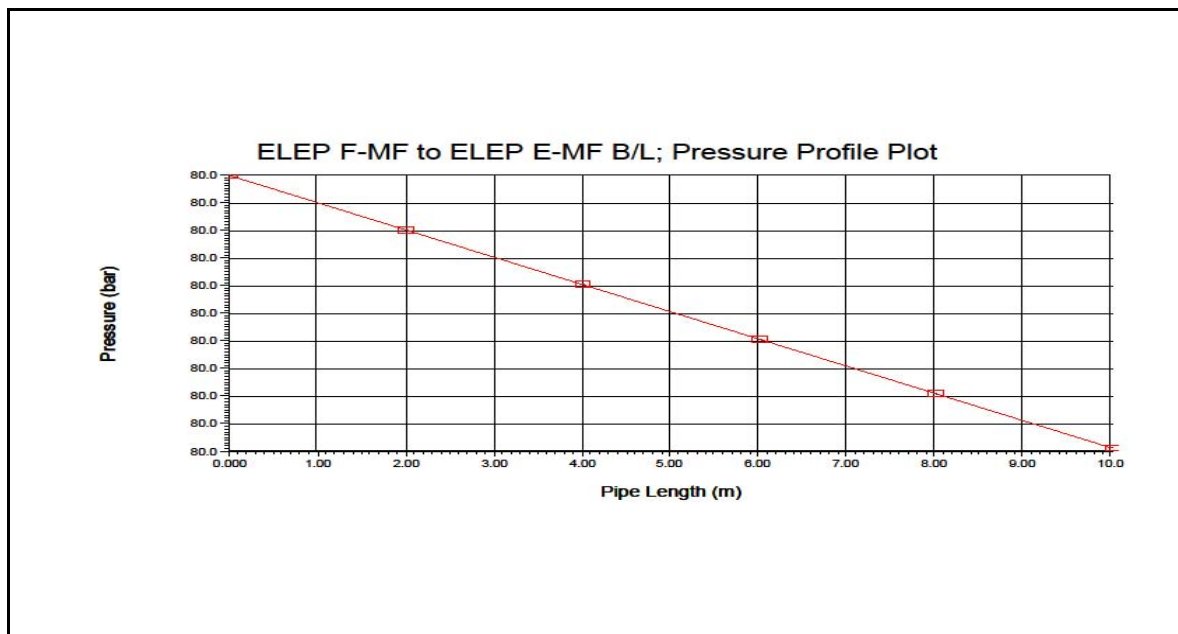


Figure 6.5: Phase-2; ELEP F-MF to ELEP E-MF Pressure Profile Plot

iv. ELEP E-MF to AKAB T/L

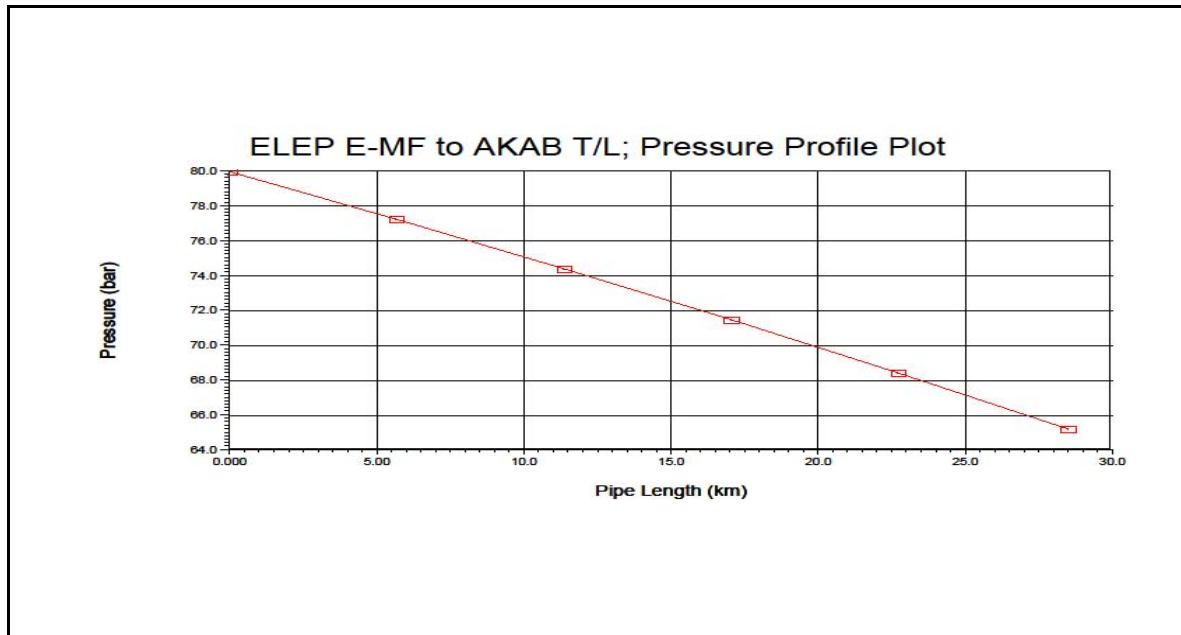


Figure 6.6: Phase-2; ELEP E-MF to AKAB Pressure Profile Plot

6.6. Velocity Profile Plot

i. NUNR F-MF to NUNR S-MF B/L

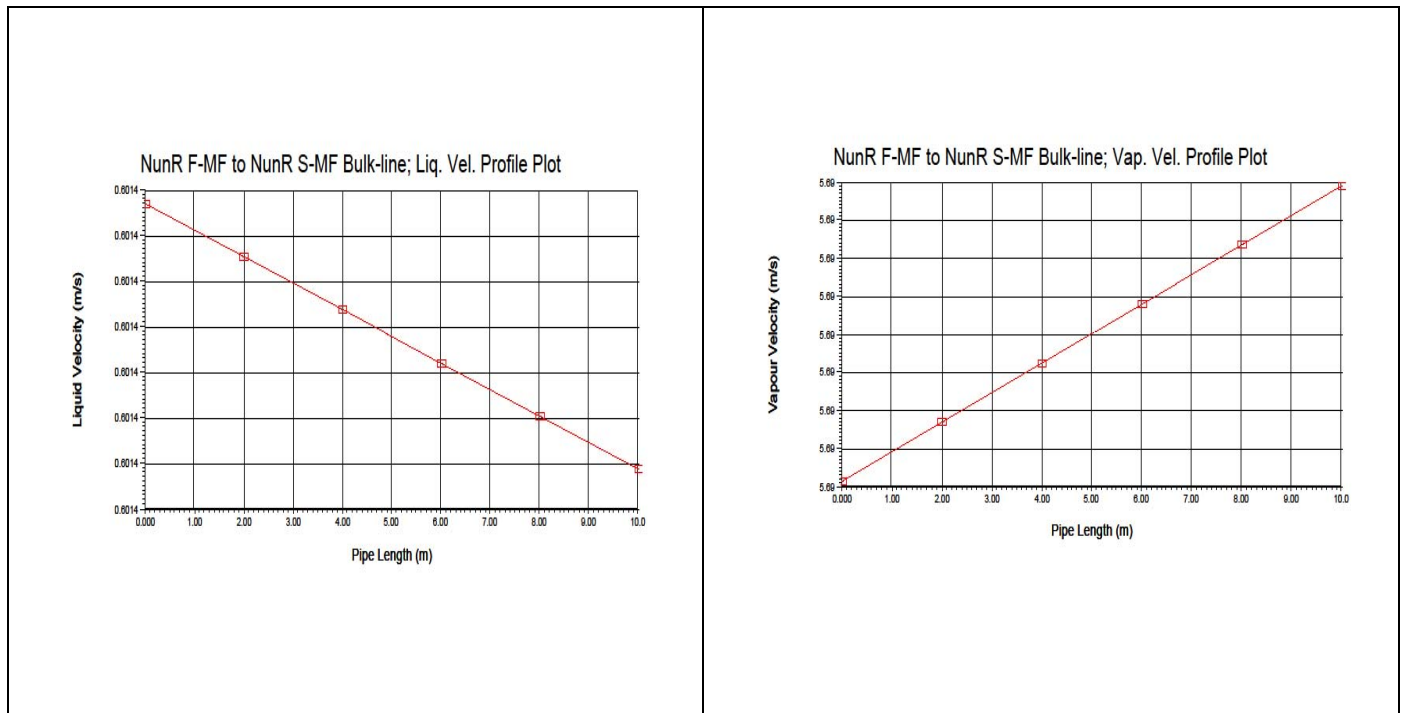


Figure 6.7: Phase-2; NunR F-MF to NunR S-MF B/L Velocity Profile Plot

ii. NUNR S-MF to ELEP E-MF T/L

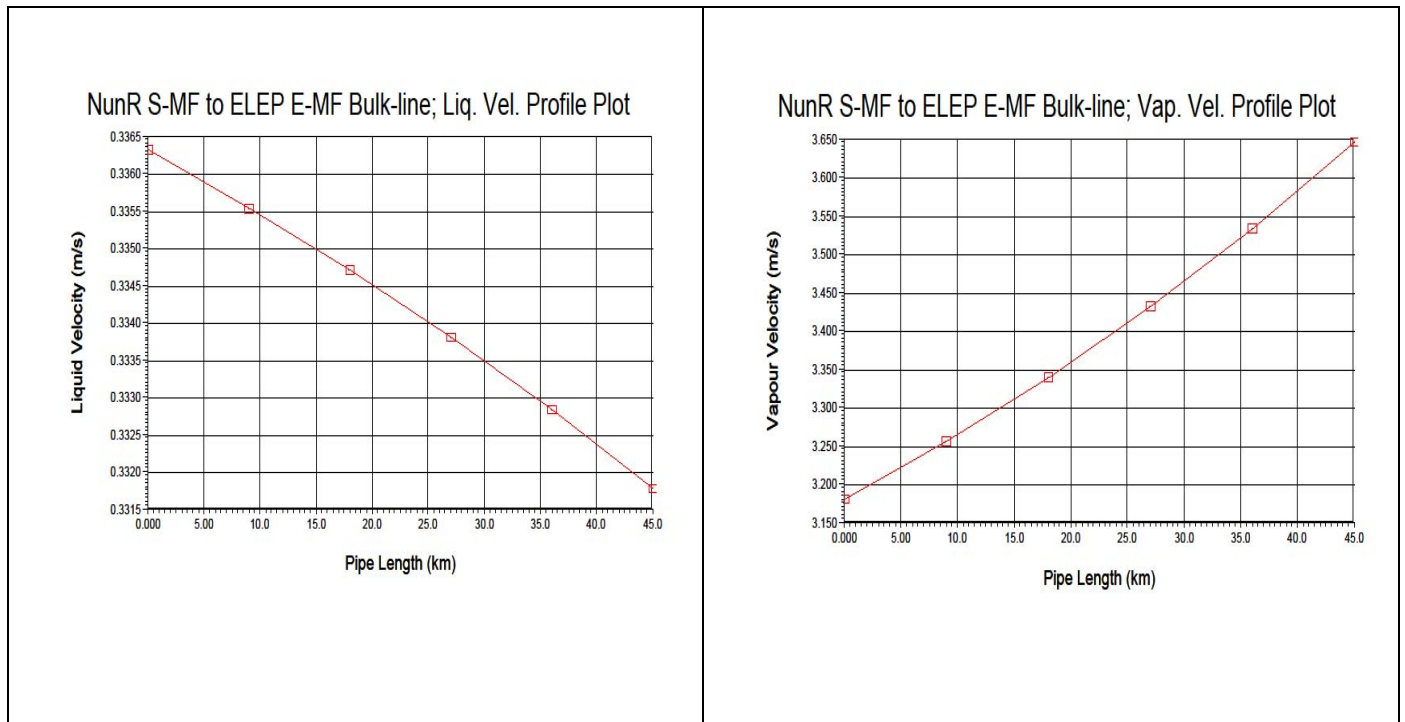


Figure 6.8: Phase-2; NUNR S-MF to ELEP E-MF B/L Velocity Profile Plot

iii. ELEP F-MF to ELEP E-MF B/L

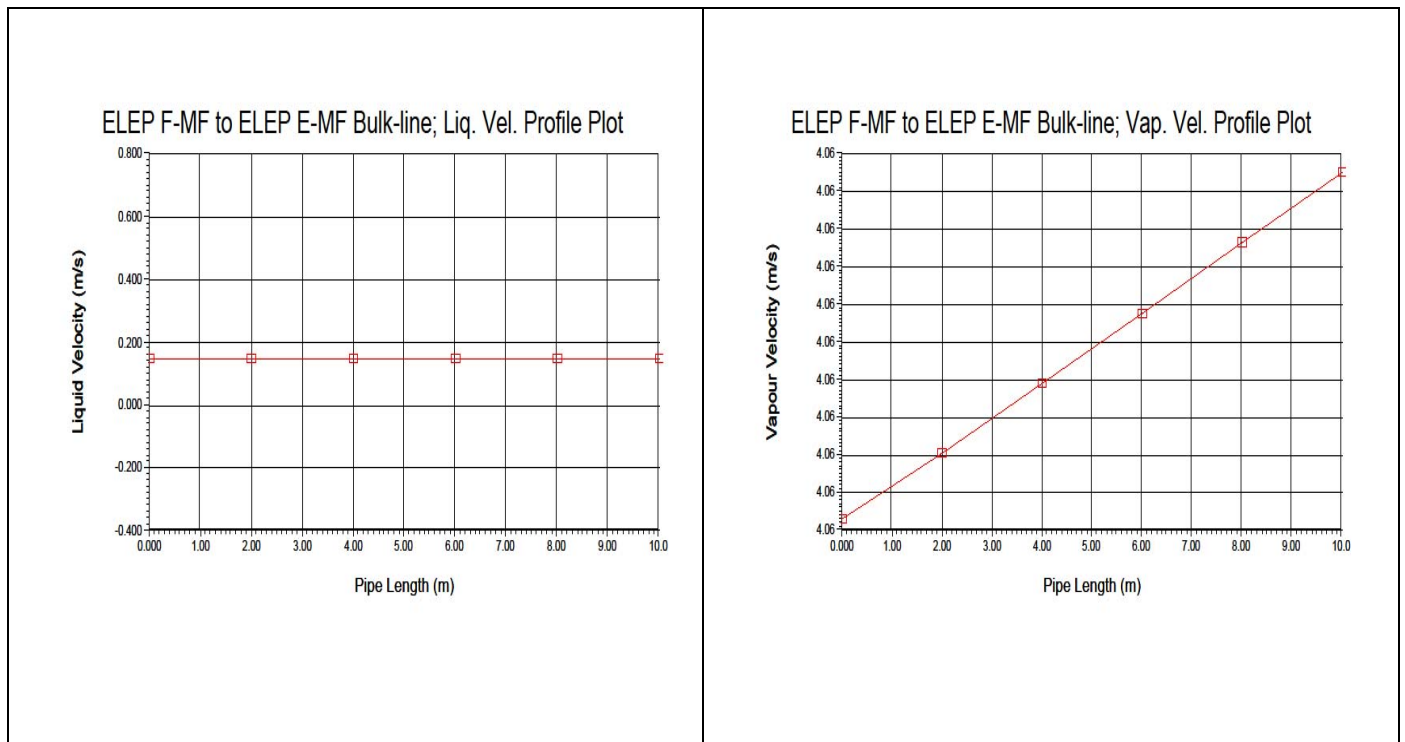


Figure 6.9: Phase-2; ELEP F-MF to ELEP E-MF B/L Velocity Profile Plot

iv. ELEP E-MF to AKAB T/L

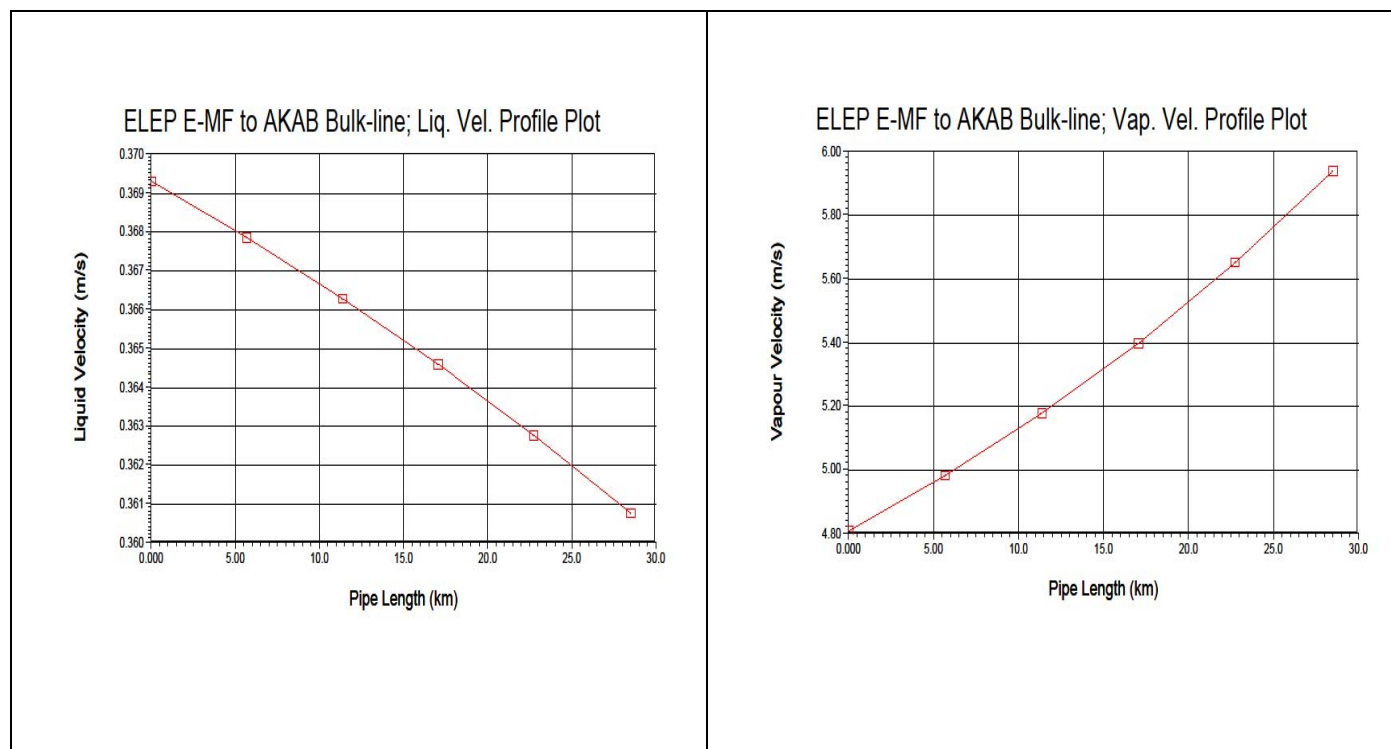


Figure 6.10: Phase-2; ELEP E-MF to AKAB T/L Velocity Profile Plot

6.7. Phase Envelopes/Hydrate Curves

i. NUNR F-MF to NUNR S-MF B/L

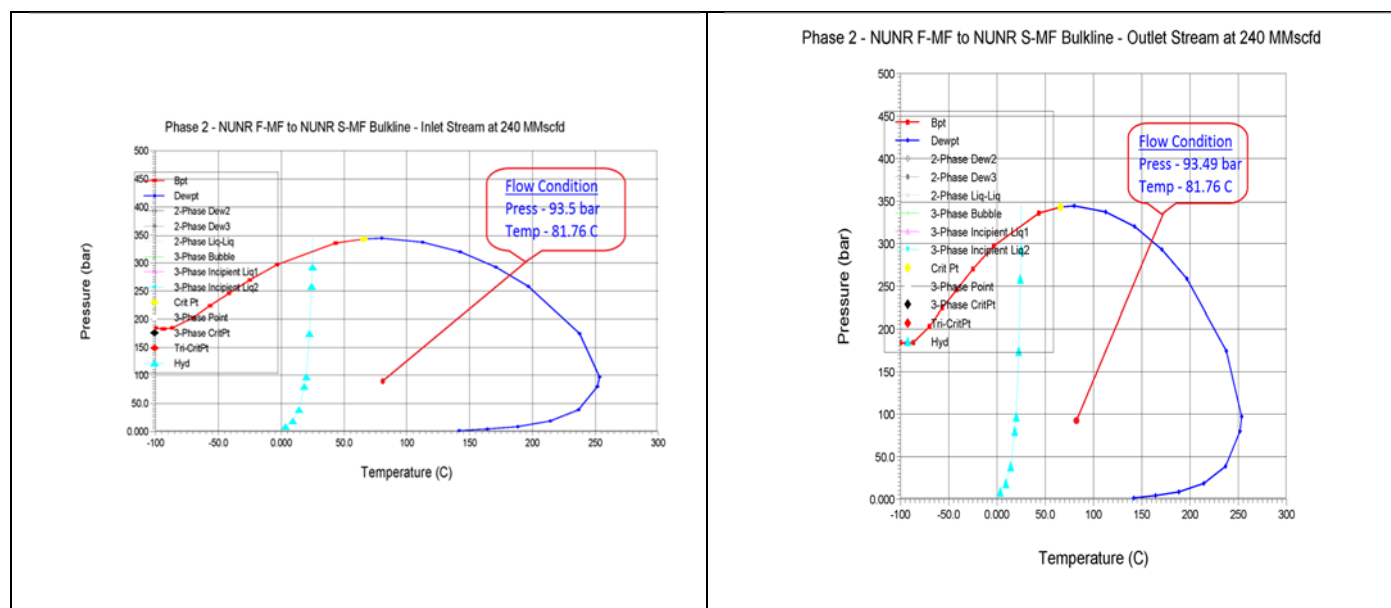


Figure 6.11: Phase-2; NunR F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

ii. NUNR S-MF to ELEP E-MF B/L

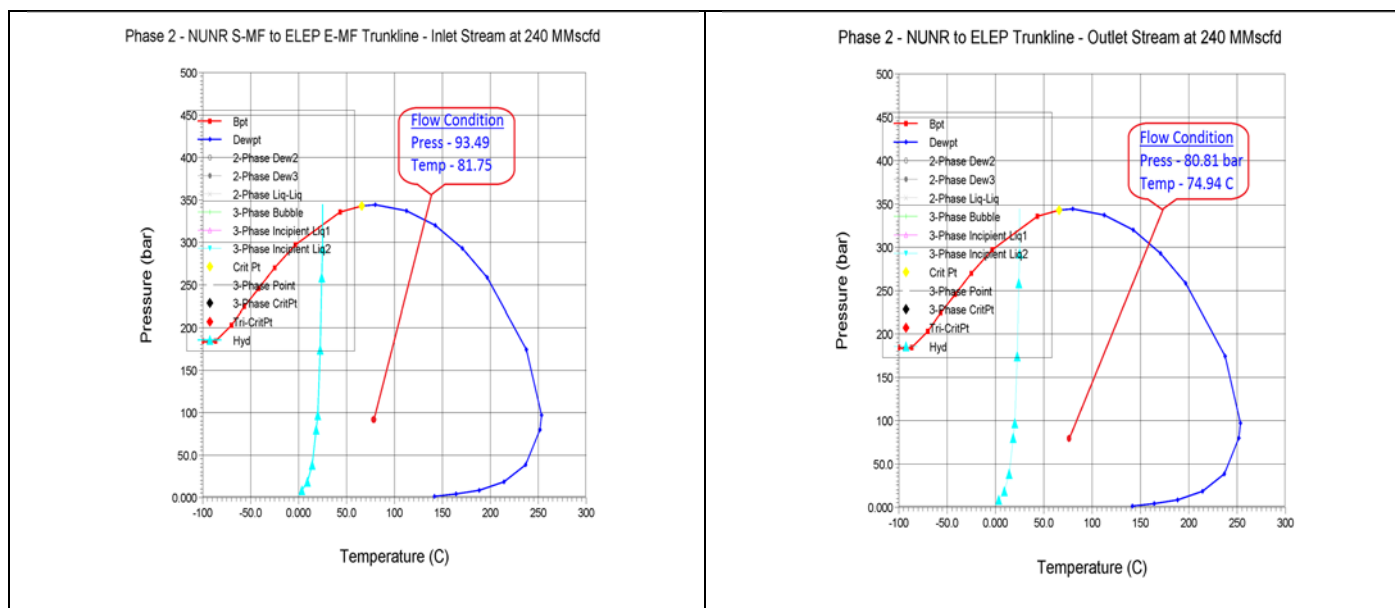


Figure 6.12: Phase-2; NUNR S-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curves

iii. ELEP F-MF to ELEP E-MF B/L

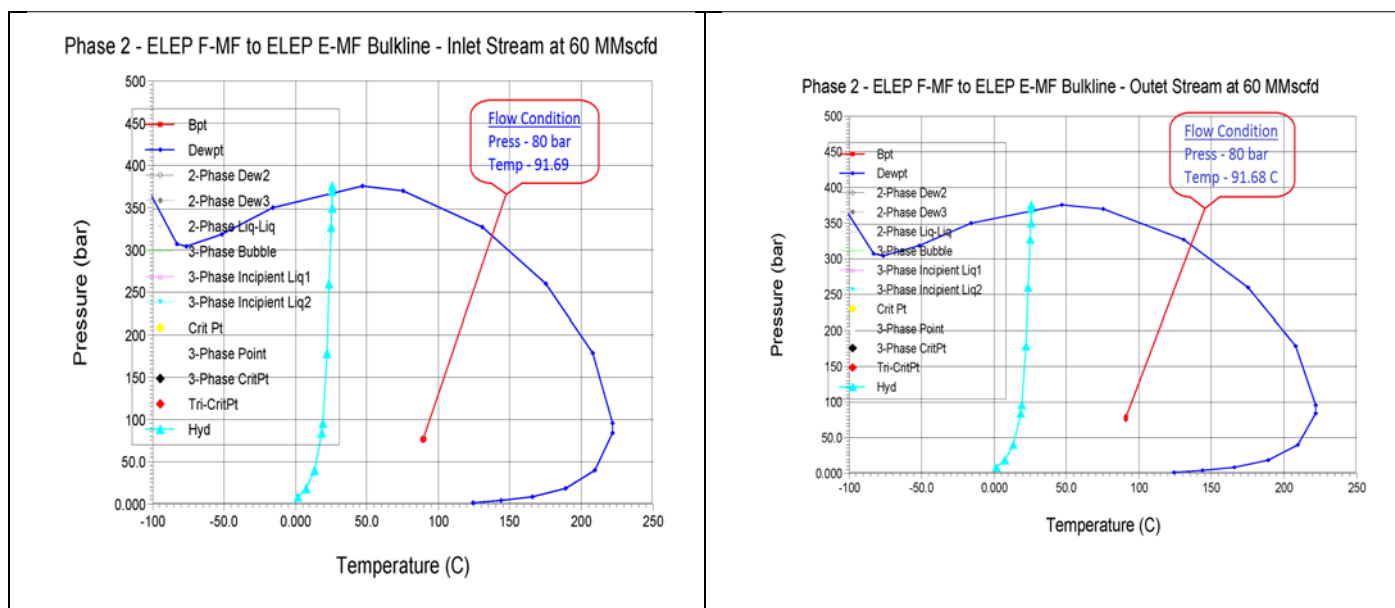


Figure 6.13: Phase-2; ELEP F-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curves

iv. ELEP E-MF to AKAB T/L

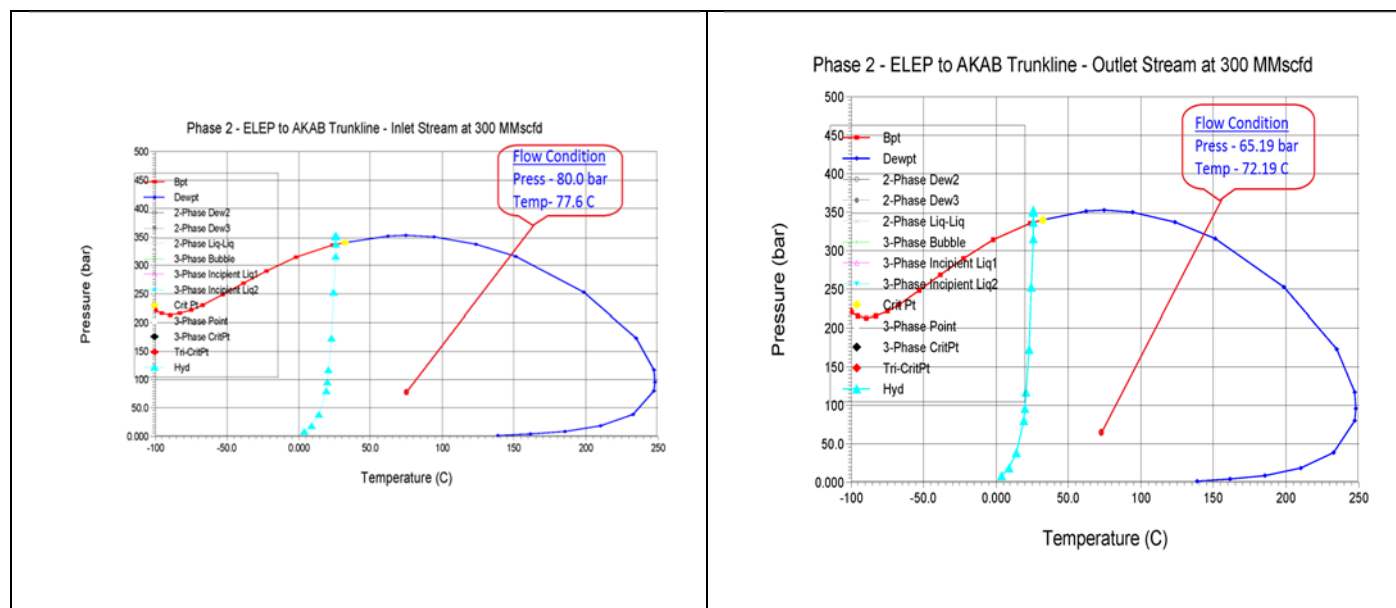


Figure 6.14: Phase-2; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curves

7. Phase-3 Gas Production Simulation Results

At phase 3 gas delivery, SEIB field has come on-stream. And the peak production anticipated; which this simulation is based on is 225 MMscfd.

The gathering is achieved from contribution of SEIB 225 MMscfd, NUNR 55 MMscfd, ELEP 20 MMscfd, to make-up a total of 300 MMscfd required at AKAB.

i. SEIB Field Flow to NUNR S-MF

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 225 MMscfd of gas flow with minimal pressure loss across the 20 km SEIB F-MF – NUNR S-MF bulk-line is 106.0 bar. The corresponding temperature is 92.8 °C.

Delivery Pressure; at NUNR S-MF is 95.4 bar, which is sufficient to support NUNR S-MF to ELEP E-MF trunk-line flow. The corresponding temperature is 88.4 °C.

The Line Size; which also supported 225 MMscfd of gas flow across the 20 km SEIB F-MF – NUNR S-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity boundary is 20". And the pressure drop is 0.5 bar.

See below; fig. 7.2, table 7.1, table 7.6, & fig. 7.3 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.7 m/s; while at the outlet is 2.9 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 4.6 m/s; while at the outlet is 5.0 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 13.2 m/s, and outlet is 13.8 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 7.2, table 7.1, table 7.6, & fig. 7.8 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 106.0 bar, the formation temperature is 20.1 °C. While the prevailing flow conditions at the inlet is 106.0 bar pressure and 92.8 °C temperature.

Similarly for outlet stream, at stream pressure of 95.4 bar, the formation temperature is 19.5 °C. While the prevailing flow conditions at the outlet are; pressure 95.4 bar and temperature 88.4 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 7.13 for more details

ii. NUNR Field Flow to NUNR S-MF

Pressure and Temperature Regime

Source Pressure at the CTP; is 95.5 bar which satisfy a good head flow for 55 MMscfd of gas across 10 m NUNR Field Flow to NUNR S-MF bulk-line. The corresponding temperature is 82.3 °C.

Delivery Pressure; at NUNR S-MF is 95.5 bar, which is sufficient to support NUNR S-MF to ELEP E-MF trunk-line flow. The corresponding temperature is 82.3 °C.

The Line Size; the existing 18" x 10 m NUNR F-MF to NUNR S-MF was employed to service the flow and deduce current flow condition under the 55 MMscfd flowrate. The pressure drop is 0 bar.

See below; fig. 7.2, table 7.2, table 7.7, & fig. 7.4 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at both inlet and outlet is 2.6 m/s.

Prevailing Fluid Velocity; at both inlet and outlet is 1.4 m/s. As a result of low volume flow on the line, the prevailing velocity is lower than the minimum calculated.

Erosional Velocity; at both inlet and outlet is 12.3 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 7.2, table 7.2, table 7.7, & fig. 7.9 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 95.5 bar, the formation temperature is 20.1 °C. While the prevailing flow conditions at the inlet is 95.5 bar pressure and 82.3 °C temperature.

Similarly for outlet stream, at stream pressure of 95.5 bar, the formation temperature is also 20.1 °C. While the prevailing flow conditions at the outlet are; pressure at 95.5 bar and temperature at 82.3 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 7.14 for more details

*iii. NunR S-MF to ELEM E-MF****Pressure and Temperature Regime***

Source Pressure at NUNR S-MF; the commingled SEIB field flow and NUNR field flow, exited NUNR S-MF at 95.4 bar pressure. The corresponding temperature is 87.0 °C.

Delivery Pressure; at ELEM E-MF is 79.3 bar, which is sufficient to support NUNR S-MF to ELEM E-MF trunk-line flow. The corresponding temperature is 79.4 °C.

The Line Size; existing rated 24" x 45 km NUNR S-MF to ELEM E-MF trunk-line was used to service the flow and deduce new flowing condition under 280 MMscfd flowrate.

See below; fig. 7.2, table 7.3, table 7.8, & fig. 7.5 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.8 m/s; while at the outlet is 3.0 m/s.

Prevailing Fluid Flow velocity; at the inlet is 4.3 m/s; while at the outlet is 5.0 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 13.5 m/s, and outlet is 14.6 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 7.2, table 7.3, table 7.8, & fig. 7.10 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 95.4 bar, the formation temperature is 19.7 °C. While the prevailing flow conditions at the inlet is 95.4 bar pressure and 87.0 °C temperature.

Similarly for outlet stream, at stream pressure of 79.3 bar, the formation temperature is 18.7 °C. While the prevailing flow conditions at the outlet are; pressure at 79.3 bar and temperature at 79.4 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 7.15 for more details

*iv. ELEM Field Flow to ELEM E-MF****Pressure and Temperature Regime***

Source Pressure at the CTP; is 80 bar which satisfy a good head flow for 20 MMscfd of gas across 10 m ELEM F-MF - ELEM E-MF bulk-line. The corresponding temperature is 90.5 °C.

Delivery Pressure; at ELEM E-MF is 80.0 bar and with a corresponding temperature of 90.5 °C.

The Line Size; existing 12" x 10 m ELEP F-MF to ELEP E-MF bulk-line was employed in reviewing the flow condition for a 60 MMscfd flow. The pressure drop is 0.0 bar.

See below; fig. 7.2, table 7.4, table 7.9, & fig. 7.6 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at both inlet and outlet is 3.3 m/s.

Prevailing fluid velocity; at both inlet and outlet is 1.4 m/s. As a result of low volume flow on the line, the prevailing velocity is lower than the minimum calculated.

Erosional Velocity; at both inlet and outlet is 16.1 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 7.2, table 7.4, table 7.9, & fig. 7.11 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.0 bar, the formation temperature is 17.9 °C. While the prevailing flow conditions at the inlet is 80.0 bar pressure and 90.5 °C temperature.

Similarly for outlet stream, at stream pressure of 80.0 bar, the formation temperature is 17.9 °C. While the prevailing flow conditions at the outlet are; pressure at 80.0 bar and temperature at 90.5 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 7.16 for more details

v. ELEP E-MF Flow to AKAB

Pressure and Temperature Regime

Source Pressure at the ELEP E-MF Outlet; the mixed SEIB field flow, NUNR field flow and ELEP field flow exited ELEP E-MF at a pressure of 79.3 bar and with a corresponding temperature of 80.1 °C.

Delivery Pressure; at AKAB is 65.2 bar and with a corresponding temperature of 74.4 °C.

The Line Size; existing 24" x 28.5 km ELEP E-MF to AKAB trunk-line was employed in reviewing the flow condition for the 300 MMscfd flow. The pressure drop is 0.49 bar/km.

See below; fig. 7.2, table 7.5, table 7.10, & fig. 7.7 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.0 m/s; while at the outlet is 3.3 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 5.4 m/s; while at the outlet is 6.5 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 14.7 m/s, and outlet is 16.1 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 7.2, table 7.5, table 7.10, & fig. 7.12 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 79.3 bar, the formation temperature is 18.6 °C. While the prevailing flow conditions at the inlet is 79.3 bar pressure and 80.1 °C temperature.

Similarly for outlet stream, at stream pressure of 65.2 bar, the formation temperature is 17.5 °C. While the prevailing flow conditions at the outlet are; pressure at 65.2 bar and temperature at 74.4 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 7.17 for more details

7.1. Hysys PFD

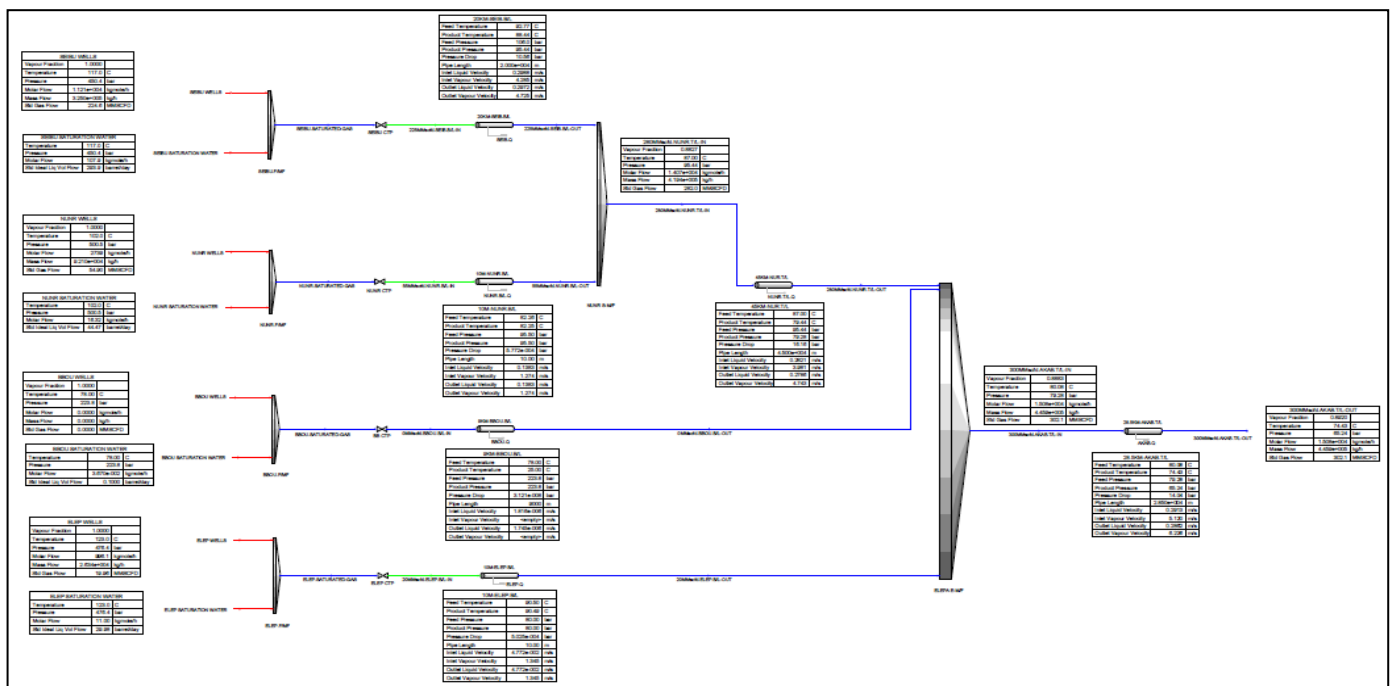


Figure 7.1: Phase-3 Gas Production; Hysys PFD

7.2. Flow Schematics

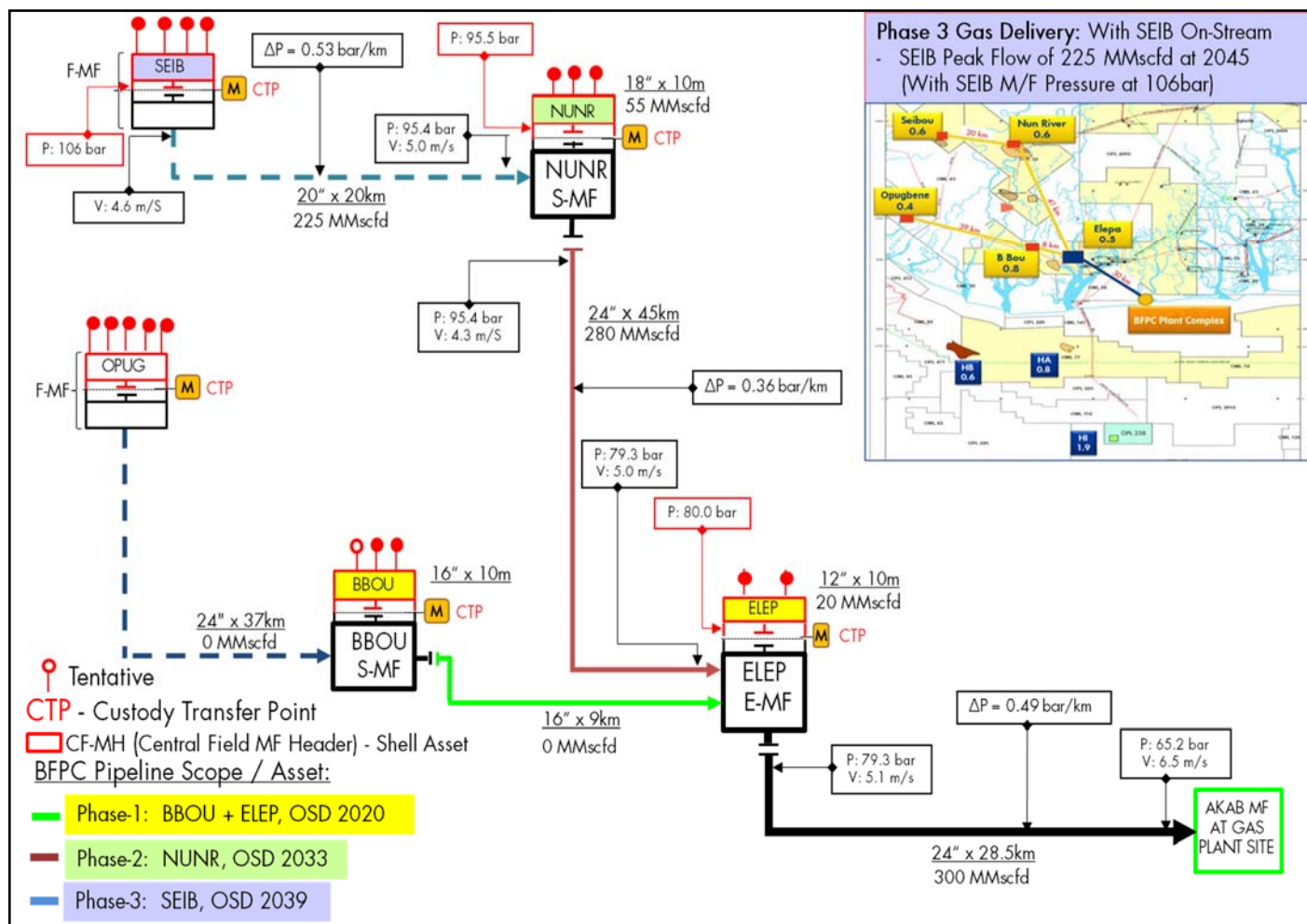


Figure 7.2: Phase-3 Gas Production; Flow Schematics

7.3. Calculation Notes

i. SEIB F-MF to NUNR S-MF B/L

Table 7.1: Phase-3; SEIB F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from SEIBU to NUNR S-MF																				
Line Dimension		in	mm	m																
20" SCH 80 (ID)		17.95	455.93	0.46																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	106.00	92.77	326,975	2,690	44.83	0.75	121.53	206,797	2,515	41.92	0.70	82.23	0.02	0.87	120,144	175.40	2.92	0.049	685.06	0.30
Outlet	95.44	88.44	326,975	2,948	49.13	0.82	110.92	205,785	2,773	46.22	0.77	74.20	0.02	0.87	121,003	174.20	2.90	0.048	694.47	0.32
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _o ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2 \pi / 4$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$p_o \cdot V_{Overall}^2$	$p_g \cdot V_o^2$	$p_L \cdot V_L^2$	$X = m/m_g \cdot (p_g/p_L)$	$Ve = c/(\sqrt{p_o})$	$Ve = c/(\sqrt{p_o})$							
Inlet	0.16	4.57	4.28	0.30	20.93	18.30	0.089	2,544	1,504	60.96	0.2013	13.15	2.72							
Outlet	0.16	5.01	4.72	0.30	25.14	22.24	0.088	2,788	1,650	60.96	0.1922	13.77	2.85							

ii. NUNR F-MF to NUNR S-MF B/L

Table 7.2: Phase-3; NUNR F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from NUNR to NUNR S-MF																				
Line Dimension 18" SCH 80 (ID)		in	mm	m																
		16.14	410.00	0.41																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
	Inlet	95.50	82.26	92,390	670	11.16	0.19	137.98	47,777	604	10.07	0.17	79.10	0.02	0.85	44,613	65.58	1.09	0.018	680.26
Outlet	95.50	82.25	92,390	670	11.16	0.19	137.98	47,776	604	10.07	0.17	79.10	0.02	0.85	44,614	65.58	1.09	0.018	680.26	0.32
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{Overall}	V _o ²	V _L ²	Overall Phase Momentu m (Pa)	Gas Phase Momentu m (Pa)	Liq. Phase Momentu m (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²π/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				p _o * V² _{Overall}	p _g * V _o ²	p _L * V _L ²	X = m _L /m _g * (p _g /p _L)	Ve = c/(Vp _o)	Ve = c/(Vp _o)							
Inlet	0.13	1.41	1.27	0.14	1.98	1.61	0.019	274	128	12.94	0.3184	12.34	2.55							
Outlet	0.13	1.41	1.27	0.14	1.98	1.61	0.019	274	128	12.94	0.3184	12.34	2.55							

iii. NUNR S-MF to ELEP E-MF T/L

Table 7.3: Phase-3; NUNR S-MF to ELEP E-MF T/L Calculation Note

Line Description																				
Trunk-line from NUNR S-MF to ELEP E-MF																				
Line Dimension	in	mm	m																	
24" SCH 80 (ID)	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	95.44	87.00	419,364	3,615	60.25	1.00	116.00	253,860	3,376	56.27	0.94	75.20	0.02	0.87	165,329	239.10	3.99	0.066	691.50	0.32
Outlet	79.28	79.44	419,364	4,259	70.98	1.18	98.47	252,389	4,022	67.03	1.12	62.74	0.02	0.87	166,504	235.80	3.93	0.066	706.20	0.36
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² ^{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² ^{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X _{LM} = m _L /m _G * (ρ _G /ρ _L)	Ve = c/(Vρ _G)	Ve = c/(Vρ _L)							
Inlet	0.24	4.26	3.97	0.28	18.11	15.80	0.079	2,101	1,188	54.79	0.2148	13.46	2.79							
Outlet	0.24	5.01	4.73	0.28	25.14	22.42	0.077	2,476	1,407	54.42	0.1966	14.61	3.02							

iv. ELEP F-MF to ELEP E-MF B/L

Table 7.4: Phase-3; ELEP F-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from ELEP to ELEP E-MF																				
Line Dimension 12" SCH 80 (ID)		in	mm	m																
		11.38	289.00	0.29																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	80.00	90.50	26,540	329	5.48	0.09	80.75	18,460	317	5.29	0.09	58.16	0.02	0.90	8,078	11.26	0.19	0.003	717.47	0.37
Outlet	80.00	90.49	26,540	329	5.48	0.09	80.76	18,460	317	5.29	0.09	58.16	0.02	0.90	8,078	11.26	0.19	0.003	717.47	0.37
Calc. Parameters																				
	X-sectional Area (m ²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V ² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d ² (π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				p _G * V ² _{Overall}	p _G * V _G ²	p _L * V _L ²	X _{LM} = (m/m _G) * (ρ _G /ρ _L)	Ve = c/(Vp _G)	Ve = c/(Vp _L)							
Inlet	0.07	1.39	1.34	0.05	1.94	1.81	0.002	156	105	1.63	0.1246	16.14	3.34							
Outlet	0.07	1.39	1.34	0.05	1.93	1.81	0.002	156	105	1.63	0.1246	16.14	3.34							

v. ELEP E-MF to AKAB T/L

Table 7.5: Phase-3; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
Trunk-line from ELEP E-MF to AKABEL																				
Line Dimension	in	mm	m																	
24" SCH 80 (ID)	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	79.28	80.08	445,905	4,589	76.48	1.27	97.17	271,067	4,342	72.37	1.21	62.43	0.02	0.88	174,346	246.60	4.11	0.069	707.11	0.36
Outlet	65.24	74.43	445,905	5,523	92.05	1.53	80.74	271,034	5,280	88.00	1.47	51.33	0.01	0.89	174,234	242.10	4.04	0.067	719.72	0.39
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2 \pi / 4$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$p_G \cdot V^2_{Overall}$	$p_G \cdot V_G^2$	$p_L \cdot V_{L2}$	$X = m_L/m_G \cdot (\rho_L/\rho_G)$	$Ve = c/(\sqrt{p_{0.1}})$	$Ve = c/(\sqrt{p_{0.1}})$							
Inlet	0.24	5.40	5.11	0.29	29.19	26.13	0.084	2,836	1,631	59.60	0.1911	14.71	3.04							
Outlet	0.24	6.50	6.22	0.29	42.28	38.64	0.081	3,413	1,983	58.46	0.1717	16.14	3.34							

7.4. Flow Parameters

i. SEIB F-MF to NUNR S-MF B/L

Table 7.6: Phase-3; SEIB F-MF to NUNR S-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	106.0	92.8	0.8927	<empty>	Transition	0.1955	0.0617	<empty>	0.00	<empty>	0.000014	<empty>	312,326	9,660,567	0.30	4.28
4,000	0	104.0	91.9	0.8929	36.53	Transition	0.1933	0.0628	2.0523	0.00	0.00	0.000015	0.000480	308,705	9,713,632	0.30	4.36
8,000	0	101.9	91.1	0.8930	36.07	Transition	0.1909	0.0639	2.0893	0.00	0.00	0.000016	0.000508	305,057	9,768,552	0.30	4.44
12,000	0	99.8	90.2	0.8933	35.60	Transition	0.1885	0.0651	2.1287	0.00	0.00	0.000016	0.000539	301,374	9,825,439	0.30	4.53
16,000	0	97.6	89.3	0.8935	35.13	Transition	0.1860	0.0664	2.1709	0.00	0.00	0.000018	0.000573	297,644	9,884,415	0.30	4.62
20,000	0	95.4	88.4	0.8938	34.65	Transition	0.1834	0.0678	2.2163	0.00	0.00	0.000019	0.000611	293,861	9,945,627	0.30	4.72

ii. NUNR F-MF to NUNR S-MF B/L

Table 7.7: Phase-3; NUNR F-MF to NUNR S-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	95.5	82.3	0.8337	<empty>	Transition	0.3305	0.0071	<empty>	0.00	<empty>	0.000000	<empty>	121,057	2,572,781	0.14	1.27
2	0	95.5	82.3	0.8337	29.48	Transition	0.3305	0.0071	0.0001	0.00	0.00	0.000000	0.000000	121,057	2,572,780	0.14	1.27
4	0	95.5	82.3	0.8337	29.48	Transition	0.3305	0.0071	0.0001	0.00	0.00	0.000000	0.000000	121,056	2,572,778	0.14	1.27
6	0	95.5	82.3	0.8337	29.48	Transition	0.3305	0.0071	0.0001	0.00	0.00	0.000000	0.000000	121,056	2,572,777	0.14	1.27
8	0	95.5	82.3	0.8337	29.48	Transition	0.3305	0.0071	0.0001	0.00	0.00	0.000000	0.000000	121,055	2,572,776	0.14	1.27
10	0	95.5	82.3	0.8337	29.48	Transition	0.3305	0.0071	0.0001	0.00	0.00	0.000000	0.000000	121,055	2,572,776	0.14	1.27

iii. NUNR S-MF to ELEP E-MF T/L

Table 7.8: Phase-3; NUNR S-MF to ELEP E-MF T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	95.4	87.0	0.8827	<empty>	Transition	0.2108	0.0406	<empty>	0.00	<empty>	0.000008	<empty>	333,231	10,214,107	0.28	3.98
9,000	0	92.4	85.5	0.8830	36.63	Transition	0.2068	0.0418	3.0719	0.00	0.00	0.000009	0.000674	326,905	10,304,027	0.28	4.10
18,000	0	89.3	84.0	0.8834	35.75	Transition	0.2025	0.0431	3.1664	0.00	0.00	0.000010	0.000743	320,456	10,399,234	0.28	4.23
27,000	0	86.1	82.5	0.8838	34.85	Transition	0.1979	0.0445	3.2725	0.00	0.00	0.000011	0.000824	313,855	10,500,294	0.28	4.38
36,000	0	82.7	81.0	0.8844	33.94	Transition	0.1931	0.0461	3.3923	0.00	0.00	0.000013	0.000923	307,070	10,607,893	0.28	4.55
45,000	0	79.3	79.4	0.8850	33.01	Transition	0.1879	0.0480	3.5289	0.00	0.00	0.000014	0.001044	300,064	10,722,878	0.28	4.74

iv. ELEP F-MF to ELEP E-MF B/L Pipe Profile

Table 7.9: Phase-3; ELEP F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	80.0	90.5	0.9300	<empty>	Segregated	0.1975	0.0061	<empty>	0.00	<empty>	0.000000	<empty>	26,434	1,468,216	0.05	1.34
2	0	80.0	90.5	0.9299	28.98	Segregated	0.1975	0.0061	0.0001	0.00	0.00	0.000000	0.000000	26,434	1,468,217	0.05	1.34
4	0	80.0	90.5	0.9299	28.98	Segregated	0.1975	0.0061	0.0001	0.00	0.00	0.000000	0.000000	26,433	1,468,218	0.05	1.34
6	0	80.0	90.5	0.9299	28.98	Segregated	0.1975	0.0061	0.0001	0.00	0.00	0.000000	0.000000	26,433	1,468,218	0.05	1.34
8	0	80.0	90.5	0.9299	28.98	Segregated	0.1975	0.0061	0.0001	0.00	0.00	0.000000	0.000000	26,432	1,468,219	0.05	1.34
10	0	80.0	90.5	0.9299	28.98	Segregated	0.1975	0.0061	0.0001	0.00	0.00	0.000000	0.000000	26,432	1,468,220	0.05	1.34

v. ELEP E-MF to AKAB T/L Pipe Profile

Table 7.10: Phase-3; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	79.3	80.1	0.8883	<empty>	Transition	0.1806	0.0551	<empty>	0.00	<empty>	0.000019	<empty>	312,898	11,509,194	0.29	5.12
5,700	0	76.7	79.0	0.8889	32.61	Transition	0.1767	0.0569	2.6506	0.00	0.00	0.000021	0.000962	307,515	11,604,188	0.29	5.29
11,400	0	74.0	77.9	0.8896	31.97	Transition	0.1726	0.0589	2.7435	0.00	0.00	0.000023	0.001070	301,955	11,704,395	0.29	5.48
17,100	0	71.2	76.8	0.8903	31.31	Transition	0.1682	0.0611	2.8479	0.00	0.00	0.000026	0.001200	296,171	11,810,470	0.29	5.70
22,800	0	68.3	75.6	0.8911	30.62	Transition	0.1635	0.0637	2.9665	0.00	0.00	0.000029	0.001359	290,147	11,923,219	0.29	5.95
28,500	0	65.2	74.4	0.8920	29.91	Transition	0.1585	0.0666	3.1025	0.00	0.00	0.000033	0.001558	283,848	12,043,650	0.29	6.23

7.5. Pressure Profile Plots

i. SEIB F-MF to NUNR S-MF B/L

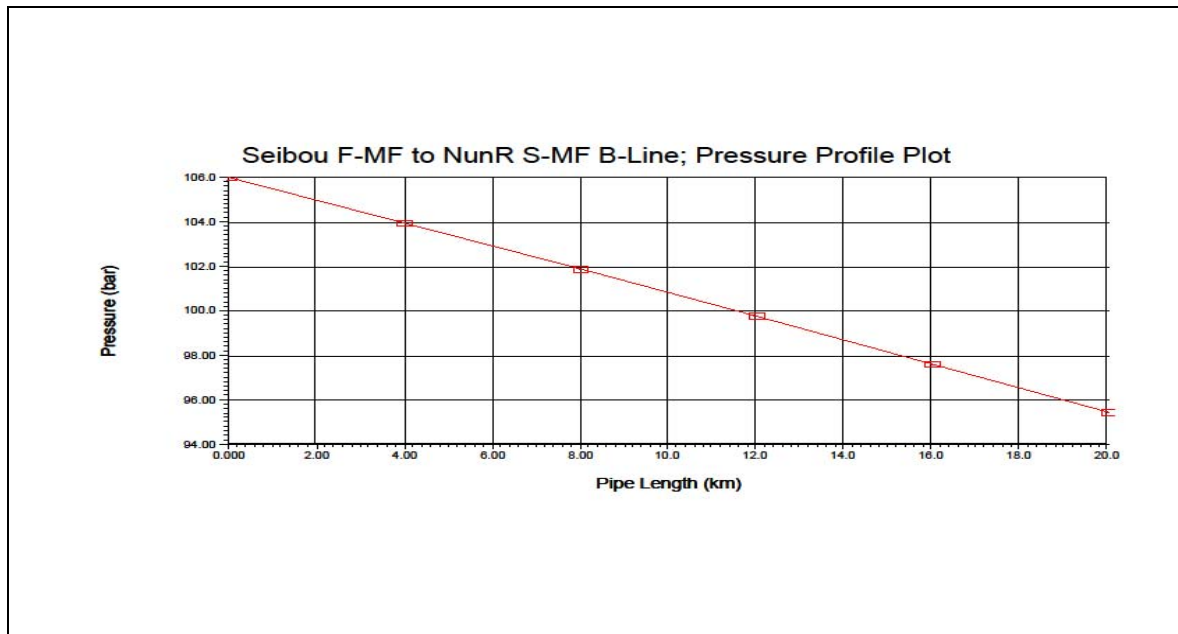


Figure 7.3: Phase-3; SEIB F-MF to NUNR S-MF B/L Pressure Profile Plot

ii. NunR F-MF to NunR S-MF B/L

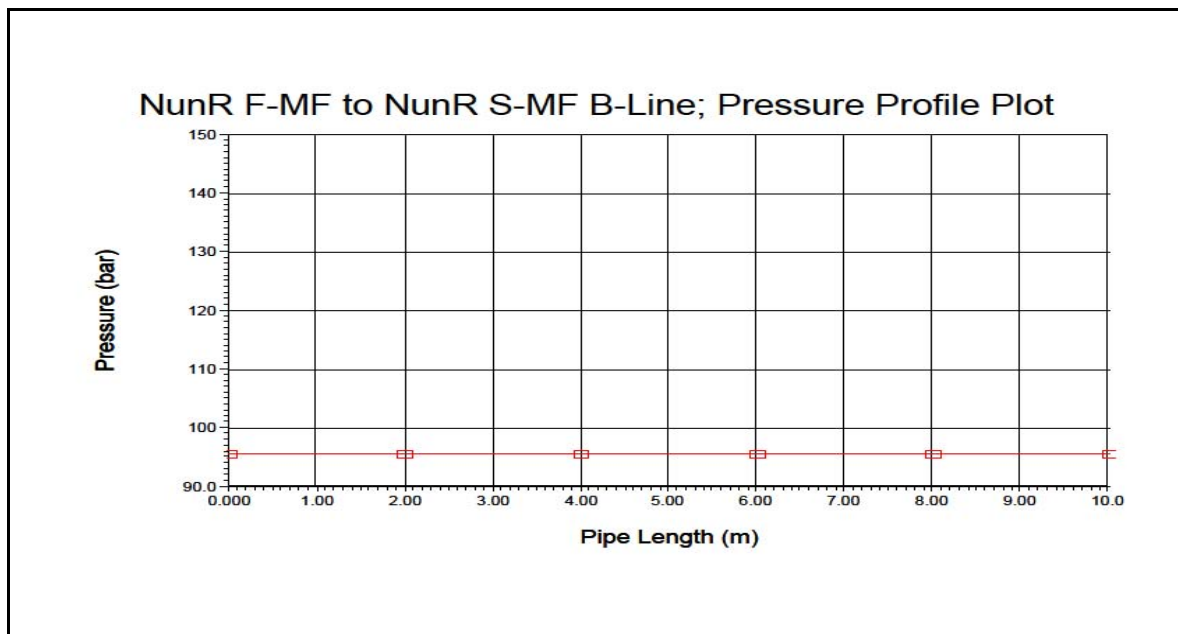


Figure 7.4: Phase-3; NunR F-MF to NunR S-MF B/L Pressure Profile Plot

iii. NUNR S-MF to ELEP E-MF T/L

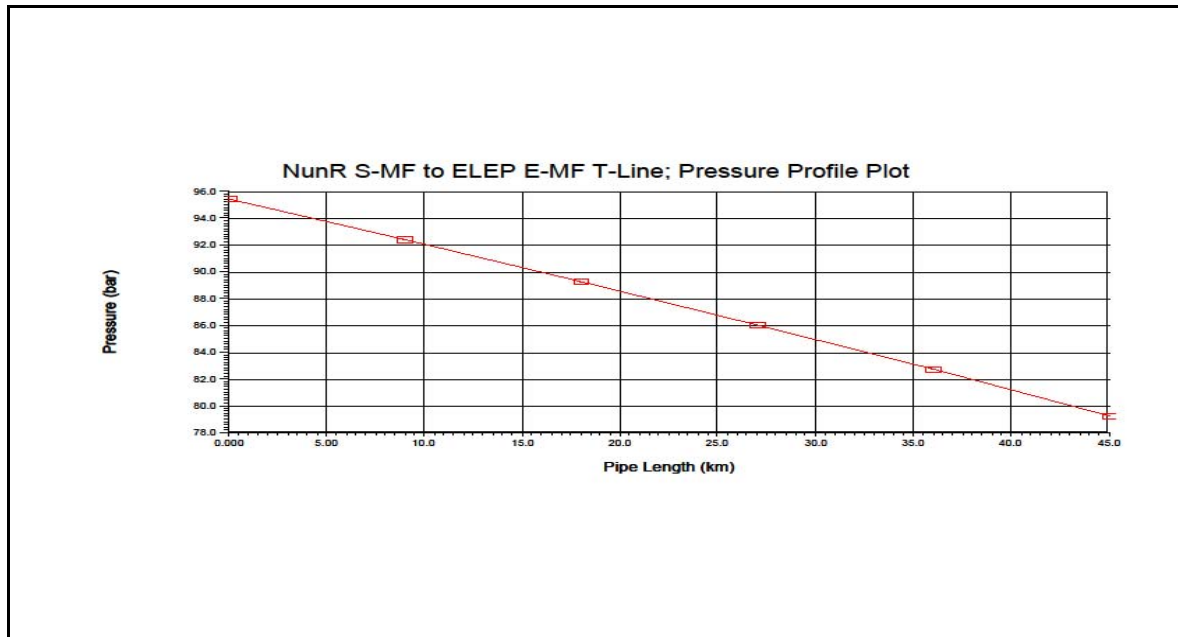


Figure 7.5: Phase-3; NUNR S-MF to ELEP E-MF T/L Pressure Profile Plot

iv. ELEP F-MF to ELEP E-MF B/L

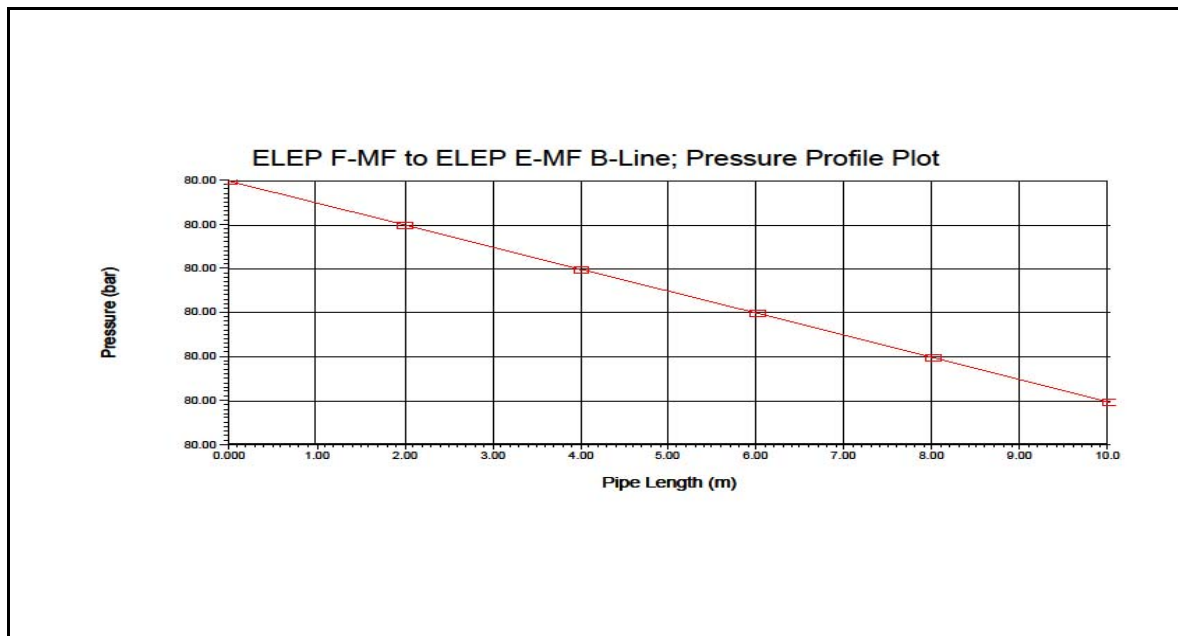


Figure 7.6: Phase-3; ELEP F-MF to ELEP E-MF B/L Pressure Profile Plot

v. ELEP E-MF to AKAB T/L

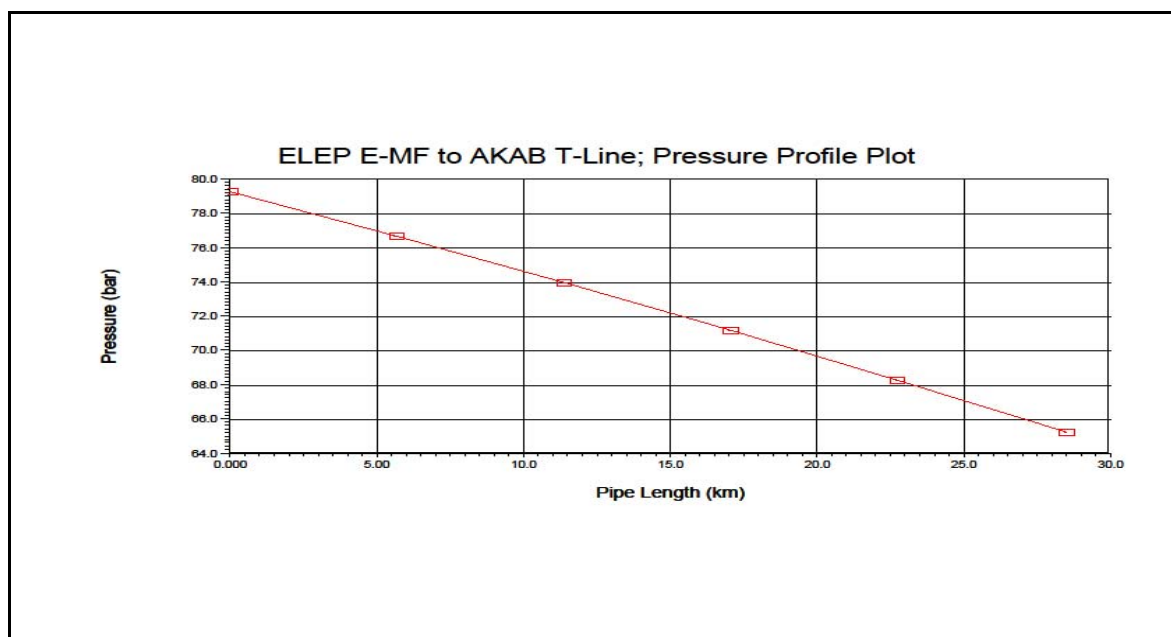


Figure 7.7: Phase-3; ELEP E-MF to AKAB T/L Pressure Profile Plot

7.6. Velocity Profile Plots

i. SEIB F-MF to NUNR S-MF B/L

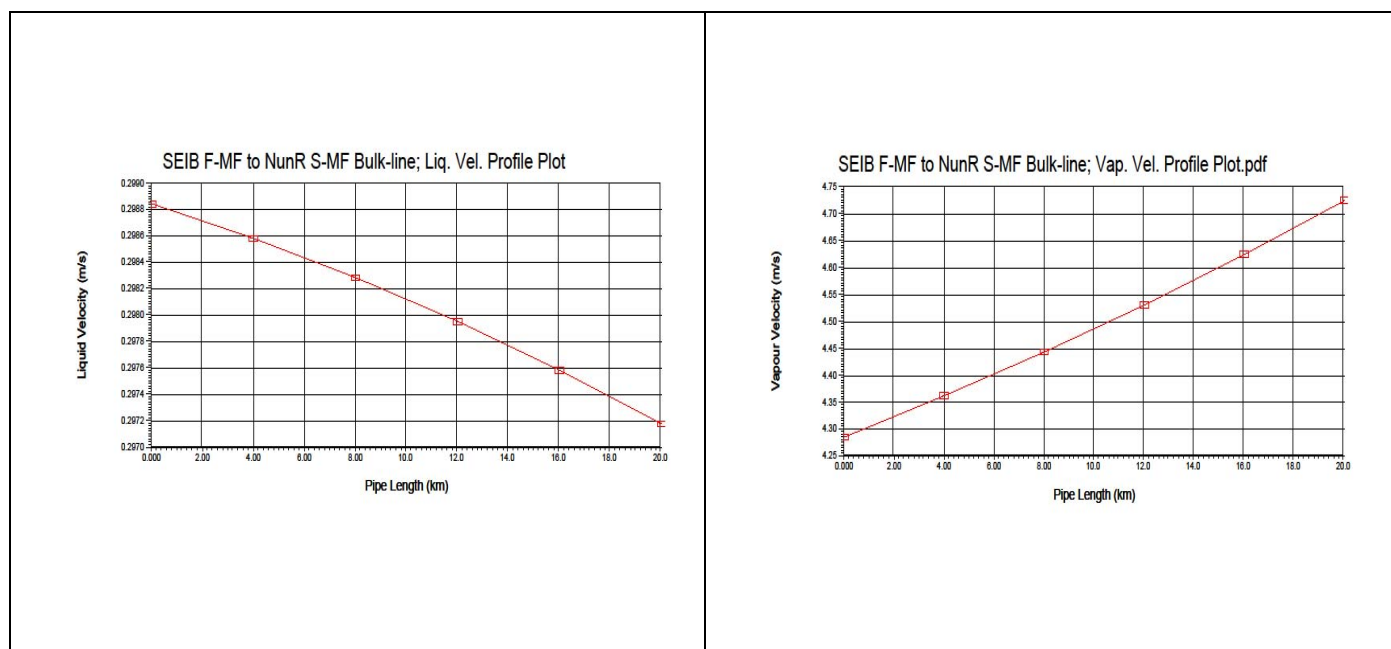


Figure 7.8: Phase-3; SEIB F-MF to NUNR S-MF B/L Velocity Profile Plot

ii. NUNR F-MF to NUNR S-MF B/L

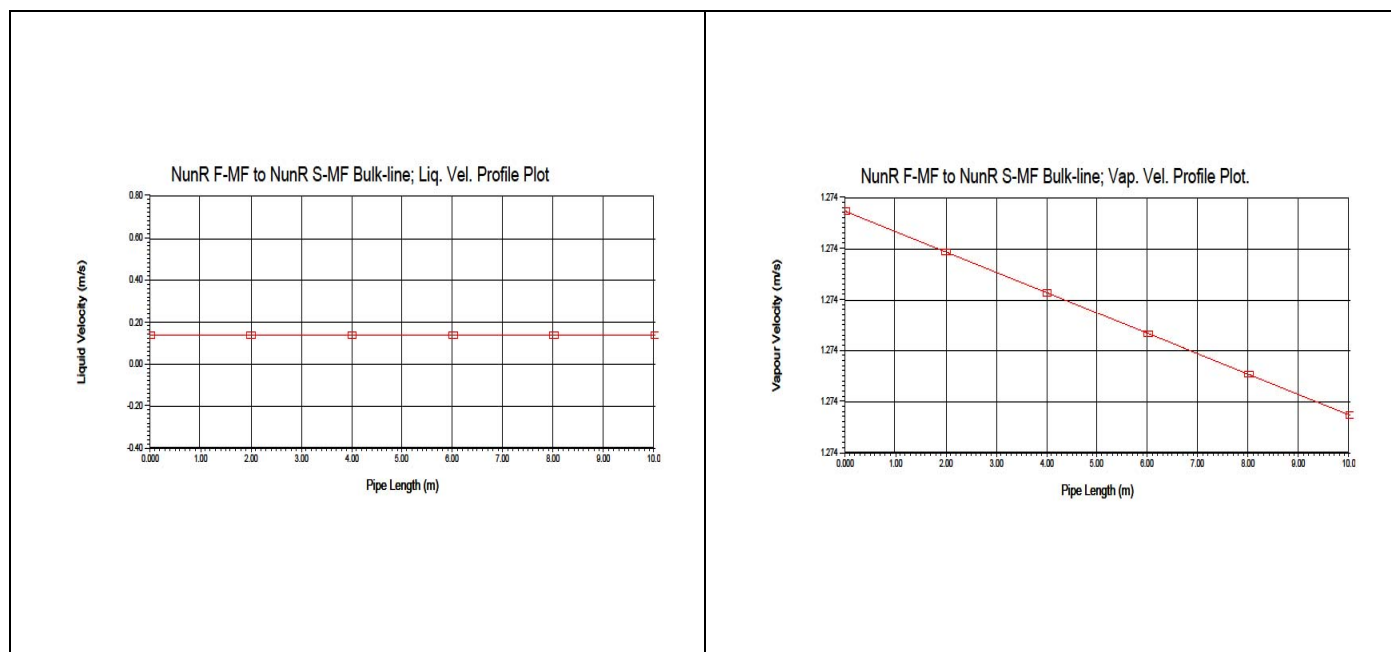


Figure 7.9: Phase-3; NUNR F-MF to NUNR S-MF B/L Velocity Profile Plot

iii. NUNR S-MF to ELEP E-MF T/L

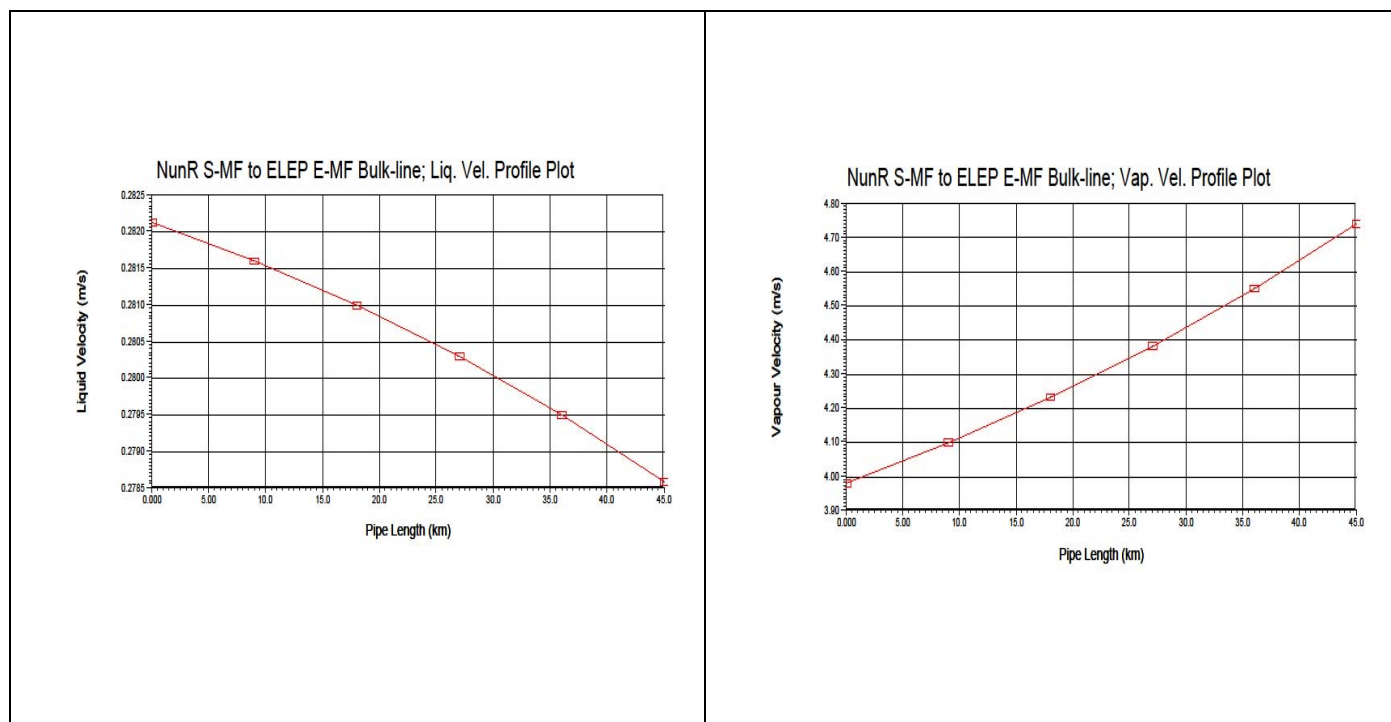


Figure 7.10: Phase-3; NUNR S-MF to ELEP E-MF T/L Velocity Profile Plot

iv. ELEP F-MF to ELEP E-MF B/L

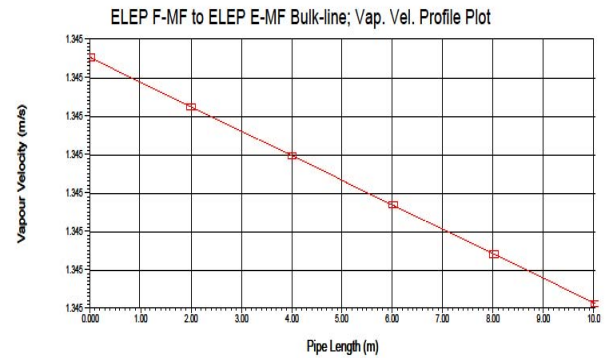
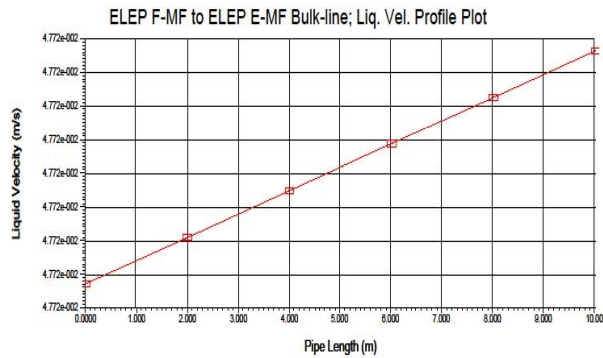


Figure 7.11: Phase-3; ELEP F-MF to ELEP E-MF B/L Velocity Profile Plot

v. ELEP E-MF to AKAB T/L

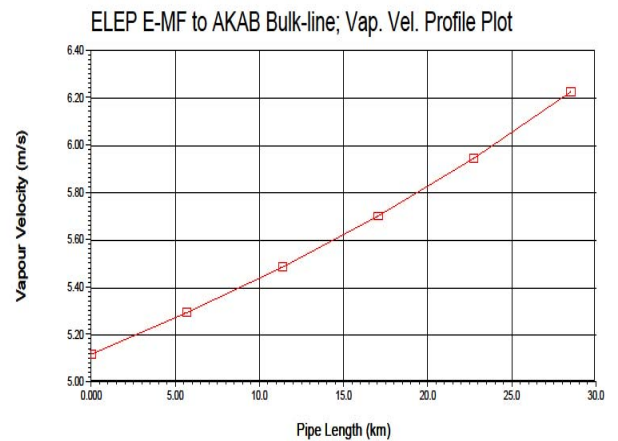
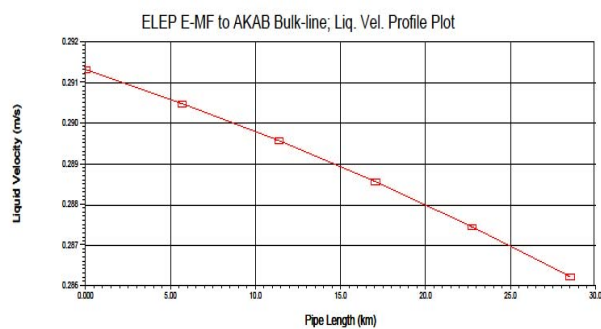
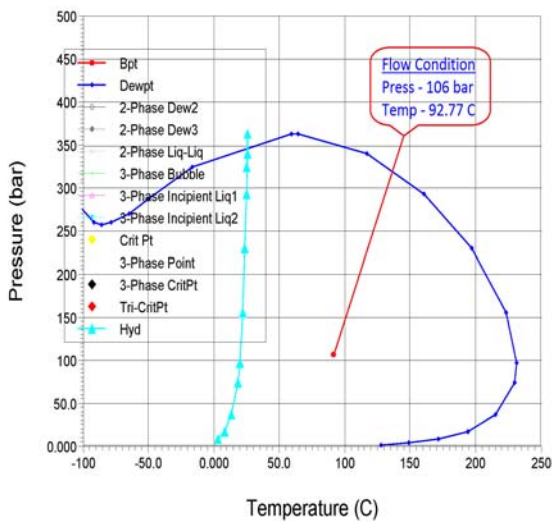


Figure 7.12: Phase-3; ELEP E-MF to AKAB T/L Velocity Profile Plot

7.7. Phase Envelopes/Hydrate Curves

i. SEIB F-MF to NUNR S-MF B/L

Phase 3 - SEIBU to NUNR Bulkline - Inlet Stream at 225 MMscfd



Phase 3 - SEIBU to NUNR Bulkline - Outlet Stream at 225 MMscfd

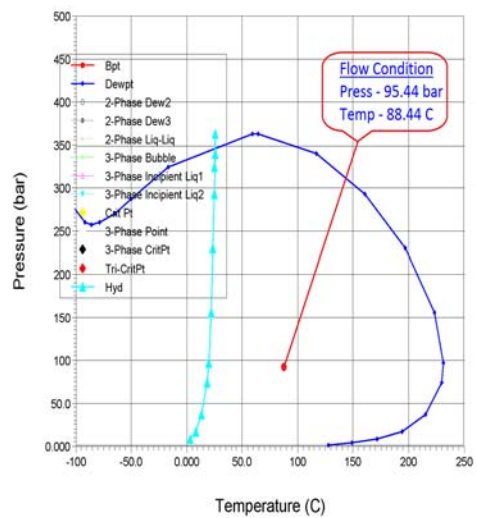
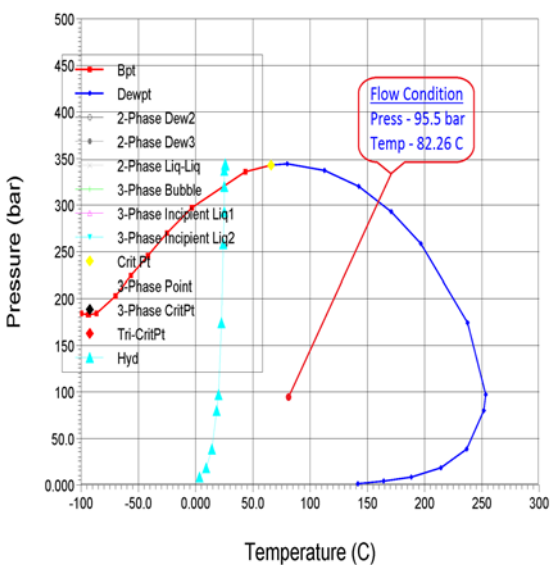


Figure 7.13: Phase-3; SEIB F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

ii. NUNR F-MF to NUNR S-MF B/L

Phase 3 - NUNR F-MF to NUNR S-MF Bulkline - Inlet Stream at 55 MMscfd



Phase 3 - NUNR F-MF to NUNR S-MF Bulkline - Outlet Stream at 55 MMscfd

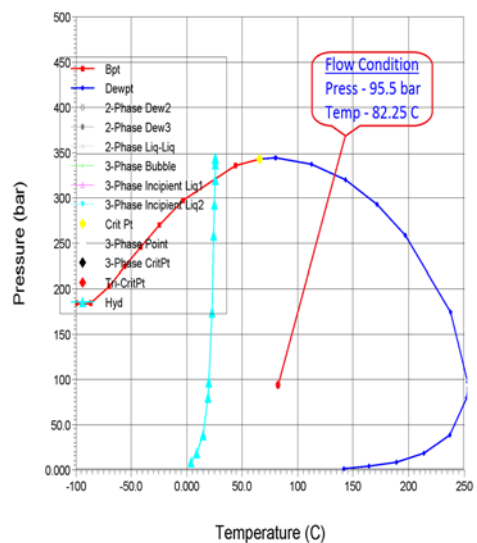


Figure 7.14: Phase-3; NUNR F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

iii. NUNR S-MF to ELEP E-MF T/L

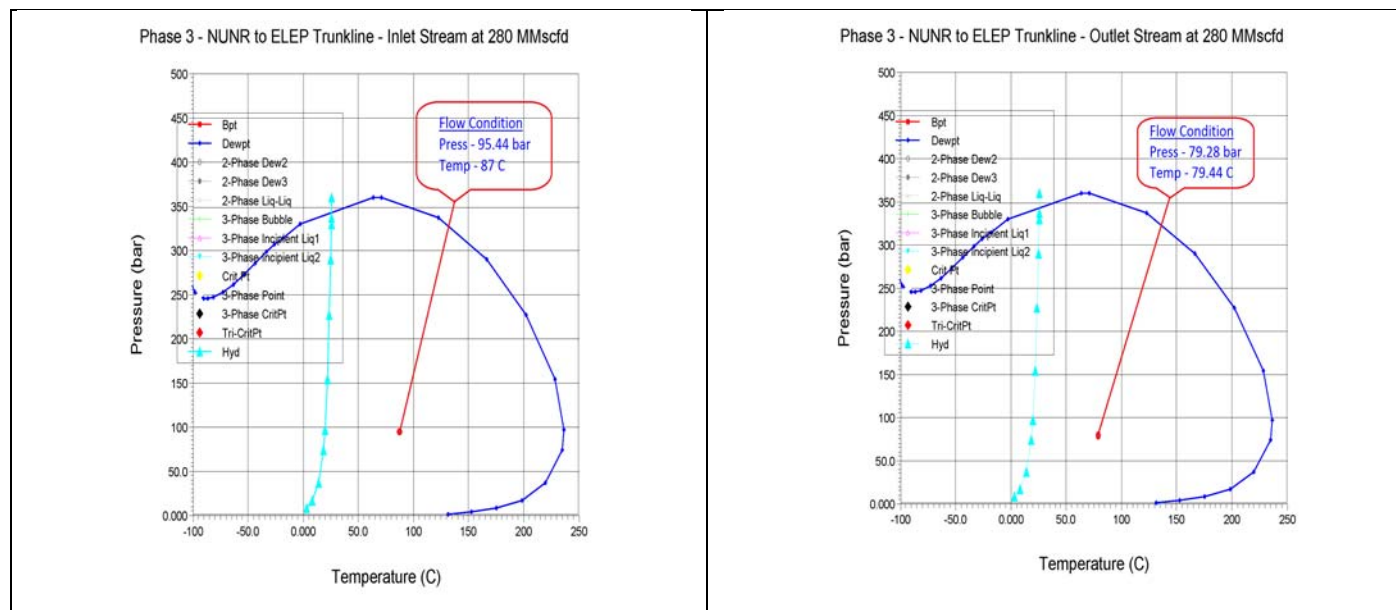


Figure 7.15: Phase-3; NunR S-MF to ELEM E-MF T/L Phase Envelope/Hydrate Curves

iv. ELEM F-MF to ELEM E-MF B/L

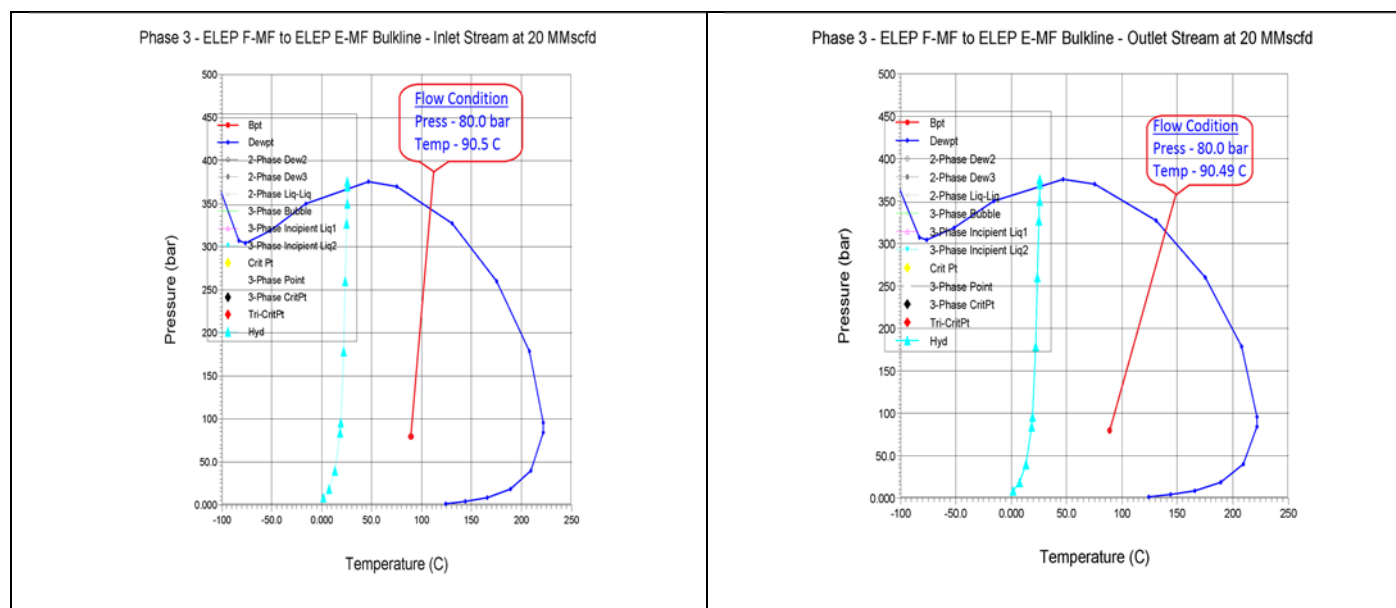
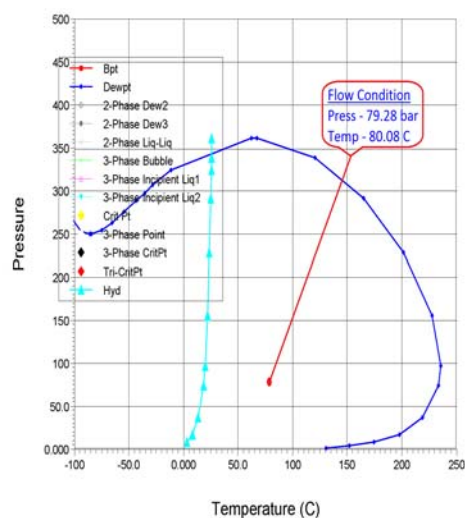


Figure 7.16: Phase-3; ELEM F-MF to ELEM E-MF Phase Envelope/Hydrate Curves

v. ELEM E-MF to AKAB T/L

Phase 3 - ELEP to AKAB Trunkline - Inlet Stream at 300 MMscfd



Phase 3 - ELEP to AKAB Trunkline - Outlet Stream at 300 MMscfd

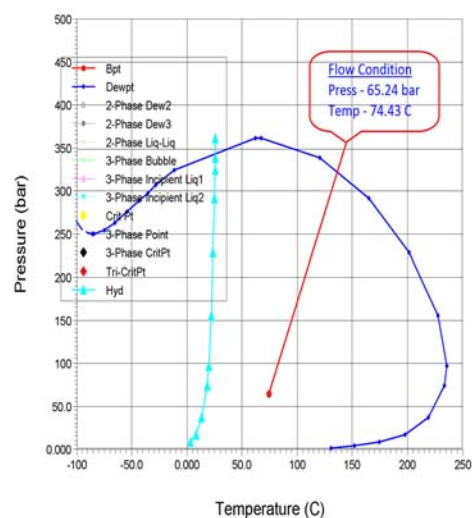


Figure 7.17: Phase-3; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curves

8. Phase-4 Gas Production Simulation Results

The final phase-4 major flow; is from Opugbene field, with a peak production of 200 MMscfd of gas flow anticipated.

As phase-4 sets-in, BBOU and ELEP field would have been depleted and as such have no flow.

Gathering plan as projected; is from 200 MMscfd OPUG field flow, 80 MMscfd SEIB field flow, and 20 MMscfd NUNR field flow.

For OPUG field flow, the pipeline from OPUG F-MF is projected to be tie-in to existing line which runs from BBOU F-MF to ELEP E-MF.

i. OPUG Field Flow to (Tie-in to BBOU – ELEP E-MF) B/L

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 200 MMscfd of gas flow with minimal pressure loss across the 37 km OPUG F-MF – BBOU F-MF bulk-line is 104.0 bar. The corresponding temperature is 101.3 °C.

Delivery Pressure; at BBOU F -MF is 96.2 bar, which is sufficient to support BBOU – ELEP E-MF B/Line flow. The corresponding temperature is 94.9 °C.

The Line Size; which also supported 200 MMscfd of gas flow across the 37 km OPUG F-MF – BBOU F-MF bulk-line with minimal pressure loss and with a velocity below erosional velocity boundary is 24". And the pressure drop is 0.2 bar.

See below; fig. 8.2, table 8.1, table 8.7, & fig. 8.3 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.3 m/s; while at the outlet is 2.4 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 2.8 m/s; while at the outlet is 3.0 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 11.3 m/s, and outlet is 11.6 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.1, table 8.7, & fig. 8.9 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 104.0 bar, the formation temperature is 20.3 °C. While the prevailing flow conditions at the inlet is 104.0 bar pressure and 101.3 °C temperature.

Similarly for outlet stream, at stream pressure of 96.2 bar, the formation temperature is 19.9 °C. While the prevailing flow conditions at the outlet are; pressure 96.2 bar and temperature 94.9 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.15 for more details

ii. OPUG Field Flow (through BBOU – ELEP E-MF) B/L

Pressure and Temperature Regime

Source Pressure at BBOU F-ME; is 96.2 bar which satisfy a good head flow for 200 MMscfd of gas across 9 km BBOU – ELEP E-MF bulk-line. The corresponding temperature is 94.9 °C.

Delivery Pressure; at ELEP E-MF is 78.3 bar, which is sufficient to support ELEP E-MF to AKAB trunk-line flow. The corresponding temperature is 91.0 °C.

The Line Size; existing 16" 9 km BBOU – ELEP E-MF bulk-line was employed in reviewing the flow condition for the 200 MMscfd flow. The pressure drop is 2 bar/km.

See below; fig. 8.2, table 8.2, table 8.8, & fig. 8.4 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.4 m/s; while at the outlet is 2.7 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 6.7 m/s; while at the outlet is 8.1 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 11.6 m/s, and outlet is 12.8 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.2, table 8.8, & fig. 8.10 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 96.18 bar, the formation temperature is 19.9 °C. While the prevailing flow conditions at the inlet is 96.2 bar pressure and 94.9 °C temperature.

Similarly for outlet stream, at stream pressure of 78.3 bar, the formation temperature is 18.7 °C. While the prevailing flow conditions at the outlet are; pressure at 78.3 bar and temperature at 91.0 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.16 for more details

iii. SEIB Field Flow to NunR S-MF

Pressure and Temperature Regime

Source Pressure at the CTP; is 82.0 bar which satisfy a good head flow for 80 MMscfd of gas across 20 km SEIB F-MF to NUNR S-MF bulk-line. The corresponding temperature is 88.1 °C.

Delivery Pressure; at NUNR S-MF is 80.5 bar, which is sufficient to support NUNR S-MF to ELEP E-MF trunk-line flow. The corresponding temperature is 81.5 °C.

The Line Size; the existing 20" x 20 km SEIB F-MF to NUNR S-MF was employed to service the flow and deduce current flow condition under the 55 MMscfd flowrate. The pressure drop is 0.1 bar.

See below; fig. 8.2, table 8.3, table 8.9, & fig. 8.5 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.1 m/s; while at the outlet is 3.1 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 2.1 m/s; while at the outlet is 2.1 m/s. As a result of low volume flow on the line, the prevailing velocity is lower than the minimum calculated.

Erosional Velocity; at inlet is 14.9 m/s, and outlet is 14.8 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.3, table 8.9, & fig. 8.11 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 82.0 bar, the formation temperature is 18.7 °C. While the prevailing flow conditions at the inlet is 82.0 bar pressure and 88.1 °C temperature.

Similarly for outlet stream, at stream pressure of 80.5 bar, the formation temperature is 18.6 °C. While the prevailing flow conditions at the outlet are; pressure 80.5 bar and temperature 81.5 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.17 for more details

*iv. NunR Field Flow to NunR S-MF****Pressure and Temperature Regime***

Source Pressure at the CTP; is 82.0 bar which satisfy a good head flow for 20 MMscfd of gas across 10 m NUNR F-MF - NUNR S-MF bulk-line. The corresponding temperature is 79.7 °C.

Delivery Pressure; at NUNR S-MF is 82.0 bar and with a corresponding temperature of 79.7 °C.

The Line Size; existing 18" x 10 m NUNR F-MF - NUNR S-MF bulk-line was employed in reviewing the flow condition for a 20 MMscfd flow. The pressure drop is 0.0 bar.

See below; fig. 8.2, table 8.4, table 8.10, & fig. 8.6 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at both inlet and outlet is 2.8 m/s.

Prevailing Fluid Velocity; at both inlet and outlet is 0.6 m/s. As a result of low volume flow on the line, the prevailing velocity is lower than the minimum calculated.

Erosional Velocity; at both inlet and outlet is 13.3 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.4, table 8.10, & fig. 8.12 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 82.0 bar, the formation temperature is 19.2 °C. While the prevailing flow conditions at the inlet is 82.0 bar pressure and 79.7 °C temperature.

Similarly for outlet stream, at stream pressure of 82 bar, the formation temperature is 19.23°C. While the prevailing flow conditions at the outlet is pressure at 82 bar and temperature at 79.67°C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.18 for more details

*v. NunR S-MF to ELEP E-MF****Pressure and Temperature Regime***

Source Pressure at NUNR S-MF; the commingled Seibou field flow and NUNR field flow, exited NUNR S-MF at 80.5 bar pressure. The corresponding temperature is 81.0 °C.

Delivery Pressure; at ELEP E-MF is 78.5 bar, which is sufficient to support NUNR S-MF to ELEP E-MF trunk-line flow. The corresponding temperature is 70.0°C.

The Line Size; existing rated 24" x 45 km NUNR S-MF to ELEP E-MF trunk-line was used to service the flow and deduce new flowing condition under 100 MMscfd flowrate.

See below; fig. 8.2, table 8.5, table 8.11, & fig. 8.7 for more details

Flow Velocity Profile

Minimum Fluid velocity; at the inlet is 3.0 m/s; while at the outlet is 3.0 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 1.8 m/s; while at the outlet is 1.7 m/s. As a result of low volume flow on the line, the prevailing velocity is lower than the minimum calculated.

Erosional Velocity; at inlet is 14.5 m/s, and outlet is 14.4 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.5, table 8.11, & fig. 8.13 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.5 bar, the formation temperature is 18.7 °C. While the prevailing flow conditions at the inlet is 80.5 bar pressure and 81.0 °C temperature.

Similarly for outlet stream, at stream pressure of 78.5 bar, the formation temperature is 18.6 °C. While the prevailing flow conditions at the outlet are; pressure at 78.5 bar and temperature at 70.0 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.19 for more details

vi. ELEP E-MF Flow to AKAB

Pressure and Temperature Regime

Source Pressure at the ELEP E-MF Outlet; the mixed Seibou field flow, NUNR field flow and OPUG field flow exited ELEP E-MF at a pressure of 78.3 bar and with a corresponding temperature of 85.0 °C.

Delivery Pressure; at Akabel is 60.8 bar and with a corresponding temperature of 79.1 °C.

The Line Size; existing rated 24" x 28.5 km ELEP E-MF to AKAB trunk-line was employed in reviewing the flow condition for the 300 MMscfd flow. The pressure drop is 0.6 bar/km.

See below; fig. 8.2, table 8.6, table 8.12, & fig. 8.8 for more details

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.8 m/s; while at the outlet is 3.1 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 5.4 m/s; while at the outlet is 6.8 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 13.3 m/s, and outlet is 15.0 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 8.2, table 8.6, table 8.12, & fig. 8.14 for more details

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 78.3 bar, the formation temperature is 18.7 °C. While the prevailing flow conditions at the inlet is 78.3 bar pressure and 85.0 °C temperature.

Similarly for outlet stream, at stream pressure of 60.8 bar, the formation temperature is 17.2 °C. While the prevailing flow conditions at the outlet are; pressure at 60.8 bar and temperature at 79.1 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 8.20 for more details

Figure 8.1: Phase-4 Gas Production; Hysys PFD

8.2. Flow Schematics

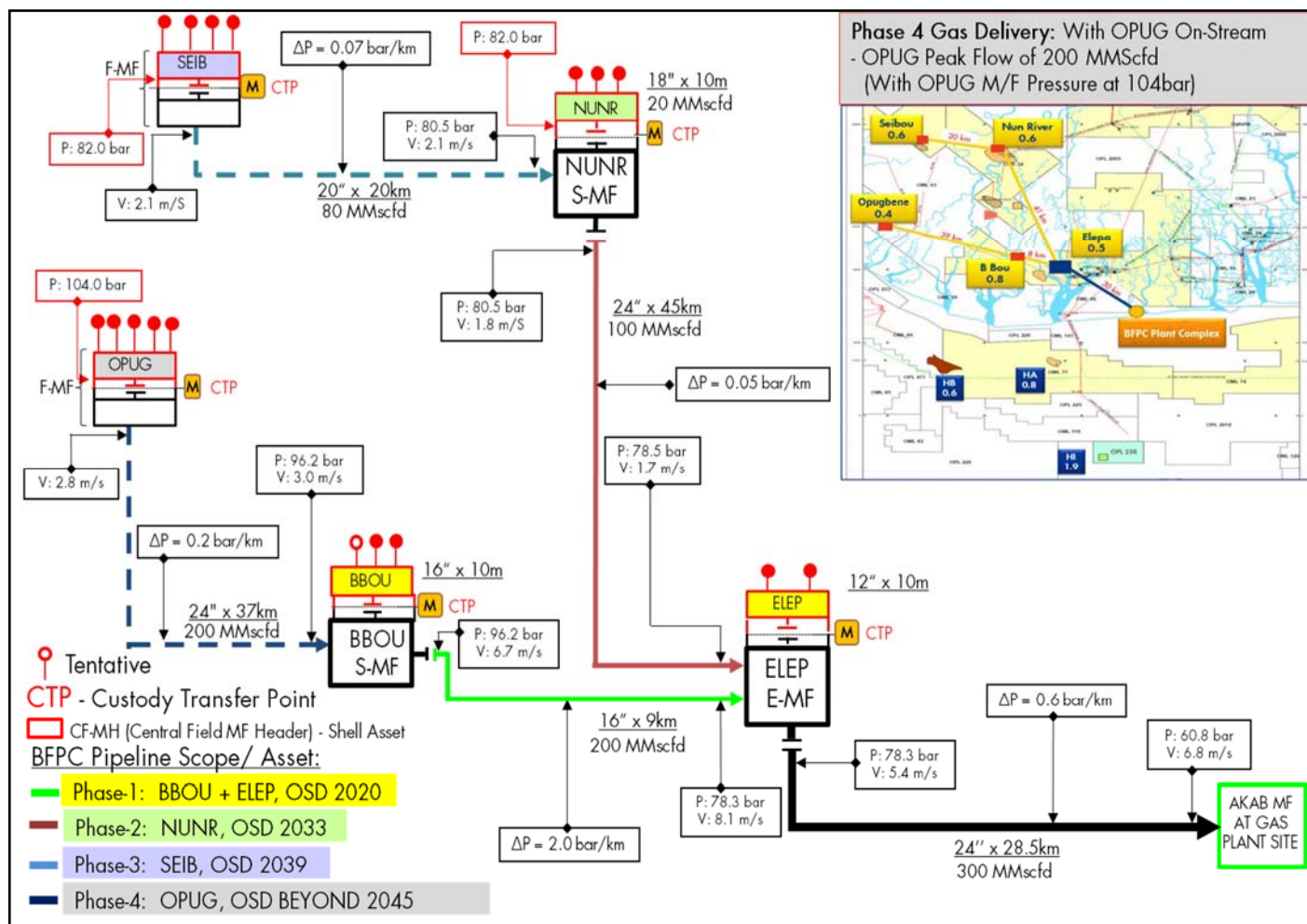


Figure 8.2: Phase-4 Gas Production; Flow Schematics

8.3. Calculation Notes

i. OPUG F-MF to BBOU F-MF B/L

Table 8.1: Phase-4; OPUG F-MF to BBOU F-MF B/L Calculation Note

Line Description																				
Flowline from OPUG to BBOU F-MF																				
Line Dimension	in	mm	m																	
24" SCH 80 (ID)	21.57	548.00	0.55																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	104.00	101.31	390,680	2,375	39.58	0.66	164.50	170,065	2,047	34.12	0.57	83.07	0.02	0.87	220,615	327.70	5.46	0.091	673.19	0.28
Outlet	96.18	94.90	390,680	2,502	41.70	0.70	156.16	168,838	2,177	36.28	0.60	77.56	0.02	0.87	221,766	324.70	5.41	0.090	682.95	0.31
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$\rho_G \cdot V^2_{\text{Overall}}$	$\rho_G \cdot V_G^2$	$\rho_L \cdot V_L^2$	$X = m/m_G(\rho_G/\rho_L)$	$Ve = c/(\sqrt{p_{sc}})$	$Ve = c/(\sqrt{p_{sc}})$							
Inlet	0.24	2.80	2.41	0.39	7.82	5.81	0.149	1,286	482	100.19	0.4557	11.31	2.34							
Outlet	0.24	2.95	2.56	0.38	8.68	6.57	0.146	1,355	509	99.79	0.4426	11.60	2.40							

ii. BBOU F-MF to ELEP E-MF B/L

Table 8.2: Phase-4; BBOU F-MF to ELEP E-MF B/L Calculation Note

Line Description																					
Flowline from BB to ELEP E-MF																					
Line Dimension 16" SCH 80 (ID)		in	mm	m																	
		14.33	363.98	0.36																	
Erosional Vel. Const.		C = 145																			
Min. Erosional Vel. Const.		C = 30																			
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase						
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity	
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP	
	Inlet	96.18	94.90	390,681	2,502	41.70	0.70	156.17	168,838	2,177	36.28	0.60	77.56	0.02	0.87	221,767	324.70	5.41	0.090	682.95	0.31
	Outlet	78.28	91.02	390,681	3,049	50.82	0.85	128.14	171,249	2,735	45.58	0.76	62.62	0.02	0.88	219,373	313.90	5.23	0.087	698.77	0.34
Calc. Parameters																					
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.								
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{Overall}	ρ _G * V _G ²	ρ _L * V _L ²	X = m₁/m₂√(ρ₁/ρ₂)	Ve = c/(Vρ _m)	Ve = c/(Vρ _m)								
Inlet	0.10	6.68	5.81	0.87	44.58	33.75	0.751	6,962	2,618	512.74	0.4426	11.60	2.40								
Outlet	0.10	8.14	7.30	0.84	66.20	53.27	0.702	8,483	3,335	490.30	0.3835	12.81	2.65								

iii. SEIB F-MF to NUNR S-MF B/L

Table 8.3: Phase-4; SEIB F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from SEIBU to NUNR S-MF																				
Line Dimension		in	mm	m																
20" SCH 80 (ID)		17.95	455.93	0.46																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	82.00	88.10	116,258	1,231	20.52	0.34	94.42	73,614	1,171	19.52	0.33	62.88	0.02	0.88	42,643	60.52	1.01	0.017	704.62	0.34
Outlet	80.54	81.55	116,258	1,218	20.30	0.34	95.44	72,737	1,157	19.28	0.32	62.88	0.02	0.88	43,374	61.28	1.02	0.017	707.84	0.35
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² ^{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				p _G * V ² _{Overall}	p _G * V _G ²	p _L * V _L ²	X _{LM} = m _L /m _G √(ρ _G /ρ _L)	Ve = c/(Vp _{min})	Ve = c/(Vp _{min})							
Inlet	0.16	2.09	1.99	0.10	4.38	3.97	0.011	414	249	7.46	0.1730	14.92	3.09							
Outlet	0.16	2.07	1.97	0.10	4.29	3.87	0.011	410	243	7.69	0.1777	14.84	3.07							

iv. NUNR F-MF to NUNR S-MF B/L Calculation Note

Table 8.4: Phase-4; NUNR F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from NUNR to NUNR S-MF																				
Line Dimension 18" SCH 80 (ID)		in	mm	m																
		16.14	410.00	0.41																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	82.00	79.68	33,596	283	4.72	0.08	118.68	17,490	260	4.33	0.07	67.32	0.02	0.86	16,106	23.27	0.39	0.006	692.21	0.34
Outlet	82.00	79.67	33,596	283	4.72	0.08	118.69	17,489	260	4.33	0.07	67.32	0.02	0.86	16,107	23.27	0.39	0.006	692.22	0.34
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$p_G * V^2_{Overall}$	$p_G * V_G^2$	$p_L * V_L^2$	$X_{LM} = m/(m_G \sqrt{(p_G/p_L)})$	$Ve = c/(\sqrt{p_m})$	$Ve = c/(\sqrt{p_m})$							
Inlet	0.13	0.60	0.55	0.05	0.35	0.30	0.002	42	20	1.66	0.2872	13.31	2.75							
Outlet	0.13	0.60	0.55	0.05	0.35	0.30	0.002	42	20	1.66	0.2872	13.31	2.75							

vi. NUNR S-MF to ELEP E-MF T/L Calculation Note

Table 8.5: Phase-4; NUNR S-MF to ELEP E-MF T/L Calculation Note

Line Description																				
Trunk-line from NUNR S-MF to ELEP E-MF																				
Line Dimension		in	mm	m																
24" SCH 80 (ID)		21.57	548.00	0.55																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	80.54	80.95	149,854	1,506	25.10	0.42	99.52	90,329	1,421	23.68	0.39	63.55	0.02	0.87	59,388	84.27	1.40	0.023	704.73	0.35
Outlet	78.47	69.96	149,854	1,469	24.48	0.41	101.99	88,444	1,383	23.05	0.38	63.96	0.01	0.86	61,044	86.09	1.43	0.024	709.07	0.38
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$p_G \cdot V^2_{Overall}$	$p_G \cdot V_G^2$	$p_L \cdot V_L^2$	$X_{GL} = m_G/m_L \sqrt{(\rho_L/\rho_G)}$	$Ve = C/(\sqrt{p_G})$	$Ve = C/(\sqrt{p_{G+L}})$							
Inlet	0.24	1.77	1.67	0.10	3.14	2.80	0.010	313	178	6.94	0.1974	14.53	3.01							
Outlet	0.24	1.73	1.63	0.10	2.99	2.65	0.010	305	170	7.28	0.2073	14.36	2.97							

vi. ELEP E-MF to AKAB T/L Calculation Note

Table 8.6: Phase-4; ELEP E-MF to AKAB T/L Calculation Note

Line Description																					
Trunk-line from ELEP E-MF to AKABEL																					
Line Dimension 24" SCH 80 (ID)		in	mm	m																	
		21.57	548.00	0.55																	
Erosional Vel. Const.		C = 145																			
Min. Erosional Vel. Const.		C = 30																			
Stream Conditions		Overall Phase						Gas Phase						Liquid Phase							
		Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
		Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet		78.28	85.01	540,535	4,548	75.80	1.26	118.85	259,934	4,149	69.15	1.15	62.65	0.02	0.88	280,155	398.70	6.65	0.111	702.71	0.35
Outlet		60.84	79.14	540,535	5,801	96.68	1.61	93.18	262,520	5,414	90.23	1.50	48.49	0.01	0.89	277,490	386.10	6.44	0.107	718.78	0.40
Calc. Parameters																					
		X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
		$A = d^2(\pi/4)$	$V = (Q/A)$	$V = (Q/A)$	$V = (Q/A)$				$p_G * V^2_{Overall}$	$p_G * V_G^2$	$p_L * V_L^2$	$X_{GL} = m/m_G \sqrt{(\rho_G/\rho_L)}$	$Ve = C/(\sqrt{p_G})$	$Ve = C/(\sqrt{p_{G+L}})$							
Inlet		0.24	5.35	4.88	0.47	28.67	23.86	0.220	3,407	1,495	154.82	0.3218	13.30	2.75							
Outlet		0.24	6.83	6.37	0.45	46.64	40.62	0.207	4,346	1,970	148.50	0.2745	15.02	3.11							

8.4. Flow Parameters

i. OPUG F-MF to BBOU F-MF B/L

Table 8.7: Phase-4; OPUG F-MF to BBOU F-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	104.0	101.3	0.7835	<empty>	Transition	0.3046	0.0252	<empty>	0.00	<empty>	0.000003	<empty>	501,921	6,500,198	0.39	2.41
7,400	0	102.5	100.0	0.7838	45.22	Transition	0.3025	0.0254	1.5382	0.00	0.00	0.000003	0.000169	495,515	6,528,784	0.39	2.44
14,800	0	100.9	98.6	0.7841	44.43	Transition	0.3003	0.0257	1.5538	0.00	0.00	0.000003	0.000175	489,152	6,558,353	0.39	2.47
22,200	0	99.4	97.3	0.7844	43.64	Transition	0.2980	0.0260	1.5702	0.00	0.00	0.000003	0.000182	482,827	6,589,009	0.38	2.50
29,600	0	97.8	96.1	0.7847	42.87	Transition	0.2956	0.0263	1.5881	0.00	0.00	0.000003	0.000189	473,794	6,620,133	0.38	2.53
37,000	0	96.2	94.9	0.7849	42.14	Transition	0.2931	0.0266	1.6071	0.00	0.00	0.000003	0.000198	466,165	6,651,114	0.38	2.57

ii. BBOU F-MF to ELEP E-MF B/L

Table 8.8: Phase-4; BBOU F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	96.2	94.9	0.7849	<empty>	Intermittent	0.2713	0.2214	<empty>	0.00	<empty>	0.000140	<empty>	702,311	10,020,351	0.87	5.83
1,800	0	92.9	94.2	0.7875	33.76	Intermittent	0.2653	0.2289	3.3676	0.00	0.00	0.000156	0.002298	687,901	10,144,785	0.86	6.06
3,600	0	89.4	93.5	0.7902	33.58	Intermittent	0.2591	0.2374	3.4913	0.00	0.00	0.000175	0.002579	673,266	10,275,841	0.86	6.32
5,400	0	85.9	92.7	0.7930	33.21	Intermittent	0.2525	0.2469	3.6311	0.00	0.00	0.000199	0.002923	658,291	10,414,456	0.85	6.61
7,200	0	82.2	91.9	0.7960	32.83	Intermittent	0.2455	0.2577	3.7906	0.00	0.00	0.000228	0.003349	642,870	10,561,785	0.85	6.94
9,000	0	78.3	91.0	0.7992	32.42	Intermittent	0.2380	0.2702	3.9748	0.00	0.00	0.000264	0.003890	626,890	10,719,294	0.84	7.32

iii. SEIB F-MF to NUNR S-MF B/L

Table 8.9: Phase-4; BBOU F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	82.0	88.1	0.9003	<empty>	Segregated	0.2279	0.0090	<empty>	0.00	<empty>	0.000000	<empty>	98,111	3,685,195	0.10	1.99
4,000	0	81.7	86.7	0.8993	33.85	Segregated	0.2286	0.0090	0.2928	0.00	0.00	0.000000	0.000014	97,471	3,686,473	0.10	1.99
8,000	0	81.4	85.4	0.8983	33.12	Segregated	0.2293	0.0089	0.2924	0.00	0.00	0.000000	0.000014	96,854	3,687,857	0.10	1.98
12,000	0	81.1	84.1	0.8973	32.40	Segregated	0.2299	0.0089	0.2920	0.00	0.00	0.000000	0.000014	96,258	3,689,341	0.10	1.98
16,000	0	80.8	82.8	0.8963	31.70	Segregated	0.2306	0.0089	0.2916	0.00	0.00	0.000000	0.000014	95,683	3,690,917	0.10	1.98
20,000	0	80.5	81.5	0.8954	31.00	Segregated	0.2312	0.0089	0.2912	0.00	0.00	0.000000	0.000014	95,126	3,692,579	0.10	1.97

iv. NUNR F-MF to NUNR S-MF B/L

Table 8.10: Phase-4; NUNR F-MF to NUNR S-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	82.0	79.7	0.8419	<empty>	Segregated	0.3603	0.0012	<empty>	0.00	<empty>	0.000000	<empty>	40,562	983,097	0.05	0.55
2	0	82.0	79.7	0.8419	28.12	Segregated	0.3603	0.0012	0.0000	0.00	0.00	0.000000	0.000000	40,562	983,095	0.05	0.55
4	0	82.0	79.7	0.8419	28.12	Segregated	0.3603	0.0012	0.0000	0.00	0.00	0.000000	0.000000	40,562	983,094	0.05	0.55
6	0	82.0	79.7	0.8419	28.12	Segregated	0.3603	0.0012	0.0000	0.00	0.00	0.000000	0.000000	40,562	983,092	0.05	0.55
8	0	82.0	79.7	0.8419	28.12	Segregated	0.3603	0.0012	0.0000	0.00	0.00	0.000000	0.000000	40,561	983,091	0.05	0.55
10	0	82.0	79.7	0.8419	28.12	Segregated	0.3603	0.0012	0.0000	0.00	0.00	0.000000	0.000000	40,561	983,089	0.05	0.55

v. NUNR S-MF to ELEP E-MF T/L

Table 8.11: Phase-4; NUNR S-MF to ELEP E-MF T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	80.5	80.9	0.8855	<empty>	Segregated	0.2540	0.0057	<empty>	0.00	<empty>	0.000000	<empty>	108,594	3,815,841	0.10	1.68
9,000	0	80.1	78.6	0.8836	32.73	Segregated	0.2556	0.0057	0.4156	0.00	0.00	0.000000	0.000015	107,523	3,816,845	0.10	1.67
18,000	0	79.7	76.3	0.8818	31.35	Segregated	0.2571	0.0056	0.4138	0.00	0.00	0.000000	0.000015	106,521	3,818,059	0.10	1.66
27,000	0	79.3	74.1	0.8800	30.01	Segregated	0.2585	0.0056	0.4121	0.00	0.00	0.000000	0.000015	105,582	3,819,456	0.10	1.65
36,000	0	78.9	72.0	0.8784	28.72	Segregated	0.2599	0.0056	0.4105	0.00	0.00	0.000000	0.000015	104,701	3,821,015	0.10	1.64
45,000	0	78.5	70.0	0.8768	27.47	Segregated	0.2612	0.0056	0.4090	0.00	0.00	0.000000	0.000015	103,874	3,822,717	0.10	1.63

vi. ELEP E-MF to AKAB T/L

Table 8.12: Phase-4; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	78.3	85.0	0.8253	<empty>	Intermittent	0.2233	0.0668	<empty>	0.00	<empty>	0.000027	<empty>	510,373	10,929,638	0.47	4.89
5,700	0	75.1	83.9	0.8271	35.56	Intermittent	0.2177	0.0695	3.2361	0.00	0.00	0.000030	0.001404	499,170	11,054,451	0.47	5.11
11,400	0	71.8	82.8	0.8291	34.91	Intermittent	0.2118	0.0725	3.3791	0.00	0.00	0.000035	0.001608	487,535	11,187,708	0.47	5.36
17,100	0	68.3	81.7	0.8312	34.23	Intermittent	0.2054	0.0761	3.5447	0.00	0.00	0.000040	0.001867	475,392	11,330,820	0.46	5.64
22,800	0	64.7	80.4	0.8335	33.51	Intermittent	0.1985	0.0803	3.7394	0.00	0.00	0.000047	0.002205	462,673	11,485,632	0.46	5.98
28,500	0	60.8	79.1	0.8360	32.76	Intermittent	0.1911	0.0853	3.9726	0.00	0.00	0.000057	0.002659	449,260	11,654,627	0.46	6.38

8.5. Pressure Profile Plots

i. OPUG F-MF to BBOU F-MF B/L

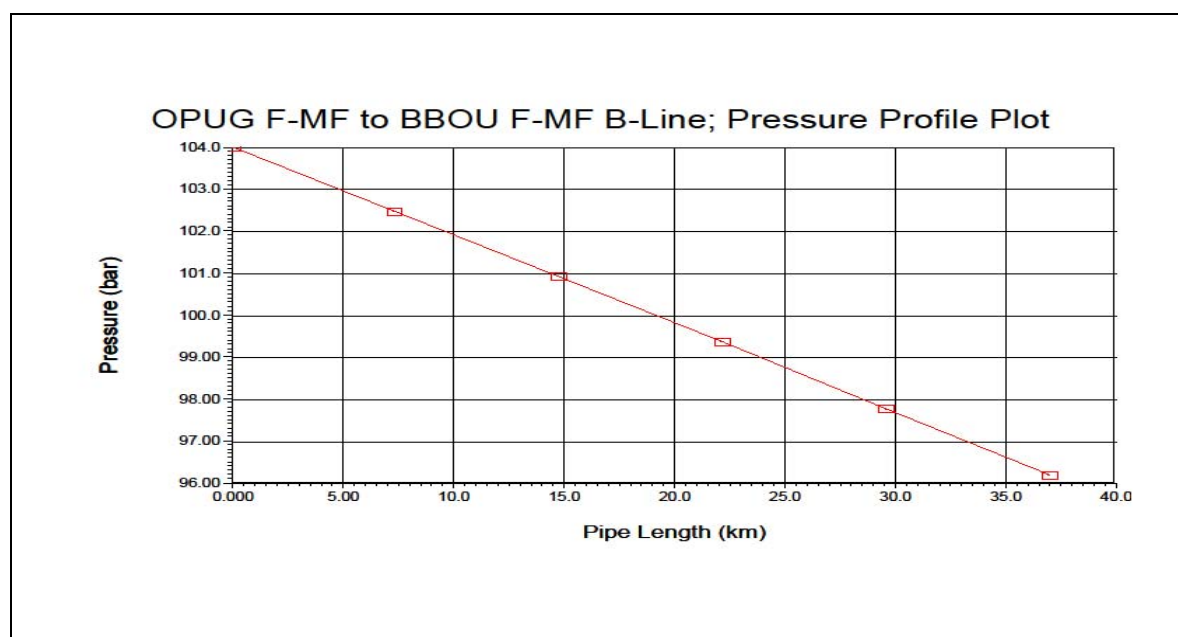


Figure 8.3: Phase-4; OPUG F-MF to BBOU F-MF B/L Pressure Profile Plot

ii. BBOU F-MF to ELEP E-MF B/L

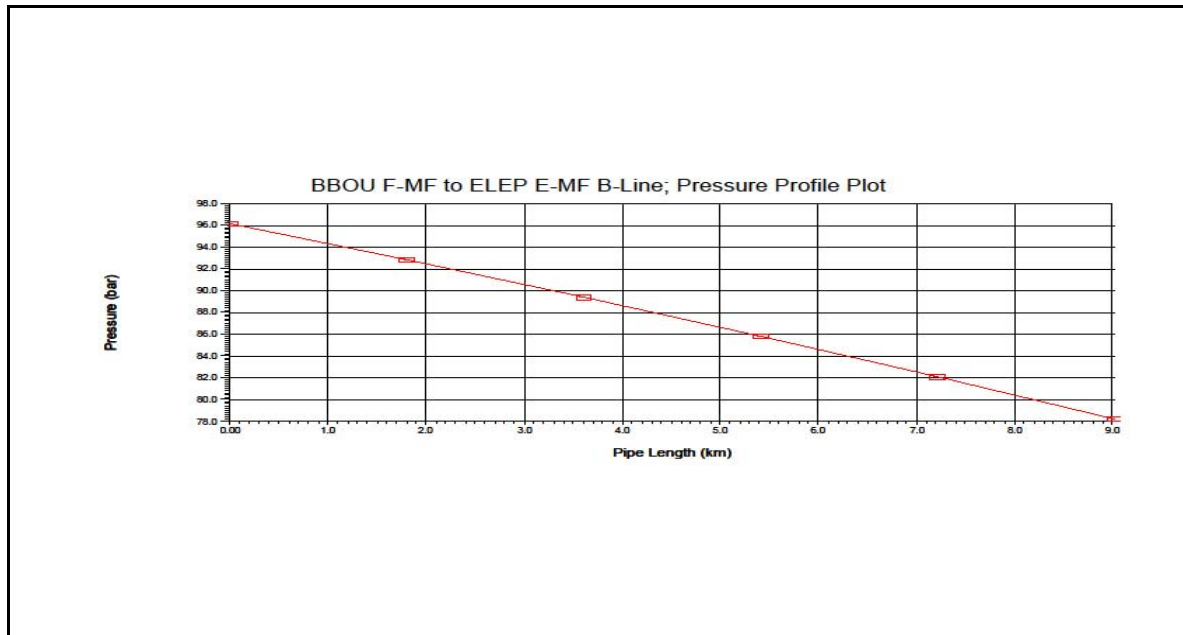


Figure 8.4: Phase-4; BBOU F-MF to ELEP E-MF B/L Pressure Profile Plot

iii. SEIB F-MF to NUNR S-MF B/L

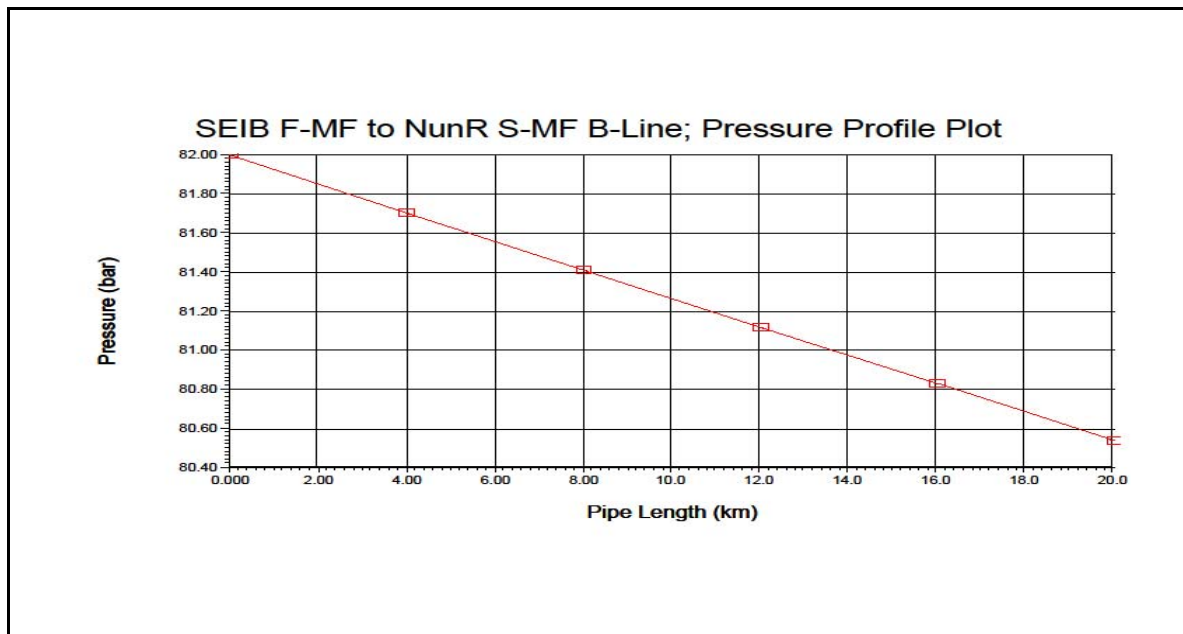


Figure 8.5: Phase-4; SEIB F-MF to NUNR S-MF B/L Pressure Profile Plot

iv. NUNR F-MF to NUNR S-MF B/L

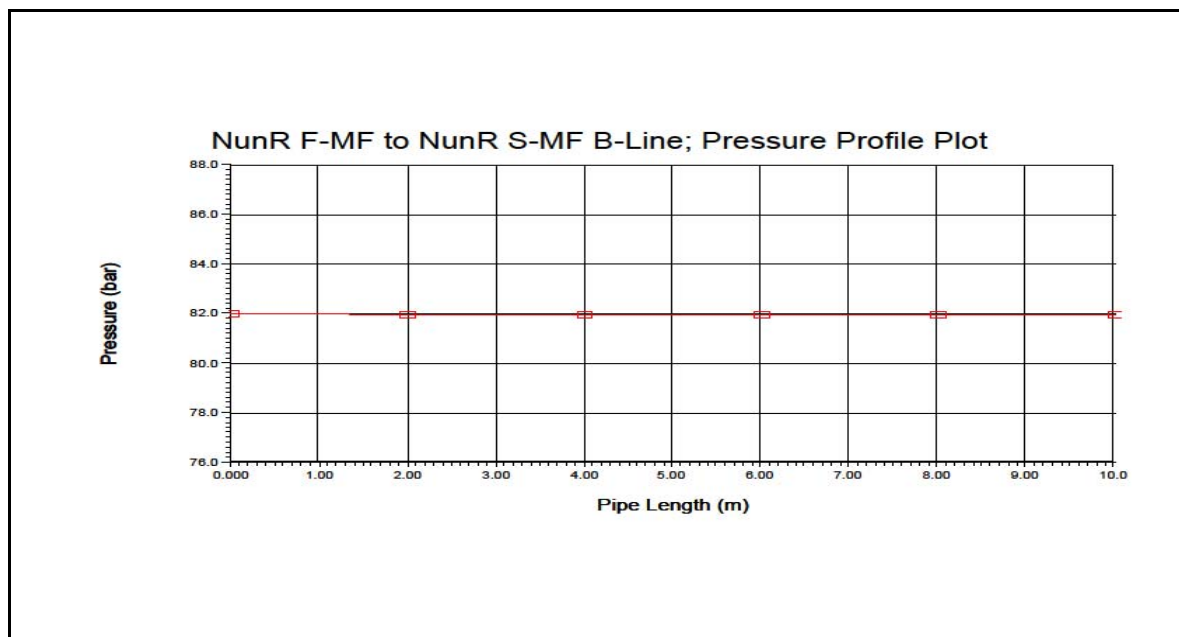


Figure 8.6: Phase-4; NUNR F-MF to NUNR S-MF B/L Pressure Profile Plot

v. NUNR S-MF to ELEP E-MF T/L

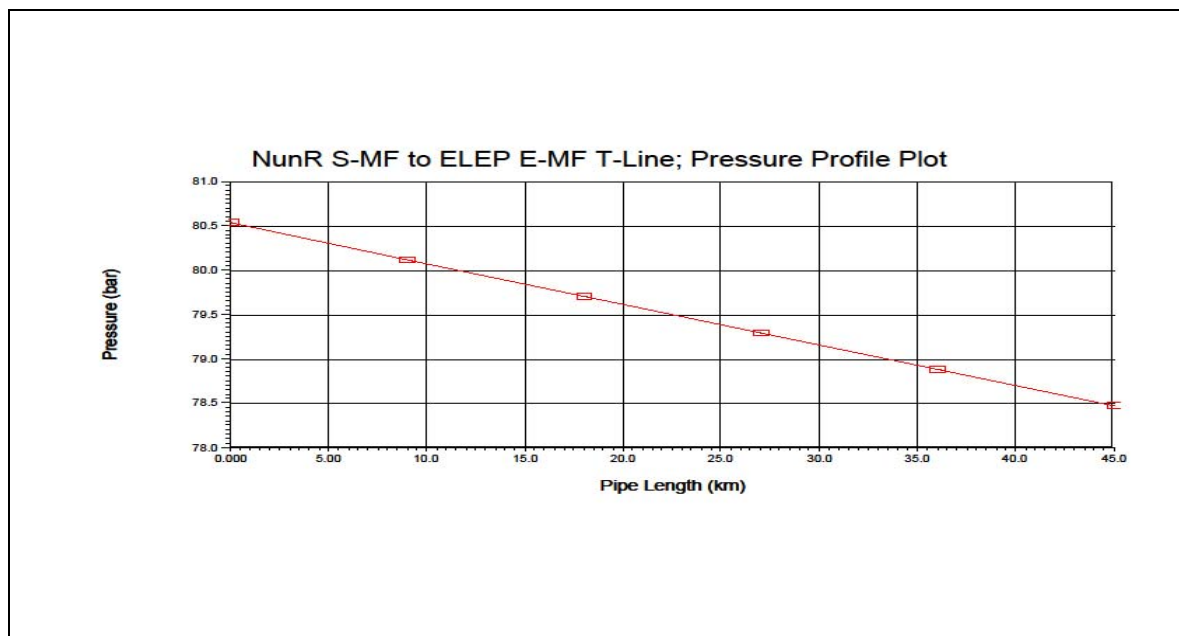


Figure 8.7: Phase-4; NUNR S-MF to ELEP E-MF T/L Pressure Profile Plot

vi. ELEP E-MF to AKAB T/L Pressure Profile Plots

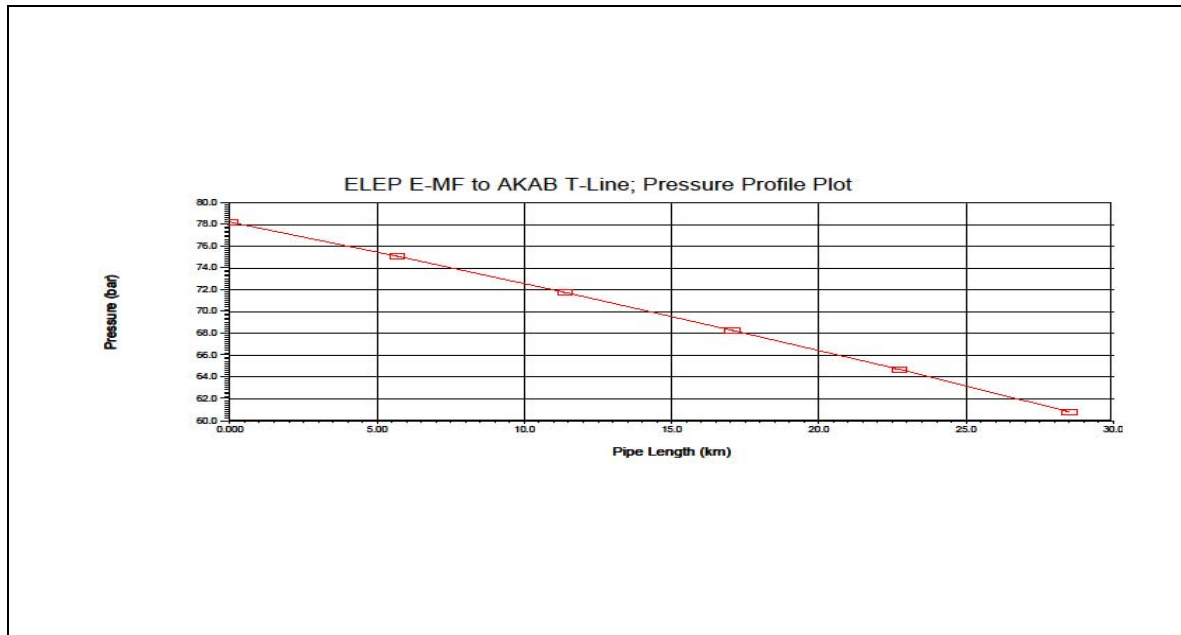


Figure 8.8: Phase-4; ELEP E-MF to AKAB T/L Pressure Profile Plot

8.6. Velocity Profile Plots

i. OPUF F-MF to BBOU F-MF B/L

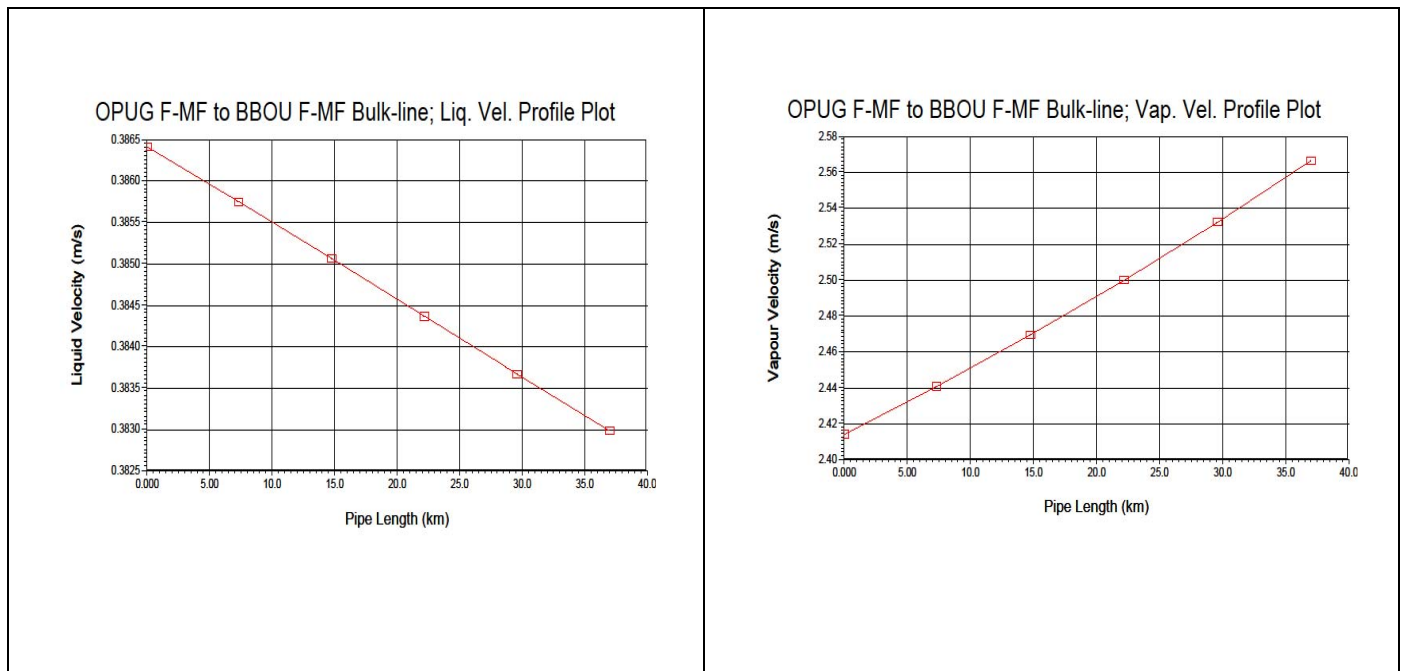


Figure 8.9: Phase-4; OPUF F-MF to BBOU F-MF B/L Velocity Profile Plot

ii. BBOU F-MF to ELEP E-MF B/L

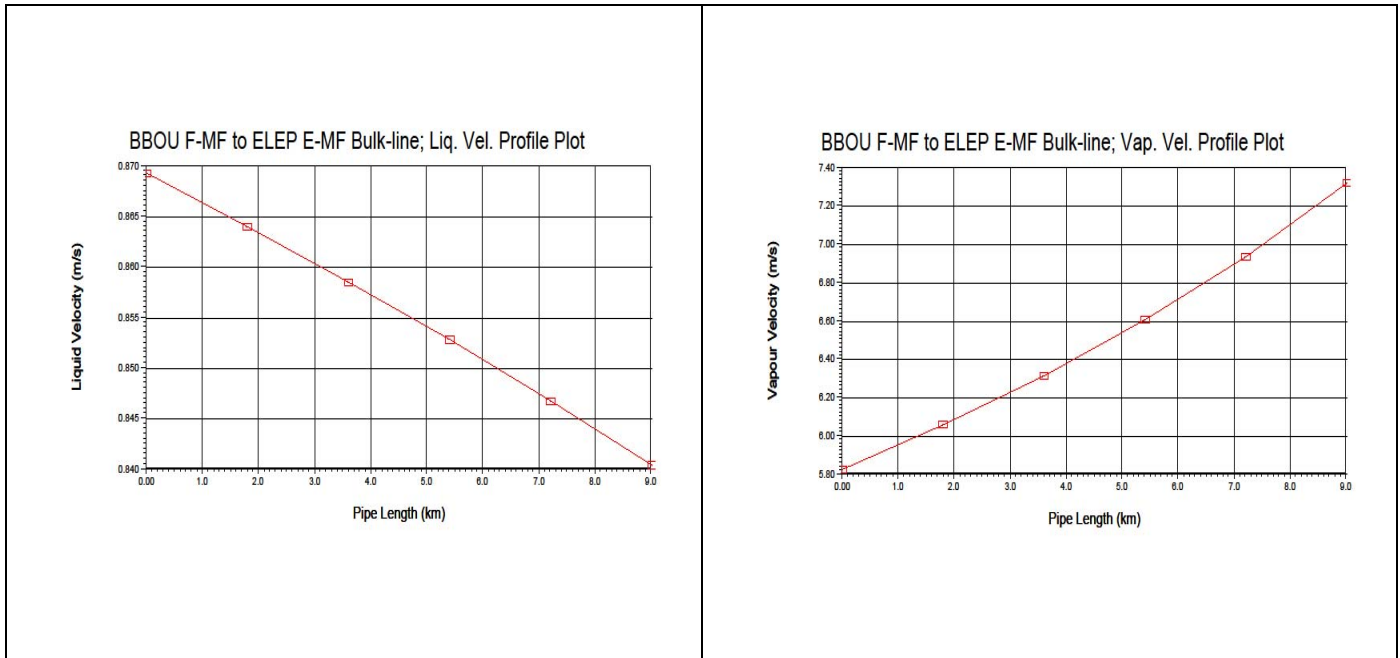


Figure 8.10: Phase-4; BBOU F-MF to ELEP E-MF B/L Velocity Profile Plot

iii. SEIB F-MF to NUNR S-MF B/L

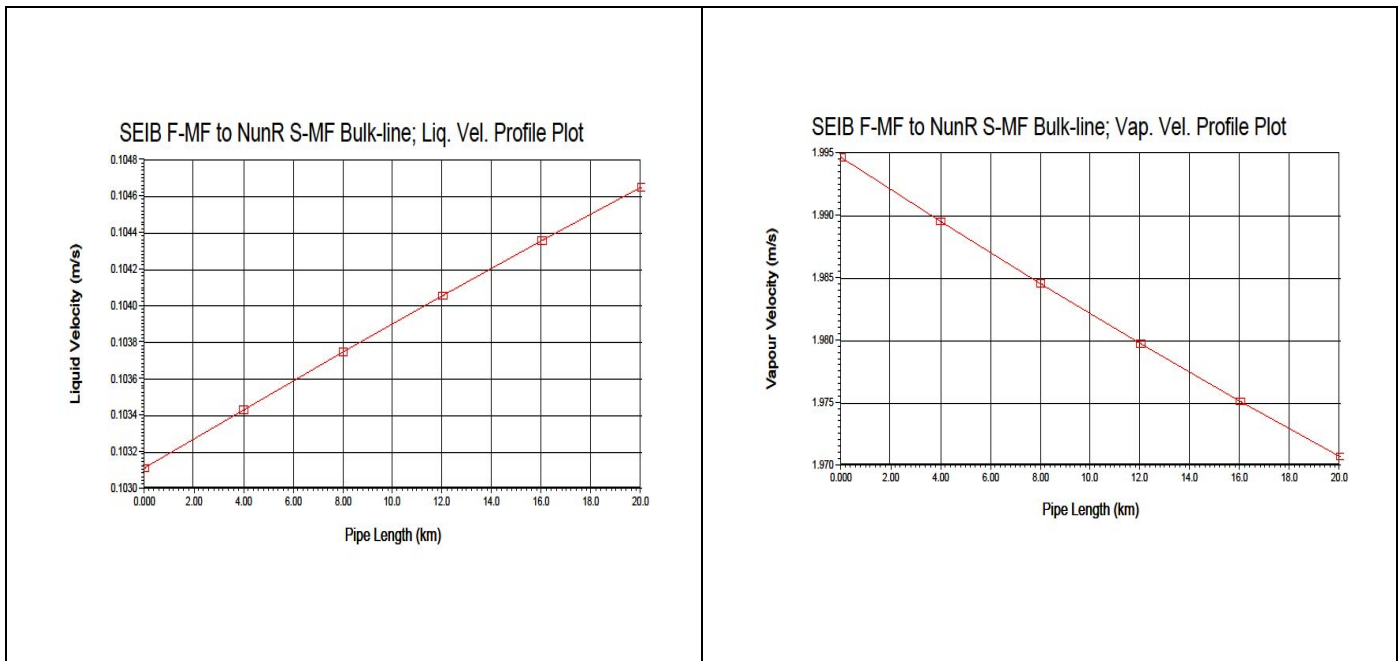


Figure 8.11: Phase-4; SEIB F-MF to NUNR S-MF B/L Velocity Profile Plot

iv. NUNR F-MF to NUNR S-MF B/L

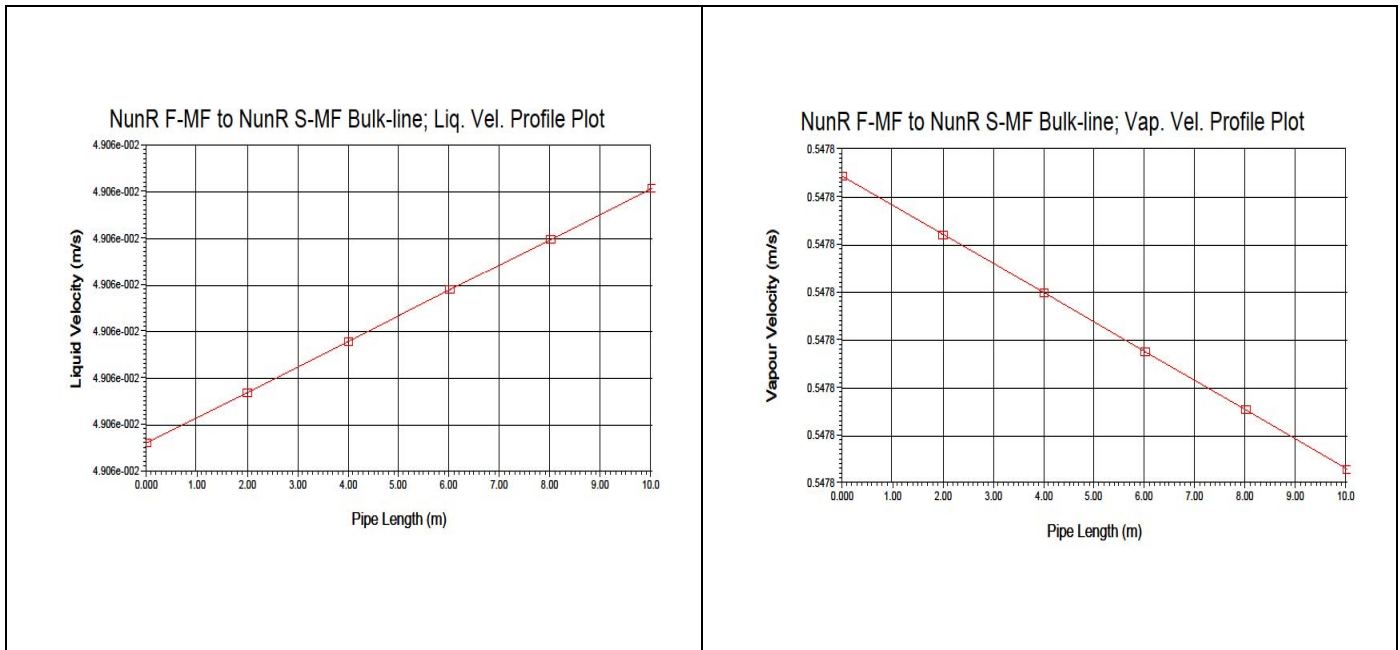


Figure 8.12: Phase-4; NUNR F-MF to NUNR S-MF B/L Velocity Profile Plot

v. NUNR S-MF to ELEP E-MF T/L

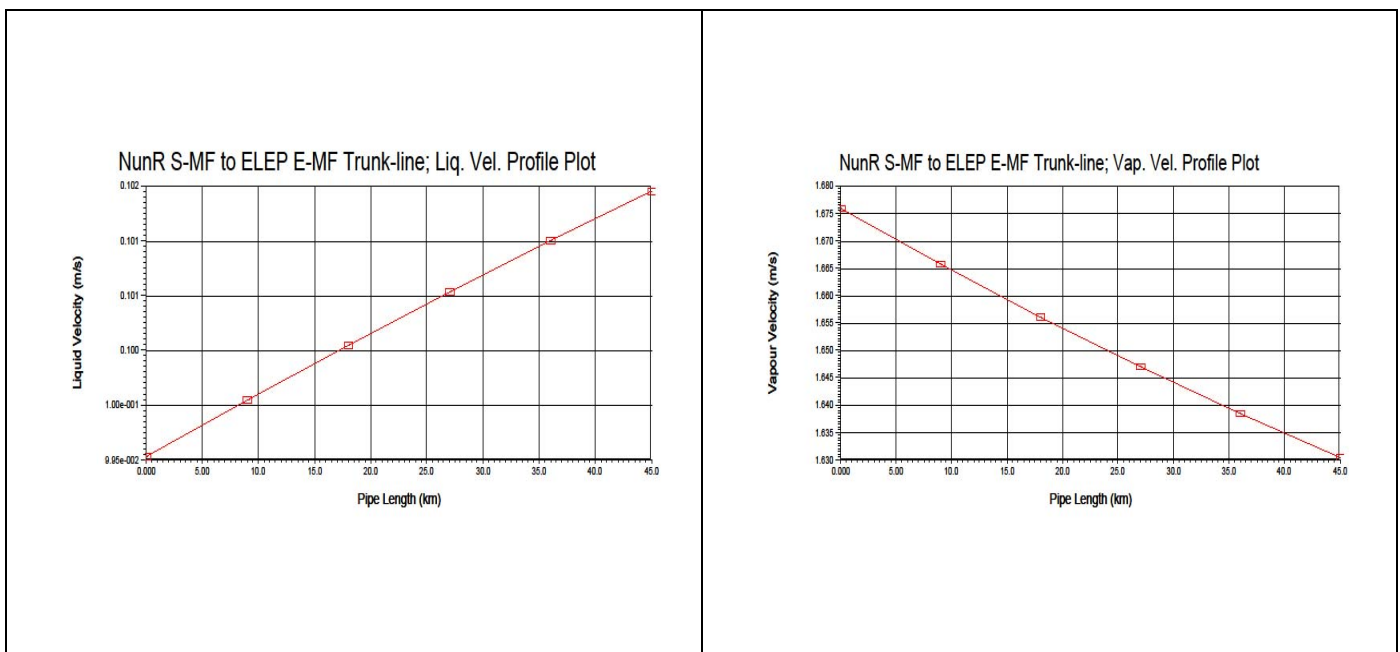


Figure 8.13: Phase-4; NUNR S-MF to ELEP E-MF T/L Velocity Profile Plot

vi. ELEP E-MF to AKAB T/L

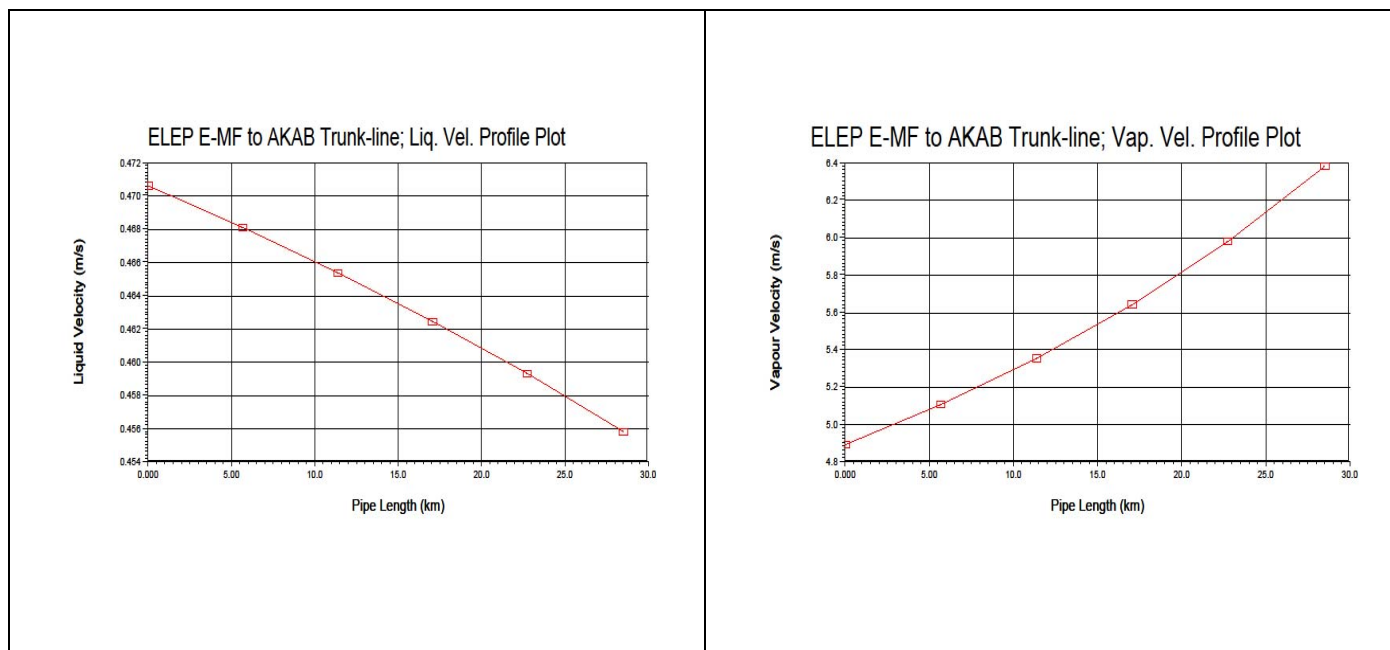


Figure 8.14: Phase-4; ELEP E-MF to AKAB T/L Velocity Profile Plot

8.7. Phase Envelopes/Hydrate Curves

i. OPUG F-MF to BBOU F-MF B/L

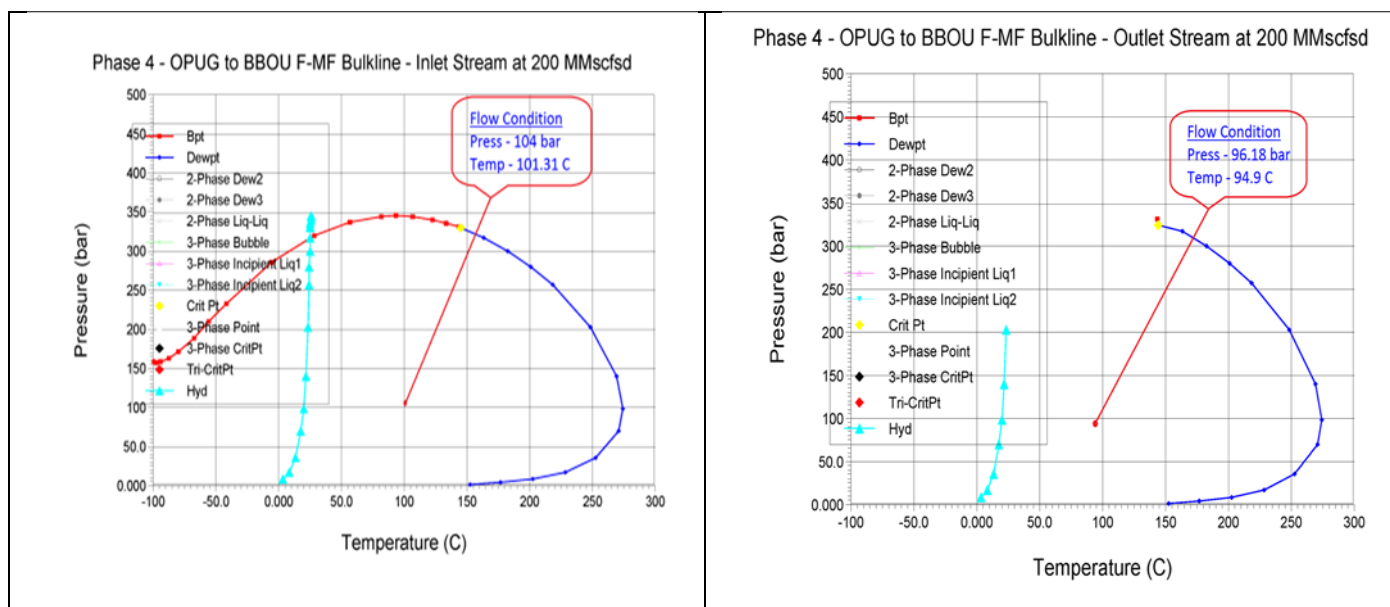


Figure 8.15: Phase-4; OPUG F-MF to BBOU F-MF B/L Phase Envelope/Hydrate Curves

ii. BBOU F-MF to ELEP E-MF B/L

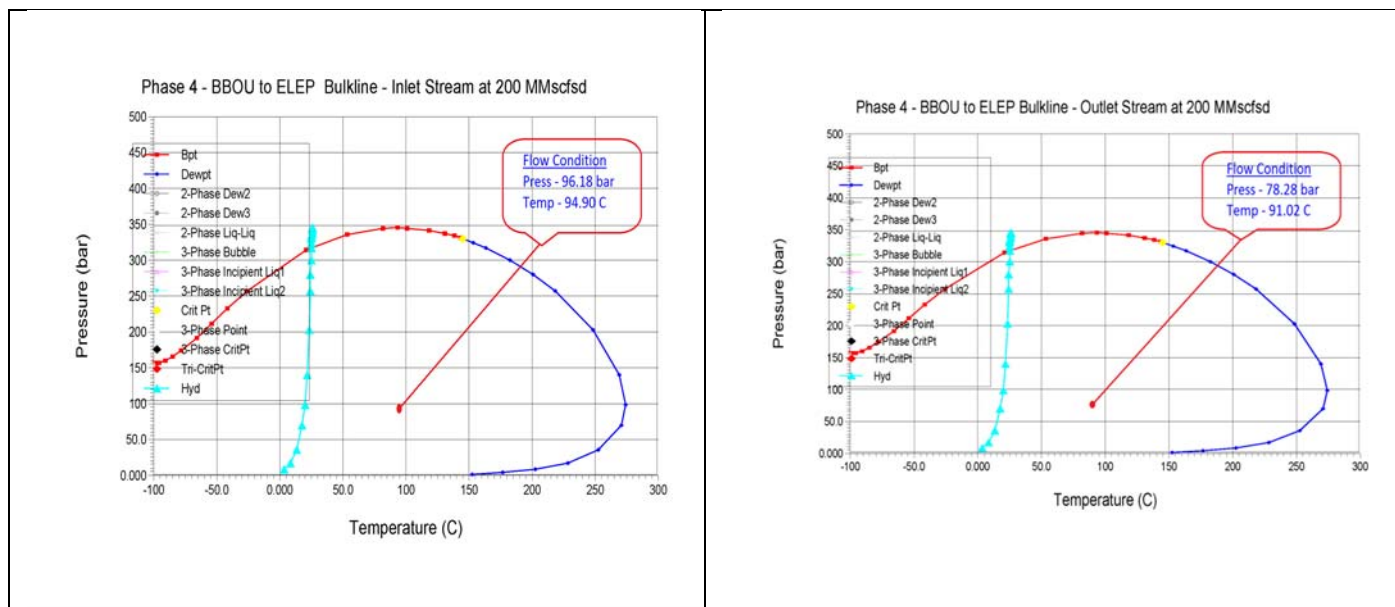


Figure 8.16: Phase-4; BBOU F-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curves

iii. SEIB F-MF to NUNR S-MF B/L

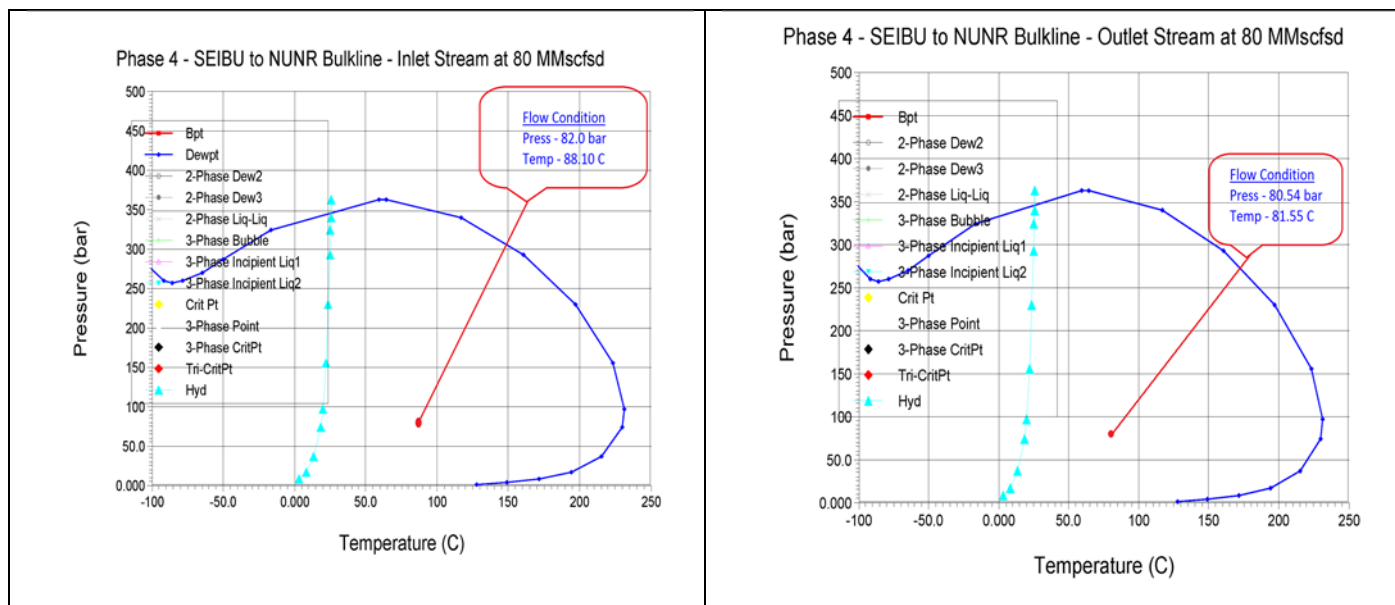


Figure 8.17: Phase-4; SEIB F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

iv. NUNR F-MF to NUNR S-MF B/L

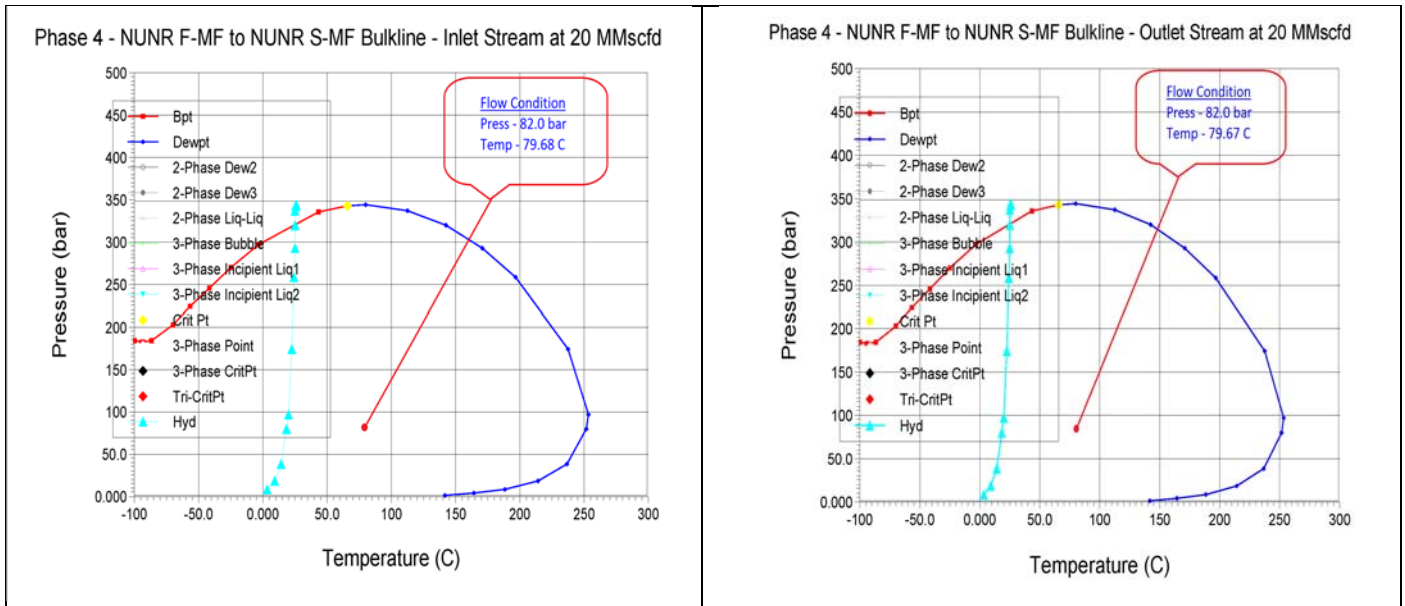


Figure 8.18: Phase-4; NUNR F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

vi. NUNR S-MF to ELEP E-MF T/L

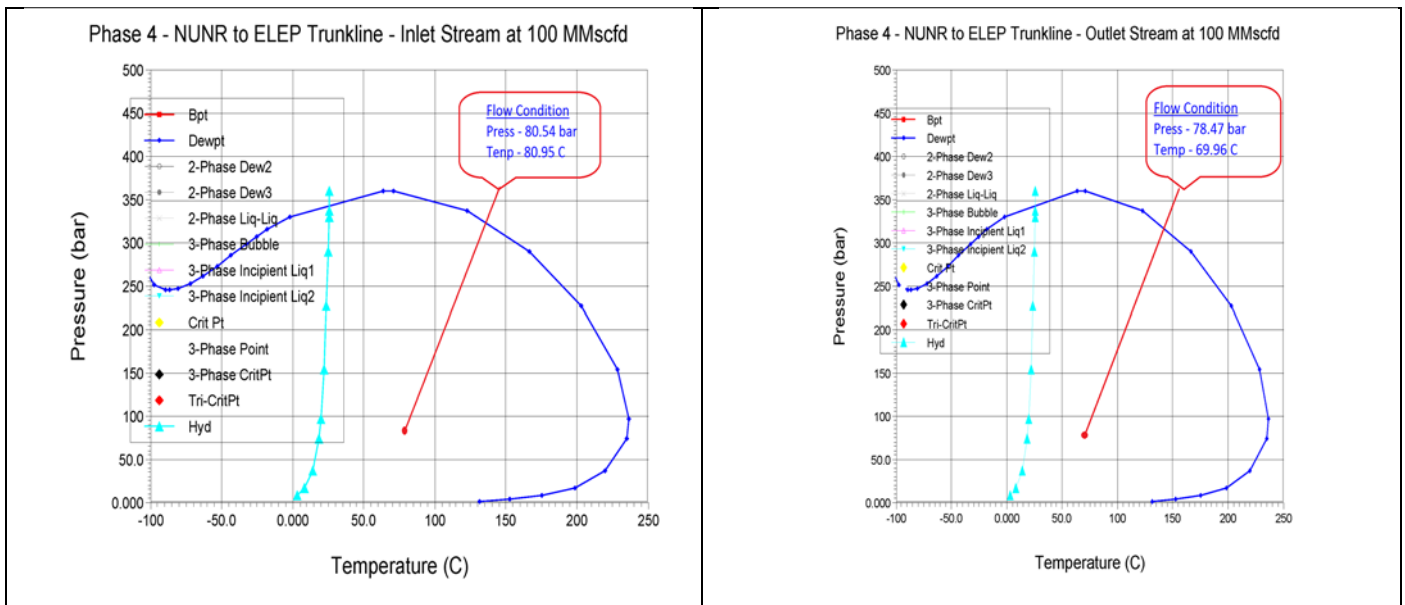


Figure 8.19: Phase-4; NUNR S-MF to ELEP E-MF T/L Phase Envelope/Hydrate Curves

vi. ELEP E-MF to AKAB T/L

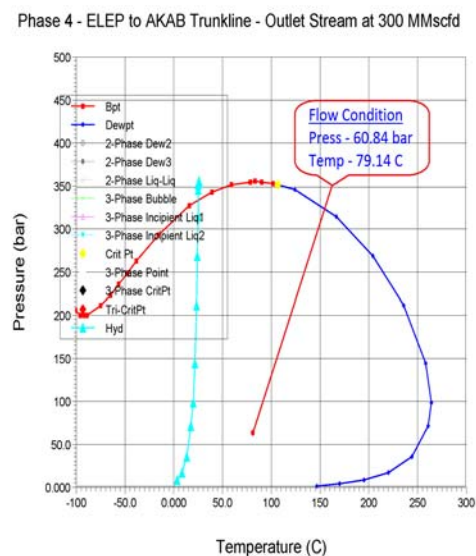
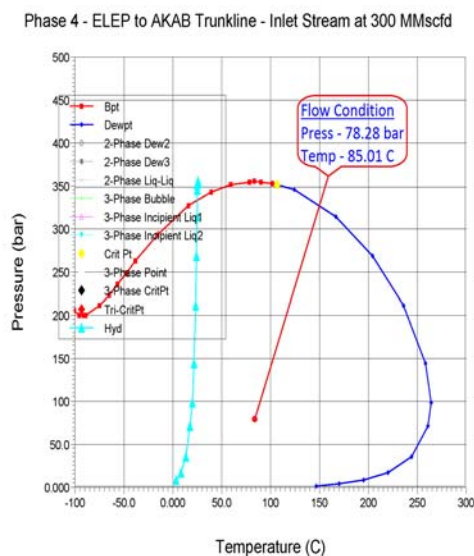


Figure 8.20: Phase-4; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curves

9. 2033 Gas Production Simulation Result (Water Production Sensitivity)

This scenario (that is, 2033 production) was undertaken to deduce the gas behaviour with free water production.

For this scenario, the existing rated 16" BBOU-ELEP E-MF bulk-line, 12" ELEP F-MF to ELEP E-MF bulk-line, 18" NUNR F-MF to NUNR S-MF, 24" NUNR S-MF to ELEP E-MF and 24" ELEP E-MF – AKAB trunk-line was employed to deduce the flow parameters (i.e. pressure, temperature, velocity, etc) for this study.

i. BBOU to ELEP E-MF (at 207 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; that supported 207 MMscfd of gas and 9,727 stb/day of produced water flow with minimal pressure loss across the 9 km BBOU - ELEP E-MF bulk-line is 101.0 bar. And the corresponding temperature is 63.6 °C.

Delivery Pressure; at ELEP E-MF is 77.3 bar, which is sufficient to support ELEP E-MF to AKAB trunk-line delivery at required pressure spec. The corresponding temperature is 58.6 °C.

The Line Size; for this phase-1/scenario-4 study, 16" rated BBOU F-MF – ELEP E-MF bulk-line was employed in reviewing the flow conditions. Pressure drop, for this case is 2.6 bar/km

See below; fig. 9.2, table 9.1, table 9.6, & fig. 9.3 for more detail

Flow Velocity Profile

Minimum Fluid Flow Velocity; that will ensure the cleanliness of the pipe at the inlet and outlet of the bulk-line is 3.1 m/s and 3.6 m/s respectively.

Prevailing Fluid Flow Velocity; that supports fluid delivery is 6.9 m/s and 9.0 m/s at both inlet and outlet of the B/L. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet and outlet of the bulk-line is 15.2 m/s and 17.3 m/s respectively. The indication is that, prevailing velocity is below erosional velocity and pipe wall erosion cannot set-in.

See below; fig. 9.2, table 9.1, table 9.6, & fig. 9.8 for more detail

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 101.0 bar, the formation temperature is 16.2 °C. While the prevailing flow conditions at the inlet are; 101.0 bar pressure and 63.6 °C temperature.

Similarly for outlet stream, at stream pressure of 77.3 bar, the formation temperature is 14.3 °C. While the prevailing flow conditions at the outlet are; pressure 77.3 bar and temperature 58.6 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 9.13 for more details

ii. ELEP F-MF to ELEP E-MF B/L (at 70 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; is 80.0 bar which satisfy a good head flow for 70 MMscfd of gas across 10m ELEP F-MF - ELEP E-MF bulk-line. The corresponding temperature is 113.4 °C.

Delivery Pressure; at ELEP E-MF is 79.99 bar and with a corresponding temperature of 113.4 °C.

The Line Size; for this phase-1/scenario-4 study, 12" rated ELEP F-MF – ELEP E-MF bulk-line was employed in reviewing the flow conditions. Pressure drop, for this case is 0.0 bar.

See below; fig. 9.2, table 9.2, table 9.7, & fig. 9.4 for more detail

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet and outlet is 2.6 m/s.

Prevailing Fluid Flow Velocity; at both inlet and outlet is 5.7 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at both inlet and outlet is 12.7 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 9.2, table 9.2, table 9.7, & fig. 9.9 for more detail

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 80.0 bar, the formation temperature is 17.9 °C. While the prevailing flow conditions at the inlet are; 80.0 bar pressure and 113.4 °C temperature.

Similarly for outlet stream, at stream pressure of 79.99 bar, the formation temperature is also 17.9 °C. The prevailing flow conditions at the outlet are; pressure 79.99 bar and temperature 113.4 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 9.14 for more details

iii. NUNR F-MF to NUNR S-MF B/L (at 22 MMscfd)

Pressure and Temperature Regime

Source Pressure at the CTP; is 77.5 bar which satisfy a good head flow for 22 MMscfd of gas across 10m NUNR F-MF to NUNR S-MF bulk-line. The corresponding temperature is 81.0 °C.

Delivery Pressure; at ELEP E-MF is 77.5 bar and with a corresponding temperature of 81.0 °C.

The Line Size; for this phase-1/scenario-4 study, 18" rated NUNR F-MF - NUNR S-MF bulk-line was employed in reviewing the flow conditions. Pressure drop, for this case is 0.0 bar.

See below; fig. 9.2, table 9.3, table 9.8, & fig. 9.5 for more detail

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet and outlet is 2.85 m/s.

Prevailing Fluid Flow Velocity; at both inlet and outlet is 0.7 m/s.

Erosional Velocity; at both inlet and outlet is 13.8 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 9.2, table 9.3, table 9.8, & fig. 9.10 for more detail

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 77.5 bar, the formation temperature is 18.9 °C. While the prevailing flow conditions at the inlet are; 77.5 bar pressure and 81.0 °C temperature.

Similarly for outlet stream, at stream pressure of 77.5 bar, the formation temperature is also 18.9 °C. The prevailing flow conditions at the outlet are; pressure 77.5 bar and temperature 81.0 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 9.15 for more details

*iv. NUNR S-MF to ELEP E-MF B/L (at 22 MMscfd)****Pressure and Temperature Regime***

Source Pressure at NUNR S-MF; is 77.5 bar which satisfy a good head flow for 22 MMscfd of gas across 45 km NUNR S-MF to ELEP E-MF bulk-line. The corresponding temperature is 81.0 °C.

Delivery Pressure; at ELEP E-MF is 77.4 bar, which is sufficient to support ELEP E-MF to AKAB bulk-line flow. The corresponding temperature is 47.8 °C.

The Line Size; for this 2033 production study, existing 24" x 45 km rated NUNR S-MF to ELEP E-MF bulk-line line was employed in reviewing the flow conditions. For this scenario, pressure drop is 0 bar/km.

See below; fig. 9.2, table 9.4, table 9.9, & fig. 9.6 for more detail

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 2.9 m/s; while at the outlet is 2.6 m/s.

Prevailing Fluid Flow Velocity; at the inlet is 0.4 m/s; while at the outlet is 0.3 m/s.

Erosional Velocity; at inlet is 13.8 m/s, and outlet is 12.6 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 9.2, table 9.4, table 9.9, & fig. 9.11 for more detail

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 77.5 bar, the formation temperature is 18.9 °C. While the prevailing flow condition at the inlet is; 77.5 bar pressure and 81.0 °C temperature.

Similarly for outlet stream, at stream pressure of 77.4 bar, the formation temperature is also 18.9 °C. While the prevailing flow conditions at the outlet is; pressure 77.5 bar and temperature 47.8 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 9.16 for more details

*v. ELEP E-MF to AKAB (at 299 MMscfd)****Pressure and Temperature Regime***

Source Pressure at the ELEP E-MF outlet; the mixed BBOU field flow and ELEP field flow exited ELEP E-MF at a pressure of 77.3 bar and with a corresponding temperature of 80.0 °C.

Delivery Pressure; at AKAB is 60.7 bar and with a corresponding temperature of 75.5 °C.

The Line Size; the existing rated 24" ELEP E-MF – AKAB trunk-line shows capacity to support 299 MMscfd flow, with minimal pressure loss and with a velocity below erosional velocity mark. For this case, the pressure drop is 0.4 bar/km.

See below; fig. 9.2, table 9.5, table 9.10, & fig. 9.7 for more detail

Flow Velocity Profile

Minimum Fluid Velocity; at the inlet is 3.2 m/s; while at the outlet is 3.6 m/s.

Prevailing Fluid Flow Velocity; at inlet is 6.2 m/s and outlet is 7.8 m/s. This indicates fluid velocity is above the minimum required for efficient flow.

Erosional Velocity; at inlet is 15.7 m/s and outlet is 17.6 m/s. The prevailing velocity is below erosional velocity, the indication is that, pipe wall erosion will not set-in.

See below; fig. 9.2, table 9.5, table 9.10, & fig. 9.12 for more detail

Phase Envelope/Hydrate Curve

Hydrate Formation; at the inlet stream pressure of 77.3 bar, the formation temperature is 16.5 °C. While the prevailing flow conditions at the inlet are; 77.3 bar pressure and 80.0 °C temperature.

Similarly for outlet stream, at stream pressure of 60.7 bar, the formation temperature is 14.9 °C. While the prevailing flow conditions at the outlet are; pressure 60.7 bar and temperature 75.5 °C.

Fluid transverse conditions (i.e. pressure & temperature) from inlet-outlet of the line is above the hydrate curve, the indication is that hydrates will not form.

See below; fig. 9.17 for more details

9.1. Hysys PFD

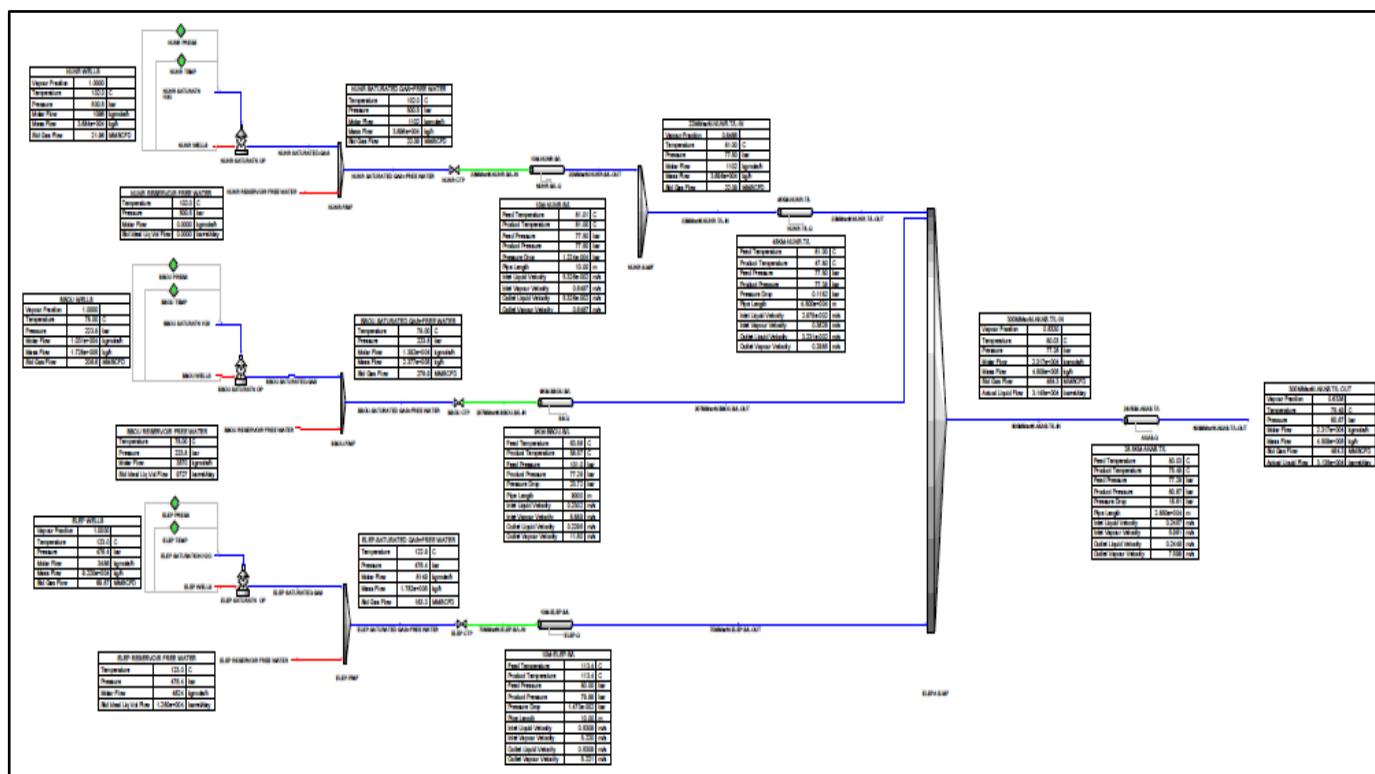


Figure 9.1: 2033 Gas Production; PFD

9.2. Flow Schematics

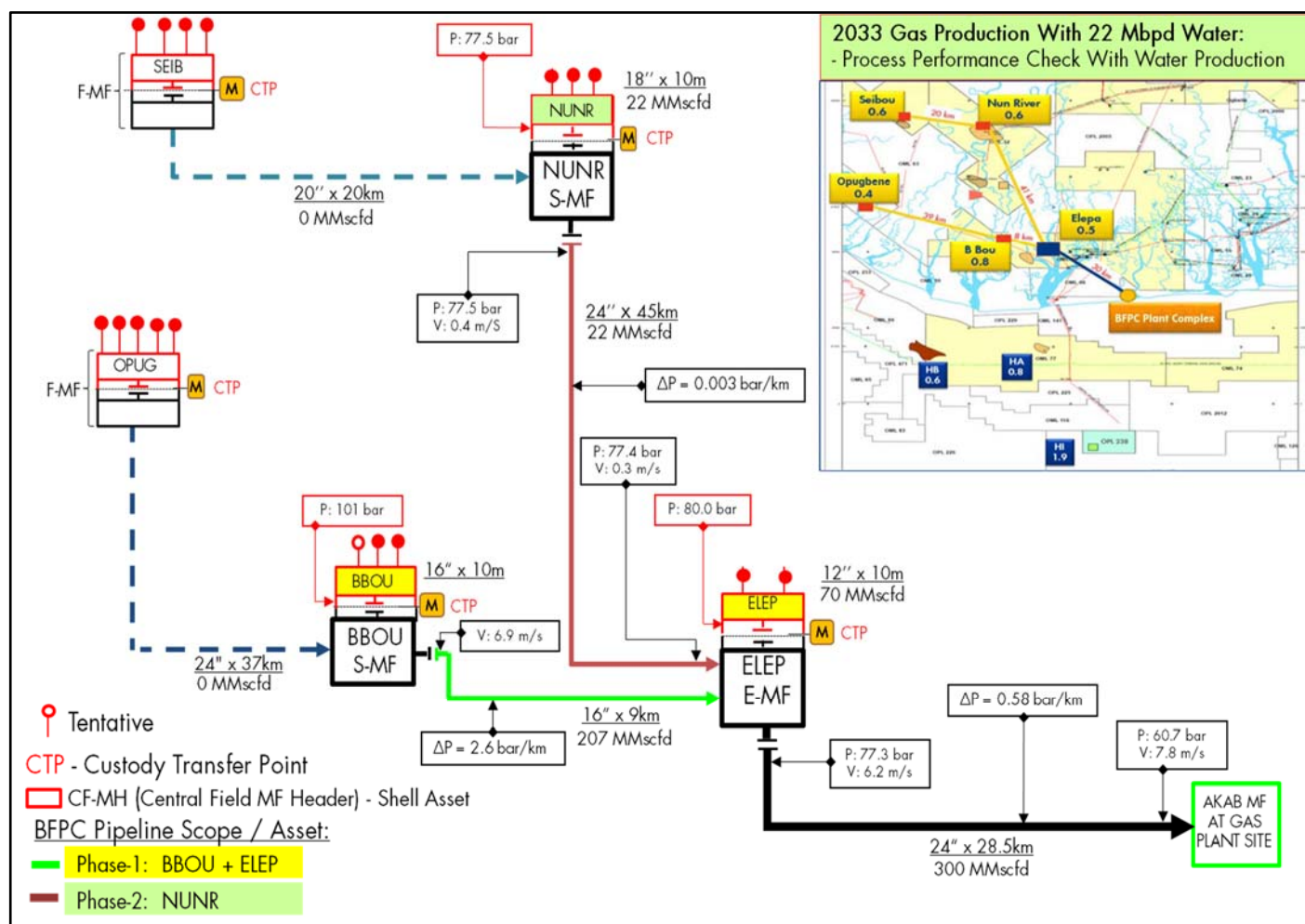


Figure 9.2: 2033 Gas Production; Flow Schematics

9.3. Calculation Notes

i. BBOU to ELEP E-MF B/L

Table 9.1: 2033 Production; BBOU to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from BB to ELEP E-MF																				
Line Dimension 16" SCH 80 (ID)		in	mm	m																
		14.33	364.00	0.36																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Aqueous Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	101.00	63.56	237,750	2,599	43.32	0.72	91.47	173,381	2,534	42.23	0.70	68.43	0.02	0.88	64,368	65.60	1.09	0.018	981.17	0.44
Outlet	77.28	58.57	237,750	3,373	56.22	0.94	70.49	173,368	3,307	55.12	0.92	52.42	0.01	0.90	64,382	65.42	1.09	0.018	984.20	0.47
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{overall}	ρ _G * V _G ²	ρ _L * V _L ²	X _{LM} = (m _L /m _G)√(ρ _L /ρ _G)	V _e = c/(√ρ _G)	V _e = c/(√ρ _L)							
Inlet	0.10	6.93	6.76	0.18	48.09	45.72	0.031	4,399	3,129	30.06	0.0980	15.16	3.14							
Outlet	0.10	9.00	8.82	0.17	81.00	77.86	0.030	5,710	4,081	29.99	0.0857	17.27	3.57							

ii. ELEP F-MF to ELEP E-MF B/L

Table 9.2: 2033 Production; ELEP F-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Flowline from ELEP to ELEP E-MF																				
Line Dimension 12" SCH 80 (ID)	in	mm	m																	
	11.38	289.00	0.29																	
Erosional Vel. Const.	C = 145																			
Min. Erosional Vel. Const.	C = 30																			
Stream Conditions	Overall Phase						Gas Phase							Liquid Phase						
	Pressure	Temp	Mass Flow	Volume Flow		Density	Mass Flow	Volume Flow		Density	Viscosity	Z factor	Mass Flow	Volume Flow		Density	Viscosity			
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	80.00	113.42	176,192	1,357	22.62	0.38	129.82	67,233	1,232	20.53	0.34	54.58	0.02	0.92	26,436	37.52	0.63	0.010	704.53	0.32
Outlet	79.99	113.41	176,192	1,357	22.62	0.38	129.80	67,233	1,232	20.53	0.34	54.57	0.02	0.92	26,437	37.52	0.63	0.010	704.54	0.32
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π/4)	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V _G ² _{overall}	ρ _G * V _G ²	ρ _L * V _L ²	X _{LM} = m _L /m _G * √(ρ _L /ρ _G)	Ve = c/(√ρ _G)	Ve = c/(√ρ _L)							
Inlet	0.07	5.74	5.21	0.16	32.99	27.19	0.025	4,283	1,484	17.77	0.1094	12.73	2.63							
Outlet	0.07	5.74	5.21	0.16	32.99	27.19	0.025	4,282	1,484	17.77	0.1094	12.73	2.63							

iii. NUNR F-MF to NUNR S-MF B/L

Table 9.3: 2033 Production; NUNR F-MF to NUNR S-MF B/L Calculation Note

Line Description																				
Flowline from NUNR to NUNR S-MF																				
Line Dimension 18" SCH 80 (ID)		in	mm	m																
		16.14	410.00	0.41																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	77.50	81.01	36,956	333	5.55	0.09	111.00	19,388	308	5.13	0.09	63.02	0.02	0.87	17,568	25.26	0.42	0.007	695.59	0.35
Outlet	77.50	81.00	36,956	333	5.55	0.09	111.01	19,387	308	5.13	0.09	63.02	0.02	0.87	17,569	25.26	0.42	0.007	695.59	0.35
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² overall	ρ _G * V _G ²	ρ _L * V _L ²	X = m ₁ /m ₂ * (ρ ₂ /ρ ₁) ^{0.5}	V _e = c/(Vρ _{ph})	Ve = c/(Vρ _{ph})							
Inlet	0.13	0.70	0.65	0.05	0.49	0.42	0.003	54	26	1.96	0.2727	13.76	2.85							
Outlet	0.13	0.70	0.65	0.05	0.49	0.42	0.003	54	26	1.96	0.2728	13.76	2.85							

iv. NUNR S-MF to ELEP E-MF B/L

Table 9.4: 2033 Production; NUNR S-MF to ELEP E-MF B/L Calculation Note

Line Description																				
Trunk-line from NUNR S-MF to ELEP E-MF																				
Line Dimension 24" SCH 80 (ID)		in	mm	m																
		21.57	548.00	0.55																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase						Liquid Phase						
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	77.50	81.00	36,956	333	5.55	0.09	111.01	19,387	308	5.13	0.09	63.02	0.02	0.87	17,569	25.26	0.42	0.007	695.59	0.35
Outlet	77.38	47.80	36,956	281	4.68	0.08	131.73	17,726	253	4.22	0.07	70.02	0.01	0.82	19,147	27.32	0.46	0.008	700.84	0.40
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² overall	Vg²	Vl²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d² [π]/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρG * V² overall	ρG * Vg²	ρL * Vl²	X= m1/m2 * (ρ2/ρ1)½	Ve = c/(Vρn)	Ve = c/(Vρn)							
Inlet	0.24	0.39	0.36	0.03	0.15	0.13	0.001	17	8	0.62	0.2728	13.76	2.85							
Outlet	0.24	0.33	0.30	0.03	0.11	0.09	0.001	14	6	0.72	0.3414	12.63	2.61							

v. ELEP E-MF to AKAB T/L

Table 9.5: 2033 Production; ELEP E-MF to AKAB T/L Calculation Note

Line Description																				
Trunk-line from ELEP E-MF to AKABEL																				
Line Dimension 24" SCH 80 (ID)		in	mm	m																
		21.57	548.00	0.55																
Erosional Vel. Const.		C = 145																		
Min. Erosional Vel. Const.		C = 30																		
Stream Conditions	Overall Phase							Gas Phase							Liquid Phase					
	Pressure	Temp	Mass Flow	Volume Flow			Density	Mass Flow	Volume Flow			Density	Viscosity	Z factor	Mass Flow	Volume Flow			Density	Viscosity
	Bar	°C	kg/h	m³/hr	m³/min	m³/s	kg/m³	kg/h	m³/hr	m³/min	m³/s	kg/m³	cP		kg/h	m³/hr	m³/min	m³/s	kg/m³	cP
Inlet	77.28	80.03	450,897	5,264	87.73	1.46	85.66	262,184	5,055	84.25	1.40	51.86	0.01	0.91	41,661	56.33	0.94	0.016	739.52	0.46
Outlet	60.67	75.49	450,897	6,653	110.88	1.85	67.78	261,703	6,445	107.42	1.79	40.61	0.01	0.92	42,172	56.19	0.94	0.016	750.53	0.51
Calc. Parameters																				
	X-sectional Area (m²)	Overall Fluid Vel. (m/s)	Gas Phase Vel. (m/s)	Liq. Phase Vel. (m/s)	V² _{overall}	V _G ²	V _L ²	Overall Phase Momentum (Pa)	Gas Phase Momentum (Pa)	Liq. Phase Momentum (Pa)	Lockhart Martinelli	Erosional Vel.	Min Vel.							
	A = d²(π)/4	V = (Q/A)	V = (Q/A)	V = (Q/A)				ρ _G * V² _{overall}	ρ _G * V _G ²	ρ _L * V _L ²	X = m ₁ /m ₂ * √(ρ ₁ /ρ ₂)	Ve = c/(√ρ _{pm})	Ve = c/(√ρ _{pm})							
Inlet	0.24	6.20	5.95	0.07	38.40	35.41	0.004	3,290	1,837	3.25	0.0421	15.67	3.24							
Outlet	0.24	7.83	7.59	0.07	61.34	57.57	0.004	4,158	2,338	3.28	0.0375	17.61	3.64							

9.4. Flow Parameters

i. BBOU to ELEP E-MF B/L

Table 9.6: 2033 Production; BBOU to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (°C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	101.0	63.6	0.7432	<empty>	Intermittent	0.1115	0.2851	<empty>	0.00	<empty>	0.000209	<empty>	163,339	12,852,216	0.23	8.89
1,800	0	96.7	62.7	0.7432	17.55	Intermittent	0.1088	0.2973	4.3735	0.00	0.00	0.000238	0.003502	161,251	12,995,009	0.23	9.28
3,600	0	92.2	61.8	0.7432	17.14	Intermittent	0.1060	0.3114	4.5808	0.00	0.00	0.000274	0.004031	159,045	13,146,212	0.23	9.72
5,400	0	87.5	60.8	0.7432	16.70	Intermittent	0.1029	0.3279	4.8228	0.00	0.00	0.000320	0.004711	156,700	13,307,080	0.23	10.24
7,200	0	82.6	59.7	0.7432	16.23	Intermittent	0.0996	0.3474	5.1102	0.00	0.00	0.000381	0.005611	154,192	13,479,220	0.23	10.86
9,000	0	77.3	58.6	0.7432	15.72	Intermittent	0.0959	0.3711	5.4589	0.00	0.00	0.000465	0.006846	151,486	13,664,758	0.23	11.60

ii. ELEP F-MF to ELEP E-MF B/L

Table 9.7: 2033 Production; ELEP F-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	80.0	113.4	0.4109	<empty>	Intermittent	0.2263	0.1801	<empty>	0.00	<empty>	0.000088	<empty>	351,536	5,153,399	0.53	5.22
2	0	80.0	113.4	0.4109	39.17	Intermittent	0.2263	0.1801	0.0029	0.00	0.00	0.000088	0.000001	351,533	5,153,434	0.53	5.22
4	0	80.0	113.4	0.4109	39.17	Intermittent	0.2263	0.1801	0.0029	0.00	0.00	0.000088	0.000001	351,531	5,153,469	0.53	5.22
6	0	80.0	113.4	0.4109	39.17	Intermittent	0.2263	0.1801	0.0029	0.00	0.00	0.000088	0.000001	351,529	5,153,505	0.53	5.22
8	0	80.0	113.4	0.4109	39.17	Intermittent	0.2263	0.1801	0.0029	0.00	0.00	0.000088	0.000001	351,526	5,153,541	0.53	5.22
10	0	80.0	113.4	0.4109	39.16	Intermittent	0.2263	0.1801	0.0029	0.00	0.00	0.000088	0.000001	351,524	5,153,576	0.53	5.22

iii. NUNR F-MF to NUNR S-MF B/L

Table 9.8: 2033 Production; NUNR F-MF to NUNR S-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	77.5	81.0	0.8466	<empty>	Segregated	0.3370	0.0015	<empty>	0.00	<empty>	0.000000	<empty>	43,635	1,101,083	0.05	0.65
2	0	77.5	81.0	0.8466	28.81	Segregated	0.3370	0.0015	0.0000	0.00	0.00	0.000000	0.000000	43,634	1,101,082	0.05	0.65
4	0	77.5	81.0	0.8466	28.81	Segregated	0.3370	0.0015	0.0000	0.00	0.00	0.000000	0.000000	43,634	1,101,081	0.05	0.65
6	0	77.5	81.0	0.8466	28.81	Segregated	0.3370	0.0015	0.0000	0.00	0.00	0.000000	0.000000	43,634	1,101,079	0.05	0.65
8	0	77.5	81.0	0.8466	28.81	Segregated	0.3370	0.0015	0.0000	0.00	0.00	0.000000	0.000000	43,634	1,101,078	0.05	0.65
10	0	77.5	81.0	0.8466	28.81	Segregated	0.3370	0.0015	0.0000	0.00	0.00	0.000000	0.000000	43,633	1,101,077	0.05	0.65

iv. NUNR S-MF to ELEP E-MF B/L

Table 9.9: 2033 Production; NUNR S-MF to ELEP E-MF B/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	77.5	81.0	0.8466	<empty>	Segregated	0.3823	0.0003	<empty>	0.00	<empty>	0.000000	<empty>	32,629	823,382	0.03	0.36
9,000	0	77.5	71.6	0.8374	30.53	Segregated	0.3975	0.0003	0.0246	0.00	0.00	0.000000	0.000000	31,756	817,482	0.03	0.35
18,000	0	77.5	64.1	0.8286	25.52	Segregated	0.4113	0.0003	0.0239	0.00	0.00	0.000000	0.000000	31,145	810,899	0.03	0.33
27,000	0	77.4	57.7	0.8208	21.38	Segregated	0.4238	0.0003	0.0232	0.00	0.00	0.000000	0.000000	30,711	804,662	0.03	0.32
36,000	0	77.4	52.3	0.8139	17.87	Segregated	0.4351	0.0003	0.0226	0.00	0.00	0.000000	0.000000	30,850	798,832	0.03	0.31
45,000	0	77.4	47.8	0.8078	14.91	Segregated	0.4451	0.0003	0.0221	0.00	0.00	0.000000	0.000000	30,822	793,474	0.03	0.30

v. ELEP E-MF to AKAB T/L

Table 9.10: 2033 Production; ELEP E-MF to AKAB T/L Flow Parameter

Length (m)	Elevation (m)	Pressure (bar)	Temperature (C)	Vap. Frac.	Heat Transferred (Btu/hr-ft)	Flow Regime	Liquid Holdup (with Slip)	Friction Gradient (in H ₂ O/ft)	Friction dP (bar)	Static Gradient (in H ₂ O/ft)	Static dP (bar)	Accel. Gradient (in H ₂ O/ft)	Accel. dP (bar)	Liquid Re	Vapour Re	Liquid Velocity (m/s)	Vapour Velocity (m/s)
0	0	77.3	80.0	0.6330	<empty>	Transition	0.1575	0.0638	<empty>	0.00	<empty>	0.000026	<empty>	231,514	11,444,304	0.25	5.96
5,700	0	74.3	79.2	0.6331	32.65	Transition	0.1535	0.0663	3.0881	0.00	0.00	0.000029	0.001372	229,067	11,528,168	0.25	6.20
11,400	0	71.1	78.3	0.6331	32.14	Transition	0.1493	0.0692	3.2223	0.00	0.00	0.000034	0.001561	226,547	11,616,531	0.25	6.48
17,100	0	67.8	77.4	0.6333	31.61	Transition	0.1447	0.0725	3.3769	0.00	0.00	0.000039	0.001800	223,939	11,710,027	0.25	6.79
22,800	0	64.3	76.5	0.6334	31.06	Transition	0.1398	0.0764	3.5577	0.00	0.00	0.000045	0.002107	221,226	11,809,466	0.25	7.16
28,500	0	60.7	75.5	0.6336	30.48	Transition	0.1345	0.0810	3.7727	0.00	0.00	0.000054	0.002514	218,382	11,915,911	0.24	7.60

9.5. Pressure Profile Plots

i. BBOU to ELEP E-MF B/L

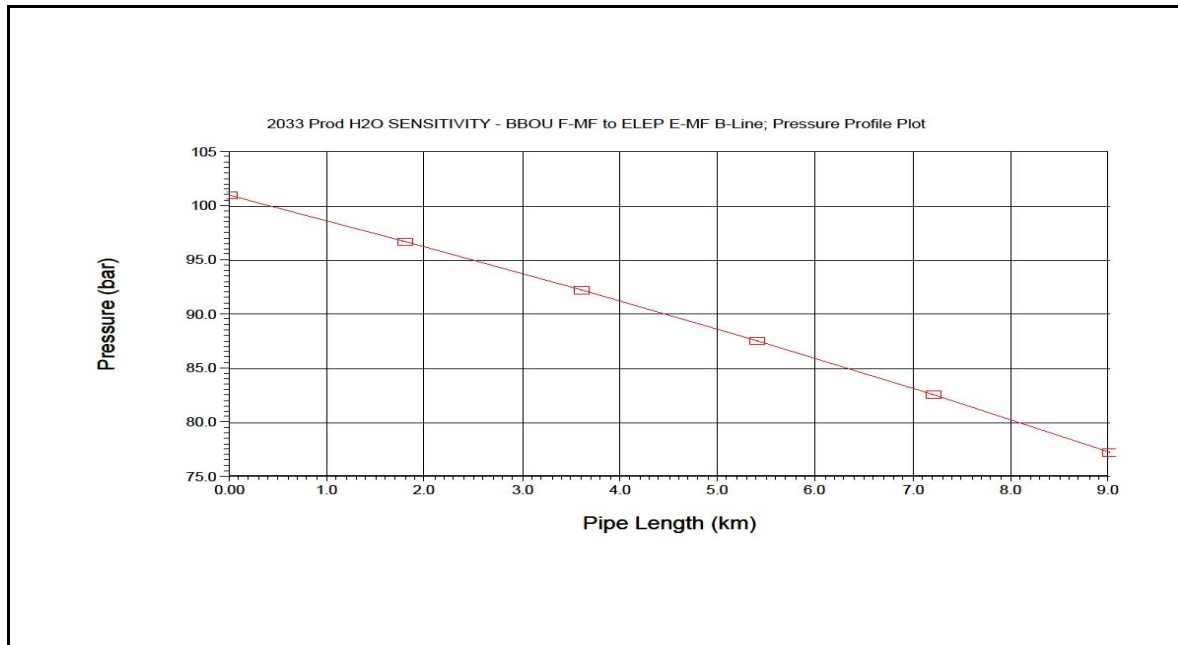


Figure 9.3: 2033 Production; BBOU to ELEP E-MF B/L Pressure Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

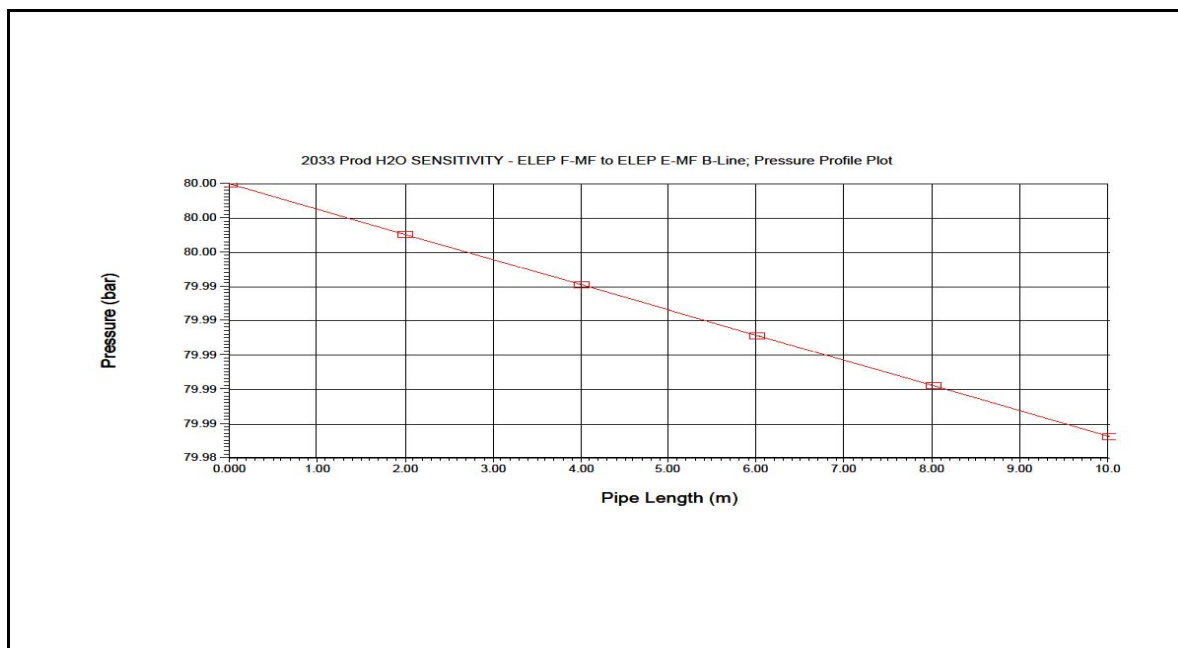


Figure 9.4: 2033 Production; ELEP F-MF to ELEP E-MF B/L Pressure Profile Plot

iii. NUNR F-MF to NUNR S-MF B/L

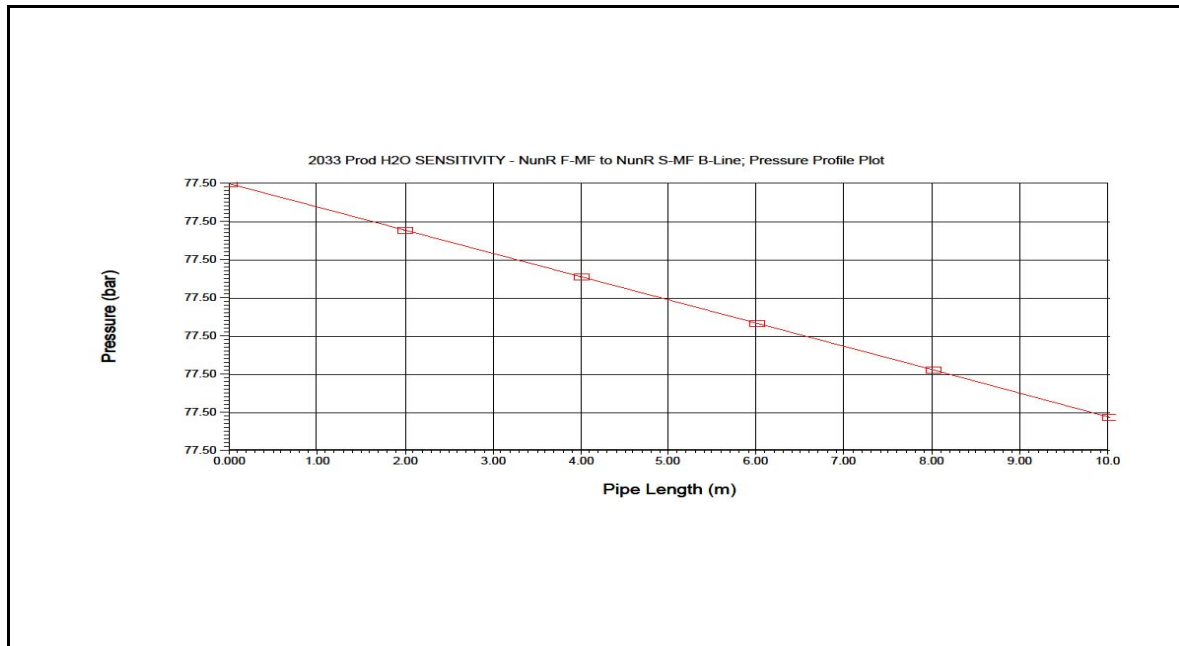


Figure 9.5: 2033 Production; NUNR F-MF to NUNR S-MF B/L Pressure Profile Plot

iv. NUNR S-MF to ELEP E-MF B/L

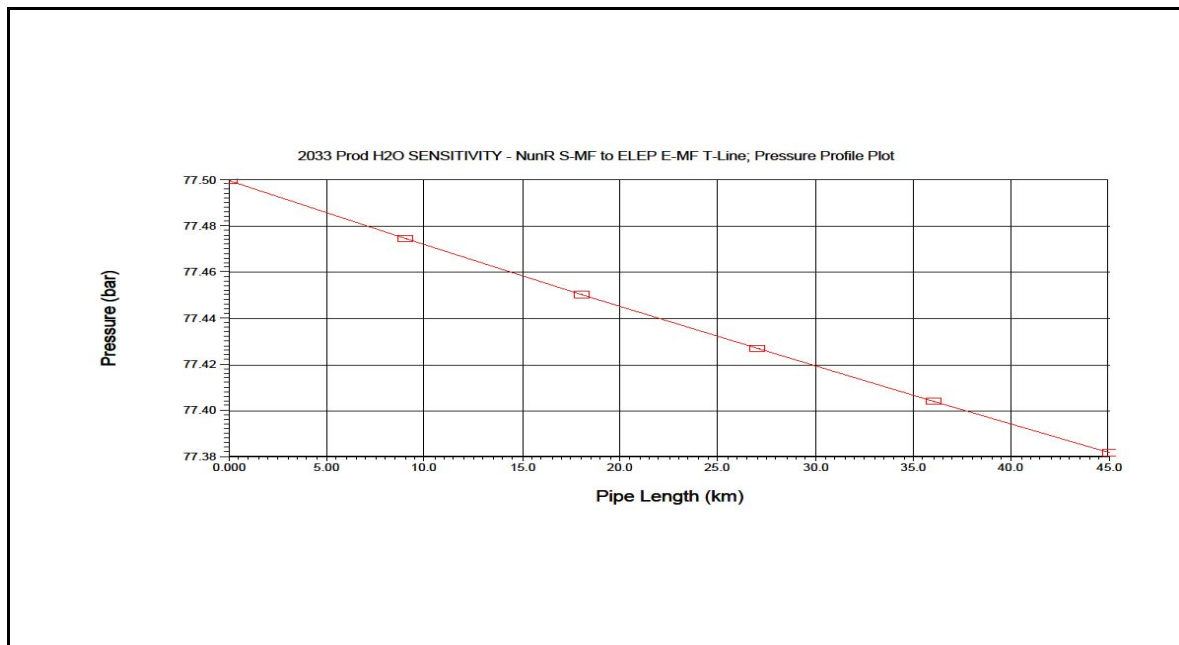


Figure 9.6: 2033 Production; NUNR S-MF to ELEP E-MF B/L Pressure Profile Plot

v. ELEP E-MF to AKAB T/L

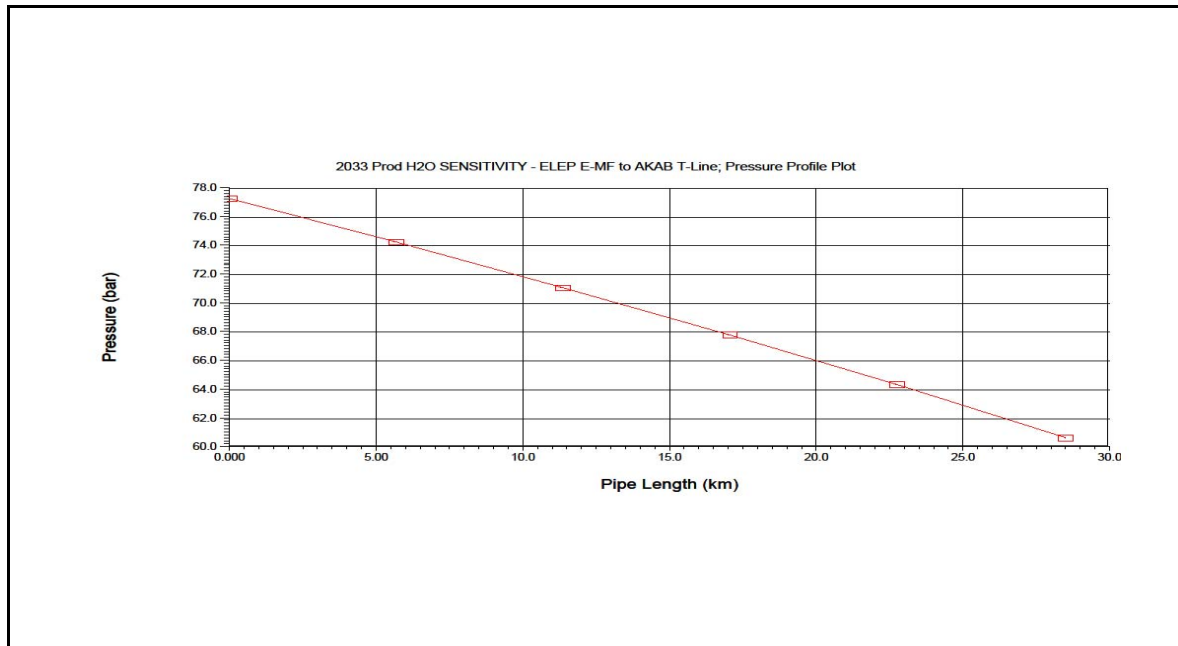


Figure 9.7: 2033 Production; ELEP E-MF to AKAB T/L Pressure Profile Plot

9.6. Velocity Profile Plot

i. BBOU to ELEP E-MF B/L

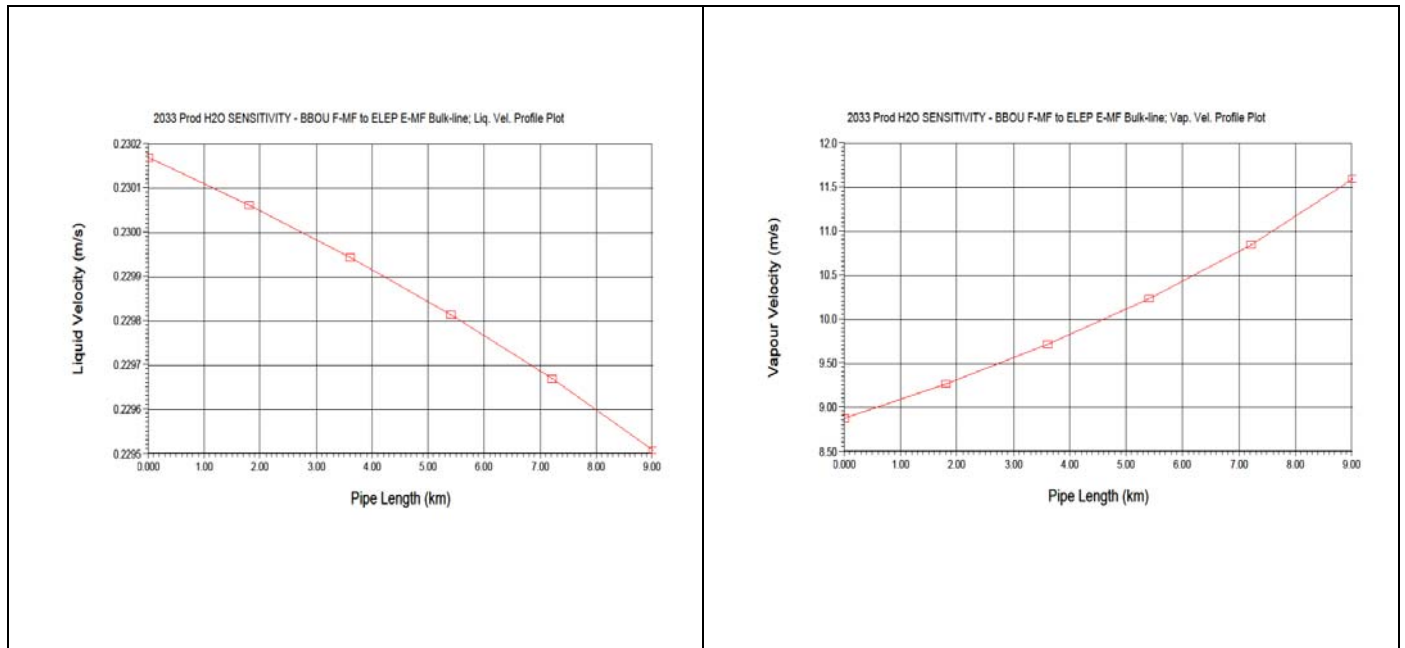


Figure 9.8: 2033 Production; BBOU to ELEP E-MF B/L Velocity Profile Plot

ii. ELEP F-MF to ELEP E-MF B/L

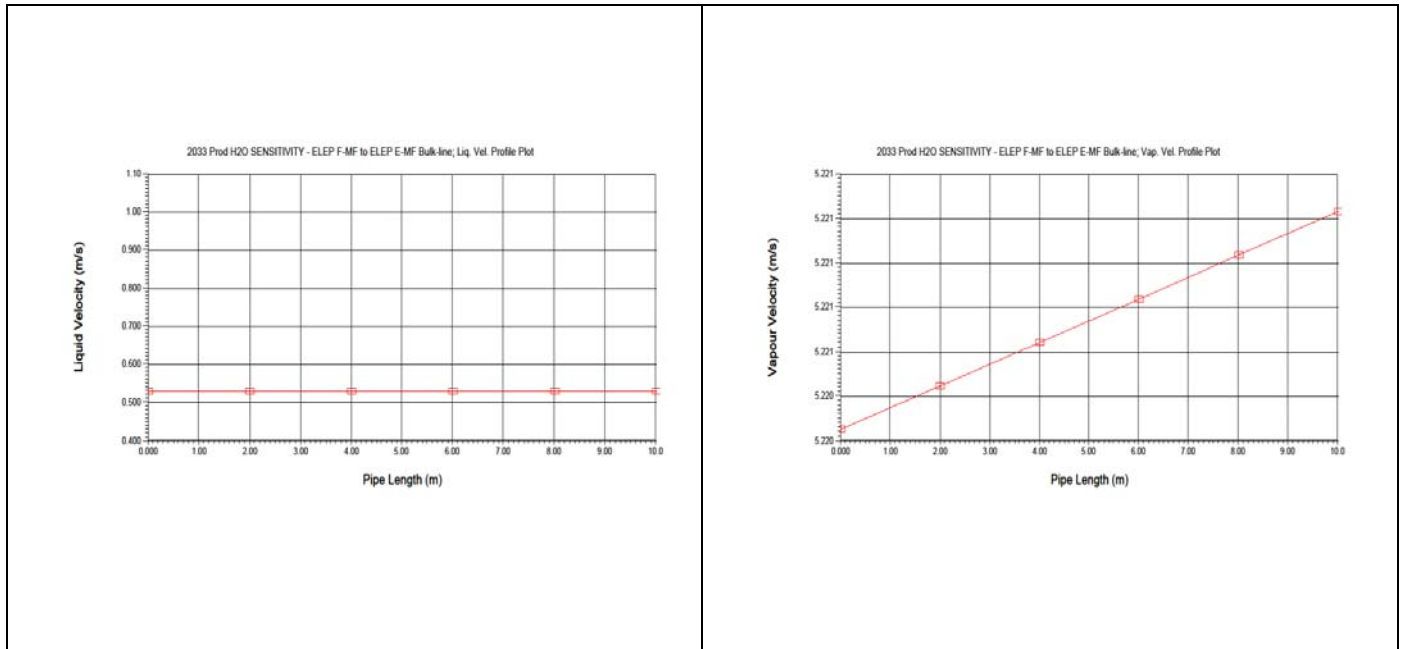


Figure 9.9: 2033 Production; ELEP F-MF to ELEP E-MF B/L Velocity Profile Plot

iii. NUNR F-MF to NUNR S-MF B/L

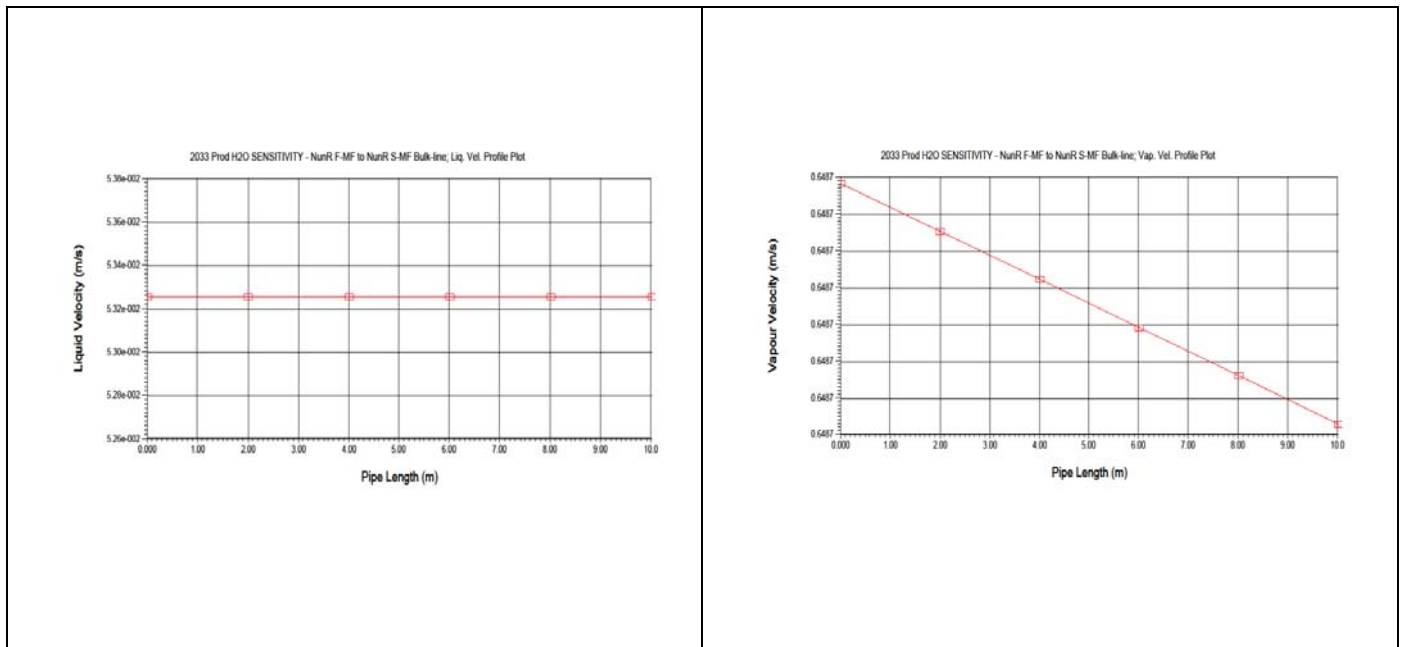


Figure 9.10: 2033 Production; NUNR F-MF to NUNR S-MF B/L Velocity Profile Plot

iv. NUNR S-MF to ELEP E-MF B/L

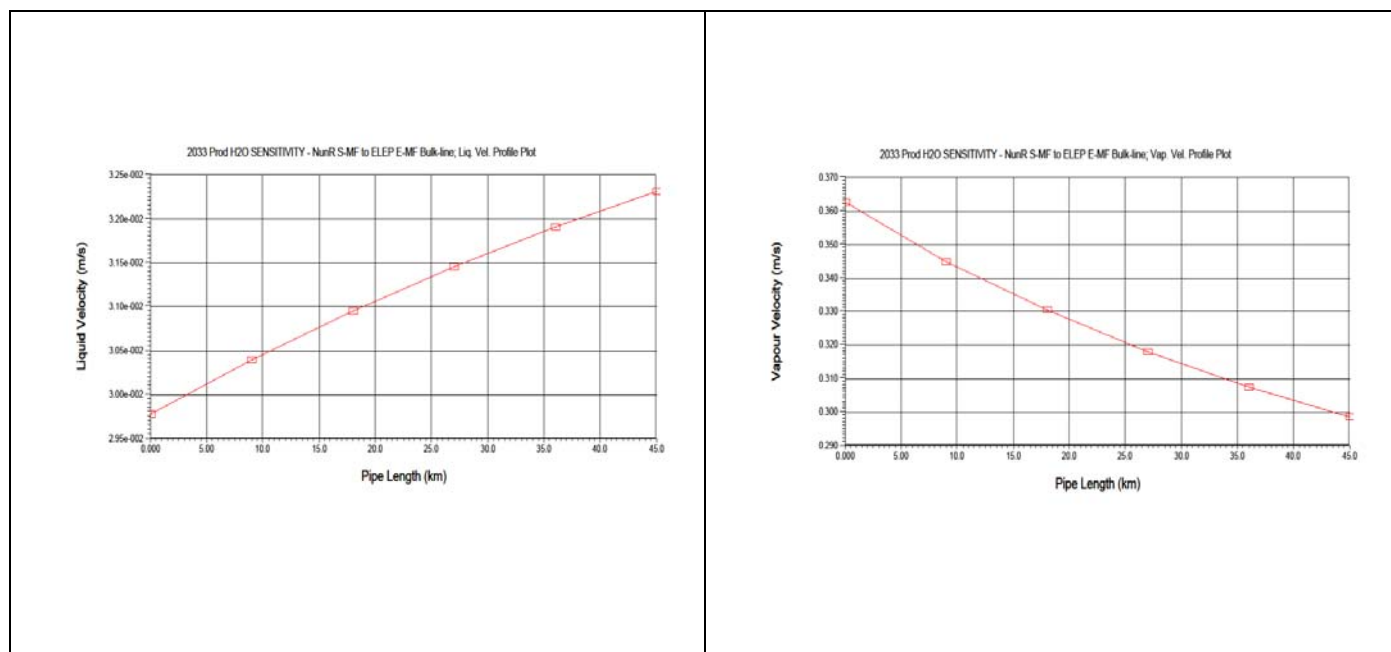


Figure 9.11: 2033 Production; NUNR S-MF to ELEP E-MF B/L Velocity Profile Plot

v. ELEP E-MF to AKAB T/L

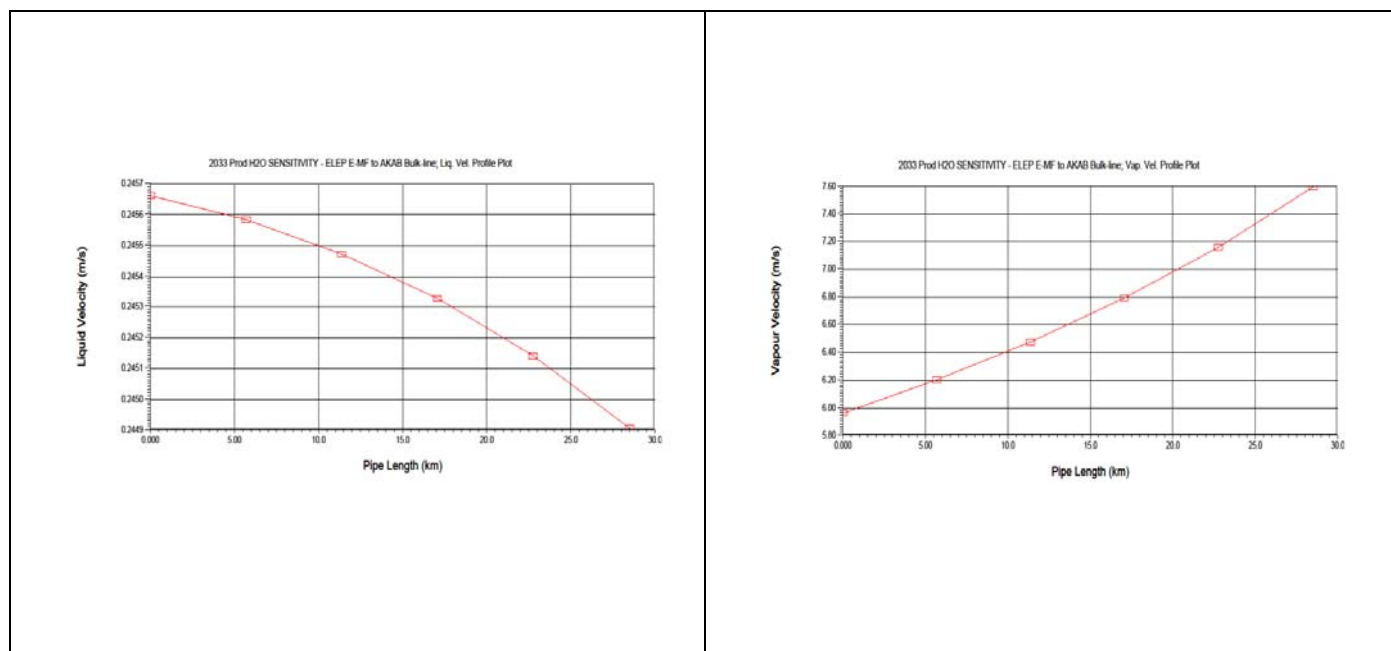


Figure 9.12: 2033 Production; ELEP E-MF to AKAB T/L Velocity Profile Plot

9.7. Phase Envelopes/Hydrate Curves

i. BBOU to ELEP E-MF B/L

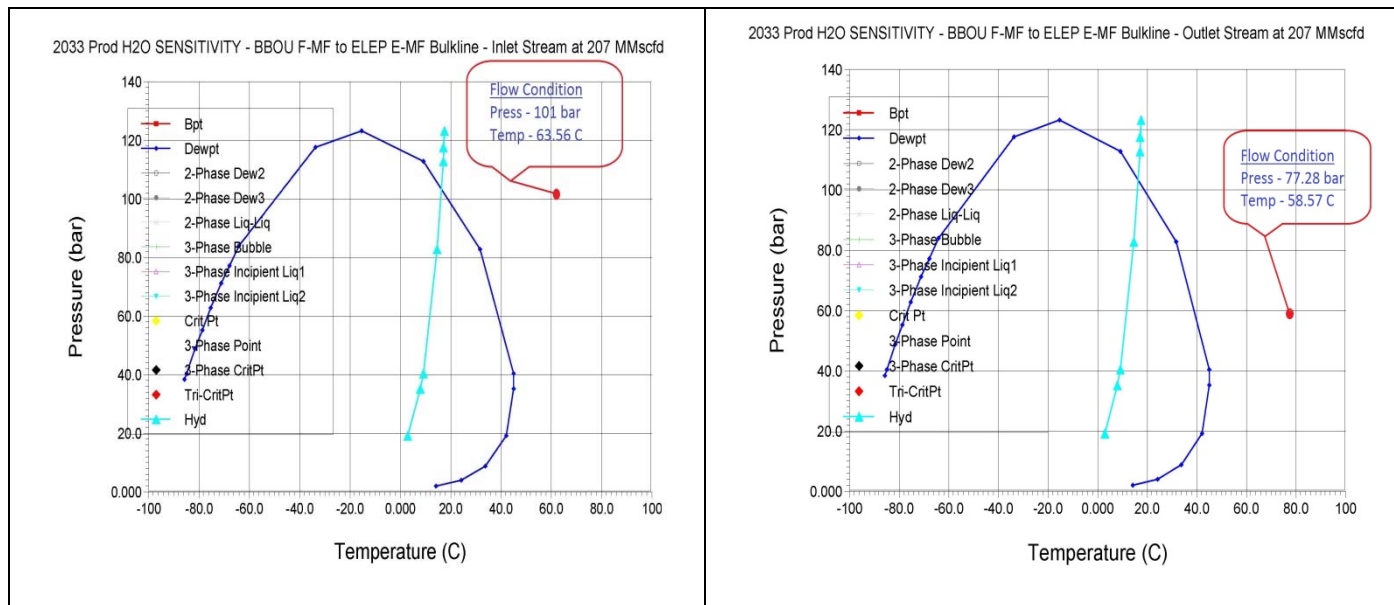


Figure 9.13: 2033 Production; BBOU to ELEP E-MF B/L Phase Envelope/Hydrate Curves

ii. ELEP F-MF to ELEP E-MF B/L

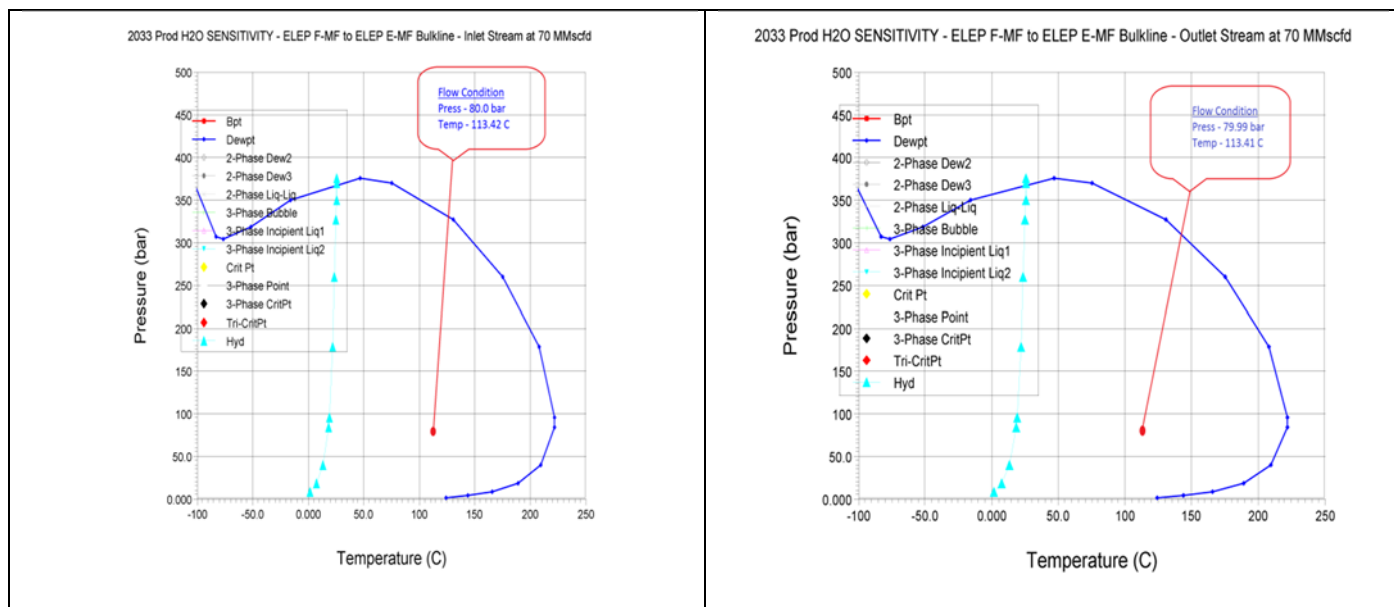


Figure 9.14: 2033 Production; ELEP F-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curves

iii. NUNR F-MF to NUNR S-MF B/L

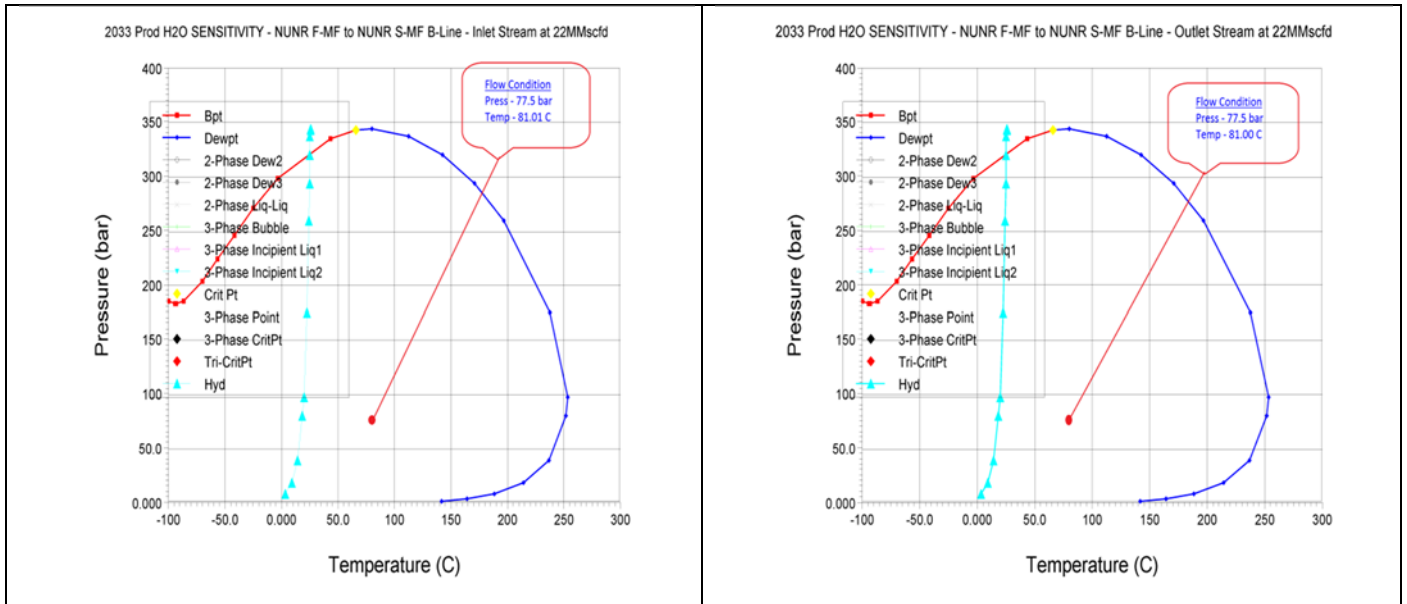


Figure 9.15: 2033 Production; NUNR F-MF to NUNR S-MF B/L Phase Envelope/Hydrate Curves

iv. NUNR S-MF to ELEP E-MF B/L

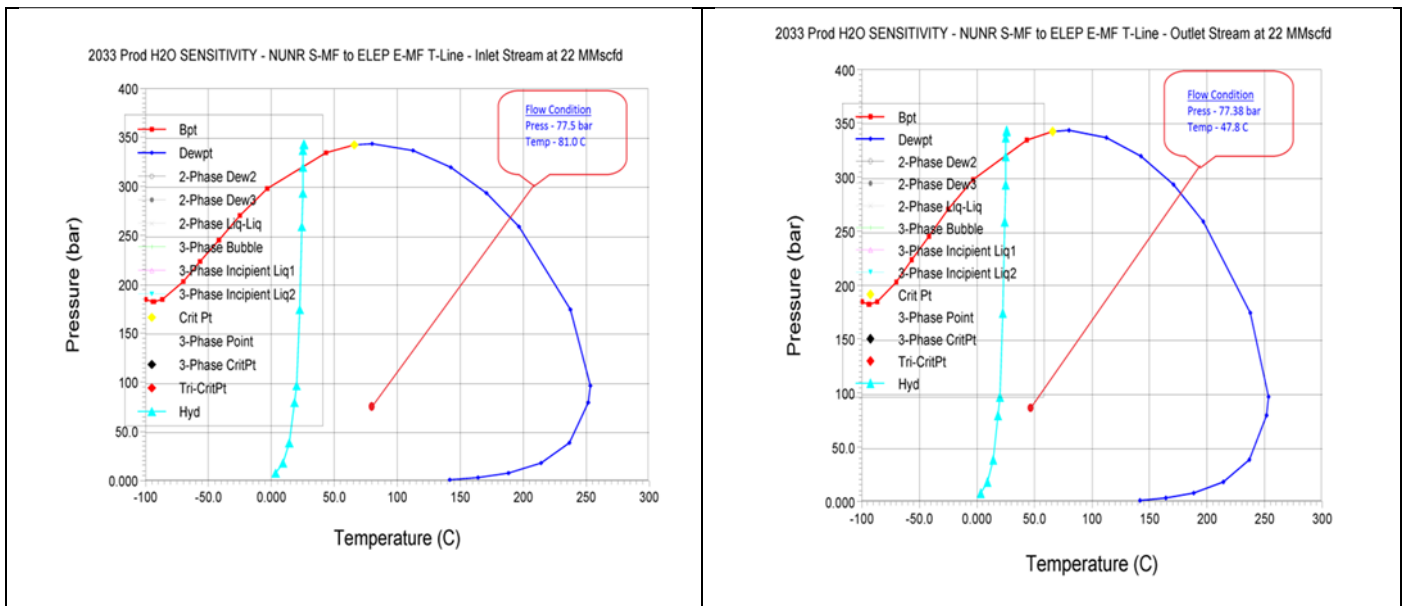


Figure 9.16: 2033 Production; NUNR S-MF to ELEP E-MF B/L Phase Envelope/Hydrate Curves

v. ELEP E-MF to AKAB T/L

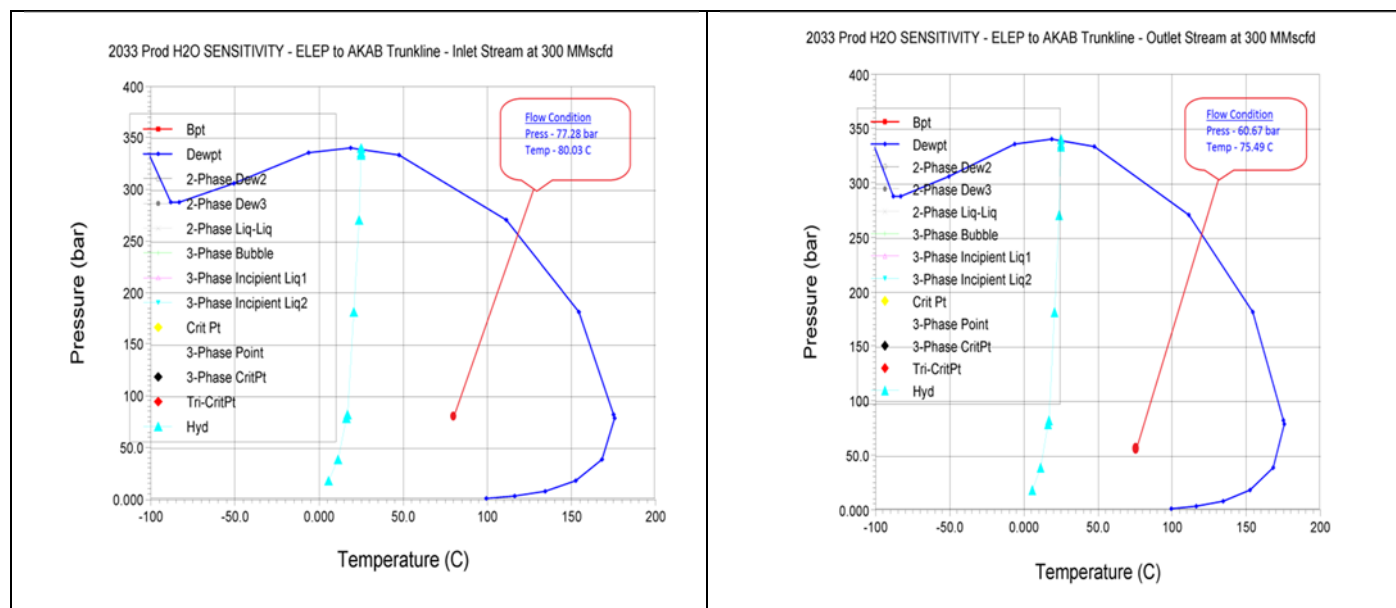


Figure 9.17: 2033 Production; ELEP E-MF to AKAB T/L Phase Envelope/Hydrate Curves

10. Conclusion

Pressure deliverability:

All scenarios studied gives satisfactory results that guarantees flow assurance and gas deliverability at required 60 – 65 barg pressure required at Akabel for gas plant operations.

Pipeline Velocity:

Notably, at the declined stage of the fields when low volume flow sets-in, the prevailing fluid velocity is below minimum velocity required to sweep the pipe clean.

Line sizes:

All the line were sized to flow at velocities below erosional velocity and also, to achieve a minimal pressure losses. This criterion guarantees that, the pipeline wall will not be eroded and also ensures adequate pressure deliverability.

Hydrate formation:

For the entire pipeline examined; at the flow conditions (pressure & temperature regimes) no indication that hydrates will be formed on the lines.

Appendix

Appendix A: Phase - 1 Scenario - 1 Schematics & Parameters

S/N	Deliverable Description	File	
A1	Hysys PFD	 A1 - Hysys PFD .pdf	
A2	Flow Schematic	 A2 - Flow Schematics	
A3	Calculation Note	 A3 - Calculation Note.pdf	
A4	Pipe Profile Data	 A4 - Pipe Profile Data.pdf	
A5	BBOU to ELEM E-MF Phase Envelope	 A5 - BBOU TO ELEM, Phase Envelope - Inle	 A5 - BBOU TO ELEM, Phase Envelope - Out
A6	ELEM E-MF to AKAB Phase Envelope	 A6 - ELEM TO AKAB , Phase Envelope - Inle	 A6 - ELEM TO AKAB , Phase Envelope - Out
A7	BBOU to ELEM Stream Datasheets	 A7 - BBOU TO ELEM Stream Datasheets -	 A7 - BBOU TO ELEM Stream Datasheets -
A8	ELEM E-MF to AKAB Stream Datasheets	 A8 - ELEM TO AKAB Stream Datasheets -	 A8 - ELEM TO AKAB Stream Datasheets -

Appendix B: Phase 1 Scenario 2 Schematics & Parameters

S/N	Deliverable Description	File	
B1	Hysys PFD	 B1 - Hysys PFD.pdf	
B2	Flow Schematic	 B2 - Flow Schematics	
B3	Calculation Note	 B3 - Calculation Notes	
B4	Pipe Profile Data	 B4 - Pipe Profile Data	
B5	BBOU to ELEP E-MF Phase Envelope	 B5 - BBOU TO ELEP, Phase Envelope - Inle	 B5 - BBOU TO ELEP, Phase Envelope - Out
B6	ELEP F-MF to ELEP E-MF Phase Envelope	 B6 - ELEP F-MF TO ELEP, Phase Envelope	 B6 - ELEP F-MF TO ELEP, Phase Envelope
B7	ELEP E-MF to AKAB Phase Envelope	 B7 - ELEP TO AKAB, Phase Envelope - Inle	 B7 - ELEP TO AKAB, Phase Envelope - Out
B8	BBOU to ELEP E-MF Stream Datasheets	 B8 - BBOU TO ELEP Stream Datasheets -	 B8 - BBOU TO ELEP Stream Datasheets -
B9	ELEP F-MF to ELEP E-MF Stream Datasheets	 B9 - ELEP F-MF TO ELEP Stream Datashe	 B9 - ELEP F-MF TO ELEP Stream Datashe
B10	ELEP E-MF to AKAB Stream Datasheets	 B10 - ELEP TO AKAB Stream Datasheets -	 B10 - ELEP TO AKAB Stream Datasheets -

Appendix C: Phase 1 Scenario 3 Schematics & Parameters

S/N	Deliverable Description	File	
C1	Hysys PFD	 C1 - Hysys PFD.pdf	
C2	Flow Schematic	 C2 - Flow Schematics	
C3	Calculation Note	 C3 - Calculation Note.pdf	
C4	Pipe Profile Data	 C4 - Pipe Profile Data.pdf	
C5	BBOU to ELEM E-MF Phase Envelope	 C5 - BBOU TO ELEM, Phase Envelope - Inle	 C5 - BBOU TO ELEM, Phase Envelope - Out
C6	ELEM F-MF to ELEM E-MF Phase Envelope	 C6 - ELEM F-MF TO ELEM, Phase Envelope	 C6 - ELEM F-MF TO ELEM, Phase Envelope
C7	ELEM E-MF to AKAB Phase Envelope	 C7 - ELEM TO AKAB, Phase Envelope - Inle	 C7 - ELEM TO AKAB, Phase Envelope - Out
C8	BBOU to ELEM E-MF Stream Datasheets	 C8 - BBOU TO ELEM Stream Datasheets -	 C8 - BBOU TO ELEM Stream Datasheets -
C9	ELEM F-MF to ELEM E-MF Stream Datasheets	 C9 - ELEM F-MF to ELEM Stream Datasht	 C9 - ELEM F-MF to ELEM Stream Datasht
C10	ELEM E-MF to AKAB Stream Datasheets	 C10 - ELEM to AKAB Stream Datasht	 C10 - ELEM TO AKAB Stream Datasheets -

Appendix D: Phase 2 Schematics & Parameters

S/N	Deliverable Description	File	
D1	Hysys PFD	 D1 - Hysys PFD.pdf	
D2	Flow Schematic	 D2: Flow Schematics	
D3	Calculation Note	 D3 - Calculation Notes.pdf	
D4	Pipe Profile Data	 D4 - Pipe Profile Data.pdf	
D5	NUNR F-MF to NUNR S-MF Phase Envelope	 D5 - NUNR F-MF to NUNR S-MF, Phase En	 D5 - NUNR F-MF to NUNR S-MF, Phase En
D6	NUNR S-MF to ELEP E-MF Phase Envelope	 D6 - NUNR S-MF to ELEP E-MF, Phase En	 D6 - NUNR S-MF to ELEP E-MF, Phase En
D7	ELEP E-MF to AKAB Phase Envelope	 D7 - ELEP E-MF to AKAB, Phase Envelop	 D7 - ELEP E-MF to AKAB, Phase Envelop
D8	NUNR F-MF to NUNR S-MF Stream Datasheets	 D8 - NUNR F-MF to NUNR S-MF, Stream I	 D8 - NUNR F-MF to NUNR S-MF, Stream I
D9	NUNR S-MF to ELEP E-MF Stream Datasheets	 D9 - NUNR S-MF to ELEP E-MF, Stream Di	 D9 - NUNR S-MF to ELEP E-MF, Stream Di
D10	ELEP E-MF to AKAB Stream Datasheets	 D10 - ELEP E-MF to AKAB, Stream Datasht	 D10 - ELEP E-MF to AKAB, Stream Datasht

Appendix E: Phase 3 Schematics & Parameters

S/N	Deliverable Description	File	
E1	Hysys PFD	 E1 - Hysys PFD.pdf	
E2	Flow Schematic	 E2 - Flow Schematics	
E3	Calculation Note	 E3 - Calculation Notes	
E4	Pipe Profile Data	 E4 - Pipe Profile Data.pdf	
E5	SEIB F-MF to NUNR S-MF Phase Envelope	 E5 - SEIBU to NUNR Bulkline, Phase Envel	 E5 - SEIBU to NUNR Bulkline, Phase Envel
E6	NUNR F-MF to NUNR S-MF Phase Envelope	 E6 - NUNR F-MF to NUNR S-MF Bulkline, f	 E6 - NUNR F-MF to NUNR S-MF Bulkline, f
E7	NUNR S-MF to ELEP E-MF Phase Envelope	 E7 - NUNR to ELEP Trunkline, Phase Env	 E7 - NUNR to ELEP Trunkline, Phase Env
E8	ELEP E-MF to AKAB Phase Envelope	 E8 - ELEP to AKAB Trunkline, Phase Env	 E8 - ELEP to AKAB Trunkline, Phase Env
E9	SEIB F-MF to NUNR S-MF Stream Datasheets	 E9 - SEIB F-MF to NUNR S-MF. Datashe	 E9 - SEIB F-MF to NUNR S-MF. Datashe
E10	NUNR F-MF to NUNR S-MF Stream Datasheets	 E10 - NUNR F-MF to NUNR S-MF. Datashe	 E10 - NUNR F-MF to NUNR S-MF. Datashe
E11	NUNR S-MF to ELEP E-MF Stream Datasheets	 E11 - NUNR S-MF to ELEP E-MF. Datashee	 E11 - NUNR S-MF to ELEP E-MF. Datashee
E12	ELEP E-MF to AKAB Stream Datasheets	 E12 - ELEP E-MF to AKAB. Datasheets - I	 E12 - ELEP E-MF to AKAB. Datasheets - C

Appendix F: Phase 4 Schematics & Parameters

S/N	Deliverable Description	File	
F1	Hysys PFD	 F1 - Hysys PFD.pdf	
F2	Flow Schematic	 F2 - Flow Schematics	
F3	Calculation Note	 F3 - Calculation Notes.pdf	
F4	Pipe Profile Data	 F4 - Pipe Profile Data.pdf	
F5	OPUG F-MF to BBOU F-MF Phase Envelope	 F5 - OPUG F-MF to BBOU F-MF. Phase Envelope	 F5 - OPUG F-MF to BBOU F-MF. Phase Envelope
F6	BBOU F-MF to ELEP E-MF Phase Envelope	 F6 - BBOU F-MF to ELEP E-MF. Phase Envelope	 F6 - BBOU F-MF to ELEP E-MF Phase Envelope
F7	SEIB F-MF to NUNR S-MF Phase Envelope	 F7 - SEIB F-MF to NUNR S-MF. Phase Envelope	 F7 - SEIB F-MF to NUNR S-MF. Phase Envelope
F8	NUNR F-MF to NUNR S-MF Phase Envelope	 F8 - NUNR F-MF to NUNR S-MF. Phase Envelope	 F8 - NUNR F-MF to NUNR S-MF. Phase Envelope
F9	NUNR S-MF to ELEP E-MF Phase Envelope	 F9 - NUNR S-MF to ELEP E-MF. Phase Envelope	 F9 - NUNR S-MF to ELEP E-MF. Phase Envelope
F10	ELEP E-MF to AKAB Phase Envelope	 F10 - ELEP E-MF to AKAB. Phase Envelope	 F10 - ELEP E-MF to AKAB. Phase Envelope
F11	OPUG F-MF to BBOU F-MF Stream Datasheets	 F11 - OPUG F-MF to BBOU F-MF. Stream Datasheets	 F11 - OPUG F-MF to BBOU F-MF. Stream Datasheets
F12	BBOU F-MF to ELEP E-MF Stream Datasheets	 F12 - BBOU F-MF to ELEP E-MF. Stream Datasheets	 F12 - BBOU F-MF to ELEP E-MF. Stream Datasheets

F13	SEIB F-MF to NUNR S-MF Stream Datasheets	 F13 - SEIB F-MF to NUNR S-MF. Stream I	 F13 - SEIB F-MF to NUNR S-MF. Stream I
F14	NUNR F-MF to NUNR S-MF Stream Datasheets	 F14 - NUNR F-MF to NUNR S-MF. Stream I	 F14 - NUNR F-MF to NUNR S-MF. Stream I
F15	NUNR S-MF to ELEM E-MF Stream Datasheets	 F15 - NUNR S-MF to ELEM E-MF. Stream D	 F15 - NUNR S-MF to ELEM E-MF. Stream D
F16	ELEM E-MF to AKAB Stream Datasheets	 F16 - ELEM E-MF to AKAB. Stream Datas	 F16 - ELEM E-MF to AKAB. Stream Datas

Appendix G: 2033 Production (Water Sensitivity Scenario) Schematics & Parameters

S/N	Deliverable Description	File	
G1	Hysys PFD	 G1 - Hysys PFD.pdf	
G2	Flow Schematic	 G2 - Flow Schematics	
G3	Calculation Note	 G3 - Calculation Notes.pdf	
G4	Pipe Profile Data	 G4 - Pipe Profile Data.pdf	
G5	BBOU to ELEP E-MF Phase Envelope	 G5 - BBOU F-MF to ELEP E-MF Phase Env	 G5 - BBOU F-MF to ELEP E-MF Phase Env
G6	ELEP F-MF to ELEP E-MF Phase Envelope	 G6 - ELEP F-MF to ELEP E-MF Phase Env	 G6 - ELEP F-MF to ELEP E-MF Phase Env
G7	NUNR F-MF to NUNR S-MF Phase Envelope	 G7 - NUNR F-MF to NUNR S-MF Phase Env	 G7 - NUNR F-MF to NUNR S-MF Phase Env
G8	NUNR S-MF to ELEP E-MF Phase Envelope	 G8 - NUNR SMF to ELEP EMF Phase Enve	 G8 - NUNR SMF to ELEP EMF Phase Enve
G9	ELEP E-MF to AKAB Phase Envelope	 G9 - ELEP to AKAB Phase Envelop	 G9 - ELEP to AKAB Phase Envelop
G10	BBOU to ELEP E-MF Stream Datasheets	 G10 - BBOU F-MF to ELEP E-MF Datasheet	 G10 - BBOU F-MF to ELEP E-MF Datasheet
G11	ELEP F-MF to ELEP E-MF Stream Datasheets	 G11 - ELEP F-MF to ELEP E-MF Datasheet	 G11 - ELEP F-MF to ELEP E-MF Datasheet
G12	NUNR F-MF to NUNR S-MF Stream Datasheets	 G12 - NUNR F-MF to NUNR S-MF Datashee	 G12 - NUNR F-MF to NUNR S-MF Datashee
G13	NUNR S-MF to ELEP E-MF Stream Datasheets	 G13 - NUNR S-MF to ELEP E-MF Datasheet	 G13 - NUNR S-MF to ELEP E-MF Datasheet
G14	ELEP E-MF to AKAB Stream Datasheets	 G14 - ELEP to AKAB Datasheet	 G14 - ELEP to AKAB Datasheet

Key References

1. Pipeline Route
2. Field's & Water Saturated Composition
3. Flow Schematics

Reference-1: Proposed Gas P/L & Alternate Routes Survey

	Document Ref	Document	Information
Ref 1a	EA SPDC-BFPC UGS-GEP-SVY-DWG-005-A01 Composite Plan for The Proposed Gas P/L Route From BBOU Thru ELEP to AKAB	 Proposed PL Route BBOU-ELEP-AKAB	Pipeline Route; BBOU Thru ELEP to AKAB

Reference-2: Prod. Forecast; Field's and Water Saturated Gas Composition

	Document Ref	Document	Information
Ref 2	Production Forecast; Field's and Water Saturated Gas Composition	 Prod Forecast Gas Comp	Production Forecast; Field's Compositions/Properties; and Water Saturated Composition

Reference-3: Flow Schematics

	Document Ref	Document	Information
Ref 3	EA/SPDC-BFPC UGS-GEP/BDM-PFS-001-A01 Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant	 WHGG Schematics & Flow Parameters	Overall Process Route- Path & Flow Parameters